Classical Field Theory

Parts I and II

Lecture notes

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About these notes

These notes are written to be read alone, one lecture at a time. Each lecture opens with its objectives and the outcomes it promises. It proves its results at the level shown by a tag beside each statement, works one example in full, and ends with self-checks. The answer to each self-check follows it.

A statement with no tag is proved in full. The tag *sketched* means that the central idea is given correctly and completely, and the routine details that are omitted are named. The tag *cited* means that the proof belongs elsewhere; the source and a precise statement are then named. The tag *imported* marks a result of a prerequisite course: *Linear Algebra for Physics*, *Theoretical Mechanics*, *Real and Complex Analysis* (Part I) or *Topology and Geometry* (Part I).

Part I

Free Fields and Maxwell

Part I asks what a classical field theory is, and answers three times: a local action whose stationary points are the solutions (Lectures 1 to 3), a Cauchy problem whose data on a time slice determine the field causally (Lectures 4 to 6), and a space of solutions with a symplectic form (Lectures 7 and 8). Gauge symmetry damages all three answers; Maxwell's theory repairs them (Lectures 9 to 13). Lecture 14 tests the three answers on the Dirac field.

Lecture 1

Local action, local equation

In *Theoretical Mechanics* a motion is a curve $q(t)$ that makes the action $\int L(q,\dot q)\,\mathrm{d}t$ stationary under variations with fixed endpoints. Now put an oscillator at every point x of space. Its displacement $\phi(t,x)$ depends on time and on the point. The coupling between neighbouring oscillators brings in the space derivatives of ϕ. Such a function of time and space is a field. Its action is an integral over a region of spacetime. This course asks what a classical field theory is. The first answer, developed in Lectures 1 to 3, is that a field theory is given by a local action.

Objectives. *We define fields, Lagrangian densities and actions on compact regions. We differentiate the action along a variation and keep the boundary term. We prove that stationary fields are exactly the solutions of the Euler–Lagrange equations.*

By the end of this lecture you will be able to:

- Compute the Euler–Lagrange expression of a first-order Lagrangian density.
- Derive the first-variation formula, with its boundary terms, on a region between two graphs.
- Prove that stationarity under compactly supported variations is equivalent to the Euler–Lagrange equations.
- Compute the boundary term for a variation that does not vanish on the boundary.

1.1 Fields and actions

Spacetime is $\mathbb{R}^{1,d}=\mathbb{R}\times\mathbb{R}^d$ with coordinates $x^0=t$ and $x=(x^1,\dots,x^d)$. Greek indices μ,ν run over $0,\dots,d$, spatial indices i,j over $1,\dots,d$, and component indices a,b over $1,\dots,N$. A repeated index is summed. The Minkowski metric is $\eta=\mathrm{diag}(1,-1,\dots,-1)$. Its inverse $\eta^{\mu\nu}$ has the same entries. We write $\partial^\mu=\eta^{\mu\nu}\partial_\nu$. The wave operator is

$$\Box=\partial_\mu\partial^\mu=\partial_t^2-\Delta,\qquad \Delta=\partial_1^2+\cdots+\partial_d^2.$$

A signal sent from the origin at the speed of light $c=1$ travels on the future light cone $|x|=t$.

We often replace space $\mathbb{R}^d$ by the torus $\mathbb{T}^d$, with $\mathbb{T}=\mathbb{R}/2\pi\mathbb{Z}$, so that space is compact. A function on $\mathbb{T}^d$ is a function on $\mathbb{R}^d$ that is 2π-periodic in every coordinate. Spacetime M is then $\mathbb{R}\times\mathbb{R}^d$ or $\mathbb{R}\times\mathbb{T}^d$. Both are oriented by the volume form

$$\mathrm{vol}=\mathrm{d}t\wedge\mathrm{d}x^1\wedge\cdots\wedge\mathrm{d}x^d,$$

and we write $\mathrm{d}^dx=\mathrm{d}x^1\cdots\mathrm{d}x^d$ for integrals over space.

Definition 1.1 (Field). A *field* with N components on M is a C^2 map

$$\phi\colon M\to\mathbb{R}^N,\qquad x\mapsto\big(\phi^1(x),\dots,\phi^N(x)\big).$$

A complex field is a field with two real components, its real and imaginary parts.

The Lagrangian of a field is a function of the point x, the value $\phi(x)$ and the first derivatives $\partial_\mu\phi(x)$. We write $u = (u^a)$ for a point of $\mathbb{R}^N$ and $v = (v^a_\mu)$ for a point of $\mathbb{R}^{N(d+1)}$, the place where the derivatives $\partial_\mu\phi^a$ will go.

Definition 1.2 (Lagrangian density)**.** A *Lagrangian density* is a C^2 function

$$L\colon M \times \mathbb{R}^N \times \mathbb{R}^{N(d+1)} \to \mathbb{R}, \qquad (x,u,v) \mapsto L(x,u,v).$$

For a field ϕ we write

$$L[\phi](x) = L\big(x, \phi(x), \partial\phi(x)\big),$$

where $\partial\phi(x)$ is the array of the derivatives $\partial_\mu\phi^a(x)$.

The density is *local*: its value at x depends only on ϕ and its first derivatives at x. The space of the arguments (x,u,v) is called the first jet space.

Definition 1.3 (Action on a region)**.** A *region* is a compact subset $D \subset M$ that is the closure of its interior. The *action* of a field ϕ on the region D is

$$S_D[\phi] = \int_D L[\phi] \operatorname{vol}.$$

The main example is the *slab*

$$D = [t_0, t_1] \times \mathbb{T}^d, \qquad t_0 < t_1,$$

which is compact because the torus is. An integral over $\mathbb{T}^d$ is taken over one period $[0, 2\pi]^d$. The boundary of the slab consists of the two *slices* $\{t_0\} \times \mathbb{T}^d$ and $\{t_1\} \times \mathbb{T}^d$.

For our running example take one component, $N = 1$, and the *Klein–Gordon density*

$$L = \frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi - \frac{1}{2}m^2\phi^2 = \frac{1}{2}\phi_t^2 - \frac{1}{2}|\nabla\phi|^2 - \frac{1}{2}m^2\phi^2, \tag{1.1}$$

with a constant $m \geq 0$. It is the oscillator Lagrangian $\frac{1}{2}\dot q^2 - \frac{1}{2}m^2q^2$ at each point, plus the coupling term $-\frac{1}{2}|\nabla\phi|^2$ between neighbouring points.

1.2 The first variation

As in mechanics, we compare a field with nearby fields and differentiate the action.

Definition 1.4 (Variation and first variation)**.** A *variation* is a C^1 map

$$h\colon M \to \mathbb{R}^N.$$

The *first variation* of the action of a field ϕ on a region D in the direction h is

$$\delta S_D[\phi]h = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\int_D L\big(x, \phi + \varepsilon h, \partial\phi + \varepsilon\partial h\big)\operatorname{vol}\Big|_{\varepsilon=0}.$$

The integrand needs only first derivatives, so the integral makes sense for the C^1 maps $\phi + \varepsilon h$. The first variation involves the partial derivatives of L in u and in v, which we name now.

Definition 1.5 (Momentum current and Euler–Lagrange expression)**.** For a field ϕ the *momentum current* has the components

$$p_a^\mu(x) = \frac{\partial L}{\partial v_\mu^a}\big(x, \phi(x), \partial\phi(x)\big),$$

and the *Euler–Lagrange expression* has the components

$$E_a[\phi](x) = \frac{\partial L}{\partial u^a}\big(x, \phi(x), \partial\phi(x)\big) - \partial_\mu p_a^\mu(x).$$

In the second term ∂_μ differentiates the composite function $x \mapsto p_a^\mu(x)$, so it also differentiates through $\phi(x)$ and $\partial\phi(x)$. Since L and ϕ are C^2, the current p_a^μ is C^1 and $E_a[\phi]$ is continuous. The time component $p_a^0 = \partial L/\partial(\partial_t \phi^a)$ is the field version of the canonical momentum $p = \partial L/\partial\dot{q}$.

Theoretical Mechanics wrote the Euler–Lagrange expressions as $\mathcal{E} = \dot{p} - \partial L/\partial q$. When there is no space ($d = 0$), this is the negative of the present expression. We use the sign of Olver, *Applications of Lie Groups to Differential Equations*, §4.1, so that the first variation below is $\int E_a h^a$ vol plus boundary terms, with no minus sign.

For the Klein–Gordon density (1.1) we get $p^\mu = \partial^\mu\phi$, so $p^0 = \phi_t$ and $p^i = -\partial_i\phi$. Hence

$$E[\phi] = -m^2\phi - \partial_\mu\partial^\mu\phi = -\big(\Box\phi + m^2\phi\big).$$

We import one result, differentiation under the integral sign. It is a special case of the theorem on parameter integrals in *Real and Complex Analysis*, Lecture 4: a continuous function on a compact set is bounded, so a constant is an integrable majorant.

Theorem 1.6 (Differentiation under the integral · *imported*)**.** *If $F(\varepsilon, x)$ and $\partial_\varepsilon F(\varepsilon, x)$ are continuous on $[-1, 1] \times D$ with D compact, then*

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\int_D F \,\mathrm{vol} = \int_D \partial_\varepsilon F \,\mathrm{vol}.$$

The boundary terms are written for a region between two graphs over the torus. For C^1 functions $\tau_0 < \tau_1$ on $\mathbb{T}^d$ we set

$$D = \{(t, x) \in \mathbb{R} \times \mathbb{T}^d : \ \tau_0(x) \le t \le \tau_1(x)\}. \tag{1.2}$$

This set is a region, and its boundary is the union of the two graphs. Constant $\tau_0 = t_0$ and $\tau_1 = t_1$ give the slab.

Proposition 1.7 (First variation)**.** *Let L be a Lagrangian density, ϕ a field and h a variation.*

(a) *For the region (1.2), with $V^\mu = p_a^\mu h^a$,*

$$\delta S_D[\phi]h = \int_D E_a[\phi]\, h^a \,\mathrm{vol} + \int_{\mathbb{T}^d} \big(V^0 - V^i\partial_i\tau_1\big)\big|_{t=\tau_1(x)}\,\mathrm{d}^d x \\ - \int_{\mathbb{T}^d} \big(V^0 - V^i\partial_i\tau_0\big)\big|_{t=\tau_0(x)}\,\mathrm{d}^d x. \tag{1.3}$$

(b) *For any region D, if the support of h lies in the interior of D, then*

$$\delta S_D[\phi]h = \int_D E_a[\phi]\, h^a \,\mathrm{vol}.$$

Proof. We differentiate under the integral, apply the product rule, and integrate one term coordinate by coordinate.

First, set $F(\varepsilon, x) = L(x, \phi + \varepsilon h, \partial\phi + \varepsilon\partial h)$. It is continuous, and by the chain rule

$$\partial_\varepsilon F(\varepsilon, x) = \frac{\partial L}{\partial u^a}(\dots)\, h^a + \frac{\partial L}{\partial v^a_\mu}(\dots)\, \partial_\mu h^a,$$

with the arguments of F in place of the dots. This is continuous, because L is C^2 and h is C^1. The imported theorem gives

$$\delta S_D[\phi]h = \int_D \partial_\varepsilon F(0, x)\,\mathrm{vol}, \qquad \partial_\varepsilon F(0, x) = \frac{\partial L}{\partial u^a}[\phi]\, h^a + p^\mu_a\, \partial_\mu h^a.$$

Next, the product rule gives $p^\mu_a \partial_\mu h^a = \partial_\mu V^\mu - (\partial_\mu p^\mu_a) h^a$, with $V^\mu = p^\mu_a h^a$. Hence

$$\delta S_D[\phi]h = \int_D E_a[\phi]\, h^a\,\mathrm{vol} + \int_D \partial_\mu V^\mu\,\mathrm{vol},$$

and the vector field V is C^1.

Finally, we integrate $\partial_\mu V^\mu$ over D by Fubini's theorem, first over t from $\tau_0(x)$ to $\tau_1(x)$. The fundamental theorem of calculus gives

$$\int_D \partial_t V^0\,\mathrm{vol} = \int_{\mathbb{T}^d} \big[V^0(\tau_1(x), x) - V^0(\tau_0(x), x)\big]\,\mathrm{d}^d x.$$

For a spatial index i, the function $G(\alpha, \beta, x) = \int_\alpha^\beta V^i(t, x)\,\mathrm{d}t$ has the partial derivatives $V^i(\beta, x)$, $-V^i(\alpha, x)$ and $\int_\alpha^\beta \partial_i V^i\,\mathrm{d}t$. The chain rule applied to $x \mapsto G(\tau_0(x), \tau_1(x), x)$ gives

$$\int_{\tau_0}^{\tau_1} \partial_i V^i\,\mathrm{d}t = \partial_i \int_{\tau_0}^{\tau_1} V^i\,\mathrm{d}t - V^i(\tau_1, x)\,\partial_i\tau_1 + V^i(\tau_0, x)\,\partial_i\tau_0.$$

The first term on the right is the derivative of a 2π-periodic function, so its integral over a period of x^i vanishes. Adding the time and spatial terms gives the boundary terms of (1.3), which proves part (a).

In part (b), the support of h is a closed subset of the compact set D, so it is compact. The field V vanishes outside it, so $\int_D \partial_\mu V^\mu\,\mathrm{vol} = \int_M \partial_\mu V^\mu\,\mathrm{vol}$. Along every coordinate line, V either vanishes outside a bounded interval or is periodic. Hence each term $\int_M \partial_\mu V^\mu\,\mathrm{vol}$ is zero by the fundamental theorem of calculus, which proves part (b). ∎

On the slab $D = [t_0, t_1] \times \mathbb{T}^d$ the boundary terms become

$$\int_{\mathbb{T}^d} p^0_a h^a\,\mathrm{d}^d x\Big|_{t=t_1} - \int_{\mathbb{T}^d} p^0_a h^a\,\mathrm{d}^d x\Big|_{t=t_0}. \tag{1.4}$$

They are the field version of the boundary term $\big[p\,\delta q\big]_{t_0}^{t_1}$ of mechanics. In Figure 1.1 the vertical edges $x = 0$ and $x = 2\pi$ are identified, so the slab over the circle has no spatial boundary. The only boundary is the pair of slices, with the outward normal $+\partial_t$ on the top slice and $-\partial_t$ on the bottom one.

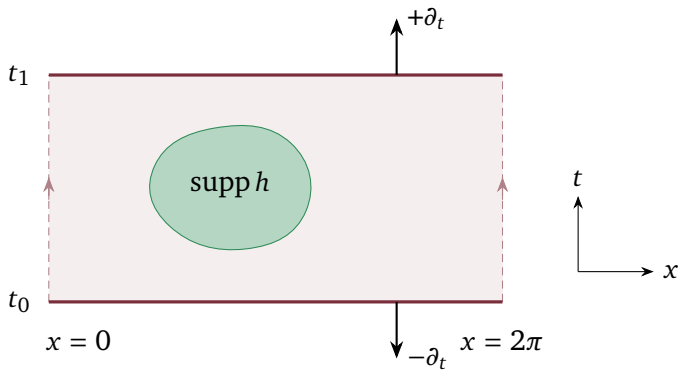


Figure 1.1: The slab $[t_0, t_1] \times \mathbb{T}$ with its two boundary slices (thick) and their outward normals. The shaded set is the support of a variation h that vanishes near the slices.

Remark 1.8 (The covariant boundary term)**.** Let $\mathrm{d}\Sigma_\mu = \iota_{\partial_\mu}\mathrm{vol}$, where ι inserts a vector into the first slot of a form. Thus $\mathrm{d}\Sigma_0 = \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^d$, and each $\mathrm{d}\Sigma_i$ contains the factor $\mathrm{d}t$. Along a graph $t = \tau(x)$ we have $\mathrm{d}t = \partial_j\tau\,\mathrm{d}x^j$, so $\mathrm{d}\Sigma_i$ pulls back to $-\partial_i\tau\,\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^d$. Hence $V^\mu \mathrm{d}\Sigma_\mu$ pulls back to $(V^0 - V^i\partial_i\tau)\,\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^d$ on either graph. Orient ∂D with the outward normal first, as in *Topology and Geometry*, Lecture 12. This orientation agrees with $\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^d$ on the top graph and is opposite on the bottom graph. So the boundary terms of (1.3) are $\int_{\partial D} V^\mu \mathrm{d}\Sigma_\mu$. Lecture 8 proves the analogous formula on $\mathbb{R}^d$ for slices that are not flat.

1.3 Stationary fields and the Euler–Lagrange equations

The slice terms in (1.4) are the reason to restrict the variations, exactly as Hamilton's principle in mechanics fixes the endpoints.

Definition 1.9 (Admissible variation, stationary field)**.** A variation h is *admissible on a region* D if its support lies in the interior of D. A field ϕ is *stationary* if $\delta S_D[\phi]h = 0$ for every region $D \subset M$ and every variation h admissible on D.

For an admissible variation, part (b) of Proposition 1.7 leaves only the integral of $E_a[\phi]h^a$. The following lemma shows that the vanishing of this integral for every admissible h forces $E[\phi] = 0$.

Lemma 1.10 (Fundamental lemma)**.** *Let $U \subset M$ be open and $g : U \to \mathbb{R}$ continuous. Suppose that*

$$\int_U g\,h\,\mathrm{vol} = 0 \quad \text{for every } C^2 \text{ function } h \text{ with compact support in } U.$$

Then $g = 0$ on U.

Proof. We argue by contradiction. Suppose $g(z) > 0$ at some point $z \in U$. We use coordinates $y = (y^0, \dots, y^d)$ centred at z, with the spatial coordinates in $(-\pi, \pi)^d$ when space is the torus, and write $|y|$ for the Euclidean length of y. By continuity there is $r \in (0, \pi)$ such that the closed ball $\bar{B}$, with $B = \{|y| < r\}$, lies in U and $g > \tfrac{1}{2}g(z)$ on $\bar{B}$. Take the bump

$$h(y) = \begin{cases} \big(r^2 - |y|^2\big)^3, & |y| < r, \\ 0, & \text{otherwise.} \end{cases}$$

It is C^2, because the cube vanishes to order three on the sphere $|y| = r$. It is positive on B. Its support is $\bar{B} \subset U$. Then

$$\int_U g\, h \,\mathrm{vol} \geq \tfrac{1}{2} g(z) \int_B h \,\mathrm{vol} > 0,$$

a contradiction. If $g(z) < 0$, apply the same argument to $-g$. ∎

Theorem 1.11 (Stationary fields). *A field ϕ is stationary if and only if it solves the* Euler–Lagrange equations

$$E_a[\phi] = \frac{\partial L}{\partial u^a}[\phi] - \partial_\mu p_a^\mu = 0 \qquad \text{on } M, \quad a = 1, \dots, N.$$

Proof. First suppose $E[\phi] = 0$. For a region D and a variation h admissible on D, part (b) of Proposition 1.7 gives $\delta S_D[\phi]h = \int_D E_a[\phi] h^a \,\mathrm{vol} = 0$.

Conversely, suppose ϕ is stationary, and fix a component a and a point $z \in M$. Let D be a closed coordinate ball about z of radius less than π, as in the proof of Lemma 1.10, and let U be its interior. For every C^2 function g with compact support in U, the variation h with $h^a = g$ and the other components zero is admissible on D. By part (b) of Proposition 1.7, $0 = \delta S_D[\phi]h = \int_U E_a[\phi]\, g \,\mathrm{vol}$. Lemma 1.10 gives $E_a[\phi] = 0$ on U, in particular at z. ∎

The theorem does not give existence. Lecture 4 constructs solutions of the Klein–Gordon equation on $\mathbb{R}^d$ from finite-energy initial data.

1.4 The Klein–Gordon field on a circle

We find the field equation of the density (1.1) and one of its solutions, and compute its energy. The data are

$$M = \mathbb{R} \times \mathbb{T}, \qquad L = \frac{1}{2}\phi_t^2 - \frac{1}{2}\phi_x^2 - \frac{1}{2}m^2\phi^2, \qquad m = 3.$$

The momentum current is $p^0 = \phi_t$ and $p^1 = -\phi_x$. The remaining derivative is $\partial L/\partial u = -m^2\phi$. Hence

$$E[\phi] = -m^2\phi - \partial_t\phi_t - \partial_x(-\phi_x) = -\big(\phi_{tt} - \phi_{xx} + m^2\phi\big).$$

By Theorem 1.11 the stationary fields are the solutions of the *Klein–Gordon equation*

$$\phi_{tt} - \phi_{xx} + m^2\phi = 0.$$

We look for a solution of the form

$$\phi(t, x) = a(t) \cos 2x.$$

Its zeros in x stay fixed while its amplitude $a(t)$ oscillates. Such a solution is called a standing wave. The function $\cos 2x$ is 2π-periodic. Since $\phi_{xx} = -4a\cos 2x$, the equation becomes

$$\big(\ddot{a} + 4a + 9a\big)\cos 2x = 0.$$

Since $\cos 2x$ is not identically zero, this holds exactly when $\ddot{a} + 13a = 0$. This is an oscillator with frequency $\nu = \sqrt{13}$. With $a(0) = 1$ and $\dot{a}(0) = 0$ we get

$$\phi(t, x) = \cos\big(\sqrt{13}\, t\big)\cos 2x,$$

with period $2\pi/\sqrt{13} \approx 1.743$. The mode with k waves around the circle oscillates with $\nu^2 = k^2 + m^2$. The coupling term contributes k^2. The mass term contributes m^2.

The energy of the field adds the kinetic term, the coupling term and the mass term of (1.1), all with the plus sign:

$$E_{\text{en}} = \int_{\mathbb{T}} \Big(\frac{1}{2}\phi_t^2 + \frac{1}{2}\phi_x^2 + \frac{1}{2}m^2\phi^2 \Big) \mathrm{d}x.$$

For the standing wave we write $s = \sin(\sqrt{13}\,t)$ and $c = \cos(\sqrt{13}\,t)$, and use $\int_{\mathbb{T}} \cos^2 2x \,\mathrm{d}x = \int_{\mathbb{T}} \sin^2 2x \,\mathrm{d}x = \pi$. The three terms give

$$E_{\text{en}} = \frac{13\pi}{2}s^2 + 2\pi c^2 + \frac{9\pi}{2}c^2 = \frac{13\pi}{2}$$

at every time. Lecture 3 derives this energy from time translations and proves that it is conserved for every solution.

1.5 A boundary term in the first variation

Theorem 1.11 uses only variations that vanish near the boundary of a region. The restriction is necessary. A solution of the Euler–Lagrange equations can have a nonzero first variation in a direction that moves its boundary values.

Take the massless field, $m = 0$, on the slab $D = [0, T] \times \mathbb{T}$ with $T > 0$, and the field $\phi(t, x) = t$. It solves $\phi_{tt} - \phi_{xx} = 0$, so $E[\phi] = 0$. Take the variation $h(t, x) = t/T$, with $h = 0$ at $t = 0$ and $h = 1$ at $t = T$. It is not admissible. By (1.3) and (1.4), with $p^0 = \phi_t = 1$,

$$\delta S_D[\phi]h = 0 + \int_{\mathbb{T}} 1 \cdot 1 \,\mathrm{d}x - \int_{\mathbb{T}} 1 \cdot 0 \,\mathrm{d}x = 2\pi.$$

A direct computation agrees:

$$S_D[\phi + \varepsilon h] = \int_0^T \int_{\mathbb{T}} \frac{1}{2}\Big(1 + \frac{\varepsilon}{T}\Big)^2 \mathrm{d}x \,\mathrm{d}t = \pi T \Big(1 + \frac{\varepsilon}{T}\Big)^2,$$

whose derivative at $\varepsilon = 0$ is 2π.

So the first variation of a solution need not vanish in a direction that moves the boundary values. Lecture 8 builds the symplectic form on the space of solutions out of this boundary term. Lecture 9 must control the same term when the field is the electromagnetic potential.

Summary. The action of a field on a compact region is the integral of a local density $L(x, \phi, \partial\phi)$. On a region between two graphs, its first variation is $\int_D E_a[\phi]h^a$ vol plus boundary terms built from the momentum current. A field is stationary exactly when $E[\phi] = 0$. For the Klein–Gordon density this is $\Box\phi + m^2\phi = 0$. Lecture 2 turns symmetries into conservation laws with the same formula.

Self-check. Find the Euler–Lagrange equation of the density

$$L = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - V(\phi)$$

for a C^2 function $V\colon \mathbb{R} \to \mathbb{R}$.

Answer. Here $p^\mu = \partial^\mu\phi$ and $\partial L/\partial u = -V'(\phi)$, so $E[\phi] = -V'(\phi) - \Box\phi$. The equation is $\Box\phi + V'(\phi) = 0$.

Self-check. In the example of Section 1.5, replace h by $h(t,x) = \sin(\pi t/T)$. Is h admissible on $D = [0,T] \times \mathbb{T}$, and what is $\delta S_D[\phi]h$?

Answer. It is not admissible: its support is all of M, which does not lie in the interior of D. Still, h vanishes on both slices, so the slice terms (1.4) are zero. Since $E[\phi] = 0$, we get $\delta S_D[\phi]h = 0$.

Self-check. In the same example take the constant variation $h = 1$. Compute the slice terms (1.4), and check the result against $S_D[\phi + \varepsilon h]$.

Answer. The slice terms are $\int_{\mathbb{T}} 1\,\mathrm{d}x - \int_{\mathbb{T}} 1\,\mathrm{d}x = 2\pi - 2\pi = 0$. Directly, $\phi + \varepsilon$ has the same derivatives as ϕ, so $S_D[\phi + \varepsilon h]$ does not depend on ε, and its derivative is 0.

Lecture 2

Noether currents

The first-variation formula separates the field equation from a divergence. A continuous symmetry gives another expression for the same variation. Comparing the expressions produces a local conservation law. Integrating that law requires a separate condition on the spatial boundary.

Objectives. *We prove Noether's identity, including transformations that move spacetime points. We compute the current for rotations of a complex field. We state and prove sufficient conditions for its spatial integral to be constant.*

By the end of this lecture you will be able to:

- Compute the characteristic of an internal symmetry or a spacetime translation.
- Derive a Noether current and its identity before imposing the field equation.
- Compute the complex Klein–Gordon current with the course normalisation.
- Prove charge conservation on the torus and under stated decay on $\mathbb{R}^d$.

2.1 Currents and infinitesimal changes

We retain the coordinates, C^2 fields and C^2 Lagrangian densities of Lecture 1. Let $I \subset \mathbb{R}$ be an open time interval. Space is either $X = \mathbb{T}^d$ or $X = \mathbb{R}^d$.

Definition 2.1 (Current and local conservation law)**.** A *current* on $I \times X$ is a C^1 map

$$j\colon I \times X \longrightarrow \mathbb{R}^{d+1}, \qquad (t,x) \longmapsto \big(j^0(t,x), \dots, j^d(t,x)\big).$$

The current satisfies a *local conservation law* if

$$\partial_\mu j^\mu = \partial_t j^0 + \partial_i j^i = 0.$$

The component j^0 is the density. The components j^i describe spatial flux. Neither the sign of j^0 nor the convergence of its spatial integral is part of this definition.

To compare a field with its transform, we must specify whether the spacetime point moves. Consider a C^2 family of local point transformations, with parameter ε near zero, given by

$$x' = f_\varepsilon(x), \qquad u' = g_\varepsilon(x,u), \qquad f_0(x) = x, \quad g_0(x,u) = u.$$

Each f_ε is a local diffeomorphism. On the image of an open set where f_ε is one-to-one, the transformed field is defined by

$$\phi_\varepsilon\big(f_\varepsilon(x)\big) = g_\varepsilon\big(x, \phi(x)\big).$$

We take the family to be C^2 jointly in its arguments. Its infinitesimal coefficients are $\xi^\mu(x) = \partial_\varepsilon f^\mu_\varepsilon(x)|_0$ and $\zeta^a(x,u) = \partial_\varepsilon g^a_\varepsilon(x,u)|_0$.

Definition 2.2 (Characteristic)**.** The *characteristic* of this family, evaluated on a field ϕ, is its change at a fixed spacetime point:

$$Q^a[\phi](x) := \partial_\varepsilon \phi^a_\varepsilon(x)\big|_0 = \zeta^a\big(x, \phi(x)\big) - \xi^\nu(x)\partial_\nu \phi^a(x). \tag{2.1}$$

A transformation with $f_\varepsilon = \mathrm{id}$ is an *internal transformation*.

The chain rule applied to the defining equation for ϕ_ε proves (2.1). For an internal transformation, the characteristic is simply $\zeta^a(x, \phi(x))$. For the translation $f_\varepsilon(x) = x + \varepsilon e_\nu$, with $g_\varepsilon(x, u) = u$, it is $Q^a = -\partial_\nu \phi^a$. Here e_ν is the coordinate vector in the x^ν direction.

Although ϕ is C^2, its characteristic need only be C^1. Lecture 1 allows C^1 variation directions. The local first-variation formula therefore applies with $h^a = Q^a$.

2.2 Symmetry up to a divergence

Write $\delta_Q L[\phi]$ for the derivative of the density at a fixed point along ϕ_ε. The chain rule gives

$$\delta_Q L[\phi] = \frac{\partial L}{\partial u^a}[\phi] Q^a + p^\mu_a \partial_\mu Q^a, \qquad p^\mu_a = \frac{\partial L}{\partial v^a_\mu}[\phi].$$

Every derivative ∂_μ of an evaluated expression below is a derivative of a composite function on spacetime. It differentiates through ϕ and its derivatives. A divergence is always understood in this sense.

Definition 2.3 (Variational and divergence symmetry)**.** The family is an infinitesimal *variational symmetry* if, for every C^2 field, there are specified local expressions $B^\mu[\phi]$ such that their evaluations are C^1 and

$$\delta_Q L[\phi] = \partial_\mu B^\mu[\phi]. \tag{2.2}$$

Allowing a nonzero divergence on the right is called a *divergence symmetry*.

This definition concerns the infinitesimal change. It requires an identity for arbitrary fields, without imposing the Euler–Lagrange equations. The expressions B^μ must be given by the transformation and density, rather than chosen separately by integrating for each field.

For a point transformation, one often knows the change of the density including the coordinate volume. Define its coefficient $\mathcal{A}[\phi]$ by

$$\mathcal{A}[\phi] = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\bigg|_0 \left\{ L[\phi_\varepsilon]\big(f_\varepsilon(x)\big) \det Df_\varepsilon(x) \right\}.$$

The derivative of the determinant at the identity is $\partial_\mu \xi^\mu$. Differentiating the other factor at its moving argument gives $\delta_Q L + \xi^\mu \partial_\mu L[\phi]$. The product rule consequently gives

$$\mathcal{A}[\phi] = \delta_Q L[\phi] + \partial_\mu(\xi^\mu L[\phi]). \tag{2.3}$$

Thus a point transformation with $\mathcal{A} = \partial_\mu b^\mu$, where the evaluated expressions b^μ are C^1, gives (2.2) with $B^\mu = b^\mu - \xi^\mu L[\phi]$. In particular, exact invariance of the density including its volume means $\mathcal{A} = 0$, and then $B^\mu = -\xi^\mu L[\phi]$. The point divergence b^μ and the fixed-point divergence B^μ differ. Exact invariance at moving points therefore need not give $\delta_Q L = 0$ at fixed points.

Theorem 2.4 (Noether's identity · *proved*)**.** *Let L and ϕ be C^2. Suppose a C^1 characteristic $Q^a[\phi]$ and C^1 expressions $B^\mu[\phi]$ satisfy (2.2) for every field. The* Noether current *is*

$$j^\mu = p^\mu_a Q^a - B^\mu. \tag{2.4}$$

It satisfies the identity

$$\partial_\mu j^\mu = -E_a[\phi]Q^a, \qquad E_a[\phi] = \frac{\partial L}{\partial u^a}[\phi] - \partial_\mu p^\mu_a. \tag{2.5}$$

On a solution, $E_a[\phi] = 0$, the current satisfies a local conservation law.

Proof. We use the product rule in the local first-variation formula:

$$\partial_\mu B^\mu = \delta_Q L = E_a Q^a + \partial_\mu (p^\mu_a Q^a).$$

Subtracting the divergence of B^μ proves (2.5). The products are C^1, so every derivative used here exists classically. On a solution the right-hand side is zero. ∎

Our Euler–Lagrange expression has the sign fixed in Lecture 1. The current sign follows from that choice and (2.4). Olver's statement writes $\operatorname{Div} P = Q^a E_a$, so his current is $P^\mu = -j^\mu$.[1]

Example 2.5 (A nonzero divergence term)**.** For one real field take the density

$$L = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi + \phi\phi_t.$$

Under the internal shift $\phi_\varepsilon = \phi + \varepsilon$, the characteristic is $Q^1 = 1$. Its density change is $\delta_Q L = \phi_t$. Thus $B^0 = \phi$ and $B^i = 0$. The momentum has components $p^0 = \phi_t + \phi$ and $p^i = -\partial_i\phi$. The current is therefore $j^\mu = \partial^\mu\phi$. The divergence term removes the extra ϕ from its time component.

2.3 Rotations of a complex Klein–Gordon field

Let $\Phi = \phi_1 + \mathrm{i}\phi_2$, with real C^2 components. Take $m \geq 0$ and the density

$$L = \frac{1}{2}\partial_\mu\overline{\Phi}\,\partial^\mu\Phi - \frac{1}{2}m^2|\Phi|^2 = \frac{1}{2}\sum_{a=1}^{2}\left(\partial_\mu\phi_a\,\partial^\mu\phi_a - m^2\phi_a^2\right). \tag{2.6}$$

There is no factor $1/\sqrt{2}$ in the definition of Φ. The displayed factor $1/2$ gives each real component the normalisation of Lecture 1. We differentiate with respect to those two real components.

The group $\mathrm{U}(1) = \{z \in \mathbb{C} : |z| = 1\}$ acts by internal rotations $\Phi_\varepsilon = e^{\mathrm{i}\varepsilon}\Phi$. Its real-component form gives

$$\begin{pmatrix}\phi_{1,\varepsilon}\\ \phi_{2,\varepsilon}\end{pmatrix} = \begin{pmatrix}\cos\varepsilon & -\sin\varepsilon\\ \sin\varepsilon & \cos\varepsilon\end{pmatrix}\begin{pmatrix}\phi_1\\ \phi_2\end{pmatrix}, \qquad (Q^1, Q^2) = (-\phi_2, \phi_1).$$

Since the parameter is constant, the derivatives rotate by the same matrix. That matrix preserves the Euclidean scalar product in the two components. Both terms in (2.6) are invariant. Hence $\delta_Q L = 0$ and we take $B^\mu = 0$.

[1] Olver, *Applications of Lie Groups to Differential Equations*, §4.4, Theorem 4.29 and equations (4.38)–(4.41), pp. 272–274. The course notation bible records this translation.

The momenta are $p_a^\mu = \partial^\mu \phi_a$. Substitution in (2.4) gives the current

$$j^\mu = \phi_1 \partial^\mu \phi_2 - \phi_2 \partial^\mu \phi_1 = \operatorname{Im}(\overline{\Phi}\, \partial^\mu \Phi). \tag{2.7}$$

Its components are $j^0 = \operatorname{Im}(\overline{\Phi}\Phi_t)$ and $j^i = -\operatorname{Im}(\overline{\Phi}\partial_i \Phi)$. The minus sign in the spatial components comes from the Minkowski metric.

The field equations are $(\Box + m^2)\phi_a = 0$ for $a = 1, 2$. We can also verify Noether's identity directly. The two products of first derivatives cancel, so

$$\partial_\mu j^\mu = \phi_1 \Box \phi_2 - \phi_2 \Box \phi_1 = \phi_1(\Box + m^2)\phi_2 - \phi_2(\Box + m^2)\phi_1 = -E_a Q^a.$$

On solutions each term containing $(\Box + m^2)\phi_a$ is zero, so the divergence vanishes.

For a specific solution on $\mathbb{R} \times \mathbb{T}$, take $m = 3$ and $\Phi(t, x) = Ae^{-\mathrm{i}\sqrt{13}t + 2\mathrm{i}x}$, where $A \in \mathbb{C}$ is constant. Its second derivatives give $(\Box + 9)\Phi = (-13 + 4 + 9)\Phi = 0$. Its current has the constant values

$$j^0 = -\sqrt{13}|A|^2, \qquad j^1 = -2|A|^2.$$

We call time dependence $e^{-\mathrm{i}\nu t}$ with $\nu > 0$ *positive frequency*. Thus this positive-frequency solution has negative density for the chosen rotation $\Phi \mapsto e^{\mathrm{i}\varepsilon}\Phi$. Reversing the rotation reverses the current.

2.4 From a local law to a charge

Definition 2.6 (Charge)**.** For a current whose time component is integrable on each slice, its *charge* is the function

$$Q \colon I \to \mathbb{R}, \qquad Q(t) = \int_X j^0(t, x)\, \mathrm{d}^d x.$$

The unindexed $Q(t)$ denotes this integral. The indexed Q^a denotes a characteristic.

On the torus the integral always exists. Differentiation under the integral is justified by continuity on each compact slab, as in Lecture 1. For a conserved current periodicity gives

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = -\sum_{i=1}^{d} \int_{\mathbb{T}^d} \partial_i j^i \, \mathrm{d}^d x = 0.$$

Indeed, integrating in x^i gives j^i at 2π minus its value at zero. Those values agree. The charged Klein–Gordon solution above therefore has constant charge $Q = -2\pi\sqrt{13}|A|^2$.

For $\mathbb{R}^d$ an integrability hypothesis is necessary. Suppose $j^0(t, \cdot)$ is in $L^1(\mathbb{R}^d)$ for every $t \in I$. Suppose also that, for every compact interval $[t_0, t_1] \subset I$, the spatial flux satisfies

$$\int_{t_0}^{t_1} \int_{\mathbb{R}^d} \sum_{i=1}^{d} |j^i(t, x)| \, \mathrm{d}^d x \, \mathrm{d}t < \infty. \tag{2.8}$$

These conditions imply that the charge of a conserved current is constant. We prove this without differentiating an integral over all space. Choose a C^2 function $\chi \colon \mathbb{R}^d \to [0, 1]$ equal to one for $|x| \le 1$ and zero for $|x| \ge 2$. Put $\chi_R(x) = \chi(x/R)$ for $R \ge 1$. Such a function can be made from a C^2 polynomial transition between the two constant values, with its first two derivatives zero at both joins. Define $Q_R(t) = \int \chi_R j^0 \, \mathrm{d}^d x$. Compact support allows differentiation and integration by parts. The local conservation law gives

$$Q_R(t_1) - Q_R(t_0) = \int_{t_0}^{t_1} \int_{\mathbb{R}^d} (\partial_i \chi_R) j^i \, \mathrm{d}^d x \, \mathrm{d}t.$$

The absolute value of this expression is at most $\|\nabla\chi\|_\infty / R$ times the finite integral in (2.8). It tends to zero. At either endpoint,

$$|Q_R(t) - Q(t)| \le \int_{|x|>R} |j^0(t,x)|\,\mathrm{d}^d x \longrightarrow 0$$

by integrability. Taking the limit proves $Q(t_1) = Q(t_0)$. This argument states precisely how the spatial boundary contribution vanishes.

One explicit sufficient decay condition is that, on each compact time interval, there are constants $C > 0$ and $\sigma > 0$ with

$$|j^0(t,x)| + \sum_{i=1}^{d} |j^i(t,x)| \le C(1+|x|)^{-d-\sigma}.$$

The integral at large radius is bounded by a constant times $\int_1^\infty r^{-1-\sigma}\,\mathrm{d}r$, which converges.

2.5 Translations and the field equation

Suppose the density has no explicit dependence on x^ν. The point translation $x' = x + \varepsilon e_\nu$ preserves the density including its volume. Thus $b^\mu = 0$ in (2.3). Its characteristic and fixed-point divergence are $Q^a = -\partial_\nu \phi^a$ and $B^\mu = -\delta^\mu_\nu L$. Its current is

$$j^\mu = -p^\mu_a \partial_\nu \phi^a + \delta^\mu_\nu L = -T^\mu{}_\nu, \qquad T^\mu{}_\nu := p^\mu_a \partial_\nu \phi^a - \delta^\mu_\nu L.$$

Lecture 3 studies this canonical stress tensor. It chooses the negative translation $x' = x - \varepsilon e_\nu$, which gives $Q^a = \partial_\nu \phi^a$, $B^\mu = \delta^\mu_\nu L$ and current $T^\mu{}_\nu$ itself.

The sign of a translation parameter is separate from the orientation of a boundary. With $\mathrm{d}\Sigma_\mu = \iota_{\partial_\mu}\mathrm{vol}$, the flux form is $j^\mu \mathrm{d}\Sigma_\mu$. Its outward-first integrals on the top and bottom of a slab are $Q(t_1)$ and $-Q(t_0)$. Lecture 3 proves the regional balance on rectangles from the fundamental theorem of calculus.

Noether's identity holds *off shell*, meaning for arbitrary fields. The conservation law holds *on shell*, meaning on solutions. For example, keep the complex Klein–Gordon density but take $\phi_1 = 1$ and $\phi_2 = t^2$ on $\mathbb{R}\times\mathbb{T}$. Formula (2.7) gives $j^0 = 2t$ and $j^1 = 0$. Its divergence is 2. The Euler–Lagrange expressions are $E_1 = -m^2$ and $E_2 = -2 - m^2 t^2$, so

$$-E_a Q^a = -\big((-m^2)(-t^2) + (-2 - m^2 t^2)\cdot 1\big) = 2.$$

The identity remains true, but the charge $Q(t) = 4\pi t$ is not constant.

Summary. A variational symmetry satisfies $\delta_Q L = \partial_\mu B^\mu$ for arbitrary fields. Its current is $j^\mu = p^\mu_a Q^a - B^\mu$, and $\partial_\mu j^\mu = -E_a Q^a$. On solutions the divergence vanishes. The integrated charge is constant on the torus, or on $\mathbb{R}^d$ under the integrability conditions of Section 2.4. Translations give the stress tensor studied in Lecture 3.

Self-check. For the complex field, replace $e^{\mathrm{i}\varepsilon}\Phi$ by $e^{-\mathrm{i}\varepsilon}\Phi$. What happens to the characteristic and current?

Answer. Both change sign: $(Q^1, Q^2) = (\phi_2, -\phi_1)$ and $j^\mu = -\operatorname{Im}(\overline{\Phi}\partial^\mu\Phi)$. The density is still invariant.

Self-check. For the massless real density $L = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi$, find the current of the shift $\phi \mapsto \phi+\varepsilon$.

Answer. The characteristic is $Q^1 = 1$ and $B^\mu = 0$, so $j^\mu = \partial^\mu\phi$. Its divergence is $\Box\phi = -E[\phi]$. It is conserved on solutions of the massless equation.

Self-check. For a density with no explicit time dependence, use $\phi_\varepsilon(t,x) = \phi(t+\varepsilon,x)$ to find Q^a, B^μ and j^μ.

Answer. Here $Q^a = \partial_t\phi^a$. The chain rule gives $\delta_Q L = \partial_t L$, so $B^\mu = \delta^\mu_0 L$. Thus $j^\mu = p^\mu_a\phi^a_t - \delta^\mu_0 L = T^\mu{}_0$.

Self-check. On $\mathbb{R}\times\mathbb{R}$, consider the current $j^0 = 1$ and $j^1 = 0$. Does its local conservation law define a finite charge?

Answer. Its divergence is zero, but $\int_{\mathbb{R}} j^0\,\mathrm{d}x = \infty$. The integrability hypothesis fails, so this current has no charge as defined here.

Lecture 3

Stress and energy flux

An oscillator conserves its energy when its Lagrangian has no explicit time dependence. A field can also carry energy from one place to another. Time translation gives a local balance law. The energy in a fixed rectangle is constant on a time interval exactly when the net outward flux through its faces vanishes throughout that interval.

Objectives. *We derive the scalar stress tensor from translations and prove the energy balance on rectangular regions. We compute the energy carried out by a massless pulse and compare the stress tensor with metric variation.*

By the end of this lecture you will be able to:

- Derive the scalar translation current, with its off-shell divergence.
- Compute the energy change in a rectangle from its boundary flux.
- Relate the scalar canonical tensor to the inverse-metric variation.

3.1 Translation currents

We use the coordinates and metric of Lecture 1 on $M = \mathbb{R} \times \mathbb{R}^d$ or $M = \mathbb{R} \times \mathbb{T}^d$. A density is *autonomous* if it has no explicit dependence on x. The momentum current remains $p^\mu = \partial L / \partial(\partial_\mu \phi)$.

Definition 3.1 (Canonical stress tensor)**.** For a scalar field ϕ, the *canonical stress tensor* is the array

$$T^\mu{}_\nu = p^\mu \partial_\nu \phi - \delta^\mu_\nu L[\phi], \qquad \mu, \nu = 0, \dots, d. \tag{3.1}$$

Its first index is the direction of flow. The second index specifies the translation. We raise or lower indices with η.

We find the current for each translation using Theorem 2.4. For a characteristic Q^1 with $\delta_Q L = \partial_\mu B^\mu$, that theorem gives $j^\mu = p^\mu Q^1 - B^\mu$ and $\partial_\mu j^\mu = -E[\phi] Q^1$ before imposing the field equation.

Let e_ν be the coordinate vector in the x^ν direction. For an autonomous density, take the translated field $\phi_\varepsilon(x) = \phi(x + \varepsilon e_\nu)$. Its characteristic is $Q^1 = \partial_\nu \phi$. Since the density has no explicit coordinate dependence, the chain rule gives $\delta_Q L = \partial_\nu L$. Thus we can take $B^\mu = \delta^\mu_\nu L$, and the current is

$$j^\mu = p^\mu \partial_\nu \phi - \delta^\mu_\nu L = T^\mu{}_\nu.$$

This fixed-coordinate field change corresponds to the coordinate translation $x'^\rho = x^\rho - \varepsilon \delta^\rho_\nu$, with $\phi'(x') = \phi(x)$.

We now use the real scalar density with a potential $V \in C^2(\mathbb{R})$:

$$L = \frac{1}{2} \partial_\mu \phi \, \partial^\mu \phi - V(\phi), \qquad E[\phi] = -\big(\Box \phi + V'(\phi)\big). \tag{3.2}$$

Its momentum current is $p^\mu = \partial^\mu \phi$. The canonical tensor with both indices lowered is

$$T_{\mu\nu} = \partial_\mu \phi \, \partial_\nu \phi - \eta_{\mu\nu} L. \tag{3.3}$$

This tensor is symmetric: $T_{\mu\nu} = T_{\nu\mu}$. Symmetry here follows from the scalar density, rather than from the definition of the canonical tensor for an arbitrary field theory.

Definition 3.2 (Energy, momentum and energy flux). For the scalar density (3.2), the *energy density* ρ_{en}, the *momentum density* $\mathcal{P}^i$ and the *energy flux* f^i are

$$\rho_{\text{en}} = T^0{}_0 = \frac{1}{2}\phi_t^2 + \frac{1}{2}|\nabla\phi|^2 + V(\phi), \qquad \mathcal{P}^i = T^{0i}, \qquad f^i = T^{i0} = -\phi_t \partial_i \phi. \tag{3.4}$$

For a bounded spatial region Ω, its *energy* at time t is

$$E_{\text{en},\Omega}(t) = \int_\Omega \rho_{\text{en}}(t,x)\, \mathrm{d}^d x.$$

The same definition applies to all space when the integral is finite.

The flux component f^i measures signed transport in the positive x^i direction. Since the tensor is symmetric, $\mathcal{P}^i = f^i$ for this scalar field. The mixed component has the opposite spatial sign: $T^0{}_i = -\mathcal{P}^i$. In the absence of space derivatives, $\rho_{\text{en}} = \frac{1}{2}\dot\phi^2 + V(\phi)$ is the oscillator energy. For $V(\phi) = \frac{1}{2}m^2\phi^2$, it is the Klein–Gordon energy computed in Lecture 1. If $V \geq 0$, then $\rho_{\text{en}} \geq 0$.

3.2 Local balance and boundary flux

For a rectangle we can state the boundary terms using its faces. Let

$$\Omega = \prod_{i=1}^{d} [a_i, b_i], \qquad a_i < b_i, \qquad \Omega_i = \prod_{j \neq i} [a_j, b_j].$$

The notation $\mathrm{d}^{d-1}x_{\widehat{i}}$ means integration in all coordinates except x^i. For $d = 1$, a face integral means evaluation at the endpoint. The outward normal to the face $x^i = b_i$ is $+e_i$. At $x^i = a_i$ it is $-e_i$. The *net outward flux* is therefore

$$\mathcal{F}_\Omega(t) = \sum_{i=1}^{d} \int_{\Omega_i} \left[f^i(t,x)\big|_{x^i = b_i} - f^i(t,x)\big|_{x^i = a_i} \right] \mathrm{d}^{d-1}x_{\widehat{i}}. \tag{3.5}$$

The sign refers to the outward direction on each face, even when f^i itself is positive on both faces.

Theorem 3.3 (Scalar stress and energy balance). *For $V \in C^2(\mathbb{R})$ and $\phi \in C^2(M, \mathbb{R})$ with density (3.2), translations give (3.1), with*

$$\partial_\mu T^\mu{}_\nu = -E[\phi]\partial_\nu\phi = \big(\Box\phi + V'(\phi)\big)\partial_\nu\phi. \tag{3.6}$$

If ϕ solves $\Box\phi + V'(\phi) = 0$, then

$$\partial_t \rho_{\mathrm{en}} + \partial_i f^i = 0. \tag{3.7}$$

On a fixed rectangle $\Omega \subset \mathbb{R}^d$, its energy satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t} E_{\mathrm{en},\Omega}(t) = -\mathcal{F}_\Omega(t), \qquad E_{\mathrm{en},\Omega}(t_1) - E_{\mathrm{en},\Omega}(t_0) = -\int_{t_0}^{t_1} \mathcal{F}_\Omega(t)\,\mathrm{d}t. \tag{3.8}$$

On $\mathbb{T}^d$, the total energy is constant.

Proof. The translation argument above identifies the current. We verify its divergence directly and then integrate one coordinate at a time.

First, differentiating the scalar density along x^ν gives

$$\partial_\nu L = \partial^\mu\phi\,\partial_\nu\partial_\mu\phi - V'(\phi)\partial_\nu\phi.$$

The product rule in (3.1) and equality of mixed second derivatives now give

$$\begin{aligned}\partial_\mu T^\mu{}_\nu &= (\Box\phi)\partial_\nu\phi + \partial^\mu\phi\,\partial_\mu\partial_\nu\phi - \partial_\nu L\\ &= \big(\Box\phi + V'(\phi)\big)\partial_\nu\phi.\end{aligned}$$

This proves (3.6). On a solution its right-hand side is zero. Taking $\nu = 0$ gives (3.7), since $T^i{}_0 = T^{i0} = f^i$.

Next, ρ_{en} and f^i are C^1. On each compact time interval, the parameter-integral theorem recalled in Lecture 1 permits $\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega \rho_{\mathrm{en}} = \int_\Omega \partial_t\rho_{\mathrm{en}}$. For each spatial index i, Fubini's theorem and the fundamental theorem of calculus give the following identity, with no sum over i:

$$\int_\Omega \partial_i f^i\,\mathrm{d}^d x = \int_{\Omega_i}\int_{a_i}^{b_i} \partial_i f^i\,\mathrm{d}x^i\,\mathrm{d}^{d-1}x_{\widehat{i}} = \int_{\Omega_i}\big[f^i\big|_{b_i} - f^i\big|_{a_i}\big]\,\mathrm{d}^{d-1}x_{\widehat{i}}.$$

Summing and using (3.7) proves the derivative identity in (3.8). The flux is continuous in t. Integration in time proves the finite-time identity.

Finally, on $\mathbb{T}^d$ integrate over $[0, 2\pi]^d$. Opposite face values agree by periodicity, so every difference in (3.5) vanishes. ∎

For a solution on $\mathbb{R}^d$ whose first derivatives ϕ_t and $\nabla\phi$ vanish outside one fixed compact spatial set throughout $[t_0, t_1]$, and with $V(\phi) = 0$ outside that set, a containing rectangle has zero boundary flux. Its energy is the total energy, so the total energy is constant on that interval. For more general decay, the cutoff proof in Lecture 2 applies to the energy current under the density and spatial-flux integrability hypotheses stated there.

3.3 A pulse leaving an interval

We compute the energy remaining in an interval while a massless pulse leaves through its right endpoint. Take $d = 1$, $V = 0$ and

$$\phi(t, x) = F(x - t), \qquad F(s) = \begin{cases}(1 - s^2)^3, & |s| < 1,\\ 0, & |s| \geq 1,\end{cases} \qquad I = [-2, 2], \quad t \geq 0.$$

The function F and its first two derivatives vanish at $s = \pm 1$, so F is C^2 on $\mathbb{R}$. Writing $s = x - t$, the derivatives are $\phi_t = -F'(s)$, $\phi_x = F'(s)$ and $\phi_{tt} = \phi_{xx} = F''(s)$. Thus this field solves the massless equation. Its support at time t is $[t-1, t+1]$: the pulse travels right at speed one.

Figure 3.1 shows the support moving across I. The two sloping lines mark $x = t - 1$ and $x = t + 1$. The vertical lines mark the endpoints of I. The pulse reaches $x = 2$ at $t = 1$ and finishes crossing at $t = 3$.

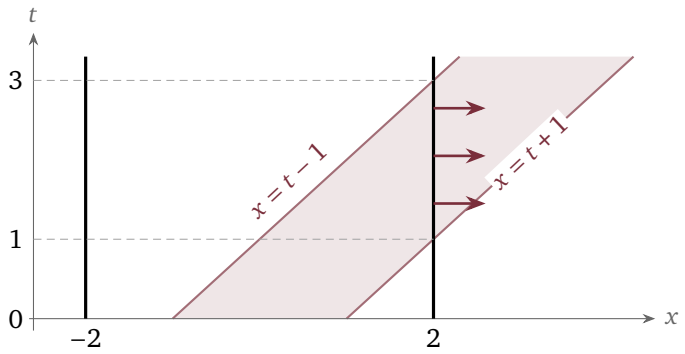


Figure 3.1: The shaded support of the right-moving pulse intersects the fixed interval $[-2, 2]$. The arrows on $x = 2$ show its outward energy flux during $1 < t < 3$.

Both the energy density and the flux are the same nonnegative function:

$$\rho_{\mathrm{en}}(t,x) = f^1(t,x) = F'(x-t)^2, \qquad F'(s)^2 = \begin{cases} 36s^2(1-s^2)^4, & |s| < 1, \\ 0, & |s| \geq 1. \end{cases} \tag{3.9}$$

The flux is positive for $0 < |x - t| < 1$ and zero otherwise. The total energy is an elementary integral:

$$E_{\mathrm{en},\mathbb{R}} = 36 \int_{-1}^{1} s^2(1-s^2)^4 \, \mathrm{d}s = 72 \left(\frac{1}{3} - \frac{4}{5} + \frac{6}{7} - \frac{4}{9} + \frac{1}{11} \right) = \frac{1024}{385}.$$

To compute the energy remaining in I, use the polynomial primitive

$$P(s) = 12s^3 - \frac{144}{5}s^5 + \frac{216}{7}s^7 - 16s^9 + \frac{36}{11}s^{11}, \qquad P'(s) = 36s^2(1-s^2)^4.$$

Its endpoint values are $P(1) = 512/385$ and $P(-1) = -512/385$. For $0 \leq t \leq 1$ the whole pulse lies in I. For $1 \leq t \leq 3$, substitution $s = x - t$ leaves the part $-1 \leq s \leq 2 - t$ in I. For $t \geq 3$, the interval contains no energy. Consequently

$$E_{\mathrm{en},I}(t) = \begin{cases} 1024/385, & 0 \leq t \leq 1, \\ P(2-t) + 512/385, & 1 \leq t \leq 3, \\ 0, & t \geq 3. \end{cases} \tag{3.10}$$

The endpoint flux confirms this result independently. At the left endpoint, $f^1(t,-2) = F'(-2-t)^2 = 0$ for all $t \geq 0$. At the right endpoint, $f^1(t,2) = F'(2-t)^2$. Thus

$$\mathcal{F}_I(t) = f^1(t,2) - f^1(t,-2) = F'(2-t)^2.$$

For $1 < t < 3$, differentiating (3.10) gives $E'_{\mathrm{en},I}(t) = -P'(2-t) = -\mathcal{F}_I(t)$. Outside this time interval both sides vanish. The total outward flux is

$$\int_0^3 \mathcal{F}_I(t) \, \mathrm{d}t = \int_1^3 F'(2-t)^2 \, \mathrm{d}t = \int_{-1}^{1} F'(s)^2 \, \mathrm{d}s = \frac{1024}{385} = E_{\mathrm{en},I}(0) - E_{\mathrm{en},I}(3).$$

The field conserves its total energy while the interval loses all its energy. This example guards the boundary hypothesis: local conservation does not imply constant energy in a fixed region. Lecture 5 uses boundary flux to study propagation. Lecture 13 uses outgoing flux to measure radiation, and Lecture 16 must distinguish energy in a region from total energy.

3.4 Variation of a background metric

The scalar tensor also describes how the action changes when the metric changes. This description will be used for the gravitational field in Lecture 28. Here the metric is prescribed, and the scalar field stays fixed during the variation.

Let $g_{\mu\nu}(x)$ be a C^1 symmetric nondegenerate matrix of signature $(+,-,\dots,-)$, with inverse $g^{\mu\nu}$. Write $g = \det(g_{\mu\nu})$ and $\mathrm{vol}_g = \sqrt{|g|}\,\mathrm{vol}$. On a compact coordinate region D the scalar action is

$$S_D[\phi, g] = \int_D L_g \,\mathrm{vol}_g, \qquad L_g = \frac{1}{2} g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi - V(\phi).$$

We vary the inverse metric by a symmetric C^1 array $k^{\mu\nu}$ with compact support in the interior of D. For sufficiently small ε, $g^{\mu\nu} + \varepsilon k^{\mu\nu}$ remains nondegenerate with the same signature.

Definition 3.4 (Hilbert stress tensor)**.** The *Hilbert stress tensor* is the continuous symmetric array $T^{\mathrm{H}}_{\mu\nu}$ defined by the response to inverse-metric variation:

$$\delta_g S_D[\phi, g]k = \frac{1}{2}\int_D T^{\mathrm{H}}_{\mu\nu} k^{\mu\nu}\,\mathrm{vol}_g. \tag{3.11}$$

The variations range over symmetric arrays. The factor $\frac{1}{2}$ is part of the convention, including for off-diagonal entries. If two continuous symmetric arrays give the same response for every such variation, apply the fundamental lemma of Lecture 1 to each diagonal entry and to each symmetric pair of entries. Their difference is zero, so this definition determines at most one tensor.

Lemma 3.5 (Scalar metric variation)**.** *The scalar action has Hilbert tensor*

$$T^{\mathrm{H}}_{\mu\nu} = \partial_\mu\phi\,\partial_\nu\phi - g_{\mu\nu}L_g. \tag{3.12}$$

At $g_{\mu\nu} = \eta_{\mu\nu}$ it agrees with the lowered canonical tensor. The equality holds for every field, without the field equation.

Proof. We differentiate the inverse and the determinant before differentiating the action. Differentiation of $g_{\mu\alpha}g^{\alpha\nu} = \delta^\nu_\mu$ gives

$$\delta g_{\mu\nu} = -g_{\mu\alpha}g_{\nu\beta}k^{\alpha\beta}.$$

For an invertible matrix G and an increment H, the determinant identity $\det(G + \varepsilon H) = \det G \det(\mathrm{id} + \varepsilon G^{-1}H)$, where id is the identity matrix, gives $\delta \det G = \det G\ \mathrm{tr}(G^{-1}H)$. Every non-identity permutation moves at least two indices, so its term in $\det(\mathrm{id} + \varepsilon G^{-1}H)$ has at least two factors of ε. The linear part of the identity term selects one diagonal entry of $\varepsilon G^{-1}H$ at a time. Their sum is $\varepsilon\,\mathrm{tr}(G^{-1}H)$. Applying this identity and then the inverse variation yields

$$\delta\sqrt{|g|} = \frac{1}{2}\sqrt{|g|}\,g^{\mu\nu}\delta g_{\mu\nu} = -\frac{1}{2}\sqrt{|g|}\,g_{\mu\nu}k^{\mu\nu}.$$

The absolute value causes no additional sign because g keeps its sign for small variations. Also, $\delta L_g = \frac{1}{2}\partial_\mu\phi\,\partial_\nu\phi\,k^{\mu\nu}$. The product rule therefore gives

$$\delta_g S_D[\phi, g]k = \frac{1}{2}\int_D \bigl(\partial_\mu\phi\,\partial_\nu\phi - g_{\mu\nu}L_g\bigr)k^{\mu\nu}\,\mathrm{vol}_g.$$

This is (3.11) with (3.12). Substitution of $g_{\mu\nu} = \eta_{\mu\nu}$ gives (3.3). ∎

For a variation $h_{\mu\nu} = \delta g_{\mu\nu}$ of the metric itself, $k^{\mu\nu} = -g^{\mu\alpha}g^{\nu\beta}h_{\alpha\beta}$. The equivalent convention is

$$\delta_g S_D = -\frac{1}{2}\int_D (T^{\mathrm{H}})^{\mu\nu}h_{\mu\nu}\,\mathrm{vol}_g.$$

Thus the functional derivative with respect to $g_{\mu\nu}$ carries a minus sign, whereas the derivative with respect to $g^{\mu\nu}$ carries the plus sign in (3.11).

Summary. For the autonomous scalar density, translations give $T^\mu{}_\nu = \partial^\mu\phi\,\partial_\nu\phi - \delta^\mu_\nu L$. On solutions its divergence vanishes. The energy density is $T^0{}_0$, and its spatial flux is $T^{i0} = -\phi_t\partial_i\phi$. The energy in a fixed rectangle changes by minus the net outward flux. Periodicity cancels that flux on the torus. Inverse-metric variation gives the same lowered symmetric tensor for the scalar field. Lecture 4 constructs Klein–Gordon solutions with finite energy.

Self-check. For a left-moving massless field $\phi(t, x) = F(x + t)$, compute the energy density and energy flux. Which sign has the flux?

Answer. Here $\phi_t = \phi_x = F'(x + t)$. Thus $\rho_{\mathrm{en}} = F'(x + t)^2$ and $f^1 = -F'(x + t)^2 \le 0$. Energy flows toward decreasing x.

Self-check. For the pulse of Section 3.3, compute the energy in I at $t = 2$ and the outward flux at that instant. Does a zero instantaneous flux imply that the interval's energy is constant over a time interval?

Answer. Since $P(0) = 0$, the remaining energy is $512/385$. The flux is $F'(0)^2 = 0$. It is positive at nearby times other than $t = 2$, so the energy is not constant on any neighbourhood of that time. Zero flux at one instant only gives zero derivative at that instant.

Self-check. Replace the autonomous scalar potential by $W(x, \phi)$, with W of class C^2. Compute $\partial_\mu T^\mu{}_\nu$ and identify when spatial translation conservation can fail even on a solution.

Answer. The chain rule adds the explicit derivative $-\partial_\nu^{\mathrm{explicit}}W$ to $\partial_\nu L$. Consequently $\partial_\mu T^\mu{}_\nu = -E[\phi]\partial_\nu\phi + \partial_\nu^{\mathrm{explicit}}W$. Spatially varying W can therefore give a nonzero divergence on a solution. Time-independent W still has zero explicit time derivative, so the energy balance survives.

Self-check. At a point where $k^{\mu\nu} = \lambda g^{\mu\nu}$, compute the pointwise volume variation $\delta\mathrm{vol}_g$ in $1 + d$ dimensions.

Answer. The contraction is $g_{\mu\nu}k^{\mu\nu} = \lambda(d + 1)$. Hence $\delta\mathrm{vol}_g = -\frac{1}{2}\lambda(d + 1)\mathrm{vol}_g$. In four spacetime dimensions the answer is $-2\lambda\mathrm{vol}_g$.

Lecture 4

The linear Cauchy problem

The field equation from Lecture 1 does not by itself construct a field from initial data. We now solve the Klein–Gordon equation on all of $\mathbb{R}^d$. Fourier transformation replaces the spatial derivatives by oscillator frequencies.

Objectives. *We construct the solution from finite-energy data and prove uniqueness, continuous dependence and conservation of total energy, including zero mass.*

By the end of this lecture you will be able to:

- Define the energy space by Fourier transformation.
- Construct and identify the unique energy-class Klein–Gordon solution.
- Control its norm on bounded time intervals, including at zero frequency.

4.1 Data and the solution class

The unitary Fourier transform from *Analysis* I, Lecture 7, has kernel $(2\pi)^{-d/2}e^{-ik\cdot x}$. It maps $L^2(\mathbb{R}^d)$ onto itself, preserves norms, and takes a Schwartz derivative to $ik_j\widehat{f}$. Write $\langle k\rangle = (1+|k|^2)^{1/2}$.

Definition 4.1 (Fourier energy space)**.** The *Sobolev space* $H^1(\mathbb{R}^d)$ and its norm are

$$H^1(\mathbb{R}^d) = \{f \in L^2(\mathbb{R}^d) : \langle k\rangle \widehat{f} \in L^2(\mathbb{R}^d)\}, \qquad \|f\|_{H^1} = \|\langle k\rangle \widehat{f}\|_2.$$

The *energy space* is $\mathcal{E} = H^1(\mathbb{R}^d) \times L^2(\mathbb{R}^d)$, with norm $\|(f,g)\|_{\mathcal{E}} = \|f\|_{H^1} + \|g\|_2$.

For $f \in H^1$, define $\partial_j f$ as the inverse transform of $ik_j\widehat{f}$. Plancherel gives

$$\|f\|_{H^1}^2 = \|f\|_2^2 + \sum_{j=1}^{d} \|\partial_j f\|_2^2.$$

These derivatives agree with distributional derivatives: Fourier inversion on Schwartz tests gives $\int (\partial_j f)\zeta = -\int f\partial_j\zeta$. All pairings here are bilinear. The integrals exist by Cauchy–Schwarz. The weighted L^2 definition also proves completeness of H^1. Indeed, if $\langle k\rangle\widehat{f_n} \to h$ in L^2, then $\widehat{f_n} \to h/\langle k\rangle$ in L^2, whose inverse transform belongs to H^1 and is the required limit.

We consider real data. Fourier transforms may be complex, with $\widehat{f}(-k) = \overline{\widehat{f}(k)}$. Fix a mass $m \geq 0$ and an open time interval I containing zero.

Definition 4.2 (Energy-class solution and well-posedness)**.** An *energy-class solution* with data $(f,g)\in\mathcal{E}$ is a map $v\in C(I;H^1)\cap C^1(I;L^2)$ satisfying $v(0)=f$ and $v_t(0)=g$, and the equation

$$\int_I\int_{\mathbb{R}^d} v(t,x)\bigl(\psi_{tt}-\Delta\psi+m^2\psi\bigr)(t,x)\,\mathrm{d}^dx\,\mathrm{d}t=0\quad\text{for every }\psi\in C_c^\infty(I\times\mathbb{R}^d). \tag{4.1}$$

The Cauchy problem on $\mathbb{R}$ is *well-posed* in $\mathcal{E}$ if each datum has exactly one energy-class solution with $I=\mathbb{R}$ and the map from data to $(v,v_t)\in C([-T,T];\mathcal{E})$ is continuous for every $T>0$.

The time derivative in this definition is a derivative in the L^2 norm. Equation (4.1) defines the distributional meaning of $v_{tt}-\Delta v+m^2v=0$. It requires neither pointwise second derivatives nor products of distributions.

4.2 Construction by Fourier multipliers

The oscillator with initial position a and velocity b solves $q''+\nu^2q=0$. Its solution is $q(t)=\cos(t\nu)a+S_t(\nu)b$, where the *sine multiplier* is the continuous function

$$S_t(\nu)=\begin{cases}\sin(t\nu)/\nu, & \nu>0,\\ t, & \nu=0.\end{cases}\qquad |S_t(\nu)|\le|t|,\qquad |\nu S_t(\nu)|\le 1. \tag{4.2}$$

The integral formula $S_t(\nu)=\int_0^t\cos(s\nu)\,\mathrm{d}s$ proves the first bound and the value at zero. The second bound follows from $|\sin|\le 1$. The transformed Klein–Gordon frequency is $\nu(k)=(|k|^2+m^2)^{1/2}$.

Theorem 4.3 (Klein–Gordon Cauchy problem)**.** *For every $(f,g)\in\mathcal{E}$ and $m\ge 0$, there is exactly one energy-class solution on $\mathbb{R}$. It is given by*

$$\widehat{v}(t,k)=\cos(t\nu(k))\widehat{f}(k)+S_t(\nu(k))\widehat{g}(k). \tag{4.3}$$

The problem is well-posed. The total energy is constant:

$$E_{\mathrm{en}}(t)=\frac{1}{2}\bigl(\|v_t(t)\|_2^2+\|\nabla v(t)\|_2^2+m^2\|v(t)\|_2^2\bigr)=E_{\mathrm{en}}(0). \tag{4.4}$$

Here $\|\nabla v\|_2^2=\sum_j\|\partial_j v\|_2^2$.

Proof. We first construct the solution and justify its time regularity. For $|t|\le T$, the multiplier bounds give

$$\begin{aligned}\|v(t)\|_2&\le\|f\|_2+T\|g\|_2,\\ \|v(t)\|_{H^1}&\le\|f\|_{H^1}+(T+1)\|g\|_2,\\ \|p(t)\|_2&\le\max(1,m)\|f\|_{H^1}+\|g\|_2,\end{aligned} \tag{4.5}$$

where $\widehat{p}=-\nu\sin(t\nu)\widehat{f}+\cos(t\nu)\widehat{g}$. For the middle bound, use $\langle k\rangle|S_t|\le(1+|k|)|S_t|\le T+1$. For the last bound, use $\nu\le\max(1,m)\langle k\rangle$.

Every multiplier is pointwise continuous in time. The same bounds supply square-integrable majorants for $\langle k\rangle\widehat{v}$ and $\widehat{p}$ on each bounded time interval. Dominated convergence, imported from *Analysis* I, Lecture 3, gives $v\in C(\mathbb{R};H^1)$ and $p\in C(\mathbb{R};L^2)$. At each k, the

difference quotient of $\widehat{v}$ tends to $\widehat{p}$. Its absolute value, for $|h| \le 1$, is bounded by $\nu|\widehat{f}| + |\widehat{g}|$, by the scalar mean-value estimate. Dominated convergence in L^2 therefore gives $v_t = p$. The values at zero are f and g. Evenness and reality of the multipliers preserve the Fourier reality condition.

To verify the equation, choose an even real $\chi \in C_c^\infty(\mathbb{R}^d)$ equal to one on $|k| \le 1$, supported in $|k| \le 2$, with $0 \le \chi \le 1$. Write $\chi_R(k) = \chi(k/R)$ for $R \ge 1$. Multiply both data transforms by χ_R and construct v_R by the same formula. Compact frequency support and Cauchy–Schwarz make every polynomial in k times either truncated transform integrable. Differentiation under the inverse integral is therefore valid to every order, also in time on compact intervals. The oscillator equation proves that v_R is a smooth classical solution. It satisfies (4.1) by integration by parts against the compact test. The estimates (4.5) and dominated convergence give

$$v_R \longrightarrow v \text{ in } C([-T,T];H^1), \qquad (v_R)_t \longrightarrow v_t \text{ in } C([-T,T];L^2). \tag{4.6}$$

Cauchy–Schwarz on the test's compact time support passes the weak equation to the limit. This also establishes the smooth approximation used in Lecture 5.

Next we prove uniqueness within the stated solution class. Let u be the difference of two solutions with the same data. Set $w_R = \mathcal{F}^{-1}(\chi_R \widehat{u})$. We justify the equation for this cutoff before using an oscillator argument. The weak equation permits tests $\eta(t)\zeta(x)$ with $\eta \in C_c^\infty(I)$ and ζ Schwartz: multiply ζ by a spatial cutoff tending to one. The test and its first two spatial derivatives converge in L^2, so Cauchy–Schwarz passes to the limit. The Fourier transform takes Schwartz space onto itself. Testing with inverse transforms of χ_R times arbitrary Schwartz functions consequently gives, as an L^2-valued distribution in time,

$$(w_R)_{tt} = -Aw_R, \qquad \widehat{Aw_R}(k) = (|k|^2 + m^2)\widehat{w}_R(k).$$

Indeed, scalar equality against the dense Schwartz tests identifies the L^2 vectors after integration against each η. The operator A is bounded on functions supported in $|k| \le 2R$ in Fourier space. Thus $-Aw_R$ is continuous in L^2.

This equation implies $w_R \in C^2(I;L^2)$. To see this, integrate $-Aw_R$ from zero in the L^2 norm. The integral exists as a limit of Riemann sums because L^2 is complete. Subtract this primitive from $(w_R)_t$. Each scalar pairing of the difference has zero distributional derivative, so is constant. The scalar assertion follows by writing every test of integral zero as a derivative of a compactly supported test. Evaluation at zero identifies the constant. Thus $(w_R)_t$ is norm differentiable, with derivative $-Aw_R$. The cutoff now has classical time derivatives, and differentiation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(\|(w_R)_t\|_2^2 + \|\nu\widehat{w}_R\|_2^2\big) = 2\,\mathrm{Re}\big(((w_R)_t, -Aw_R) + (Aw_R, (w_R)_t)\big) = 0.$$

The inner product is antilinear in its first slot. The initial value is zero, so $(w_R)_t = 0$ and then $w_R = w_R(0) = 0$. As $R \to \infty$, Plancherel and dominated convergence give $u(t) = 0$. This proves uniqueness without assuming a pointwise Fourier equation for u.

For continuous dependence, apply (4.5) to the difference of two data pairs. Taking the supremum over $|t| \le T$ gives

$$\sup_{|t|\le T} \|(v-\widetilde{v}, v_t-\widetilde{v}_t)(t)\|_{\mathcal{E}} \le C_{T,m}\|(f-\widetilde{f}, g-\widetilde{g})\|_{\mathcal{E}}, \quad C_{T,m} = T + 2 + \max(1,m).$$

This is continuity in the required norm.

Finally, the formula for the solution and its derivative reads

$$\nu\widehat{v} = \cos(t\nu)\nu\widehat{f} + \sin(t\nu)\widehat{g},$$
$$\widehat{v}_t = -\sin(t\nu)\nu\widehat{f} + \cos(t\nu)\widehat{g}.$$

Expansion of the absolute squares cancels their opposite cross terms and uses $\cos^2(t\nu) + \sin^2(t\nu) = 1$. Hence, for almost every k,

$$|\widehat{v}_t|^2 + \nu^2|\widehat{v}|^2 = |\widehat{g}|^2 + \nu^2|\widehat{f}|^2.$$

Every term is integrable by (4.5). Integration and Plancherel prove (4.4). ∎

For smooth solutions this is the energy of Lecture 3 with $V(v) = \frac{1}{2}m^2v^2$. That lecture proved a local balance for C^2 fields. The multiplier proof above establishes the global identity in the larger energy class, without assuming a boundary flux at spatial infinity.

4.3 A Gaussian and the low-frequency guard

Take one space dimension and Gaussian displacement with zero velocity:

$$\widehat{f}(k) = e^{-ak^2/2}, \qquad g = 0, \qquad a > 0.$$

The Gaussian transform from *Analysis* I, Lecture 4, gives $f(x) = a^{-1/2}e^{-x^2/(2a)}$. Its solution and initial energy are

$$\widehat{v}(t,k) = \cos\bigl(t\sqrt{k^2+m^2}\bigr)e^{-ak^2/2}, \qquad E_{\mathrm{en}}(0) = \frac{\sqrt{\pi}}{4a^{3/2}} + \frac{m^2\sqrt{\pi}}{2\sqrt{a}}.$$

Indeed, the Gaussian integral and its derivative with respect to a give $\int e^{-ak^2}\mathrm{d}k = \sqrt{\pi/a}$ and $\int k^2e^{-ak^2}\mathrm{d}k = \sqrt{\pi}/(2a^{3/2})$. For $m = 0$, the identity $\cos(t|k|) = \cos(tk)$ splits the inverse transform into two translated Gaussians:

$$v(t,x) = \frac{1}{2\sqrt{a}}\left(e^{-(x-t)^2/(2a)} + e^{-(x+t)^2/(2a)}\right).$$

The two profiles travel at unit speed. Their sum has the same total energy at every time even when they overlap.

The massless energy does not bound the full energy-space norm. For an explicit low-frequency family, take $m = 0$, $g_\varepsilon = 0$ and

$$\widehat{f_\varepsilon}(k) = \varepsilon^{-1/2}e^{-k^2/(2\varepsilon^2)}, \qquad 0 < \varepsilon \le 1.$$

Substitution $k = \varepsilon q$ gives

$$\|f_\varepsilon\|_2^2 = \sqrt{\pi}, \qquad E_{\mathrm{en}}[f_\varepsilon, 0] = \frac{\sqrt{\pi}}{4}\varepsilon^2 \longrightarrow 0.$$

No constant can bound $\|f\|_2^2$ by the massless energy for all data. The bound $\|v(t)\|_2 \le \|f\|_2 + |t|\|g\|_2$ must therefore be retained. The value $S_t(0) = t$ also shows why division by the frequency cannot be estimated by a uniform bound on $1/\nu$. Lecture 15 uses this finite-time control when it adds a nonlinear forcing term.

Summary. For $m \ge 0$, data in $H^1(\mathbb{R}^d) \times L^2(\mathbb{R}^d)$ determine exactly one solution in $C(H^1) \cap C^1(L^2)$ of the distributional Klein–Gordon equation. Fourier multipliers give continuous

dependence on bounded time intervals, smooth approximation and conserved energy. At zero mass, energy omits control of the low-frequency L^2 norm. The sine bound supplies that control for finite time. Lecture 5 uses the smooth approximation to prove finite speed of propagation.

Self-check. At $m = 0$ and $k = 0$, which oscillator solution has initial position a and velocity b? How does the multiplier formula reproduce it?

Answer. The equation is $q'' = 0$, so $q(t) = a + tb$. The values $\cos 0 = 1$ and $S_t(0) = t$ give exactly this solution. A single point in frequency space has measure zero, but the continuous extension controls frequencies approaching it.

Self-check. For $m > 0$, bound $\|v(t)\|_2$ by the conserved energy. Why does the resulting bound fail at $m = 0$?

Answer. The mass term gives $\|v(t)\|_2 \le \sqrt{2E_{\mathrm{en}}(0)}/m$. At zero mass this term vanishes. The family f_ε above has fixed nonzero L^2 norm and energy tending to zero.

Self-check. For the Gaussian displacement, compute $\|f\|_{H^1}^2$ and the massless solution's value at $x = 0$.

Answer. The norm is $\sqrt{\pi}/\sqrt{a} + \sqrt{\pi}/(2a^{3/2})$. The value is $v(t, 0) = a^{-1/2} e^{-t^2/(2a)}$.

Self-check. Does the definition of an energy-class solution permit differentiating its energy directly as in Lecture 3? Which step in the uniqueness proof makes time differentiation legitimate?

Answer. The definition supplies no pointwise second derivatives. The frequency cutoff makes the spatial operator bounded in L^2. Its distributional equation then implies C^2 time regularity in L^2, so the cutoff energy can be differentiated. The total energy of the original solution follows instead from the integrable Fourier identity.

Lecture 5

Domain of dependence

Lecture 4 constructs a field from data on all of space. The local energy balance gives a stronger conclusion: data far enough from a point cannot change the field at that point. We prove this for the real Klein–Gordon field with $m \geq 0$ and propagation speed one.

Objectives. *We prove an energy inequality on shrinking balls and deduce local uniqueness and finite propagation speed. A compact initial pulse and the heat equation show what the propagation statement does and does not say.*

By the end of this lecture you will be able to:

- Derive the nonnegative energy flux through a backward cone.
- Identify the initial region determining a field at a point.
- Distinguish finite propagation speed from Huygens' principle.

5.1 The region determining a point

Write $B_r(a) = \{x \in \mathbb{R}^d : |x - a| < r\}$, where $d \geq 1$. We prescribe data at $t = 0$. A light cone has boundary slope corresponding to speed one.

Definition 5.1 (Dependence and influence regions)**.** For an event (T, a) with $T > 0$, its *backward cone* is

$$C^-_{T,a} = \{(t, x) : 0 \leq t \leq T,\ |x - a| \leq T - t\}.$$

An initial region is a *domain of dependence* for (T, a) if any two C^2 Klein–Gordon solutions with equal data there have equal values at (T, a). For a closed initial set $K \subset \mathbb{R}^d$, its *range of influence* is

$$J^+(K) = \{(t, x) : t \geq 0,\ \operatorname{dist}(x, K) \leq t\}.$$

The energy argument below proves that the base $\overline{B_T(a)}$ is a domain of dependence for Klein–Gordon solutions. Figure 5.1 shows the shrinking slices of one backward cone. Each boundary line has speed one.

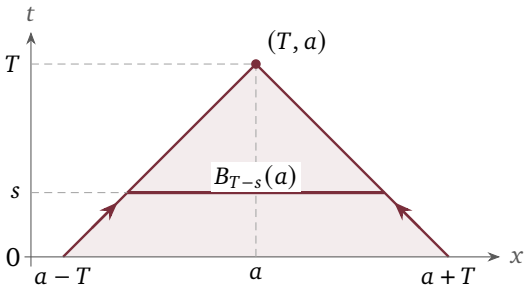


Figure 5.1: A spacetime section of $C^-_{T,a}$ through one spatial axis. The slice at time s is the ball of radius $T - s$. Its lateral boundary moves inward at speed one.

For $u \in C^2$ satisfying $(\Box + m^2)u = 0$, Lecture 3 gives the C^1 energy density and spatial energy flux

$$\rho_{\mathrm{en}} = \frac{1}{2}\bigl(u_t^2 + |\nabla u|^2 + m^2 u^2\bigr), \qquad \mathbf{f} = -u_t \nabla u, \qquad \partial_t \rho_{\mathrm{en}} + \operatorname{div} \mathbf{f} = 0. \tag{5.1}$$

These are the local identities of Theorem 3.3. Its rectangular boundary formula does not yet give a formula on a ball.

5.2 Energy on a shrinking ball

Fix a centre $a \in \mathbb{R}^d$ and $R > 0$. At time $t \in [0, R)$ let $r(t) = R - t$. Define the shrinking-ball energy by

$$E_C(t) = \int_{B_{r(t)}(a)} \rho_{\mathrm{en}}(t, x)\, \mathrm{d}^d x. \tag{5.2}$$

On the sphere $\partial B_{r(t)}(a)$, write $n = (x - a)/r(t)$ for the outward spatial unit normal. The normal derivative is $\partial_n u = n \cdot \nabla u$. The tangential gradient is $\nabla_{\tan} u = \nabla u - n \partial_n u$.

Definition 5.2 (Cone energy flux). For a C^2 field, the *outward energy flux through the lateral cone* between times s and t, where $0 \le s < t < R$, is

$$\mathcal{F}_C[s, t] = \int_s^t \int_{\partial B_{R-\tau}(a)} \bigl(\rho_{\mathrm{en}} + \mathbf{f} \cdot n\bigr)\, \mathrm{d}S\, \mathrm{d}\tau. \tag{5.3}$$

Here $\mathrm{d}S$ is spatial sphere area. In $d = 1$, a sphere integral is the sum of its two endpoint values.

The density in (5.3) includes the energy removed as the ball shrinks. It therefore differs from the flux through a fixed spatial boundary. The measure $\mathrm{d}S\, \mathrm{d}\tau$ in this definition is not Euclidean area on the sloping spacetime surface.

Theorem 5.3 (Shrinking-cone energy and local uniqueness). *Let $m \ge 0$ and let u be a real C^2 solution of $(\Box + m^2)u = 0$ on an open neighbourhood of $C^-_{R,a}$. For $0 \le s < t < R$,*

$$E_C(t) + \mathcal{F}_C[s, t] = E_C(s), \qquad E_C(t) \le E_C(s). \tag{5.4}$$

If $u(0, x) = u_t(0, x) = 0$ on $B_R(a)$, then $u = 0$ on the entire closed cone, including its vertex. No assumption on total energy is needed.

Proof. We derive the moving-boundary formula and then use the sign of its integrand.

First, polar coordinates about a express (5.2) as

$$E_C(t) = \int_{S^{d-1}} \int_0^{r(t)} \rho_{\mathrm{en}}(t, a + q\theta) q^{d-1} \, \mathrm{d}q \, \mathrm{d}S_\theta.$$

For $d = 1$, the outer integral sums the directions $\theta = -1$ and $\theta = 1$. The integrand and its time derivative are continuous on each compact truncated cone. Differentiation of this integral and the fundamental theorem of calculus at its upper limit give

$$E_C'(t) = \int_{B_r(a)} \partial_t \rho_{\mathrm{en}} \, \mathrm{d}^d x - \int_{\partial B_r(a)} \rho_{\mathrm{en}} \, \mathrm{d}S, \qquad r = R - t. \tag{5.5}$$

The minus sign comes from $r'(t) = -1$.

Next, the divergence formula for a C^1 vector field on a ball follows from coordinate slices. For $d \geq 2$, hold all coordinates except x^i fixed, writing the remaining vector as y. The endpoints of the slice are $a_i \pm h(y)$, where $h(y) = \sqrt{r^2 - |y - a_{\hat{i}}|^2}$. Fubini and the one-dimensional fundamental theorem give

$$\int_{B_r(a)} \partial_i f^i \, \mathrm{d}^d x = \int_{|y - a_{\hat{i}}| < r} \left[f^i(a_i + h(y), y) - f^i(a_i - h(y), y) \right] \mathrm{d}^{d-1} y.$$

On these two sphere graphs, $n_i \, \mathrm{d}S = \pm \mathrm{d}^{d-1} y$. Indeed, their area factor is $r/h(y)$ and their normal component is $\pm h(y)/r$. The equator has sphere area zero. Summing over i proves the ball divergence identity

$$\int_{B_r} \operatorname{div} \mathbf{f} \, \mathrm{d}^d x = \int_{\partial B_r} \mathbf{f} \cdot n \, \mathrm{d}S. \tag{5.6}$$

For $d = 1$ this identity is the endpoint fundamental theorem itself. Substitution of (5.1) into (5.5) now yields

$$E_C'(t) = -\int_{\partial B_{R-t}(a)} \left(\rho_{\mathrm{en}} + \mathbf{f} \cdot n \right) \mathrm{d}S. \tag{5.7}$$

The orthogonal decomposition of ∇u into normal and tangential parts gives the sign:

$$\rho_{\mathrm{en}} + \mathbf{f} \cdot n = \frac{1}{2} \left[(u_t - \partial_n u)^2 + |\nabla_{\tan} u|^2 + m^2 u^2 \right] \geq 0. \tag{5.8}$$

Integrating (5.7) proves (5.4).

Finally, zero initial data make $E_C(0) = 0$. Continuity and positivity force $u_t = 0$ throughout every open shrinking ball. For a point with $|x - a| < R - t$, the whole vertical segment (τ, x), $0 \leq \tau \leq t$, stays inside the cone. Thus $u(t, x) = u(0, x) + \int_0^t u_t(\tau, x) \, \mathrm{d}\tau = 0$. This argument also works for $m = 0$, when energy alone does not detect a constant field. Continuity gives zero on the lateral boundary and at the vertex. ∎

Apply the theorem to the difference of two solutions. Agreement of their initial displacement and velocity on $B_R(a)$ implies agreement throughout the cone. In particular, the data on $\overline{B_T(a)}$ determine the value at (T, a).

> **Corollary 5.4** (Local uniqueness). *Two C^2 Klein–Gordon solutions on a common open time slab $I \times \mathbb{R}^d$ that have the same data on a complete slice $t = t_0$, with $t_0 \in I$, agree throughout the slab. They need not have finite total energy.*

Proof. For any later event, apply Theorem 5.3 to the difference on its backward cone, translating the initial time. Reverse time for earlier events. ∎

5.3 Energy solutions and finite propagation

Lecture 4 constructs solutions in $C(\mathbb{R}; H^1) \cap C^1(\mathbb{R}; L^2)$. Their spatial gradients are the L^2 Fourier derivatives, and their energy density is defined almost everywhere by (5.1). Values at a single point need not exist. Agreement on a ball therefore means agreement almost everywhere there.

Corollary 5.5 (Energy-class dependence). *For the energy solutions of Theorem 4.3, $E_C(t) \le E_C(s)$ for $0 \le s < t < R$. If two such solutions have equal initial data on $B_R(a)$, their values agree almost everywhere on $B_{R-t}(a)$ at each such time.*

Proof. We use the smooth Fourier approximations u_j of Lecture 4, equation (4.6). For every $T > 0$, they satisfy

$$u_j \longrightarrow u \text{ in } C([-T,T]; H^1), \qquad \partial_t u_j \longrightarrow u_t \text{ in } C([-T,T]; L^2).$$

Restriction to any ball has L^2 operator norm at most one. Consequently each integral of the squared terms in the energies at times s and t converges. The smooth inequality passes to the limit. No trace of u_t on the cone boundary is required.

For the difference of solutions with equal local data, its initial energy on $B_R(a)$ is zero. Here its gradient also vanishes there: an H^1 function that is zero on an open ball has zero weak derivatives on that ball, by integration against compactly supported test functions. Thus its velocity vanishes on each shrinking ball at every time. Fix a final ball $B_{R-t}(a)$. All earlier balls contain it. The L^2 fundamental theorem $u(t) = u(0) + \int_0^t u_t(s)\,\mathrm{d}s$, restricted to this fixed ball, proves $u(t) = 0$ there, including when $m = 0$. ∎

If the initial data vanish outside a closed set K, either version of the dependence result implies

$$\operatorname{supp} u(t,\cdot) \subset \{x : \operatorname{dist}(x,K) \le |t|\}. \tag{5.9}$$

For energy solutions, support means the complement of the largest open set where the function is zero almost everywhere. To prove (5.9) for $t \ge 0$, take x with $\operatorname{dist}(x,K) > t$ and choose $t < R < \operatorname{dist}(x,K)$. The base ball $B_R(x)$ has zero data. The solution therefore vanishes on $B_{R-t}(x)$ at time t. Time reversal proves the claim for $t < 0$. This is *finite propagation speed*: changing initial data in K can change the field only inside $J^+(K)$ in the future.

5.4 A compact initial velocity pulse

Consider the massless equation in one space dimension, with zero initial displacement and the compact initial velocity

$$u(0,x) = 0, \qquad u_t(0,x) = g(x), \qquad g(x) = \begin{cases} (1-x^2)^3, & |x| < 1, \\ 0, & |x| \ge 1. \end{cases}$$

The function g is C^2, since its endpoint derivatives through order two vanish. Define

$$u(t,x) = \frac{1}{2} \int_{x-t}^{x+t} g(q)\,\mathrm{d}q. \tag{5.10}$$

The fundamental theorem gives

$$u_t = \frac{1}{2}[g(x+t)+g(x-t)], \qquad u_x = \frac{1}{2}[g(x+t)-g(x-t)], \qquad u_{tt} = u_{xx} = \frac{1}{2}[g'(x+t)-g'(x-t)].$$

These identities verify the equation and both initial data. At (T, a), formula (5.10) uses only $[a - T, a + T]$, exactly the base of its backward cone. For $t \geq 0$, if $|x| > 1 + t$, that interval misses $[-1, 1]$, so $u(t, x) = 0$. This verifies finite speed directly for the pulse.

Inside the cone, the answer can remain nonzero after both endpoints have passed the initial support. For $t > |x| + 1$, the interval contains $[-1, 1]$, and hence

$$u(t,x) = \frac{1}{2}\int_{-1}^{1} (1-q^2)^3 \,\mathrm{d}q = \left[q - q^3 + \frac{3}{5}q^5 - \frac{1}{7}q^7\right]_0^1 = \frac{16}{35}.$$

For example, $u(2, 0) = 16/35$, although both initial data vanish near the endpoints -2 and 2. Finite speed allows dependence on the interior of the base ball. *Huygens' principle* is the stronger assertion that the value at an event depends only on initial data arbitrarily near the base sphere. The massless equation in one space dimension fails that assertion.

5.5 The heat equation has instantaneous tails

Finite speed is a property of the equation. It does not follow merely from solvability of its Cauchy problem. Consider instead $v_t = v_{xx}$ on $\mathbb{R}$, with the same nonnegative compact datum g as initial value. For $t > 0$, define the heat kernel and its solution by

$$K_{\mathrm{heat}}(t,z) = \frac{e^{-z^2/(4t)}}{\sqrt{4\pi t}}, \qquad v(t,x) = \int_{\mathbb{R}} K_{\mathrm{heat}}(t, x-y) g(y)\,\mathrm{d}y.$$

Direct differentiation gives

$$\partial_t K_{\mathrm{heat}} = \left(-\frac{1}{2t} + \frac{z^2}{4t^2}\right) K_{\mathrm{heat}} = \partial_z^2 K_{\mathrm{heat}}.$$

On any compact subset of $t > 0$, derivatives can pass through the integral because g has compact support. Thus $v_t = v_{xx}$. Substitution $y = x - \sqrt{t}\, z$ gives

$$v(t,x) = \int_{\mathbb{R}} \frac{e^{-z^2/4}}{\sqrt{4\pi}} g(x - \sqrt{t}\, z)\,\mathrm{d}z \longrightarrow g(x).$$

The Gaussian has integral one. Boundedness and continuity of g justify the limit by dominated convergence, so this solution has the prescribed data.

For every $t > 0$ and $z \in \mathbb{R}$, the kernel $K_{\mathrm{heat}}(t, z)$ is strictly positive. The datum g is positive on $(-1, 1)$. Their integral is therefore positive for every $x \in \mathbb{R}$, even when $|x| > 1 + t$. The heat equation develops instantaneous tails. Lecture 6 constructs a wave kernel with support confined to the future cone.

Summary. For a real C^2 Klein–Gordon field with $m \geq 0$, energy on a ball shrinking at unit speed decreases by its nonnegative lateral cone flux. Zero local initial data imply zero field throughout the cone, without a global energy assumption. Smooth approximation extends the inequality and local dependence to Lecture 4's energy solutions. Compact initial support expands by at most $|t|$. This conclusion allows interior dependence and does not assert Huygens' principle. The heat equation has no such finite speed bound.

Self-check. At a cone boundary point, let $u_t = 2$, $\partial_n u = 1$, $|\nabla_{\tan} u| = 3$, $m = 2$ and $u = 1$. Compute the cone flux density.

Answer. The density is $\frac{1}{2}[(2-1)^2 + 3^2 + 2^2] = 7$. Omitting the tangential or mass term would give an incorrect smaller flux.

Self-check. For massless $u = C$ with a nonzero constant C, what is its cone energy? Why does this not contradict the zero-data conclusion?

Answer. Every derivative vanishes, so its energy is zero. Its initial displacement is C, whereas the theorem requires zero displacement as well as zero velocity. The vertical integral in the proof retains that displacement.

Self-check. For data supported in $[-1, 1]$, identify the domain of dependence of $(3, 5)$ and decide whether the field can be nonzero there.

Answer. The initial interval is $[2, 8]$. It misses the data support, so the field vanishes at $(3, 5)$ for a C^2 solution. An energy solution vanishes in a neighbourhood of that spatial point at time 3.

Self-check. For the compact velocity pulse, compare $u(2, 0)$ with the data near the base sphere. Which propagation statement does this test?

Answer. The value is $16/35$, while both data vanish near $x = -2$ and $x = 2$. This disproves Huygens' principle in one dimension. It respects finite speed, since the initial support lies inside the base interval $[-2, 2]$.

Lecture 6

The retarded kernel

Lecture 5 proves that a wave cannot propagate faster than unit speed. A source requires a further choice: its field must vanish before the source acts. We construct this solution for the wave operator in three spatial dimensions, with the convention $\Box = \partial_t^2 - \Delta$.

Objectives. *We prove the distributional identity for the retarded kernel, including its contribution at the cone vertex. We use the kernel to solve the wave equation for a smooth compactly supported source and determine where its field can occur.*

By the end of this lecture you will be able to:

- Verify the kernel identity by spherical averaging and integration by parts.
- Construct the unique past-vanishing C^2 solution for a smooth compact source.
- Bound the arrival times of the response to a source supported in a ball.

6.1 The support condition

We work on $\mathbb{R}^{1+3}$. A *test function* is a function $\psi \in C_c^\infty(\mathbb{R}^{1+3})$. A distribution is a linear functional on test functions, continuous when their supports stay in a fixed compact set and all their derivatives converge uniformly. Its pairing $\langle T, \psi\rangle$ is bilinear, without complex conjugation. Its support is the complement of the largest open set on whose tests it vanishes. The point distribution is given by $\langle \delta^4, \psi\rangle = \psi(0,0)$.

Definition 6.1 (Retarded fundamental solution)**.** A *fundamental solution* for $\Box$ is a distribution G such that

$$\langle G, \Box\psi\rangle = \psi(0,0) \qquad \text{for every } \psi \in C_c^\infty(\mathbb{R}^{1+3}).$$

It is *retarded* if its support lies in the future cone $J^+ = \{(t,x) : t \geq |x|\}$. It is *advanced* if its support lies in $J^- = \{(t,x) : t \leq -|x|\}$.

Distributional differentiation is defined by $\langle \partial_\mu T, \psi\rangle = -\langle T, \partial_\mu \psi\rangle$. Two derivatives remove this minus sign, so the defining identity says $\Box G = \delta^4$. The support condition selects the direction of time.

Theorem 6.2 (Retarded wave kernel)**.** *The distribution G_{ret} given by*

$$\langle G_{\mathrm{ret}}, \psi\rangle = \int_{\mathbb{R}^3} \frac{\psi(|x|,x)}{4\pi|x|}\,\mathrm{d}^3x, \qquad \psi \in C_c^\infty(\mathbb{R}^{1+3}), \tag{6.1}$$

is a retarded fundamental solution for $\Box$. Its support lies on $\{(t,x) : t = |x|\}$. For every $f \in C_c^\infty(\mathbb{R}^{1+3})$, the convolution $u = G_{\mathrm{ret}} * f$ is smooth and satisfies $\Box u = f$. It is the unique C^2 solution on all spacetime which vanishes for all sufficiently negative times.

The expression $\delta(t-|x|)/(4\pi|x|)$ is shorthand for (6.1). The displayed pairing gives its meaning at $x=0$. We prove the kernel identity first and the source assertions in Section 6.3.

6.2 Spherical averaging and the vertex

Write $\mathbb{S}^2=\{\theta\in\mathbb{R}^3:|\theta|=1\}$ and let $\mathrm{d}\Omega$ be its area measure, with total area 4π. Define the spherical mean by

$$M(t,r)=\frac{1}{4\pi}\int_{\mathbb{S}^2}\psi(t,r\theta)\,\mathrm{d}\Omega,\qquad t\in\mathbb{R},\quad r\geq 0.$$

Polar coordinates turn (6.1) into $\langle G_{\mathrm{ret}},\psi\rangle=\int_0^\infty rM(r,r)\,\mathrm{d}r$. The factor r makes the integral finite at the origin. If the spatial projection of $\operatorname{supp}\psi$ lies in $B_A(0)$, its absolute value is at most $A^2\sup|\psi|/2$. This bound proves continuity on tests with fixed compact support. Thus the formula defines a distribution. It vanishes on every test supported off $t=|x|$, which proves the asserted support condition.

To compute the wave operator, we must retain the angular part of the spatial Laplacian. The ball divergence formula (5.6) in Lecture 5, applied to the smooth vector field $\nabla\psi(t,\cdot)$, gives, for $r>0$,

$$4\pi r^2M_r(t,r)=\int_{B_r(0)}\Delta\psi(t,x)\,\mathrm{d}^3x.$$

Indeed, the outward normal derivative on ∂B_r is $\nabla\psi(t,r\theta)\cdot\theta$, whose integral is the left side. Differentiation in r by polar coordinates therefore gives

$$\frac{1}{4\pi}\int_{\mathbb{S}^2}\Delta\psi(t,r\theta)\,\mathrm{d}\Omega=\frac{1}{r^2}\partial_r(r^2M_r)=M_{rr}+\frac{2}{r}M_r.\tag{6.2}$$

This identity applies to every test function, whether or not it is radial. Time differentiation passes under the compact angular integral.

Set $H(t,r)=rM(t,r)$. The averaged wave operator then obeys

$$r\left(M_{tt}-M_{rr}-\frac{2}{r}M_r\right)=H_{tt}-H_{rr}.$$

Along the diagonal $t=r$, the chain rule cancels the mixed derivatives:

$$\frac{\mathrm{d}}{\mathrm{d}r}\big(H_t(r,r)-H_r(r,r)\big)=H_{tt}(r,r)-H_{rr}(r,r).$$

For $0<\varepsilon<A$, with A sufficiently large to contain the test's support, integration by parts yields

$$\int_\varepsilon^A r\left(M_{tt}-M_{rr}-\frac{2}{r}M_r\right)(r,r)\,\mathrm{d}r=\big[H_t(r,r)-H_r(r,r)\big]_\varepsilon^A.\tag{6.3}$$

The endpoint at A is zero, since the test and its derivatives vanish there.

At the other endpoint, smoothness of ψ makes M smooth for $r\geq 0$, with bounded derivatives near $(0,0)$. The integral of θ over the sphere is zero, so $M_r(t,0)=0$. Also $M(t,0)=\psi(t,0)$. Consequently

$$H_t(\varepsilon,\varepsilon)-H_r(\varepsilon,\varepsilon)=\varepsilon M_t(\varepsilon,\varepsilon)-M(\varepsilon,\varepsilon)-\varepsilon M_r(\varepsilon,\varepsilon)\longrightarrow-\psi(0,0).$$

The left integrand of (6.3) is the spherical mean of $r\Box\psi$ and is integrable at zero. Letting $\varepsilon\downarrow 0$ proves

$$\langle G_{\mathrm{ret}},\Box\psi\rangle=\psi(0,0).$$

The source is precisely the lower endpoint term. Solving the homogeneous equation away from the vertex would not establish this identity.

Example 6.3 (A normalisation check). Evaluate the pairing with the wave operator applied to $\psi(t,x) = e^{-(t-1/3)^2-|x|^2}$. This function is rapidly decreasing rather than compactly supported. The same integral and endpoint calculation apply because the function and its derivatives decay exponentially along the cone. Its spherical mean is $M(t,r) = e^{-(t-1/3)^2-r^2}$. Direct differentiation gives

$$\begin{aligned} r(\Box\psi)(r,r\theta) &= \left(\frac{40}{9}r - \frac{8}{3}r^2\right) e^{-(r-1/3)^2-r^2} \\ &= \frac{\mathrm{d}}{\mathrm{d}r}\left[\left(\frac{2}{3}r - 1\right) e^{-(r-1/3)^2-r^2}\right]. \end{aligned}$$

The upper endpoint is zero. The lower endpoint is $-e^{-1/9}$, so $\langle G_{\mathrm{ret}}, \Box\psi\rangle = e^{-1/9} = \psi(0,0)$. Jackson's kernel $\delta(t-r)/r$ would give $4\pi e^{-1/9}$. Our fundamental solution includes the divisor 4π.

6.3 The field produced by a source

For $f \in C_c^\infty(\mathbb{R}^{1+3})$, define its convolution with the kernel by

$$u(t,x) = \langle G_{\mathrm{ret}}(s,y), f(t-s,x-y)\rangle = \int_{\mathbb{R}^3} \frac{f(t-|y|,x-y)}{4\pi|y|}\,\mathrm{d}^3y. \tag{6.4}$$

For fixed (t,x), the function paired with G_{ret} is a test function of (s,y). This defines convolution without requiring a product of distributions.

We finish the proof of Theorem 6.2. Let (t,x) range over a compact set. Since the source has bounded spatial support, the integrand in (6.4) vanishes outside a fixed bounded set of y. Every derivative of f is bounded. Each differentiated integrand is therefore dominated by $C|y|^{-1}$ on a fixed ball. This is integrable in three dimensions. Differentiation under the integral proves $u \in C^\infty$. Second derivatives in (t,x) agree with second derivatives in (s,y) of $f(t-s,x-y)$. The kernel identity thus gives

$$\Box u(t,x) = \langle G_{\mathrm{ret}}, \Box_{s,y} f(t-s,x-y)\rangle = \langle \Box G_{\mathrm{ret}}, f(t-s,x-y)\rangle = f(t,x).$$

Choose $a \in \mathbb{R}$ with $f(s,z) = 0$ for $s < a$. For $t < a$ and every y, the retarded argument $t-|y|$ is below a. Hence $u(t,x) = 0$ for $t < a$.

For uniqueness, let v be another C^2 solution which vanishes for all sufficiently negative times. The difference $w = v - u$ solves $\Box w = 0$. Choose a slice $t = t_0$ before both fields can be nonzero. Both w and w_t vanish on that slice. The local cone uniqueness of Corollary 5.4, applied after translating time, gives $w = 0$ at each later point. For complex fields, apply that real-field result to the real and imaginary parts. Before this slice the fields already vanish. No finite total energy assumption is needed. This completes the theorem's proof.

Changing variables to the source position $z = x - y$ gives the spatial form

$$u(t,x) = \int_{\mathbb{R}^3} \frac{f(t-|x-z|,z)}{4\pi|x-z|}\,\mathrm{d}^3z. \tag{6.5}$$

The *retarded time* at the source position z, for observation at (t,x), is $t_{\mathrm{ret}} = t - |x-z|$. The travel time is the distance because the wave speed is one. The integral uses the source only at events on the observation point's past light cone, where $t - t_{\mathrm{ret}} = |x-z|$.

The cone in Figure 6.1 shows which source events can contribute.

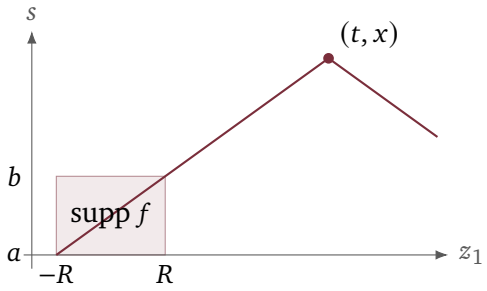


Figure 6.1: A spacetime section through one spatial axis of the source region and an observation point's past light cone. Only source events on the cone can enter the retarded integral.

Example 6.4 (A source in a ball)**.** Suppose a source is supported in

$$\operatorname{supp} f \subset [a,b] \times \overline{B_R(0)}, \qquad a < b, \quad R > 0.$$

For a fixed observer position x, the distance to a source position satisfies

$$\max\{|x| - R, 0\} \le |x - z| \le |x| + R.$$

A nonzero integrand in (6.5) requires $a \le t - |x - z| \le b$. Consequently the field can be nonzero only at times

$$a + \max\{|x| - R, 0\} \le t \le b + |x| + R. \tag{6.6}$$

For $a = 0$, $b = 2$, $R = 1$ and $|x| = 4$, the field vanishes when $t < 3$ and when $t > 7$. A wave from the near side needs at least three time units to arrive. A wave emitted at time two from the far side can arrive as late as time seven. These are support bounds, not claims that the field is nonzero at every intervening time.

The upper bound uses the largest distance to the source ball. Using its smallest distance would discard later arrivals from the far side. This finite response interval follows from the particular cone-supported kernel in three spatial dimensions.

6.4 Retardation is an additional condition

If $h \in C^2(\mathbb{R}^{1+3})$ solves $\Box h = 0$, then $u + h$ still solves $\Box(u + h) = f$. For example, $h = 1$ gives a field which does not vanish before the source. The equation alone therefore does not select the retarded solution. Local uniqueness shows that a nonzero homogeneous addition cannot vanish for all sufficiently negative times. The advanced kernel $G_{\text{adv}} = \delta(t + |x|)/(4\pi|x|)$ is also a fundamental solution, with support in the past cone.

The outgoing Helmholtz kernel of *Analysis*, Lecture 25, imposes a condition at spatial infinity for a fixed frequency. Retardation imposes a condition in time. Lectures 10 and 13 use the latter condition for fields produced by sources.

Summary. For $\Box = \partial_t^2 - \Delta$ on $\mathbb{R}^{1+3}$, the retarded kernel is defined by (6.1). Spherical averaging reduces its wave operator to an endpoint calculation. The vertex gives δ^4 with coefficient one. Convolution with a smooth compact source gives the unique C^2 solution vanishing sufficiently early. The source is evaluated at the retarded time $t - |x - z|$. Adding a homogeneous solution preserves the equation but can violate the past condition.

Self-check. Why can the term $2M_r/r$ not be dropped when averaging the Laplacian? Test the proposed omission on $\psi(t,x) = |x|^2$ near the origin.

Answer. The spherical mean is $M = r^2$. Its second radial derivative is 2, but $\Delta|x|^2 = 6$. The missing term $2M_r/r = 4$ supplies the difference. A compact cutoff equal to one near the origin makes this a test-function example.

Self-check. What is lost if the lower endpoint is omitted from (6.3)?

Answer. The remaining endpoint is zero. The calculation would incorrectly give $\Box G_{\mathrm{ret}} = 0$. The lower endpoint has limit $-\psi(0,0)$ and is subtracted, so it supplies $+\psi(0,0)$.

Self-check. A source is supported in $[1,3] \times \overline{B_2(0)}$. At a position with $|x| = 5$, give the earliest and latest possible response times.

Answer. The smallest distance is 3 and the largest is 7. Formula (6.6) permits a response only for $4 \le t \le 10$. Cancellations or smaller source support may narrow this interval.

Self-check. Does the uniqueness proof require the unknown C^2 solution to have finite total energy on each spatial slice?

Answer. No. Its difference from the retarded solution has zero data on an early slice. Lecture 5 applies the energy estimate on each bounded backward cone. The conclusion is pointwise uniqueness without a global integrability assumption.

Lecture 7

Canonical field observables

An oscillator has a position and a conjugate momentum. A scalar field has one such pair at every spatial point. We give this statement a precise meaning by integrating fields against test functions and defining a bracket on differentiable functions of the initial data.

Objectives. *We define functional derivatives on rapidly decreasing data, prove the field bracket's Jacobi identity and recover Klein–Gordon evolution from a Hamiltonian. We identify the charge generating a spatial translation.*

By the end of this lecture you will be able to:

- Compute functional derivatives with their boundary terms.
- Verify closure and Jacobi for the stated class of functionals.
- Recover the field equation and a translation from their generators.

7.1 Functions of field data

Definition 7.1 (Schwartz phase space). The real *Schwartz space* $\mathcal{S} = \mathcal{S}(\mathbb{R}^d, \mathbb{R})$ consists of smooth functions whose seminorms

$$p_{N,\alpha}(f) = \sup_{x \in \mathbb{R}^d} (1 + |x|)^N |\partial^\alpha f(x)|$$

are finite for every integer $N \geq 0$ and every derivative multi-index α. The *Schwartz phase space* is $\mathcal{Z} = \mathcal{S} \times \mathcal{S}$, with elements $z = (\phi, \pi)$. Convergence means convergence in every seminorm of both components.

The field ϕ and its momentum π are data on one fixed time slice. For $h = (h_\phi, h_\pi) \in \mathcal{Z}$, use the real L^2 pairing

$$(z, h)_0 = \int_{\mathbb{R}^d} (\phi h_\phi + \pi h_\pi)\, \mathrm{d}^d x.$$

It is continuous in the Schwartz topology by Cauchy–Schwarz and the decay seminorms. This topology also makes differentiation and multiplication continuous, by the product rule for derivatives.

A *functional* is a map from $\mathcal{Z}$ to $\mathbb{R}$. Here a map is *smooth* if its iterated directional derivatives of all orders exist, are multilinear in the directions, and are jointly continuous in the base point and directions, using the stated topologies. For example, its first derivative is

$$DF(z)h = \frac{\mathrm{d}}{\mathrm{d}\varepsilon} F(z + \varepsilon h)\Big|_{\varepsilon = 0}.$$

The product rule follows by restricting to affine families of variations. For the chain rule, a scalar or Schwartz-valued smooth map $\mathcal{T}$ satisfies the straight-segment fundamental theorem

$$\mathcal{T}(z+h) - \mathcal{T}(z) = \int_0^1 D\mathcal{T}(z+sh)h \, \mathrm{d}s.$$

For a Schwartz-valued integrand, the integral is the limit of Riemann sums in every seminorm. Continuity on the compact parameter interval makes these sums Cauchy in each seminorm. Schwartz space is complete: the uniform limits of all weighted derivatives of a Cauchy sequence remain compatible derivatives, by the ordinary fundamental theorem of calculus. The limit therefore lies in Schwartz space. Evaluating at each spatial point proves the displayed identity from the scalar fundamental theorem.

For a smooth curve $z(t)$, take $\Delta t \in \mathbb{R} \setminus \{0\}$ and write $v_{\Delta t} = (z(t+\Delta t) - z(t))/\Delta t$. As $\Delta t \to 0$, this quotient converges to $\dot{z}(t)$ in every seminorm. The segment identity gives

$$\frac{\mathcal{T}(z(t+\Delta t)) - \mathcal{T}(z(t))}{\Delta t} = \int_0^1 D\mathcal{T}(z(t) + s\Delta t\, v_{\Delta t}) v_{\Delta t} \, \mathrm{d}s \longrightarrow D\mathcal{T}(z(t))\dot{z}(t).$$

Joint continuity makes this convergence uniform in $s \in [0,1]$, in every seminorm for a Schwartz-valued map. This proves the chain rule for the curves used below. Applying it to a map along each straight line gives the composition rule. Repeated differentiation proves the corresponding rules for the higher derivatives in the definition of smoothness.

Definition 7.2 (Admissible functional and functional derivative)**.** A smooth functional $F : \mathcal{Z} \to \mathbb{R}$ is *admissible* if there is a smooth map $\operatorname{grad} F : \mathcal{Z} \to \mathcal{Z}$ such that

$$DF(z)h = (\operatorname{grad} F(z), h)_0 = \int_{\mathbb{R}^d} \big(F_\phi[z]h_\phi + F_\pi[z]h_\pi\big) \, \mathrm{d}^d x \quad (h \in \mathcal{Z}). \tag{7.1}$$

Its components $F_\phi[z]$ and $F_\pi[z]$ are the *functional derivatives*. Write $\mathcal{A}$ for the class of admissible functionals.

These derivatives are unique. The difference of two candidate gradients can itself be chosen as h in (7.1), giving zero L^2 norm. Smoothness of F alone does not assert admissibility.

For a real test function $f \in \mathcal{S}$, define the *smeared fields* $\phi(f) : \mathcal{Z} \to \mathbb{R}$ and $\pi(f) : \mathcal{Z} \to \mathbb{R}$ by

$$\phi(f)[z] = \int f\phi \, \mathrm{d}^d x, \qquad \pi(f)[z] = \int f\pi \, \mathrm{d}^d x. \tag{7.2}$$

Their gradients are $(f, 0)$ and $(0, f)$, so both belong to $\mathcal{A}$. Polynomials in finitely many such smeared fields are also admissible, by the product rule. Local integrals can be admissible too. We will verify this directly for the Klein–Gordon Hamiltonian.

7.2 The field bracket and Jacobi

In mechanics, the canonical bracket pairs a position derivative with a momentum derivative. Define the fixed continuous linear map $\mathbb{J} : \mathcal{Z} \to \mathcal{Z}$ by $\mathbb{J}(a,b) = (b,-a)$. It is skew with respect to the pairing: $(a, \mathbb{J}b)_0 = -(\mathbb{J}a, b)_0$.

Definition 7.3 (Field bracket and Hamiltonian vector field)**.** For $F, G \in \mathcal{A}$, the *field bracket* is the functional

$$\{F, G\}[z] = (\operatorname{grad} F, \mathbb{J} \operatorname{grad} G)_0 = \int_{\mathbb{R}^d} (F_\phi G_\pi - F_\pi G_\phi)\, \mathrm{d}^d x. \tag{7.3}$$

The *Hamiltonian vector field* of F is the map $X_F : \mathcal{Z} \to \mathcal{Z}$ given by $X_F(z) = \mathbb{J} \operatorname{grad} F(z) = (F_\pi, -F_\phi)$.

All derivatives in (7.3) are evaluated at z. The integral is finite because its factors are Schwartz functions. The order of the bracket matters: the chain rule gives

$$X_F G := DG(z) X_F(z) = \{G, F\}[z]. \tag{7.4}$$

Proposition 7.4 (Canonical field bracket)**.** *The class $\mathcal{A}$ is closed under products and the field bracket. The bracket is bilinear and antisymmetric, obeys the product rule, and satisfies the Jacobi identity*

$$\{\{F, G\}, K\} + \{\{G, K\}, F\} + \{\{K, F\}, G\} = 0 \qquad (F, G, K \in \mathcal{A}). \tag{7.5}$$

For a smooth path with $\dot{z} = X_H(z)$, every $F \in \mathcal{A}$ obeys $\frac{\mathrm{d}}{\mathrm{d}t} F[z(t)] = \{F, H\}[z(t)]$.

Proof. We first identify the derivative needed to prove both closure and Jacobi. For $F \in \mathcal{A}$, let $H_F(z) = D(\operatorname{grad} F)(z)$. It is a continuous linear map from $\mathcal{Z}$ to $\mathcal{Z}$. Mixed ordinary derivatives of $F(z + sh + tk)$ commute. Differentiating (7.1) therefore gives the symmetry

$$(H_F h, k)_0 = D^2 F(z)[h, k] = (h, H_F k)_0 \qquad (h, k \in \mathcal{Z}). \tag{7.6}$$

This is symmetry for the real pairing; no bounded operator on all of L^2 is being asserted.

Write $a = \operatorname{grad} F$, $b = \operatorname{grad} G$ and $a_K = \operatorname{grad} K$ at the same point z. Differentiating the bracket, using (7.6) and skewness of $\mathbb{J}$, gives

$$D\{F, G\}(z)h = (H_F h, \mathbb{J} b)_0 + (a, \mathbb{J} H_G h)_0 = (h, H_F \mathbb{J} b - H_G \mathbb{J} a)_0.$$

Thus its gradient is

$$\operatorname{grad}\{F, G\} = H_F \mathbb{J} b - H_G \mathbb{J} a. \tag{7.7}$$

Both terms are Schwartz-valued and smooth. Indeed, derivatives of the smooth gradient maps are jointly smooth in their base point and direction, and their evaluation at the smooth directions $\mathbb{J} b$ and $\mathbb{J} a$ uses the chain rule. Repeated differentiation gives finite sums of higher derivatives of these gradient maps, evaluated on smooth directions. Every such term is jointly continuous by the definition of smoothness. This proves bracket closure. The gradient of a product is $F \operatorname{grad} G + G \operatorname{grad} F$, which proves product closure and the bracket product rule. Bilinearity and antisymmetry follow from the defining pairing.

For Jacobi, substitute (7.7) into the cyclic sum and collect terms having the same Hessian. The sum is

$$\begin{aligned} &(H_F \mathbb{J} b, \mathbb{J} a_K)_0 - (H_F \mathbb{J} a_K, \mathbb{J} b)_0 \\ &+ (H_G \mathbb{J} a_K, \mathbb{J} a)_0 - (H_G \mathbb{J} a, \mathbb{J} a_K)_0 \\ &+ (H_K \mathbb{J} a, \mathbb{J} b)_0 - (H_K \mathbb{J} b, \mathbb{J} a)_0 = 0. \end{aligned}$$

Each line vanishes by Hessian symmetry. Finally, differentiation along the path and (7.4) give the stated evolution of F. ∎

For the smeared fields, substitution of their constant gradients yields

$$\{\phi(f),\pi(g)\}=\int_{\mathbb{R}^d} fg\,\mathrm{d}^d x,\qquad \{\phi(f),\phi(g)\}=\{\pi(f),\pi(g)\}=0. \tag{7.8}$$

For example, in one dimension take $f(x)=e^{-x^2}$ and $g(x)=(1+x)e^{-2x^2}$. The odd term integrates to zero and the Gaussian integral gives $\{\phi(f),\pi(g)\}=\sqrt{\pi/3}$. This is the precise smeared meaning of canonical field pairs.

7.3 The Klein–Gordon Hamiltonian

Write v for the time derivative and $w\in\mathbb{R}^d$ for the spatial gradient. The Klein–Gordon density is

$$L(\phi,v,w)=\frac{1}{2}v^2-\frac{1}{2}|w|^2-\frac{1}{2}m^2\phi^2,\qquad m\geq 0.$$

Definition 7.5 (Legendre map and field Hamiltonian)**.** The *Legendre map* of this density is

$$\mathbb{R}\times\mathbb{R}\times\mathbb{R}^d\longrightarrow\mathbb{R}\times\mathbb{R}\times\mathbb{R}^d,$$
$$(\phi,v,w)\longmapsto(\phi,\pi,w),\qquad \pi=\frac{\partial L}{\partial v}.$$

Using its inverse to express v in terms of π, the *Hamiltonian density* is $\mathcal{H}=\pi v-L$. The *field Hamiltonian* is the functional $H:\mathcal{Z}\to\mathbb{R}$ given by

$$H[\phi,\pi]=\int_{\mathbb{R}^d}\mathcal{H}(\phi,\pi,\nabla\phi)\,\mathrm{d}^d x.$$

For this density the momentum is $\pi=v$, so the Legendre map is a bijection. The subtraction $\pi v-L$ gives

$$\mathcal{H}=\frac{1}{2}\bigl(\pi^2+|w|^2+m^2\phi^2\bigr),\qquad H[\phi,\pi]=\frac{1}{2}\int_{\mathbb{R}^d}\bigl(\pi^2+|\nabla\phi|^2+m^2\phi^2\bigr)\,\mathrm{d}^d x. \tag{7.9}$$

We compute its derivative, keeping the boundary term first. On the cube $Q_R=[-R,R]^d$, integration by parts gives

$$\int_{Q_R}\nabla\phi\cdot\nabla h_\phi\,\mathrm{d}^d x=\int_{\partial Q_R}h_\phi\,\partial_n\phi\,\mathrm{d}S-\int_{Q_R}(\Delta\phi)h_\phi\,\mathrm{d}^d x.$$

Schwartz decay makes the face integrals tend to zero as $R\to\infty$. For example, choose a decay exponent larger than d and bound the integrand on each face by that decay bound times its area. The derivative and its gradient are therefore

$$DH(z)h=\int\bigl[\pi h_\pi+(-\Delta\phi+m^2\phi)h_\phi\bigr]\,\mathrm{d}^d x,\qquad \operatorname{grad}H=(-\Delta\phi+m^2\phi,\pi). \tag{7.10}$$

The gradient is a continuous linear map on $\mathcal{Z}$, so $H\in\mathcal{A}$.

The *Hamilton equations* are $\dot z=X_H(z)$, namely

$$\phi_t=\pi,\qquad \pi_t=\Delta\phi-m^2\phi. \tag{7.11}$$

They are equivalent to $(\Box+m^2)\phi=0$ with $\pi=\phi_t$. Existence is supplied by the actual Fourier solution of Lecture 4. For Schwartz data it gives a smooth path in $\mathcal{Z}$: the multipliers $\cos(t\nu)$

and $\sin(t\nu)/\nu$ are smooth in k with polynomially bounded k-derivatives on each bounded time interval, as are their time derivatives. At $m = 0$ and $k = 0$, their power series in $|k|^2$ remove the apparent singularity. Away from that point the bounds follow by differentiation. The product rule and the Schwartz Fourier automorphism from *Analysis* I, Lecture 7, then give the assertion in every seminorm. Thus Hamilton evolution agrees with the unique energy solution of Lecture 4. Antisymmetry also gives $\dot H = \{H, H\} = 0$ along this path.

7.4 A charge generating translations

Lecture 3 associates a translation in direction i with the characteristic $Q^1 = \partial_i\phi$. The superscript 1 labels the single field component. Its charge density is the mixed stress component $T^0{}_i = \pi\partial_i\phi$. Its spatial translation charge $J_i : \mathcal{Z} \to \mathbb{R}$ is

$$J_i[\phi, \pi] = \int_{\mathbb{R}^d} \pi\partial_i\phi \, \mathrm{d}^d x, \qquad i = 1, \dots, d. \tag{7.12}$$

This is the charge with a lower spatial index. The physical momentum $\int T^{0i}$ has the opposite sign in our metric convention. Integration by parts retains $\int_{\partial Q_R} \pi h_\phi n_i \, \mathrm{d}S$ before Schwartz decay removes it. Hence

$$\operatorname{grad} J_i = (-\partial_i\pi, \partial_i\phi), \qquad X_{J_i} = (\partial_i\phi, \partial_i\pi).$$

The gradient is continuous linear, so this charge is admissible. Its Hamiltonian vector field is exactly the derivative at zero of $(\phi(x + \varepsilon e_i), \pi(x + \varepsilon e_i))$. In terms of admissible observables, this assertion reads

$$\{\phi(f), J_i\} = \int f\partial_i\phi \, \mathrm{d}^d x, \qquad \{\pi(g), J_i\} = \int g\partial_i\pi \, \mathrm{d}^d x. \tag{7.13}$$

The Noether charge therefore generates its characteristic, in the stated bracket order. It is conserved under Klein–Gordon evolution. Indeed, integration by parts and decay give

$$\{H, J_i\} = \int \left[(-\Delta\phi + m^2\phi)\partial_i\phi + \pi\partial_i\pi\right] \mathrm{d}^d x = \frac{1}{2}\int \partial_i\left(|\nabla\phi|^2 + m^2\phi^2 + \pi^2\right) \mathrm{d}^d x = 0.$$

Consequently $\dot J_i = \{J_i, H\} = 0$.

Corollary 7.6 (Klein–Gordon and translation generators). *For the Klein–Gordon density, the Legendre map is bijective. The functionals H and J_i belong to $\mathcal{A}$. The equation $\dot z = X_H(z)$ is the Klein–Gordon system (7.11). The vector field X_{J_i} is the infinitesimal generator of the spatial translation $(\phi, \pi) \mapsto (\phi(\,\cdot + \varepsilon e_i), \pi(\,\cdot + \varepsilon e_i))$. Both charges are conserved under Klein–Gordon evolution.*

Proof. The momentum computation gives $\pi = \nu$. The gradients in (7.10) and the translation computation above give the two vector fields. Along the smooth Klein–Gordon path constructed above, Proposition 7.4, antisymmetry and $\{H, J_i\} = 0$ prove conservation. ∎

7.5 Point values fall outside this class

Fix $x_0 \in \mathbb{R}^d$. The functional $F[z] = \phi(x_0)$ is smooth on $\mathcal{Z}$, but it is not admissible. Its derivative is $DF(z)h = h_\phi(x_0)$. If a Schwartz function a represented this derivative, it would satisfy

$$\int a(x)h(x) \, \mathrm{d}^d x = h(x_0) \qquad (h \in \mathcal{S}).$$

Tests supported away from x_0 force $a = 0$ there by the fundamental lemma. Continuity forces $a(x_0) = 0$ too. A test with $h(x_0) = 1$ gives a contradiction. The formal derivative is instead the delta distribution, defined by $\langle \delta_{x_0}, h \rangle = h(x_0)$. It is not a Schwartz function. Thus a formal pointwise bracket kernel does not enlarge the class in Proposition 7.4. Sharp spatial boundaries can also produce distributions when a functional is varied. Lectures 11 and 12 must specify the functionals and their boundary conditions before using their brackets.

Summary. On real Schwartz pairs, admissible functionals have smooth Schwartz-valued functional derivatives. Their canonical bracket is closed and satisfies Jacobi because the Hessians are symmetric and $\mathbb{J}$ is constant. The Klein–Gordon Hamiltonian gives $\phi_t = \pi$ and $\pi_t = \Delta\phi - m^2\phi$. The lower-index translation charge generates $\partial_i\phi$ and $\partial_i\pi$. Point evaluation is smooth but is excluded from this bracket class. Lecture 8 describes the corresponding pairing directly on solutions.

Self-check. Compute the functional derivatives of $F[\phi, \pi] = \int \phi^3 \, \mathrm{d}^d x$. Does F belong to the stated class?

Answer. The gradient is $(3\phi^2, 0)$. Multiplication is continuous on Schwartz space, so this gradient is a smooth Schwartz-valued polynomial map. Thus $F \in \mathcal{A}$.

Self-check. Compute $\{\pi(g), \phi(f)\}$ and $\{\phi(f), H\}$.

Answer. The values are $-\int fg \, \mathrm{d}^d x$ and $\int f\pi \, \mathrm{d}^d x$, respectively. The second is the time derivative of the smeared displacement.

Self-check. Which hypothesis makes the two terms in each line of the Jacobi computation cancel? Would skewness of a field-dependent replacement for $\mathbb{J}$ suffice?

Answer. Hessian symmetry cancels each pair. Constancy of $\mathbb{J}$ ensures that its derivative contributes no extra terms. Skewness alone does not remove such terms for a field-dependent replacement.

Self-check. Use the physical momentum $P^i = \int T^{0i} = -J_i$ as generator. Which spatial translation does it produce?

Answer. Its vector field is $-(\partial_i\phi, \partial_i\pi)$, so it generates $(\phi(x - \varepsilon e_i), \pi(x - \varepsilon e_i))$. This sign follows from raising the spatial stress index.

Lecture 8

The symplectic form on solutions

Lecture 7 describes a field by its displacement and momentum on one time slice. A solution contains these data on every slice. We show that their antisymmetric pairing is independent of a compact deformation of the slice. The boundary term in the first variation determines its sign.

Objectives. *We construct the Klein–Gordon symplectic current, prove conservation of its integral on spacelike graphs, and prove nondegeneracy using compact Cauchy data. A finite spatial interval shows why side flux must be retained.*

By the end of this lecture you will be able to:

- Derive the symplectic current from the first-variation boundary term.
- Compute its integral on a spacelike graph with the future orientation.
- Prove slice independence and nondegeneracy for compact Cauchy data.

8.1 Solutions and their variations

Fix $m \geq 0$ on $\mathbb{R}^{1+d}$. The Klein–Gordon equation is $(\Box + m^2)\phi = 0$.

Definition 8.1 (Space of solutions)**.** The *space of solutions* $\mathcal{S}_m$ is the real vector space

$$\mathcal{S}_m = \{\phi \in C^\infty(\mathbb{R}^{1+d}, \mathbb{R}) : (\Box + m^2)\phi = 0, \quad (\phi(0), \phi_t(0)) \in C_c^\infty(\mathbb{R}^d)^2\}.$$

A *linearised solution* at ϕ is the derivative h of a smooth family of solutions through ϕ. Here the allowed variations are $h \in \mathcal{S}_m$, realised by the family $\phi + \varepsilon h$.

Linearity makes the linearised equation identical to the original equation: $(\Box + m^2)h = 0$. Theorem 4.3 constructs the solution of every compact smooth Cauchy datum. Such data have rapidly decreasing Fourier transforms, so the multiplier formula can be differentiated to every order. The resulting solution is smooth.

Theorem 5.3 gives a common compact spatial support on each bounded time interval. More precisely, if a datum is supported in $B_R(0)$, its solution vanishes outside $|x| \leq R + |t|$. Derivatives also vanish outside this set. These facts justify the integrals below.

We write δ for differentiation along solution families and d for spacetime exterior differentiation. For example, $\delta\phi(h) = h$ and $\delta p^\mu(h) = \partial^\mu h$, since $p^\mu = \partial^\mu \phi$ for the scalar field. These two differentiations commute because they act on separate variables. All field-space derivatives here can be evaluated on finite-dimensional affine families of solutions.

8.2 The boundary term and its antisymmetric variation

Recall $\mathrm{d}\Sigma_\mu = \iota_{\partial_\mu}\mathrm{vol}$ from Lecture 1. For any vector field A, the identity $d(A^\mu \mathrm{d}\Sigma_\mu) = (\partial_\mu A^\mu)\mathrm{vol}$ follows by differentiating its coefficients.

Definition 8.2 (Presymplectic potential and symplectic current)**.** The *presymplectic potential* is the spacetime d-form, linear in a variation h, given by

$$\theta_\phi(h) = p^\mu h \,\mathrm{d}\Sigma_\mu, \qquad p^\mu = \partial^\mu \phi. \tag{8.1}$$

The *symplectic current* is $\omega = -\delta\theta$. On constant variation directions h_1 and h_2, this means

$$\omega(h_1, h_2) = \omega^\mu(h_1, h_2)\mathrm{d}\Sigma_\mu, \qquad \omega^\mu = h_1\partial^\mu h_2 - h_2\partial^\mu h_1. \tag{8.2}$$

The potential is the boundary expression in the pointwise first variation:

$$\delta(L\mathrm{vol})(h) = E[\phi]h\,\mathrm{vol} + d\theta_\phi(h).$$

This identity follows from the product rule, as in Lecture 1. Its validity does not require extending that lecture's torus boundary theorem to other regions.

To check the minus sign in the current, vary the coefficient of $\theta_\phi(h_2)$ along h_1 and subtract the reverse variation:

$$(\delta\theta)(h_1, h_2) = \big[(\partial^\mu h_1)h_2 - (\partial^\mu h_2)h_1\big]\mathrm{d}\Sigma_\mu = -\omega(h_1, h_2).$$

For solutions of the linearised equation, the current is closed in spacetime. The product rule and the symmetry of the metric give

$$\partial_\mu \omega^\mu = h_1\Box h_2 - h_2\Box h_1 = -m^2 h_1 h_2 + m^2 h_2 h_1 = 0, \qquad d\omega = 0. \tag{8.3}$$

The mass terms cancel. No boundary assertion has yet been used.

8.3 A graph and its orientation

Definition 8.3 (Spacelike graph and its pairing)**.** A *spacelike graph* is a surface $\Sigma_\tau = \{(\tau(x), x) : x \in \mathbb{R}^d\}$ with $\tau \in C^1(\mathbb{R}^d)$ and $|\nabla\tau| < 1$. We orient it as the boundary of its past, with the future normal first. When the current is integrable on the graph, its pairing is

$$\Omega_{\Sigma_\tau}(h_1, h_2) = \int_{\Sigma_\tau} \omega(h_1, h_2). \tag{8.4}$$

For a constant graph $\tau = t_0$, write Ω_{t_0}.

A tangent vector to the graph has the form $(\nabla\tau \cdot v, v)$. Its Minkowski square is $(\nabla\tau \cdot v)^2 - |v|^2 < 0$ when $v \neq 0$. The future unit normal and the positive induced area measure satisfy

$$N_\mu = \frac{(1, -\nabla\tau)}{\sqrt{1 - |\nabla\tau|^2}}, \qquad dS = \sqrt{1 - |\nabla\tau|^2}\,\mathrm{d}^d x, \qquad N_\mu dS = (1, -\nabla\tau)\,\mathrm{d}^d x.$$

The area factor is the square root of the determinant of $I - \nabla\tau \otimes \nabla\tau$.

The pullback can also be read directly from the differential forms. The pullback of $\mathrm{d}\Sigma_0$ is $\mathrm{d}^d x$. In $\mathrm{d}\Sigma_i$, replace $\mathrm{d}t$ by $\partial_j\tau\,\mathrm{d}x^j$. Only the term $j = i$ survives, and its sign is negative. Consequently

$$\Omega_{\Sigma_\tau}(h_1, h_2) = \int_{\mathbb{R}^d} \big[\omega^0 - \omega^i\partial_i\tau\big](\tau(x), x)\,\mathrm{d}^d x. \tag{8.5}$$

Both surfaces in a comparison carry this future orientation. As a boundary of the region between them, the lower face instead carries the opposite sign.

Theorem 8.4 (Slice independence). *For $h_1, h_2 \in \mathcal{S}_m$ and $t_0 \in \mathbb{R}$, let $\tau \in C^1(\mathbb{R}^d)$ satisfy $|\nabla\tau| < 1$ and $\tau = t_0$ outside a compact set. Then*

$$\Omega_{\Sigma_\tau}(h_1, h_2) = \Omega_{t_0}(h_1, h_2) = \int_{\mathbb{R}^d} (h_1 \partial_t h_2 - h_2 \partial_t h_1)(t_0, x)\, \mathrm{d}^d x. \tag{8.6}$$

The value is also independent of t_0. This pairing is a nondegenerate closed alternating form on the solution space. On a constant slice, with $\pi = \phi_t$, it is $\Omega = -\delta\Theta = \int \delta\phi \wedge \delta\pi$, where $\Theta = \int \pi\, \delta\phi$.

Proof. We integrate the divergence by coordinates, including when the graph crosses $t = t_0$. Since $\tau - t_0$ has compact support, τ is bounded. Choose a bounded time interval containing both slices. Finite propagation places the current's support in one spatial rectangle Q throughout that interval. Choose Q large enough also to contain the deformation of the graph.

Define the signed time integrals

$$A^i(x) = \int_{t_0}^{\tau(x)} \omega^i(t, x)\, \mathrm{d}t.$$

They are C^1 and vanish near every face of Q. The fundamental theorem and the chain rule give, with summation over i,

$$\partial_i A^i = \int_{t_0}^{\tau(x)} \partial_i \omega^i(t, x)\, \mathrm{d}t + \omega^i(\tau(x), x)\partial_i \tau.$$

Use $\partial_t \omega^0 = -\partial_i \omega^i$ and integrate in time. Rearranging yields

$$\omega^0(\tau(x), x) - \omega^i(\tau(x), x)\partial_i \tau - \omega^0(t_0, x) = -\partial_i A^i(x).$$

Fubini and coordinate FTC on Q show that the integral of the right side is zero: its opposite face values vanish. Formula (8.5) proves the equality of pairings. This is the graph form of the boundary theorem, proved here on $\mathbb{R}^d$. For a constant graph $\tau = t_1$, the same argument works because finite propagation still makes A^i vanish near the faces. Thus all constant-time pairings agree.

On the constant slice, evaluating the wedge gives $(\delta\phi \wedge \delta\pi)(h_1, h_2) = h_1 \partial_t h_2 - h_2 \partial_t h_1$. Also $-\delta(\pi\, \delta\phi) = \delta\phi \wedge \delta\pi$. The pairing is alternating. It is closed under δ because its coefficients are independent of the background solution, so their derivatives along every affine family are zero.

Finally, suppose $\Omega_{t_0}(h, k) = 0$ for every $k \in \mathcal{S}_m$. Write $f = h(t_0)$ and $g = h_t(t_0)$, both compact smooth functions. Arbitrary compact smooth data at time t_0 produce a solution in $\mathcal{S}_m$: apply Lecture 4 after translating time and Lecture 5 backwards to time zero. Choose k with data $(0, f)$. The pairing is $\int f^2 = 0$, so $f = 0$. Choose k with data $(g, 0)$. It gives $-\int g^2 = 0$, so $g = 0$. Cauchy uniqueness gives $h = 0$. This is nondegeneracy. ∎

The term *symplectic* here means a closed alternating form with this nondegeneracy property. It does not assert that every linear functional has a representing vector. For a functional with compact smooth gradients, the equation $\iota_{X_F}\Omega = \delta F$ gives $X_F = (F_\pi, -F_\phi)$ and $X_F G = \{G, F\}$, with the functional derivatives and bracket order of Lecture 7. Compactness of these gradients ensures that this vector is tangent to the solution space used here.

8.4 Two compact waves on a deformed slice

Consider $d = 1$ and $m = 0$. Choose an even smooth cutoff χ with $\chi = 1$ on $[-3, 3]$ and $\chi = 0$ outside $[-4, 4]$. Define two right-moving solutions by

$$h_1(t, x) = \chi(q), \qquad h_2(t, x) = (q + q^2)\chi(q), \qquad q = x - t.$$

Both solve $h_{tt} - h_{xx} = 0$ and have compact smooth Cauchy data. The derivatives of the cutoff cancel in the antisymmetric product:

$$W(q) = \chi(q)\big[(q + q^2)\chi(q)\big]' - (q + q^2)\chi(q)\chi'(q) = (1 + 2q)\chi(q)^2, \qquad \omega^0 = \omega^1 = -W(q).$$

Set $A = \int_{\mathbb{R}} \chi(q)^2 \, \mathrm{d}q > 0$. Evenness of χ^2 makes the term $2q\chi^2$ odd, so $\Omega_0(h_1, h_2) = -A \neq 0$.

Deform the slice by the compact graph

$$\tau(x) = \frac{2}{5} b(x/2), \qquad b(s) = \begin{cases} (1 - s^2)^4, & |s| < 1, \\ 0, & |s| \geq 1. \end{cases}$$

This graph is C^3. For $|s| \leq 1$, its slope obeys

$$|\tau'(x)| = \frac{8}{5}|s|(1 - s^2)^3 \leq \frac{8}{5}|s|(1 - s^2) \leq \frac{16}{15\sqrt{3}} < 1, \qquad s = x/2.$$

The middle bound follows by maximising $s(1 - s^2)$ on $[0, 1]$. Thus the graph is spacelike. It agrees with $t = 0$ outside $[-2, 2]$.

Figure 8.1 shows the compact comparison. During $0 \leq t \leq 1/2$, both waves vanish outside $[-4, 9/2]$. The vertical sides $x = \pm 5$ therefore carry no current.

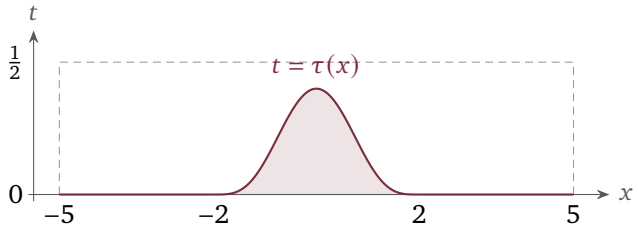


Figure 8.1: The deformed slice $t = \tau(x)$ and the constant slice $t = 0$ inside a compact comparison cylinder. The shaded region lies between the slices. The vertical sides lie outside the waves' support on the displayed interval.

The graph formula gives

$$\Omega_{\Sigma_\tau}(h_1, h_2) = -\int_{\mathbb{R}} W(x - \tau(x))(1 - \tau'(x)) \, \mathrm{d}x.$$

The change of variable $q = x - \tau(x)$ has derivative $1 - \tau'(x) > 0$ and agrees with $q = x$ outside a compact set. It is therefore a bijection of $\mathbb{R}$ onto $\mathbb{R}$. Substitution gives $\Omega_{\Sigma_\tau} = -\int W = -A$. This explicitly verifies the conserved, nonzero pairing on the two slices.

8.5 Side flux changes the pairing

Conservation of the spacetime current does not conserve its integral over a fixed spatial subregion. In one dimension, coordinate FTC gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_\alpha^\beta \omega^0(t, x) \, \mathrm{d}x = -\big[\omega^1(t, \beta) - \omega^1(t, \alpha)\big]. \tag{8.7}$$

The boundaries $x = \alpha$ and $x = \beta$ are timelike sides: they extend in the time direction at fixed spatial positions.

For an explicit local example, take $m = 0$, $h_1 = 1$ and $h_2 = (x+t)^2$. These are smooth wave solutions, although their Cauchy data are not compact. Their current and its integral on $I = [0, 1]$ are

$$\omega^0 = 2(x+t), \qquad \omega^1 = -2(x+t), \qquad \Omega_I(t) = \int_0^1 \omega^0 \, \mathrm{d}x = 1 + 2t.$$

The net outward side flux is $\omega^1(t,1) - \omega^1(t,0) = -2$. Thus $\Omega_I' = 2$, in agreement with (8.7). The current is closed, but the side contribution is nonzero. Boundary conditions must control this contribution when a field is restricted to a spatial region, as in Lectures 12 and 27.

8.6 General Cauchy surfaces

A *Cauchy surface* meets every inextendible timelike curve exactly once. A timelike curve has tangent of positive Minkowski square. For the same smooth Klein–Gordon solutions with compact Cauchy data, the pairing is also finite and independent of any smooth spacelike Cauchy surface in full Minkowski spacetime, oriented by its future normal. This extension is *cited*, not proved here. It uses compactness of the solution support on general Cauchy surfaces in addition to current closure. See Khavkine, *Covariant phase space, constraints, gauge and the Peierls formula* (2014), §3.3.3, Lemma 3.11, arXiv:1402.1282. The coordinate proof above establishes the compact graph deformations needed in this course.

Summary. For smooth real Klein–Gordon solutions with compact Cauchy data, $\theta = p^\mu \delta\phi \, \mathrm{d}\Sigma_\mu$ gives $\omega = -\delta\theta$ and $d\omega = 0$. The future-oriented graph element is $(1, -\nabla\tau)\mathrm{d}^d x$. Its integral agrees with the constant-slice form $\Omega = \int \delta\phi \wedge \delta\pi$ for a compact spacelike deformation. Arbitrary compact Cauchy test data prove nondegeneracy. On a bounded spatial region, side flux can change this integral.

Self-check. For the compact waves above, reverse the order of h_1 and h_2. What is the pairing on either slice?

Answer. Antisymmetry changes $-A$ to $+A$. The order of the arguments and the sign $\omega = -\delta\theta$ must both be retained.

Self-check. On a graph $t = vx$ in one spatial dimension, with $|v| < 1$, what is the pullback density for a right-moving current with $\omega^0 = \omega^1 = -W(x-t)$?

Answer. The density is $-(1-v)W((1-v)x)\,\mathrm{d}x$. Its sign uses the future graph element $(1, -v)\mathrm{d}x$. This local formula does not require the affine graph to be a compact deformation.

Self-check. Suppose h pairs to zero with every compact-data solution. Which two choices of Cauchy test data force its displacement and velocity to vanish?

Answer. Writing the data of h as (f, g), choose test data $(0, f)$ and $(g, 0)$. The pairings are $\int f^2$ and $-\int g^2$. Their vanishing gives $f = g = 0$, and uniqueness then gives $h = 0$.

Self-check. For the interval example, compare $\Omega_I(1) - \Omega_I(0)$ with the integrated outward side flux from $t = 0$ to $t = 1$.

Answer. The change is $3 - 1 = 2$. The integrated outward flux is -2. Their sum is zero, as required by the local boundary balance.

Lecture 9

Maxwell: variation and identity

An electromagnetic potential has four components, but its field equations cannot determine all four independently. We derive the equations from an action and identify the differential relation between them.

Objectives. *We fix the electromagnetic signs, vary the Maxwell action and derive charge conservation as a compatibility condition. We prove that gauge transformations leave the field unchanged and explain the resulting identity between equations.*

By the end of this lecture you will be able to:

- Derive Gauss's and Ampère's laws from the action with the stated signs.
- Prove the off-shell identity for the Maxwell equations.
- Identify the restriction on a prescribed current and the freedom in its potential.

9.1 Potential and electromagnetic field

We work on $\mathbb{R}^{1+3}$ with metric $\eta = \operatorname{diag}(1,-1,-1,-1)$ and orientation $\mathrm{vol} = \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$. All fields in this lecture are real.

Definition 9.1 (Gauge potential and gauge transformation)**.** An electromagnetic *gauge potential* is a C^2 one-form

$$A_{\mathrm{em}} = -\phi_{\mathrm{em}}\,\mathrm{d}t + A_x\,\mathrm{d}x + A_y\,\mathrm{d}y + A_z\,\mathrm{d}z.$$

Its *field strength* is the two-form $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}$. A *gauge transformation*, with smooth scalar function χ, is

$$A_{\mathrm{em}} \longmapsto A_{\mathrm{em}} + \mathrm{d}\chi, \qquad \phi_{\mathrm{em}} \longmapsto \phi_{\mathrm{em}} - \chi_t, \qquad \mathbf{A} \longmapsto \mathbf{A} + \nabla\chi, \tag{9.1}$$

where $\mathbf{A} = (A_x, A_y, A_z)$ is the spatial potential.

Recall from *Topology and Geometry*, Part I, that the exterior derivative satisfies $\mathrm{d}^2 = 0$ and the graded product rule $\mathrm{d}(\alpha \wedge \beta) = \mathrm{d}\alpha \wedge \beta + (-1)^p \alpha \wedge \mathrm{d}\beta$ when α has degree p. Thus the transformed potential has the same field strength. This assertion uses equality of mixed derivatives, not a field equation.

Expanding $\mathrm{d}A_{\mathrm{em}}$ defines the electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$ by

$$\begin{aligned} F_{\mathrm{em}} &= E_x\,\mathrm{d}x \wedge \mathrm{d}t + E_y\,\mathrm{d}y \wedge \mathrm{d}t + E_z\,\mathrm{d}z \wedge \mathrm{d}t \\ &\quad + B_x\,\mathrm{d}y \wedge \mathrm{d}z + B_y\,\mathrm{d}z \wedge \mathrm{d}x + B_z\,\mathrm{d}x \wedge \mathrm{d}y, \\ \mathbf{E} &= -\nabla\phi_{\mathrm{em}} - \partial_t \mathbf{A}, \qquad \mathbf{B} = \operatorname{curl}\mathbf{A}. \end{aligned} \tag{9.2}$$

The spatial dot product and curl use the Euclidean spatial metric. Since $\mathrm{d}F_{\mathrm{em}} = 0$, these fields obey $\operatorname{div}\mathbf{B} = 0$ and $\partial_t \mathbf{B} + \operatorname{curl}\mathbf{E} = 0$. For example, the coefficient of $\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$

is $\partial_x B_x + \partial_y B_y + \partial_z B_z$. The coefficient of $\mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z$ is $\partial_t B_x + \partial_y E_z - \partial_z E_y$. The other two coefficients follow by cyclic permutation.

9.2 The spacetime Hodge star

The action needs a map from two-forms to two-forms. We construct the needed constant-metric Hodge star here. This also recalls the convention of *Topology and Geometry*, Lecture 15. For basis wedges of degree p, the metric pairing is the determinant

$$\langle \mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_p}, \mathrm{d}x^{j_1} \wedge \cdots \wedge \mathrm{d}x^{j_p} \rangle = \det(\eta^{i_a j_b})_{a,b=1}^{p}.$$

Here $x^0 = t$, $x^1 = x$, $x^2 = y$ and $x^3 = z$.

Definition 9.2 (Spacetime Hodge star)**.** The *Hodge star* is the linear map from p-forms to $(4-p)$-forms determined by

$$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle \mathrm{vol} \qquad \text{for all forms } \alpha, \beta \text{ of degree } p. \tag{9.3}$$

An increasing basis wedge pairs under wedge product only with its complementary wedge. Its metric norm is the product of its diagonal metric signs. These two facts determine its star uniquely and construct the map. For example, $\mathrm{d}t \wedge \mathrm{d}x$ has metric norm squared -1, while $(\mathrm{d}t \wedge \mathrm{d}x) \wedge (\mathrm{d}y \wedge \mathrm{d}z) = \mathrm{vol}$. Hence $\star(\mathrm{d}t \wedge \mathrm{d}x) = -\mathrm{d}y \wedge \mathrm{d}z$. Applying the same rule gives all the entries needed below:

α	$\star\alpha$	β	$\star\beta$
$\mathrm{d}t \wedge \mathrm{d}x$	$-\mathrm{d}y \wedge \mathrm{d}z$	$\mathrm{d}y \wedge \mathrm{d}z$	$\mathrm{d}t \wedge \mathrm{d}x$
$\mathrm{d}t \wedge \mathrm{d}y$	$-\mathrm{d}z \wedge \mathrm{d}x$	$\mathrm{d}z \wedge \mathrm{d}x$	$\mathrm{d}t \wedge \mathrm{d}y$
$\mathrm{d}t \wedge \mathrm{d}z$	$-\mathrm{d}x \wedge \mathrm{d}y$	$\mathrm{d}x \wedge \mathrm{d}y$	$\mathrm{d}t \wedge \mathrm{d}z$

$$\star\mathrm{d}t = \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z, \quad \star\mathrm{d}x = \mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z,$$
$$\star\mathrm{d}y = \mathrm{d}t \wedge \mathrm{d}z \wedge \mathrm{d}x, \quad \star\mathrm{d}z = \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y.$$

The table gives $\star^2 = -1$ on two-forms. Symmetry of the determinant pairing also gives $\alpha \wedge \star\beta = \beta \wedge \star\alpha$ for equal degrees. In particular, varying $F_{\mathrm{em}} \wedge \star F_{\mathrm{em}}$ gives twice $\delta F_{\mathrm{em}} \wedge \star F_{\mathrm{em}}$.

9.3 Action and its variation

Let the prescribed charge density ρ and spatial current $\mathbf{J}$ be smooth functions of spacetime. Define the current one-form and current three-form by

$$\begin{aligned} j &= \rho\,\mathrm{d}t - J_x\,\mathrm{d}x - J_y\,\mathrm{d}y - J_z\,\mathrm{d}z, \qquad \mathcal{J} = \star j, \\ \mathcal{J} &= \rho\,\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z - J_x\,\mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z - J_y\,\mathrm{d}t \wedge \mathrm{d}z \wedge \mathrm{d}x - J_z\,\mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y. \end{aligned} \tag{9.4}$$

Let $D \subset \mathbb{R}^{1+3}$ be a compact region, the closure of its interior, with the fields defined on an open neighbourhood of D. Only variations compactly supported in its interior will be used.

Definition 9.3 (Maxwell action)**.** The *Maxwell action* with prescribed current is

$$S_D[A_{\mathrm{em}}] = -\frac{1}{2}\int_D F_{\mathrm{em}} \wedge \star F_{\mathrm{em}} + \int_D A_{\mathrm{em}} \wedge \mathcal{J}, \qquad F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}. \tag{9.5}$$

The current is held fixed when the potential varies.

Proposition 9.4 (Maxwell variation and differential identity)**.** *For a C^2 potential and every smooth compactly supported one-form variation a in the interior of D,*

$$\delta S_D[A_{\mathrm{em}}]a = \int_D a \wedge (\mathcal{J} - \mathrm{d}{\star}F_{\mathrm{em}}). \tag{9.6}$$

Stationarity is equivalent to $\mathrm{d}{\star}F_{\mathrm{em}} = \mathcal{J}$. The off-shell identity $\mathrm{d}(\mathrm{d}{\star}F_{\mathrm{em}}) \equiv 0$ makes $\mathrm{d}\mathcal{J} = 0$ necessary for a solution. The action is invariant under every compactly supported gauge transformation exactly when $\mathrm{d}\mathcal{J} = 0$ in the interior of D. Gauge transformations leave the field equations unchanged and prevent them from determining the potential uniquely.

Proof. Since the metric is fixed, its star does not vary. The symmetry just proved and $\delta F_{\mathrm{em}} = \mathrm{d}a$ give

$$\delta S_D[A_{\mathrm{em}}]a = -\int_D \mathrm{d}a \wedge {\star}F_{\mathrm{em}} + \int_D a \wedge \mathcal{J}.$$

The graded product rule gives $\mathrm{d}(a \wedge {\star}F_{\mathrm{em}}) = \mathrm{d}a \wedge {\star}F_{\mathrm{em}} - a \wedge \mathrm{d}{\star}F_{\mathrm{em}}$. Thus the boundary contribution is the exact term $-\int_D \mathrm{d}(a \wedge {\star}F_{\mathrm{em}})$. It is zero: extending its compactly supported coefficients by zero, each coordinate derivative integrates to zero by Fubini and the fundamental theorem of calculus. The remaining terms give (9.6). The pairing of one-forms with three-forms by wedge product is nondegenerate. Testing each coefficient with the fundamental lemma of Lecture 1 proves the stationarity assertion. Its smooth-test version follows by choosing a smooth nonnegative bump in a ball where the coefficient keeps its sign.

Applying d to the field equation gives $\mathrm{d}\mathcal{J} = 0$ because $\mathrm{d}^2 = 0$. For a C^2 potential the coefficients of $\mathrm{d}{\star}F_{\mathrm{em}}$ are continuous. Its further derivative is interpreted distributionally, where mixed derivatives still commute. For a smooth potential the same identity holds pointwise. No equation for the potential was used to obtain the identity.

A gauge transformation leaves F_{em} unchanged. Its exact change in the action is therefore

$$S_D[A_{\mathrm{em}} + \mathrm{d}\chi] - S_D[A_{\mathrm{em}}] = \int_D \mathrm{d}\chi \wedge \mathcal{J} = -\int_D \chi\, \mathrm{d}\mathcal{J}, \qquad \chi \in C_c^\infty(\operatorname{int} D). \tag{9.7}$$

Here the discarded exact term is $\int_D \mathrm{d}(\chi\mathcal{J})$. The fundamental lemma makes invariance for every χ equivalent to $\mathrm{d}\mathcal{J} = 0$. Finally, adding any $\mathrm{d}\chi \neq 0$ preserves the field strength and hence every field equation, which proves the nonuniqueness. ∎

9.4 Gauss and Ampère from the scalar density

Example 9.5 (The component field equations)**.** Compute the Euler–Lagrange equations of the Maxwell action for ϕ_{em} and $\mathbf{A}$. The Hodge table and the complementary wedge signs give

$$F_{\mathrm{em}} \wedge {\star}F_{\mathrm{em}} = (|\mathbf{B}|^2 - |\mathbf{E}|^2)\mathrm{vol}, \qquad A_{\mathrm{em}} \wedge \mathcal{J} = (-\rho\phi_{\mathrm{em}} + \mathbf{J}\cdot\mathbf{A})\mathrm{vol}.$$

Thus the scalar density is

$$L = \frac{1}{2}(|\mathbf{E}|^2 - |\mathbf{B}|^2) - \rho\phi_{\mathrm{em}} + \mathbf{J}\cdot\mathbf{A}.$$

The scalar potential occurs through $\mathbf{E} = -\nabla\phi_{\mathrm{em}} - \mathbf{A}_t$. Its density derivatives are $\partial L/\partial\phi_{\mathrm{em}} = -\rho$ and $\partial L/\partial(\partial_i\phi_{\mathrm{em}}) = -E_i$. The Euler–Lagrange expression is consequently

$$E_{\phi_{\mathrm{em}}} = \operatorname{div}\mathbf{E} - \rho.$$

For the spatial component A_x, the nonzero derivative coefficients are

$$\frac{\partial L}{\partial A_x} = J_x, \qquad \frac{\partial L}{\partial(\partial_t A_x)} = -E_x, \qquad \frac{\partial L}{\partial(\partial_y A_x)} = B_z, \qquad \frac{\partial L}{\partial(\partial_z A_x)} = -B_y.$$

They give $E_{A_x} = J_x + \partial_t E_x - \partial_y B_z + \partial_z B_y$. Cyclic permutation gives the other two components. Stationarity therefore produces Gauss's and Ampère's laws with the specified signs:

$$\operatorname{div}\mathbf{E} = \rho, \qquad \operatorname{curl}\mathbf{B} - \partial_t \mathbf{E} = \mathbf{J}. \tag{9.8}$$

These equations agree with the form equation. Indeed, the spatial coefficient of $\mathrm{d}{\star}F_{\mathrm{em}}$ is $\operatorname{div}\mathbf{E}$. Its $\mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z$ coefficient is $\partial_t E_x - (\operatorname{curl}\mathbf{B})_x$. Comparison with (9.4) gives the same source signs, and cyclic permutation supplies the rest.

9.5 The identity and the source restriction

The components of (9.4) give the conservation condition

$$\mathrm{d}\mathcal{J} = (\partial_t \rho + \operatorname{div}\mathbf{J})\mathrm{vol} = 0.$$

To see the identity directly in the equations, set $\mathcal{M}_0 = \operatorname{div}\mathbf{E}$ and $\boldsymbol{\mathcal{M}} = \partial_t \mathbf{E} - \operatorname{curl}\mathbf{B}$. Commutation of spatial derivatives gives $\operatorname{div}\operatorname{curl}\mathbf{B} = 0$, so

$$\partial_t \mathcal{M}_0 - \operatorname{div}\boldsymbol{\mathcal{M}} = \partial_t \operatorname{div}\mathbf{E} - \operatorname{div}(\partial_t \mathbf{E} - \operatorname{curl}\mathbf{B}) \equiv 0. \tag{9.9}$$

For C^2 potentials this equality is distributional. On a solution, $\mathcal{M}_0 = \rho$ and $\boldsymbol{\mathcal{M}} = -\mathbf{J}$, giving charge conservation.

This identity also follows from gauge invariance of the source-free action. For a smooth potential, its variation along $\delta\phi_{\mathrm{em}} = -\chi_t$ and $\delta\mathbf{A} = \nabla\chi$ is

$$0 = \int_D (-\mathcal{M}_0 \chi_t + \boldsymbol{\mathcal{M}} \cdot \nabla\chi)\,\mathrm{vol} = \int_D \chi(\partial_t \mathcal{M}_0 - \operatorname{div}\boldsymbol{\mathcal{M}})\,\mathrm{vol}.$$

Arbitrariness of the compactly supported function χ forces the identity. For C^2 potentials the same integration by parts defines its distributional meaning. This is the first instance of Noether's second theorem: a symmetry depending on an arbitrary function gives a differential relation between field equations. Lecture 19 develops the general statement.

Example 9.6 (An incompatible prescribed current). Take $\rho(t,x) = t$ and $\mathbf{J}(t,x) = 0$ on an open spacetime region. Its continuity expression is $\partial_t \rho + \operatorname{div}\mathbf{J} = 1$. If a potential solved Maxwell's equations there, the divergence of Ampère's law would give $\partial_t \operatorname{div}\mathbf{E} = 0$. The time derivative of Gauss's law would give $\partial_t \operatorname{div}\mathbf{E} = 1$. These contradict each other, even when interpreted distributionally. A nonconserved prescribed current is therefore inconsistent with these field equations. Lectures 10 and 13 require conserved sources.

Summary. The Maxwell action has density $\frac{1}{2}(|\mathbf{E}|^2 - |\mathbf{B}|^2) - \rho\phi_{\mathrm{em}} + \mathbf{J}\cdot\mathbf{A}$. Compactly supported variation gives $\mathrm{d}{\star}F_{\mathrm{em}} = \mathcal{J}$. The off-shell identity $\mathrm{d}^2{\star}F_{\mathrm{em}} = 0$ requires a conserved current. That condition also makes the source-coupled action gauge invariant for compactly supported gauge functions. Potentials related by $\mathrm{d}\chi$ give the same field. Lecture 10 fixes this freedom when solving the initial-data problem.

Self-check. Compute $\star(\mathrm{d}t \wedge \mathrm{d}x)$ and its star again. Which sign would an Euclidean metric incorrectly give in the first step?

Answer. The first result is $-\mathrm{d}y \wedge \mathrm{d}z$, and the second is $-\mathrm{d}t \wedge \mathrm{d}x$. An Euclidean metric would give $+\mathrm{d}y \wedge \mathrm{d}z$ in the first step.

Self-check. Start with the zero potential and use $\chi = tx^2 + yz$. Compute the transformed potential components and both fields.

Answer. The new components are $\phi_{\mathrm{em}} = -x^2$ and $\mathbf{A} = (2tx, z, y)$. The gradient and time derivative cancel in $\mathbf{E}$, so $\mathbf{E} = 0$. The curl is also zero, so $\mathbf{B} = 0$, although the potential is nonzero.

Self-check. For a gauge function supported in the interior of D, what is the exact change of the action if $\mathrm{d}\mathcal{J} \neq 0$?

Answer. It is $-\int_D \chi \, \mathrm{d}\mathcal{J}$. There is no boundary contribution because χ vanishes near the boundary. For nonzero $\mathrm{d}\mathcal{J}$, some such χ makes this change nonzero.

Self-check. Does charge conservation prove that Maxwell's equations determine the potential uniquely? Which identity holds before imposing any field equation?

Answer. No. Adding $\mathrm{d}\chi$ leaves F_{em} unchanged. The identity $\mathrm{d}(\mathrm{d}\star F_{\mathrm{em}}) = 0$ holds off shell, distributionally for a C^2 potential. Conservation is necessary for the equations to have a solution.

Lecture 10

Maxwell initial data

The Maxwell equations admit many potentials for the same field. We impose a condition on the potential that turns its equations into wave equations. Two compatibility conditions on the initial data make this condition persist during evolution.

Objectives. *We construct a smooth Maxwell potential from a conserved compact source and compatible compact initial data. We prove propagation of the gauge condition and exhibit the remaining gauge freedom in a vacuum packet.*

By the end of this lecture you will be able to:

- Check the Lorenz and Gauss compatibility conditions on potential data.
- Construct the potential by retarded and homogeneous wave solutions.
- Prove constraint propagation and identify residual gauge waves.

10.1 The Lorenz condition

We retain Lecture 9's potential and fields on $\mathbb{R}^{1+3}$: $A_{\mathrm{em}} = -\phi_{\mathrm{em}}\,\mathrm{d}t + \mathbf{A}\cdot\mathrm{d}x$, $\mathbf{E} = -\nabla\phi_{\mathrm{em}} - \mathbf{A}_t$ and $\mathbf{B} = \operatorname{curl}\mathbf{A}$. The current one-form is $j = \rho\,\mathrm{d}t - \mathbf{J}\cdot\mathrm{d}x$.

Definition 10.1 (Lorenz gauge and constraints)**.** The *Lorenz function* of a potential is

$$\ell = \partial^{\mu}(A_{\mathrm{em}})_{\mu} = -\big(\partial_t\phi_{\mathrm{em}} + \operatorname{div}\mathbf{A}\big). \tag{10.1}$$

The *Lorenz gauge* is the condition $\ell = 0$. A *constraint* is a condition on initial data required by the field equations. *Constraint propagation* means that an initially satisfied constraint remains satisfied under evolution.

The Lorenz function relates the wave operator to the inhomogeneous Maxwell equations. For every C^2 potential,

$$\operatorname{div}\mathbf{E} = \Box\phi_{\mathrm{em}} + \ell_t, \qquad \operatorname{curl}\mathbf{B} - \mathbf{E}_t = \Box\mathbf{A} - \nabla\ell. \tag{10.2}$$

The first equality follows by expanding $\operatorname{div}\mathbf{E}$ and ℓ_t. For the second, use $\operatorname{curl}\operatorname{curl}\mathbf{A} = \nabla\operatorname{div}\mathbf{A} - \Delta\mathbf{A}$. For example, its x component is $\partial_x\partial_y A_y + \partial_x\partial_z A_z - \partial_y^2 A_x - \partial_z^2 A_x$, which is the x component of the stated right side. Cyclic permutation proves the other components. Substitution of the electric field gives (10.2).

Consequently, in Lorenz gauge the reduced potential equations are

$$\Box A_{\mathrm{em}} = -j, \qquad \Box\phi_{\mathrm{em}} = \rho, \qquad \Box\mathbf{A} = \mathbf{J}. \tag{10.3}$$

The wave operator acts on each coefficient of the one-form. Its time coefficient is $-\phi_{\mathrm{em}}$, which explains the scalar source sign.

10.2 Compatible data and their evolution

Write the potential data at time zero as

$$\phi_{\mathrm{em}}(0) = a_\phi, \qquad \partial_t \phi_{\mathrm{em}}(0) = b_\phi, \qquad \mathbf{A}(0) = a_A, \qquad \mathbf{A}_t(0) = b_A.$$

The initial Lorenz condition is $b_\phi + \operatorname{div} a_A = 0$. The initial electric field is $-\nabla a_\phi - b_A$, so Gauss's law requires $-\Delta a_\phi - \operatorname{div} b_A = \rho(0)$.

Theorem 10.2 (Maxwell evolution in Lorenz gauge). *Let ρ and $\mathbf{J}$ be smooth real functions of compact spacetime support, with $\partial_t \rho + \operatorname{div} \mathbf{J} = 0$. Let a_ϕ, b_ϕ, a_A and b_A be smooth compactly supported spatial data, scalar for the first two and vector-valued for the last two. Assume the compatibility conditions*

$$b_\phi + \operatorname{div} a_A = 0, \qquad -\Delta a_\phi - \operatorname{div} b_A = \rho(0, \cdot). \tag{10.4}$$

There is a unique smooth global potential solving (10.3) with these data. It satisfies Lorenz gauge and all Maxwell equations. It is also the unique C^2 Lorenz-gauge Maxwell potential with these potential data.

Proof. We construct the wave solution first and then propagate its Lorenz condition. Set $R = G_{\mathrm{ret}} * (-j)$, taking the convolution coefficient by coefficient. Theorem 6.2 gives a smooth one-form with $\Box R = -j$. Its scalar-potential component is $G_{\mathrm{ret}} * \rho$, and its spatial vector component is $G_{\mathrm{ret}} * \mathbf{J}$.

The data $R(0)$ and $R_t(0)$ are smooth and compactly supported in space. Indeed, enclose the source in $[s_-, s_+] \times \overline{B_L(0)}$. The support bound of Lecture 6 puts $R(t)$ inside $\overline{B_{L+t-s_-}(0)}$ when $t \geq s_-$, and gives zero when $t < s_-$. Each time derivative has the same support bound because it is the retarded convolution with the corresponding derivative of the source.

Let $a = -a_\phi \, \mathrm{d}t + a_A \cdot \mathrm{d}x$ and $b = -b_\phi \, \mathrm{d}t + b_A \cdot \mathrm{d}x$ denote the one-form data. Use the massless construction of Theorem 4.3 to solve

$$\Box H = 0, \qquad H(0) = a - R(0), \qquad H_t(0) = b - R_t(0).$$

The corrected data are smooth and compactly supported, hence Schwartz. The multiplier argument of Lecture 7 gives a smooth spacetime solution for every component. Thus

$$A_{\mathrm{em}} = G_{\mathrm{ret}} * (-j) + H \tag{10.5}$$

solves the reduced wave equations and has exactly the prescribed data. The subtraction is needed whenever the retarded field has nonzero data at time zero. If the source starts later, both corrections vanish.

Next, differentiation commutes with the wave operator on this smooth solution. Current conservation gives

$$\Box \ell = -\partial_t \Box \phi_{\mathrm{em}} - \operatorname{div} \Box \mathbf{A} = -\partial_t \rho - \operatorname{div} \mathbf{J} = 0.$$

The first compatibility condition gives $\ell(0) = 0$. Using $\partial_t^2 \phi_{\mathrm{em}} = \Delta \phi_{\mathrm{em}} + \rho$ in its time derivative gives

$$\ell_t(0) = -\big(\Delta a_\phi + \rho(0, \cdot) + \operatorname{div} b_A\big) = 0$$

by the Gauss condition. Corollary 5.4 now implies $\ell = 0$ on all spacetime. It applies to this smooth scalar wave without a total-energy assumption. Equation (10.2) supplies Gauss's and Ampère's laws. The other two Maxwell equations follow from $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}$ as in Lecture 9. This proves propagation of the gauge condition and Gauss constraint.

Finally, two C^2 reduced wave solutions with the same data have a homogeneous difference with zero data. The same local uniqueness corollary makes every component of the difference zero. Every C^2 Lorenz-gauge Maxwell potential obeys the reduced wave equations by (10.2), so this also proves the final uniqueness assertion. ∎

The compact-current assumption restricts total charge. Define $Q(t) = \int_{\mathbb{R}^3} \rho(t,x)\,\mathrm{d}^3x$. The current has uniformly bounded spatial support, so differentiation and coordinate integration by parts give

$$Q'(t) = -\int_{\mathbb{R}^3} \operatorname{div}\mathbf{J}(t,x)\,\mathrm{d}^3x = 0.$$

The source vanishes sufficiently early, hence $Q(t) = 0$ for every t. The theorem does not cover a source with nonzero total charge. A Coulomb potential with tail $Q/(4\pi|x|)$ for $Q \neq 0$ is not in L^2, hence not in H^1 at spatial infinity: its squared radial integral contains a nonzero constant integrated over $1 < r < \infty$. Sheet 10 treats that different data class.

10.3 A transverse packet and a residual gauge wave

Example 10.3 (A compact vacuum packet)**.** Take $\rho = 0$ and $\mathbf{J} = 0$. Choose a smooth compactly supported cutoff ϑ equal to one near the origin, and put $f(x,y,z) = (1 - x^2 - y^2)\vartheta(x,y,z)$. Let u be the massless scalar solution of Lecture 4 with $u(0) = f$ and $u_t(0) = 0$. Define

$$\phi_{\mathrm{em}} = 0, \qquad \mathbf{A} = \operatorname{curl}(ue_z) = (u_y, -u_x, 0), \qquad e_z = (0,0,1).$$

The initial vector potential is $(f_y, -f_x, 0)$ and its initial velocity is zero. Its divergence vanishes by commutation of mixed derivatives. Both compatibility conditions therefore hold. Differentiation of $\Box u = 0$ shows that every component of $\mathbf{A}$ solves the wave equation. The potential is in Lorenz gauge because $\phi_{\mathrm{em}} = 0$ and $\operatorname{div}\mathbf{A} = 0$ for every time. Its fields are

$$\mathbf{E} = (-u_{ty}, u_{tx}, 0), \qquad \mathbf{B} = (u_{xz}, u_{yz}, -u_{xx} - u_{yy}).$$

At time zero, $\mathbf{E} = 0$ and $B_z(0,0) = 4$. The packet is nonzero. If $\operatorname{supp} f \subset \overline{B_L(0)}$, finite propagation in Lecture 5 puts these fields inside $\overline{B_{L+|t|}(0)}$ at each time.

A gauge transformation preserves Lorenz gauge exactly when its function satisfies a wave equation. Direct substitution into (10.1) gives

$$\ell \longmapsto \ell + \Box\chi. \tag{10.6}$$

A *residual gauge wave* is a smooth solution of $\Box\chi = 0$. For the packet, choose such a wave with $\chi(0) = 0$ and $\chi_t(0) = p \in C_c^\infty(\mathbb{R}^3)$, where $p \neq 0$. The transformed potential has data

$$a'_\phi = -p, \qquad b'_\phi = 0, \qquad a'_A = (f_y, -f_x, 0), \qquad b'_A = \nabla p.$$

Its Lorenz condition is unchanged, and its Gauss condition reads $-\Delta(-p) - \operatorname{div}\nabla p = 0$. The transformed fields are the same because $\mathrm{d}^2\chi = 0$. Residual gauge waves change potential data while preserving electric and magnetic fields.

10.4 Spatial transversality alone is insufficient

The condition $\operatorname{div} \mathbf{A} = 0$ is often called spatial transversality. It does not impose the scalar condition or Gauss's law. In vacuum, take $a_A = b_A = 0$ and $a_\phi = b_\phi = h$, where $h \in C_c^\infty(\mathbb{R}^3)$ is nonzero and $\Delta h \neq 0$. The reduced wave solution has $\mathbf{A} = 0$ for every time, hence is spatially transverse. At time zero, however,

$$\ell(0) = -h \neq 0, \qquad \operatorname{div} \mathbf{E}(0) = -\Delta h \neq 0.$$

Both required conditions fail. The scalar potential and vector velocity must enter the compatibility test. Lectures 11 and 12 distinguish these conditions in the Hamiltonian formulation.

Summary. A smooth conserved current of compact spacetime support and compatible smooth compact potential data determine a unique Lorenz-gauge Maxwell potential. The construction adds a retarded source field to a homogeneous wave with corrected initial data. The Lorenz function obeys the homogeneous wave equation. Its zero displacement and velocity are the Lorenz and Gauss compatibility conditions. Residual gauge waves preserve the fields while changing potential data. Spatial transversality alone supplies neither condition.

Self-check. For data $(a_\phi, b_\phi, a_A, b_A)$, compute $\ell(0)$ and $\ell_t(0)$ using the reduced wave equation. Which condition makes each one zero?

Answer. The values are $-(b_\phi + \operatorname{div} a_A)$ and $-(\Delta a_\phi + \rho(0) + \operatorname{div} b_A)$. The initial Lorenz condition kills the first. Initial Gauss compatibility kills the second.

Self-check. The retarded source field may have nonzero data at time zero. What goes wrong if the homogeneous field is assigned the original data a and b without correction?

Answer. The sum then has data $a + R(0)$ and $b + R_t(0)$. Its wave equation is correct, but its data are wrong. Subtracting both retarded data gives the required a and b.

Self-check. Why does compact spacetime support, together with conservation, force zero total charge? Does conservation alone force it?

Answer. Conservation makes the total charge constant, and compact time support makes it zero sufficiently early. Conservation alone allows a nonzero constant charge when the spatial integral and vanishing boundary flux are defined.

Self-check. A smooth gauge function satisfies $\Box\chi \neq 0$. Does it preserve the electric and magnetic fields? Does it preserve Lorenz gauge?

Answer. It preserves both fields because $\mathrm{d}^2\chi = 0$. It changes the Lorenz function by $\Box\chi$, so it does not preserve Lorenz gauge. A residual gauge wave has $\Box\chi = 0$.

Lecture 11

Maxwell's constraints

The Klein–Gordon Legendre map determines the velocity from the momentum. The Maxwell density contains no time derivative of the scalar potential. Its momentum cannot determine that velocity. We retain the undetermined velocity and require the momentum relation to hold throughout evolution. This requirement produces Gauss's law.

Objectives. *We derive Maxwell's primary and secondary constraints and their preservation. We prove that the smeared Gauss constraint generates a spatial gauge transformation and distinguish it from the full time-dependent transformation.*

By the end of this lecture you will be able to:

- Identify the singular velocity Hessian and its primary constraint.
- Derive and propagate Gauss's law with a prescribed conserved source.
- Compute the constraint brackets and the generated gauge variation.

11.1 The failed Legendre inverse

Definition 11.1 (Singular Lagrangian and primary constraint)**.** For velocity coordinates v^a, the *velocity Hessian* of a density is $W_{ab} = \partial^2 L/\partial v^a \partial v^b$. The density is *singular* where this matrix is not invertible. A *primary constraint* is a relation on the canonical variables that holds on the image of the Legendre map.

Use the component Maxwell density derived in Lecture 9:

$$L = \frac{1}{2}|\mathbf{A}_t + \nabla\phi_{\mathrm{em}}|^2 - \frac{1}{2}|\operatorname{curl}\mathbf{A}|^2 - \rho\phi_{\mathrm{em}} + \mathbf{J}\cdot\mathbf{A}, \qquad \mathbf{E} = -\mathbf{A}_t - \nabla\phi_{\mathrm{em}}, \qquad \mathbf{B} = \operatorname{curl}\mathbf{A}. \tag{11.1}$$

The prescribed source is held fixed in every variation. The canonical momenta are

$$\pi_\phi = \frac{\partial L}{\partial(\partial_t\phi_{\mathrm{em}})} = 0, \qquad \mathbf{\Pi} = \frac{\partial L}{\partial \mathbf{A}_t} = \mathbf{A}_t + \nabla\phi_{\mathrm{em}} = -\mathbf{E}. \tag{11.2}$$

In the velocity order $(\partial_t\phi_{\mathrm{em}}, A_{x,t}, A_{y,t}, A_{z,t})$, the Hessian is

$$W = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{11.3}$$

Its kernel contains $(1, 0, 0, 0)$ and its rank is three. The relation $\pi_\phi = 0$ is primary. Only the spatial velocity is determined: $\mathbf{A}_t = \mathbf{\Pi} - \nabla\phi_{\mathrm{em}}$. Changing $\partial_t\phi_{\mathrm{em}}$ leaves all four momenta unchanged. No notation for an inverse matrix can recover this missing velocity. Adding a term $\frac{1}{2}(\partial_t\phi_{\mathrm{em}})^2$ would remove that kernel by changing the density and its equations.

11.2 Extended canonical variables

We work on $\mathbb{R}^3$ with real Schwartz fields. The extended phase space is

$$\mathcal{Z}_{\mathrm{em}} = \{(\phi_{\mathrm{em}}, \mathbf{A}; \pi_\phi, \mathbf{\Pi}) : \text{all eight components lie in } \mathcal{S}(\mathbb{R}^3, \mathbb{R})\}.$$

The scalar momentum is retained as an independent coordinate before imposing its constraint. This space includes the two canonical pairs $(\phi_{\mathrm{em}}, \pi_\phi)$ and $(\mathbf{A}, \mathbf{\Pi})$, the latter with three components. Admissibility means the same as in Lecture 7: a smooth real functional has functional derivatives forming a smooth Schwartz-valued gradient map. Variations pair with those derivatives by the sum of the eight real L^2 pairings.

Definition 11.2 (Extended field bracket)**.** For admissible F and G on $\mathcal{Z}_{\mathrm{em}}$, their bracket is

$$\{F, G\} = \int_{\mathbb{R}^3} \left[F_{\phi_{\mathrm{em}}} G_{\pi_\phi} - F_{\pi_\phi} G_{\phi_{\mathrm{em}}} + F_{\mathbf{A}} \cdot G_{\mathbf{\Pi}} - F_{\mathbf{\Pi}} \cdot G_{\mathbf{A}}\right] \mathrm{d}^3x. \tag{11.4}$$

This is the sum of four brackets of Lecture 7. The same constant canonical matrix and symmetric-Hessian proof establish closure and Jacobi. All bracket statements below concern these admissible functionals or their smeared versions, rather than point evaluations.

Let the source ρ and $\mathbf{J}$ be smooth time-dependent Schwartz functions on an open time interval I, with

$$\rho_t + \operatorname{div} \mathbf{J} = 0. \tag{11.5}$$

We consider compatible initial data in this phase space. They necessarily have zero total charge: integrating $\operatorname{div} \mathbf{\Pi} + \rho = 0$ and using Schwartz decay gives $\int \rho = 0$. Vacuum is included. Nonzero total charge requires a larger spatial class, as in the Coulomb-tail question of Sheet 10.

11.3 The Hamiltonian and preservation of constraints

The Legendre subtraction on the primary surface gives the *canonical Hamiltonian* $H_c : \mathcal{Z}_{\mathrm{em}} \to \mathbb{R}$:

$$H_c = \int_{\mathbb{R}^3} \left[\frac{1}{2}|\mathbf{\Pi}|^2 + \frac{1}{2}|\mathbf{B}|^2 - \mathbf{\Pi} \cdot \nabla\phi_{\mathrm{em}} + \rho\phi_{\mathrm{em}} - \mathbf{J} \cdot \mathbf{A}\right] \mathrm{d}^3x. \tag{11.6}$$

Indeed, substitute $\mathbf{A}_t = \mathbf{\Pi} - \nabla\phi_{\mathrm{em}}$ in $\mathbf{\Pi} \cdot \mathbf{A}_t - L$. We extend this formula off the primary surface by leaving it independent of π_ϕ.

The boundary term in the scalar-potential term is visible on a cube Q_R:

$$-\int_{Q_R} \mathbf{\Pi} \cdot \nabla\phi_{\mathrm{em}} \, \mathrm{d}^3x = \int_{Q_R} \phi_{\mathrm{em}} \operatorname{div} \mathbf{\Pi} \, \mathrm{d}^3x - \int_{\partial Q_R} \phi_{\mathrm{em}} \mathbf{\Pi} \cdot n \, \mathrm{d}S.$$

Schwartz decay makes the face integral tend to zero. Thus

$$\begin{aligned} H_c &= H_0 + \int \phi_{\mathrm{em}} (\operatorname{div} \mathbf{\Pi} + \rho) \, \mathrm{d}^3x - \int \mathbf{J} \cdot \mathbf{A} \, \mathrm{d}^3x, \\ H_0 &= \frac{1}{2} \int (|\mathbf{\Pi}|^2 + |\mathbf{B}|^2) \, \mathrm{d}^3x. \end{aligned} \tag{11.7}$$

The coefficient of the scalar potential will be the secondary constraint.

Definition 11.3 (Total Hamiltonian and secondary constraint)**.** For an arbitrary prescribed smooth Schwartz multiplier $\lambda(t,x)$, the *total Hamiltonian* is

$$H_T(t) = H_c(t) + \int_{\mathbb{R}^3} \lambda(t,x)\pi_\phi(x)\,\mathrm{d}^3x. \tag{11.8}$$

Evolution obeys $\dot F = \partial_t F + \{F, H_T\}$. A *secondary constraint* is a new relation required to preserve a primary constraint during this evolution.

The extra term assigns $\partial_t \phi_{\mathrm{em}} = \lambda$ without imposing a new relation between that velocity and its zero momentum. The Hamiltonians above have smooth Schwartz gradients for the stated fields, sources and multiplier.

Proposition 11.4 (Gauss constraint and its generators)**.** *The primary constraint $\pi_\phi = 0$ is preserved exactly when the* Gauss constraint *holds:*

$$\mathcal{D} := \operatorname{div}\mathbf{\Pi} + \rho = 0. \tag{11.9}$$

This secondary constraint propagates under (11.5)*. For fixed real $\zeta, \chi \in \mathcal{S}$, define the smeared constraints*

$$P[\zeta] = \int \zeta\pi_\phi\,\mathrm{d}^3x, \qquad C[\chi] = -\int \chi(\operatorname{div}\mathbf{\Pi} + \rho)\,\mathrm{d}^3x. \tag{11.10}$$

Their mutual brackets vanish identically. The generator $C[\chi]$ shifts $\mathbf{A}$ by $\nabla\chi$ and leaves $\mathbf{\Pi}$ unchanged.

Proof. We compute the Hamilton equations before restricting to the constraints. Varying the magnetic term gives

$$\delta\left(\frac{1}{2}\int_{Q_R} |\mathbf{B}|^2\,\mathrm{d}^3x\right) = \int_{Q_R} (\operatorname{curl}\mathbf{B})\cdot\delta\mathbf{A}\,\mathrm{d}^3x + \int_{\partial Q_R} n\cdot(\delta\mathbf{A}\times\mathbf{B})\,\mathrm{d}S.$$

This follows by integrating the coordinate curl derivatives by parts. The boundary integral vanishes as $R\to\infty$ by Schwartz decay. Differentiating the other terms of H_T gives

$$\partial_t\phi_{\mathrm{em}} = \lambda, \qquad (\pi_\phi)_t = -\mathcal{D}, \qquad \mathbf{A}_t = \mathbf{\Pi} - \nabla\phi_{\mathrm{em}}, \qquad \mathbf{\Pi}_t = -\operatorname{curl}\mathbf{B} + \mathbf{J}. \tag{11.11}$$

These are equations for smooth Schwartz paths; equivalently, each follows by testing the bracket against arbitrary Schwartz smearings. The second equation proves the primary-preservation assertion. The last equation and commutation of spatial derivatives give

$$\mathcal{D}_t = \operatorname{div}\mathbf{\Pi}_t + \rho_t = \operatorname{div}\mathbf{J} + \rho_t = 0, \tag{11.12}$$

since the divergence of a curl is zero. There is no further constraint. The equations for $\mathbf{A}$ and $\mathbf{\Pi}$, with $\mathbf{E} = -\mathbf{\Pi}$, are Ampère's law and the potential relation of Lecture 9. The secondary constraint is Gauss's law.

Next, integrate the divergence in $C[\chi]$ by parts. Its only nonzero phase derivative is $C_{\mathbf{\Pi}} = \nabla\chi$. The only nonzero derivative of $P[\zeta]$ is $P_{\pi_\phi} = \zeta$. They are constant Schwartz gradient maps. Formula (11.4) gives

$$\{P[\zeta], P[\vartheta]\} = 0, \qquad \{P[\zeta], C[\chi]\} = 0, \qquad \{C[\chi], C[\vartheta]\} = 0. \tag{11.13}$$

For the generator assertion, its Hamiltonian vector has spatial components $\delta\mathbf{A} = C_{\mathbf{\Pi}} = \nabla\chi$ and $\delta\mathbf{\Pi} = -C_{\mathbf{A}} = 0$. Its scalar components are both zero. ∎

The source is explicitly time dependent in $C[\chi]$. Its preservation must include that dependence. For fixed smearings the same calculation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}P[\zeta] = \{P[\zeta], H_T\} = C[\zeta], \qquad \frac{\mathrm{d}}{\mathrm{d}t}C[\chi] = \partial_t C[\chi] + \{C[\chi], H_T\} = -\int \chi(\rho_t + \operatorname{div}\mathbf{J})\,\mathrm{d}^3x = 0.$$

Ignoring $\partial_t C$ would lose the ρ_t term.

Definition 11.5 (Weak equality and constraint classes)**.** The *constraint surface* Γ_t is the common zero set of all $P[\zeta]$ and $C[\chi]$ at time t. The notation $F \approx G$ means equality on Γ_t; this is *weak equality*. A *strong equality* holds on the entire extended phase space. A constraint is *first class* if its bracket with every defining constraint vanishes weakly. For a finite list of constraints, the list is *second class* if its bracket matrix is invertible on the constraint surface.

Since π_ϕ and $\mathcal{D}$ are Schwartz, their squared-norm tests show that Γ_t is exactly the set $\pi_\phi = \mathcal{D} = 0$. All Maxwell constraints above are first class, because (11.13) holds strongly. In contrast, $\{P[\zeta], H_T\} = C[\zeta]$ vanishes only weakly in general. No inverse bracket matrix is asserted for field constraints here.

11.4 The smeared gauge transformation

For $\chi \in \mathcal{S}$, consider the admissible functional

$$G[\chi] = \int_{\mathbb{R}^3} \mathbf{\Pi} \cdot \nabla\chi \,\mathrm{d}^3x. \tag{11.14}$$

On Q_R, its relation to the Gauss constraint includes the boundary term:

$$C_{Q_R}[\chi] = G_{Q_R}[\chi] - \int_{Q_R} \chi\rho \,\mathrm{d}^3x - \int_{\partial Q_R} \chi\mathbf{\Pi} \cdot n \,\mathrm{d}S.$$

The last integral vanishes for our Schwartz data. Hence $G[\chi] = C[\chi] + \int \chi\rho \,\mathrm{d}^3x$. The source term is constant as a function of the canonical variables. These functionals have the same Hamiltonian vector field, although $G[\chi]$ is not generally zero on Γ_t.

For a real vector test function $f \in \mathcal{S}^3$, write $\mathbf{A}(f) = \int f \cdot \mathbf{A}\,\mathrm{d}^3x$ and similarly for $\mathbf{\Pi}(f)$. Their brackets compute the transformation entirely inside the admissible class:

$$\{\mathbf{A}(f), G[\chi]\} = \int f \cdot \nabla\chi \,\mathrm{d}^3x, \qquad \{\mathbf{\Pi}(f), G[\chi]\} = 0, \qquad \{G[\chi], G[\vartheta]\} = 0. \tag{11.15}$$

The flow is exactly $(\mathbf{A}, \mathbf{\Pi}) \mapsto (\mathbf{A} + \varepsilon\nabla\chi, \mathbf{\Pi})$. The curl of a gradient is zero, so both electromagnetic fields are unchanged when the scalar potential is also fixed and χ is time independent.

For $\chi \in C^\infty(I; \mathcal{S})$, the secondary constraint alone still leaves ϕ_{em} fixed. The complete potential transformation of Lecture 9 also requires $\delta\phi_{\mathrm{em}} = -\chi_t$. On the extended phase space its generator is

$$C[\chi] - P[\chi_t].$$

The primary term gives $\delta\phi_{\mathrm{em}} = -\chi_t$ by the canonical bracket. The spatial shift remains $\delta\mathbf{A} = \nabla\chi$ and both momenta stay fixed.

To map Hamiltonian solutions, also transform the multiplier as $\lambda' = \lambda - \partial_t^2\chi$. Indeed, $\partial_t\phi'_{\mathrm{em}} = \partial_t\phi_{\mathrm{em}} - \partial_t^2\chi = \lambda'$. The spatial equation is preserved since $\partial_t\mathbf{A}' = \partial_t\mathbf{A} + \nabla\partial_t\chi = \mathbf{\Pi} - \nabla\phi'_{\mathrm{em}}$; the momentum equations and constraints are unchanged. Thus a spatial generator, a fixed-time phase transformation and a transformation of Hamiltonian solutions have distinct scopes.

Summary. Maxwell's velocity Hessian has rank three, with primary constraint $\pi_\phi = 0$ and spatial momentum $\mathbf{\Pi} = -\mathbf{E}$. The total Hamiltonian retains an arbitrary scalar-potential velocity. Primary preservation gives $\operatorname{div}\mathbf{\Pi} + \rho = 0$, whose propagation requires source conservation. The smeared constraints commute strongly and are first class. The Gauss generator shifts $\mathbf{A}$ by $\nabla\chi$; its difference from $G[\chi]$ is the prescribed source term. Lecture 12 removes this spatial gauge freedom in vacuum.

Self-check. Two potentials have the same spatial velocities and spatial derivatives, but their scalar-potential velocities differ. Can their canonical momenta coincide? Which term in H_T retains this freedom?

Answer. Yes. The density contains no $\partial_t\phi_{\mathrm{em}}$, so all four momenta coincide. The term $\int \lambda\pi_\phi$ assigns the arbitrary velocity $\partial_t\phi_{\mathrm{em}} = \lambda$.

Self-check. Is $\{P[\zeta], H_T\} = 0$ a strong identity? What must hold on the constraint surface?

Answer. The bracket equals $C[\zeta]$, which is not identically zero on phase space. It vanishes on Γ_t. This is weak preservation of the primary constraint.

Self-check. At a fixed time, suppose $\int \chi\rho\,\mathrm{d}^3x = 3$. Compare $C[\chi]$ and $G[\chi]$ on Γ_t. Do they generate different spatial variations?

Answer. The constraint has value zero and $G[\chi] = 3$. Their phase derivatives coincide because the source is prescribed, so both generate $\delta\mathbf{A} = \nabla\chi$ and $\delta\mathbf{\Pi} = 0$.

Self-check. For time-dependent χ, which term supplements $C[\chi]$ to generate $\delta\phi_{\mathrm{em}} = -\chi_t$? Does it alter $\mathbf{\Pi}$?

Answer. The term is $-P[\chi_t] = -\int \chi_t\pi_\phi\,\mathrm{d}^3x$. It shifts the scalar potential and leaves $\mathbf{\Pi}$ unchanged.

Lecture 12

Transverse reduction

The Gauss constraint restricts Maxwell's momentum. Gauge transformations identify different spatial potentials. We impose the restriction and make the identification explicitly, using Fourier transformation away from zero frequency. The remaining variables have two independent directions at each nonzero wave vector.

Objectives. *We construct the transverse projector, identify the null directions of the constrained canonical form and derive the reduced bracket. Constant fields on a torus show why zero frequency requires a separate analysis.*

By the end of this lecture you will be able to:

- Find the unique transverse representative of an annulus-supported gauge orbit.
- Prove that the constrained form becomes nondegenerate after the gauge quotient.
- Compute the transverse smeared bracket and retain torus harmonic modes.

12.1 A class separated from zero frequency

Fix $0 < k_- < k_+ < \infty$ and the open Fourier annulus $\mathcal{U} = \{k \in \mathbb{R}^3 : k_- < |k| < k_+\}$. We use the unitary Fourier transform of Lecture 4.

Definition 12.1 (Annulus data class). The real *annulus data class* is

$$\mathcal{X} = \{f \in \mathcal{S}(\mathbb{R}^3, \mathbb{R}) : \widehat{f} \in C_c^\infty(\mathcal{U}, \mathbb{C})\}.$$

Its vector version is $\mathcal{X}^3$. All data, smearing functions and gauge parameters in the main reduction belong to these classes.

Reality means $\widehat{f}(-k) = \overline{\widehat{f}(k)}$. Inverse transforms of smooth compact frequency profiles are Schwartz functions, by the Fourier automorphism recalled in Lecture 7. Smooth multipliers on a neighbourhood of $\mathcal{U}$ preserve frequency support. The multipliers used below also preserve Fourier reality, so they preserve this real class. No division by $|k|$ below reaches $k = 0$.

We work in vacuum, $\rho = 0$. Lecture 11 gives $\mathbf{\Pi} = -\mathbf{E}$ and the Gauss constraint $\operatorname{div} \mathbf{\Pi} = 0$. The primary constraint sets $\pi_\phi = 0$, while its generator $P[\zeta]$ in (11.10) shifts the scalar potential arbitrarily. Removing this pair leaves the spatial data space and constraint surface given by

$$\mathcal{Z}_{\mathcal{U}} = \mathcal{X}^3 \times \mathcal{X}^3, \qquad C = \{(\mathbf{A}, \mathbf{\Pi}) \in \mathcal{Z}_{\mathcal{U}} : \operatorname{div} \mathbf{\Pi} = 0\}.$$

The constraint is equivalently $k \cdot \widehat{\mathbf{\Pi}}(k) = 0$. For $\chi \in \mathcal{X}$, its gauge action is $(\mathbf{A}, \mathbf{\Pi}) \mapsto (\mathbf{A} + \nabla\chi, \mathbf{\Pi})$. Its generator has the positive-gradient convention of (11.15): $G[\chi] = \int \mathbf{\Pi} \cdot \nabla\chi \, \mathrm{d}^3x$.

Definition 12.2 (Gauge orbit and reduced phase space)**.** The *gauge orbit* of $(\mathbf{A}, \mathbf{\Pi}) \in C$ is

$$[(\mathbf{A}, \mathbf{\Pi})] = \{(\mathbf{A} + \nabla\chi, \mathbf{\Pi}) : \chi \in \mathcal{X}\}.$$

The *reduced phase space* $C/\mathcal{G}$ is the set of these orbits, where $\mathcal{G}$ denotes the additive group of gauge parameters.

The canonical form from Lecture 8, summed over the three spatial components, is the bilinear pairing

$$\Omega((a, p), (b, q)) = \int_{\mathbb{R}^3} (a \cdot q - b \cdot p)\, \mathrm{d}^3x. \tag{12.1}$$

Its restriction to C can have a kernel. A vector belongs to this kernel when its pairing with every allowed tangent vector is zero. Here tangent vectors to C are pairs (a, p) with $\operatorname{div} p = 0$.

12.2 Projection and the quotient

Definition 12.3 (Transverse projector)**.** For $k \neq 0$, the *transverse projector* is the matrix

$$P_{\mathrm{T}}(k) = I - \frac{kk^{\mathsf{T}}}{|k|^2}, \qquad P_{\mathrm{T}}(k)_{ij} = \delta_{ij} - \frac{k_i k_j}{|k|^2}. \tag{12.2}$$

For $v \in \mathcal{X}^3$, define its transverse part by $\widehat{v_T}(k) = P_{\mathrm{T}}(k)\widehat{v}(k)$. We also write $P_{\mathrm{T}}v = v_T$ for this spatial operator. A vector field is *transverse* when $\operatorname{div} v = 0$.

The matrix projects a vector onto the plane perpendicular to k. Figure 12.1 separates this transverse part from the longitudinal part parallel to k.

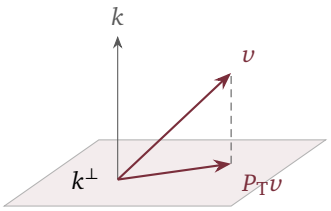


Figure 12.1: The transverse plane $k^{\perp}$. A vector splits as $v = P_{\mathrm{T}}v + (I - P_{\mathrm{T}})v$, with the second component parallel to k.

For real smearing functions $f, g \in \mathcal{X}^3$, the reduced linear observables are

$$A_T(f) = \int f \cdot \mathbf{A}_T\, \mathrm{d}^3x, \qquad \Pi_T(g) = \int g \cdot \mathbf{\Pi}_T\, \mathrm{d}^3x.$$

We consider finite polynomials in these observables. Their gradients remain in the annulus class. This specifies the observable algebra used for the reduced bracket.

Theorem 12.4 (Transverse reduction). *In the stated annulus class, every orbit in $\mathcal{C}$ has exactly one representative with* $\operatorname{div}\mathbf{A}_T = 0$. *The quotient is identified with*

$$\mathcal{Z}_T = \{(\mathbf{A}_T, \mathbf{\Pi}_T) \in \mathcal{X}^3 \times \mathcal{X}^3 : \operatorname{div}\mathbf{A}_T = \operatorname{div}\mathbf{\Pi}_T = 0\}.$$

The kernel of the constrained canonical form consists exactly of the gauge directions $(\nabla\chi, 0)$. *The form therefore descends to the nondegenerate canonical form* (12.1) *on* $\mathcal{Z}_T$. *The reduced bracket on smeared generators is*

$$\{A_T(f), \Pi_T(g)\}_{\mathrm{red}} = \int f \cdot P_{\mathrm{T}} g \, \mathrm{d}^3x, \qquad \{A_T(f), A_T(g)\}_{\mathrm{red}} = \{\Pi_T(f), \Pi_T(g)\}_{\mathrm{red}} = 0. \tag{12.3}$$

The bracket on their polynomial algebra follows by bilinearity and the product rule. At each $k \neq 0$, *the transverse position and momentum each have two independent components.*

Proof. First, put $K = kk^{\mathsf{T}}/|k|^2$. Since $k^{\mathsf{T}}k = |k|^2$, $K^2 = K$. Thus $P_{\mathrm{T}}^2 = (I - K)^2 = I - K = P_{\mathrm{T}}$. Also $P_{\mathrm{T}}^{\mathsf{T}} = P_{\mathrm{T}}$, $P_{\mathrm{T}}k = 0$ and $\operatorname{tr} P_{\mathrm{T}} = 3 - 1 = 2$. These identities show that P_{T} is the orthogonal projection onto $k^{\perp}$. It is real and even in k, so it preserves Fourier reality.

The transverse part of a potential is gauge invariant because $P_{\mathrm{T}}(\widehat{\mathbf{A}} + ik\widehat{\chi}) = P_{\mathrm{T}}\widehat{\mathbf{A}}$. To reach that representative, choose

$$\widehat{\chi}(k) = \frac{i\,k \cdot \widehat{\mathbf{A}}(k)}{|k|^2}.$$

This is a smooth compact annulus profile with the required reality condition. It gives $\widehat{\mathbf{A}} + ik\widehat{\chi} = P_{\mathrm{T}}\widehat{\mathbf{A}}$. If two transverse representatives differ by $ik\widehat{\chi}$, taking the dot product with k gives $i|k|^2\widehat{\chi} = 0$. Thus they coincide. The constraint already gives $\mathbf{\Pi} = \mathbf{\Pi}_T$.

Next, a gauge direction pairs to zero with every $(b, q) \in \mathcal{C}$:

$$\Omega((\nabla\chi, 0), (b, q)) = \int \nabla\chi \cdot q \, \mathrm{d}^3x = -\int \chi \operatorname{div} q \, \mathrm{d}^3x = 0.$$

Schwartz decay removes the face terms, as in Lecture 7. Conversely, suppose (a, p) is in the kernel. Testing against $(p, 0)$ gives $-\int |p|^2 = 0$, so $p = 0$. Testing against $(0, P_{\mathrm{T}}a)$ gives $\int a \cdot P_{\mathrm{T}}a = \int |P_{\mathrm{T}}a|^2 = 0$, by symmetry and idempotence of the multiplier and Plancherel. Hence $P_{\mathrm{T}}a = 0$, and the preceding gauge construction expresses a as a gradient. Removing these null directions gives a well-defined form on orbits. For a transverse pair (a, p), its pairing with $(-p, a)$ is $\|a\|_2^2 + \|p\|_2^2$. Thus the quotient form is nondegenerate. It is closed because its coefficients are constant.

Finally, symmetry of P_{T} in the real L^2 pairing gives $A_T(f) = \int (P_{\mathrm{T}}f) \cdot \mathbf{A}$ and $\Pi_T(g) = \int (P_{\mathrm{T}}g) \cdot \mathbf{\Pi}$. Their canonical gradients are $(P_{\mathrm{T}}f, 0)$ and $(0, P_{\mathrm{T}}g)$. The bracket of Lecture 7 therefore gives

$$\{A_T(f), \Pi_T(g)\} = \int P_{\mathrm{T}}f \cdot P_{\mathrm{T}}g \, \mathrm{d}^3x = \int f \cdot P_{\mathrm{T}}g \, \mathrm{d}^3x.$$

These observables are constant on each gauge orbit, so this is their reduced bracket. Their Hamiltonian vectors are transverse and agree with those obtained directly from the quotient form. The same construction and the product rule apply to their finite polynomials. ∎

12.3 Two polarisations and the Fourier bracket

A *polarisation vector* at $k \neq 0$ is a unit vector in $k^{\perp}$. Choose an orthonormal pair $\varepsilon_1(k)$ and $\varepsilon_2(k)$ at that fixed k. The two components of $\widehat{\mathbf{A}}_T$ along this pair, and the two corresponding momentum components, describe the transverse data there. This pointwise choice does not require a smooth basis on the whole annulus.

The Fourier transform of a real field couples k to $-k$ through its reality condition. On smooth frequency tests supported in $\mathcal{U}$, the bracket identity can be written in the formal kernel notation

$$\{\widehat{A}_{T,i}(k), \widehat{\Pi}_{T,j}(k')\}_{\mathrm{red}} = P_{\mathrm{T}}(k)_{ij}\, \delta(k + k'). \tag{12.4}$$

This notation means that testing with profiles $\alpha_i(k)$ and $\beta_j(k')$ gives $\int \alpha_i(k) P_{\mathrm{T}}(k)_{ij} \beta_j(-k)\, \mathrm{d}^3k$. Complex tests use the complex bilinear extension of the real bracket. This specifies only the tested identity on the chosen annulus class. The sum $k + k'$ comes from the two forward Fourier phases:

$$(2\pi)^{-3} \int_{\mathbb{R}^3} e^{-i(k+k')\cdot x}\, \mathrm{d}^3x = \delta(k + k').$$

As in Lecture 7, the kernel notation does not make mode point values admissible observables.

Example 12.5 (The bracket along a coordinate axis)**.** Take $k = (0, 0, k_3) \in \mathcal{U}$, with $k_3 \neq 0$. The projector and transverse Fourier data are

$$P_{\mathrm{T}}(k) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \widehat{\mathbf{A}}_T = (\widehat{A}_1, \widehat{A}_2, 0), \qquad \widehat{\mathbf{\Pi}}_T = (\widehat{\Pi}_1, \widehat{\Pi}_2, 0).$$

Choose $\varepsilon_1 = e_1$ and $\varepsilon_2 = e_2$. In the tested notation of (12.4), the two-component bracket matrix at this wave vector is

$$\big(\{\widehat{A}_\alpha(k), \widehat{\Pi}_\beta(k')\}_{\mathrm{red}}\big)_{\alpha,\beta=1}^{2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \delta(k + k').$$

The Gauss constraint removes $\widehat{\Pi}_3$, and gauge equivalence removes $\widehat{A}_3$. It leaves two canonical pairs, with Fourier reality relating the coefficients at the opposite wave vector.

12.4 The zero mode on the torus

The annulus assumption excludes harmonic data. On $\mathbb{T}^3$, a constant vector field is *harmonic*: every component has zero Laplacian. Take constant data $\mathbf{A} = a$ and $\mathbf{\Pi} = p$, with $a, p \in \mathbb{R}^3$. The Gauss constraint holds automatically. Conversely, $\Delta v = 0$ on the torus gives $|n|^2 \widehat{v}_n = 0$, so all harmonic Fourier components have $n = 0$ and are constant. A periodic gauge parameter obeys $\int_{\mathbb{T}^3} \nabla\chi\, \mathrm{d}^3x = 0$, by opposite-face cancellation. Consequently it cannot change the mean a. These constant modes survive the quotient, with canonical form

$$\Omega_{\mathrm{harm}} = (2\pi)^3 \sum_{i=1}^{3} \delta a_i \wedge \delta p_i.$$

There are three harmonic pairs. Applying the nonzero-k count to $k = 0$ would incorrectly remove one of them. Here gauge parameters are single-valued periodic real functions, as in the gauge action above. Lectures 20 and 21 must retain harmonic and global data when the spatial topology changes.

12.5 The bounded projector and the energy-space scope

For any $v \in L^2(\mathbb{R}^3;\mathbb{R}^3)$, define $P_{\mathrm{T}}v$ directly by the multiplier (12.2) for $k \neq 0$. Its value at the single point $k = 0$ has no effect on an L^2 function. Orthogonality gives

$$\|P_{\mathrm{T}}v\|_2^2 = \int |P_{\mathrm{T}}(k)\widehat{v}(k)|^2 \,\mathrm{d}^3k \leq \int |\widehat{v}(k)|^2 \,\mathrm{d}^3k = \|v\|_2^2.$$

Thus this is a bounded orthogonal projector on all of L^2, not merely on one fixed annulus. The same calculation with Fourier weight $1 + |k|^2$ gives its bounded action on H^1.

The corresponding *inhomogeneous energy-space reduction* also follows from this projector. Take $\mathbf{A} \in H^1$ and $\mathbf{\Pi} \in L^2$ with $\operatorname{div}\mathbf{\Pi} = 0$ distributionally, and quotient by distributional gradients $\nabla\chi \in H^1$. The reduced space is

$$H^1_{\mathrm{trans}}(\mathbb{R}^3;\mathbb{R}^3) \times L^2_{\mathrm{trans}}(\mathbb{R}^3;\mathbb{R}^3),$$

where the subscript means distributional divergence zero. Indeed, set $\widehat{\chi} = -i(k \cdot \widehat{\mathbf{A}})/|k|^2$ for $k \neq 0$. This defines a tempered distribution: near zero, Cauchy–Schwarz uses $\int_{|k|<1} |k|^{-2}\mathrm{d}^3k < \infty$ to prove local integrability; at infinity its pairing with every Schwartz test converges by Cauchy–Schwarz. It preserves Fourier reality and gives $\nabla\chi = (I - P_{\mathrm{T}})\mathbf{A} \in H^1$. Conversely, an H^1 gradient is longitudinal: its Fourier curl is zero, so $k \times \widehat{\nabla\chi} = 0$ almost everywhere and its transverse projection vanishes. Thus each orbit has exactly one transverse representative. The canonical form descends because longitudinal gradients are L^2-orthogonal to transverse momenta, by Plancherel.

Its form (12.1) is weakly nondegenerate. If (a, p) pairs to zero with every transverse pair, testing with $(0, a)$ first gives $a = 0$. For the momentum test take $(h, 0)$, where $\widehat{h} = (1 + |k|^2)^{-1}\widehat{p}$. Then h is transverse and $\|h\|_{H^1}^2 = \int (1 + |k|^2)^{-1}|\widehat{p}|^2\mathrm{d}^3k \leq \|p\|_2^2$, and the pairing is $-\int (1 + |k|^2)^{-1}|\widehat{p}|^2\mathrm{d}^3k$. Its vanishing gives $p = 0$. The form is closed because it is constant. Here *weak* means nondegenerate without requiring every continuous functional to have a representing vector. For the resulting transverse Sobolev phase model, see Emberton, *An introduction to free quantum field theory through Klein–Gordon theory*, §7, p. 14, in the University of Chicago notes. Their momentum is $\mathbf{E}$, while ours is $\mathbf{\Pi} = -\mathbf{E}$. The fixed-annulus theorem above remains independent of this broader argument. Sheet 12 develops the completion and gauge-parameter details further. Requiring only finite physical energy controls $\operatorname{curl}\mathbf{A}$ and $\mathbf{\Pi}$, but need not put $\mathbf{A}$ in L^2. It would not by itself justify the canonical integral used here.

Summary. For vacuum annulus-supported Schwartz data, the Gauss constraint sets the longitudinal momentum to zero and gauge equivalence removes the longitudinal potential. Their quotient has the nondegenerate canonical form on $(\mathbf{A}_T, \mathbf{\Pi}_T)$ and the smeared bracket $\int f \cdot P_{\mathrm{T}}g$. At nonzero k, the projector has rank two. The tested Fourier bracket contains $\delta(k+k')$. On the torus, three constant canonical pairs survive periodic gauge transformations. The projector itself is bounded on all of L^2.

Self-check. At $k = (0, 0, 2)$, project $v = (3, -1, 5)$. Which canonical components does the Gauss constraint remove, and which does gauge equivalence remove?

Answer. The projection is $(3, -1, 0)$. The constraint removes the third momentum component. Gauge equivalence removes the third potential component.

Self-check. For annulus data, find the gauge parameter that takes $\mathbf{A}$ to $\mathbf{A}_T$. Why does it preserve the real-field condition?

Answer. Its transform is $\widehat{\chi} = i(k \cdot \widehat{\mathbf{A}})/|k|^2$. At $-k$, its value is $-i(k \cdot \overline{\widehat{\mathbf{A}}})/|k|^2$, the complex conjugate of its value at k.

Self-check. For a real nonzero Fourier mode, should the bilinear canonical bracket pair $\widehat{A}(k)$ with $\widehat{\Pi}(k)$ or with $\widehat{\Pi}(-k)$?

Answer. It pairs opposite wave vectors. The tested kernel contains $\delta(k + k')$, as follows from the two forward Fourier phases. A conjugated Hilbert inner product has a different frequency bookkeeping rule.

Self-check. On $\mathbb{T}^3$, can a periodic real gauge parameter remove a nonzero constant potential? What is the symplectic coefficient for each constant pair?

Answer. No. Its gradient has zero mean, so the constant potential is unchanged. Each pair has coefficient $\mathrm{vol}(\mathbb{T}^3) = (2\pi)^3$ in the canonical form.

Lecture 13

Radiation

A compact source sends energy through increasingly large spheres. The observation time must increase with the radius to follow that energy. We compute the limiting outward flux of the retarded Maxwell field.

Objectives. *We derive the energy balance directly from Maxwell's equations and prove the far-field power formula. We apply it to a slowly varying neutral dipole source and distinguish an absolute error from a relative one.*

By the end of this lecture you will be able to:

- Derive the Poynting balance with its source-work sign.
- Compute the radiated power from a smooth compact conserved current.
- Obtain the dipole coefficient and control the slow-time error.

13.1 Observation time and energy flux

Use the real fields and units of Lectures 9 and 10. Write $r = |x|$ and $n = x/r \in \mathbb{S}^2$ when $x \neq 0$. The sphere area measure $\mathrm{d}\Omega$ has total area 4π.

Definition 13.1 (Far zone and radiated power)**.** The *retarded time* of an observation is $u = t - r$. The *far-zone limit* is $r \to \infty$ at fixed u and n. For the outward energy flux $\mathbf{E} \times \mathbf{B}$, the *radiated power* is the limit, when it exists,

$$P(u) = \lim_{r\to\infty} r^2 \int_{\mathbb{S}^2} (\mathbf{E} \times \mathbf{B})(r + u, rn) \cdot n \, \mathrm{d}\Omega. \tag{13.1}$$

For completeness, recall the Maxwell stress tensor in the convention of *Topology and Geometry*, Lecture 19. Writing $F_{\mathrm{em}} = \frac{1}{2} F_{\mu\nu} \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu$, it is

$$T_{\mu\nu} = -F_{\mu\alpha} F_\nu{}^\alpha + \frac{1}{4} \eta_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}.$$

The coefficients $F_{0i} = -E_i$ and $F_{ij} = \epsilon_{ijk} B_k$ give its energy components by contraction:

$$\rho_{\mathrm{en}} = T^{00} = \frac{1}{2}(|\mathbf{E}|^2 + |\mathbf{B}|^2), \qquad T^{i0} = (\mathbf{E} \times \mathbf{B})_i.$$

Here ϵ_{ijk} is the alternating symbol, with $\epsilon_{123} = 1$. We prove the required balance without importing a stress-tensor conservation theorem.

Maxwell's equations give $\mathbf{E}_t = \operatorname{curl} \mathbf{B} - \mathbf{J}$ and $\mathbf{B}_t = -\operatorname{curl} \mathbf{E}$. The component product rule gives $\operatorname{div}(\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot \operatorname{curl} \mathbf{E} - \mathbf{E} \cdot \operatorname{curl} \mathbf{B}$. Taking the two scalar products yields the *Poynting identity*

$$\partial_t \rho_{\mathrm{en}} = \mathbf{E} \cdot (\operatorname{curl} \mathbf{B} - \mathbf{J}) - \mathbf{B} \cdot \operatorname{curl} \mathbf{E} = -\operatorname{div}(\mathbf{E} \times \mathbf{B}) - \mathbf{J} \cdot \mathbf{E}. \tag{13.2}$$

Thus $\mathbf{J}\cdot\mathbf{E}$ is the rate of energy transfer from the field to the prescribed source. Outside its support, the source-work term in the Poynting identity vanishes.

13.2 The far-field limit

Assume ρ and $\mathbf{J}$ are smooth, compactly supported in spacetime, and satisfy $\rho_t + \operatorname{div}\mathbf{J} = 0$. Use the retarded potentials from Lecture 6:

$$\phi_{\mathrm{em}}(t,x) = \frac{1}{4\pi}\int \frac{\rho(t-|x-y|,y)}{|x-y|}\,\mathrm{d}^3y, \qquad \mathbf{A}(t,x) = \frac{1}{4\pi}\int \frac{\mathbf{J}(t-|x-y|,y)}{|x-y|}\,\mathrm{d}^3y.$$

Derivatives pass to the smooth source as proved there. Their Lorenz function is $\ell = -G_{\mathrm{ret}} * (\rho_t + \operatorname{div}\mathbf{J}) = 0$. The wave equations and the reduction in Lecture 10 therefore give Maxwell's equations.

Theorem 13.2 (Radiated power). *For this retarded field define the vector and scalar amplitudes by*

$$\mathcal{A}(u,n) = \frac{1}{4\pi}\int \mathbf{J}(u+n\cdot y,y)\,\mathrm{d}^3y, \qquad \mathcal{V}(u,n) = \frac{1}{4\pi}\int \rho(u+n\cdot y,y)\,\mathrm{d}^3y. \tag{13.3}$$

Primes denote derivatives in u. With $P_{\mathrm{T}}(n) = \mathbb{1} - nn^{\mathsf{T}}$,

$$\begin{aligned}
\mathbf{A}(r+u,rn) &= \frac{\mathcal{A}(u,n)}{r} + O(r^{-2}), \qquad \phi_{\mathrm{em}}(r+u,rn) = \frac{\mathcal{V}(u,n)}{r} + O(r^{-2}),\\
\mathbf{E}(r+u,rn) &= -\frac{P_{\mathrm{T}}(n)\mathcal{A}'(u,n)}{r} + O(r^{-2}),\\
\mathbf{B}(r+u,rn) &= -\frac{n\times\mathcal{A}'(u,n)}{r} + O(r^{-2}).
\end{aligned} \tag{13.4}$$

The errors are uniform in $u\in\mathbb{R}$ and $n\in\mathbb{S}^2$. The limit (13.1) exists and equals

$$P(u) = \int_{\mathbb{S}^2} |n\times\mathcal{A}'(u,n)|^2\,\mathrm{d}\Omega. \tag{13.5}$$

Proof. We estimate the potentials and their derivatives before computing the flux. Enclose the spatial source support in $B_R(0)$ and take $r > 2R$. For $|y|\le R$, put $d = |rn-y|$ and $e = (rn-y)/d$. Taylor expansion of $\sqrt{1-2n\cdot y/r+|y|^2/r^2}$ on this bounded set gives

$$d = r - n\cdot y + O(r^{-1}), \qquad d^{-1} = r^{-1} + O(r^{-2}), \qquad e = n + O(r^{-1}). \tag{13.6}$$

These bounds are uniform in n and y. Figure 13.1 shows why the retarded source time approaches $u + n\cdot y$.

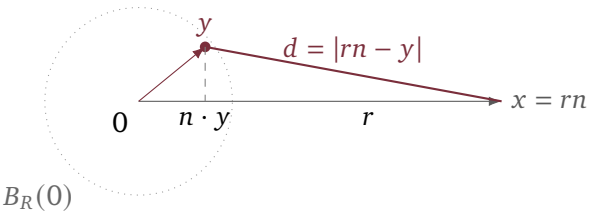


Figure 13.1: The source point y and observer rn have separation d. The projection $n\cdot y$ shortens the leading travel distance, so the source is sampled at time $u + n\cdot y$ in the limit.

Let V_f be the retarded potential of any one source component f, and let $\mathcal{V}_f(u,n) = (4\pi)^{-1}\int f(u+n\cdot y,y)\mathrm{d}^3y$. All derivatives of f are bounded, and the integration region is fixed. The mean-value estimate and (13.6) first give $V_f(r+u,rn) = \mathcal{V}_f(u,n)/r + O(r^{-2})$. Differentiating the exact integral at fixed (t,x) gives

$$\begin{aligned}\partial_t V_f(t,rn) &= \frac{1}{4\pi}\int \frac{f_t(t-d,y)}{d}\,\mathrm{d}^3y = \frac{\mathcal{V}_f'(u,n)}{r} + O(r^{-2}),\\ \partial_{x_i} V_f(t,rn) &= -\frac{1}{4\pi}\int e_i\left(\frac{f_t(t-d,y)}{d} + \frac{f(t-d,y)}{d^2}\right)\mathrm{d}^3y = -\frac{n_i\mathcal{V}_f'(u,n)}{r} + O(r^{-2}),\end{aligned} \tag{13.7}$$

where the asymptotic equalities are evaluated at $t = r+u$. Bounded second time derivatives justify the same mean-value estimates here. In particular, the radial derivative at fixed t has leading term $-\mathcal{V}_f'/r$. For a unit tangent vector v to the direction sphere, $D_n d[v] = -r(v\cdot y)/d = O(1)$. Differentiating the integral in this direction gives $D_n V_f = O(r^{-1})$. Its contribution to the Cartesian gradient has an additional factor $1/r$ and is therefore $O(r^{-2})$. Thus no angular derivative of the leading amplitude has been discarded at order $1/r$.

Current conservation relates the amplitudes. The total y-divergence of the compactly supported vector $\mathbf{J}(u+n\cdot y,y)$ integrates to zero. The chain rule gives

$$0 = \int \left[\operatorname{div}_y \mathbf{J} + n\cdot\partial_t\mathbf{J}\right](u+n\cdot y,y)\,\mathrm{d}^3y.$$

Combining this equality with $\rho_t = -\operatorname{div}_y\mathbf{J}$ proves $\mathcal{V}' = n\cdot\mathcal{A}'$. Substitution into (13.7) gives $\mathbf{E} = (n\mathcal{V}' - \mathcal{A}')/r + O(r^{-2})$ and $\mathbf{B} = -n\times\mathcal{A}'/r + O(r^{-2})$. These are (13.4).

Finally, set $b = \mathcal{A}'(u,n)$ and $E_* = -P_{\mathrm{T}}(n)b$. The magnetic leading coefficient is $n\times E_* = -n\times b$. Since $E_*\cdot n = 0$, the vector triple-product identity gives

$$r^2(\mathbf{E}\times\mathbf{B})(r+u,rn)\cdot n = \left[E_*\times(n\times E_*)\right]\cdot n + O(r^{-1}) = |n\times b|^2 + O(r^{-1}).$$

The error is uniform on the sphere. Its integral tends to zero, proving (13.5). ∎

13.3 A slowly varying neutral source

For a compact charge distribution of zero total charge, the *electric dipole moment* is $\mathbf{p}(t) = \int y\rho(t,y)\mathrm{d}^3y$. Choose fixed real $f\in C_c^\infty(\mathbb{R}^3)$ with $\int f\mathrm{d}^3y = 1$ and fixed $\mathbf{p}_0\in C_c^\infty(\mathbb{R};\mathbb{R}^3)$. For $0<\varepsilon\le 1$, consider

$$\mathbf{p}(t) = \mathbf{p}_0(\varepsilon t), \qquad \rho(t,y) = -\mathbf{p}(t)\cdot\nabla f(y), \qquad \mathbf{J}(t,y) = \dot{\mathbf{p}}(t)f(y). \tag{13.8}$$

This source is smooth and compactly supported in spacetime for each ε. Direct differentiation gives $\rho_t + \operatorname{div}\mathbf{J} = 0$. Coordinate integration by parts gives $\int\rho = 0$ and $\int y_i\rho\mathrm{d}^3y = p_i(t)\int f\mathrm{d}^3y = p_i(t)$, so the prescribed vector is indeed its dipole moment.

Example 13.3 (Dipole power with an absolute error)**.** For the source (13.8), compute the leading radiated power as $\varepsilon\downarrow 0$. Its amplitude derivative is

$$\mathcal{A}'(u,n) = \frac{1}{4\pi}\int \ddot{\mathbf{p}}(u+n\cdot y)f(y)\,\mathrm{d}^3y.$$

Since $\|\ddot{\mathbf{p}}\|_\infty = O(\varepsilon^2)$ and $\|\mathbf{p}^{(3)}\|_\infty = O(\varepsilon^3)$, the mean-value estimate gives

$$\mathcal{A}'(u,n) = \frac{\ddot{\mathbf{p}}(u)}{4\pi} + R_\varepsilon(u,n), \qquad |R_\varepsilon| \le \frac{\varepsilon^3\|\mathbf{p}_0^{(3)}\|_\infty}{4\pi}\int |y|\,|f(y)|\,\mathrm{d}^3y.$$

This estimate is uniform in u and n. The cross term in the squared amplitude is $O(\varepsilon^5)$, and its remainder square is $O(\varepsilon^6)$. To integrate the leading term, rotate a fixed vector $\mathbf{a}$ to the polar axis:

$$\int_{\mathbb{S}^2} |n \times \mathbf{a}|^2 \, \mathrm{d}\Omega = 2\pi |\mathbf{a}|^2 \int_{-1}^{1} (1 - s^2) \, \mathrm{d}s = \frac{8\pi}{3} |\mathbf{a}|^2.$$

The power formula therefore yields the rationalised *Larmor formula* with its slow-time error:

$$P(u) = \frac{|\ddot{\mathbf{p}}(u)|^2}{6\pi} + O(\varepsilon^5) \qquad \text{uniformly in } u \in \mathbb{R}. \tag{13.9}$$

The implied constant depends on the fixed profiles f and $\mathbf{p}_0$. The estimate is absolute. Dividing it by $|\ddot{\mathbf{p}}(u)|^2$ is invalid at a zero of the acceleration.

13.4 The limits of the source model

Retardation is a condition on the solution, as in Lecture 6. Adding a homogeneous Maxwell field preserves the source equations and can change the flux through infinity. Formula (13.5) describes the retarded field of the stated smooth source. The planned QFT course returns to choices of propagator and boundary conditions.

A point charge is outside these smooth-source hypotheses. The Abraham–Lorentz–Dirac self-force equation has runaway solutions. We do not derive or use it here. See Spohn, *Dynamics of Charged Particles and their Radiation Field* (2004), Chapter 9, and his study of the Lorentz–Dirac equation. The prescribed smooth current in this lecture is not a point particle acted on by its own singular field.

Summary. The Poynting vector is $\mathbf{E} \times \mathbf{B}$ and its balance includes $-\mathbf{J} \cdot \mathbf{E}$. A smooth compact conserved current has retarded far-field power $\int |n \times \mathcal{A}'|^2 \mathrm{d}\Omega$. Conservation removes the longitudinal electric coefficient. For the neutral slow-time dipole source, the leading power is $|\ddot{\mathbf{p}}|^2/(6\pi)$ with a uniform absolute $O(\varepsilon^5)$ error. This formula does not describe a singular self-force problem.

Self-check. At a fixed direction, suppose $\mathcal{A}'(u, n)$ is parallel to n. What are the leading radiation fields and flux in that direction?

Answer. Both $P_{\mathrm{T}}(n)\mathcal{A}'$ and $n \times \mathcal{A}'$ vanish. The $1/r$ radiation fields and the limiting flux are zero in that direction. The scalar amplitude is needed for this cancellation in the electric field.

Self-check. Why does differentiating an angle-dependent amplitude not add another $1/r$ term to the Cartesian gradient of $\mathcal{V}_f(u, n)/r$?

Answer. Changing x tangentially changes $n = x/r$ at rate $1/r$. The amplitude already has its factor $1/r$, so an angular derivative enters at order r^{-2}. The exact derivative estimate in the proof bounds this term uniformly.

Self-check. For a dipole acceleration $\mathbf{a}$, compute the angular integral of its leading power density $|n \times \mathbf{a}|^2/(16\pi^2)$.

Answer. The integral of $|n \times \mathbf{a}|^2$ is $8\pi|\mathbf{a}|^2/3$. Division by $16\pi^2$ gives $|\mathbf{a}|^2/(6\pi)$. A missing 4π in the retarded potential would change this coefficient.

Self-check. Let f be radial, nonnegative, supported in $|y| \le 1$, with integral one. Put $M_2 = \int |y|^2 f(y) \mathrm{d}^3 y$ and take $\mathbf{p}_0(s) = \mathbf{a}s^4$ for $|s| \le 1$, with a smooth compact cutoff outside this interval and $\mathbf{a} \in \mathbb{R}^3 \setminus \{0\}$. Compute $P(0)$ for $0 < \varepsilon \le 1$, although $\ddot{\mathbf{p}}(0) = 0$.

Answer. Radial symmetry gives $\int (n \cdot y)^2 f(y) \mathrm{d}^3 y = M_2/3$. The sampled acceleration is $12\varepsilon^4 (n \cdot y)^2 \mathbf{a}$, so $\mathcal{A}'(0, n) = \varepsilon^4 M_2 \mathbf{a}/\pi$. The angular integral gives $P(0) = 8\varepsilon^8 M_2^2 |\mathbf{a}|^2/(3\pi) > 0$. A relative error with a zero leading value could not allow this result.

Lecture 14

The Dirac field

A field equation can be first order in both time and space. The Dirac field provides such an example. We construct its evolution and recover a conserved current and a real symplectic form. Its conserved positive norm does not make its classical energy positive.

Objectives. *We derive and solve the two-component Dirac equation in one space dimension. We prove finite propagation speed and identify its symplectic form on the underlying real solution space.*

By the end of this lecture you will be able to:

- Derive the Dirac equation and its adjoint from the action.
- Construct the unique distributional evolution for arbitrary L^2 data.
- Compute mode currents and distinguish positive charge from energy.

14.1 Matrices, field and action

Definition 14.1 (Clifford relation and Dirac matrices)**.** With $\eta = \operatorname{diag}(1,-1)$, the *Clifford relation* is $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu} I$ for $\mu,\nu = 0,1$. Our *Dirac matrices* are

$$\gamma^0 = \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \gamma^1 = \mathrm{i}\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Here I is the two-dimensional identity matrix.

Their squares are I and $-I$, and their mixed anticommutator is zero. Define the Hermitian matrices

$$\alpha = \gamma^0\gamma^1 = -\sigma_3 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \beta = \gamma^0.$$

They obey $\alpha^2 = \beta^2 = I$ and $\alpha\beta + \beta\alpha = 0$.

Definition 14.2 (Spinor field and Dirac adjoint)**.** A *spinor field* here is a two-component column $\psi : \mathbb{R}^{1+1} \to \mathbb{C}^2$. Its entries are ordinary commuting complex numbers. Its *Dirac adjoint* is the row $\bar\psi = \psi^\dagger\gamma^0$, where $\dagger$ denotes conjugate transpose.

For smooth fields and $m \geq 0$, the Dirac density is

$$L = \bar\psi(\mathrm{i}\gamma^\mu\partial_\mu - m)\psi = \mathrm{i}\psi^\dagger\psi_t + \mathrm{i}\psi^\dagger\alpha\psi_x - m\psi^\dagger\beta\psi. \tag{14.1}$$

During variation, treat ψ and $\psi^\dagger$ as independent complex variables and then restore conjugacy. On a compact spacetime rectangle, integrating the derivatives of $\delta\psi$ gives the boundary term $\int_{\partial D} \mathrm{i}\bar\psi\gamma^\mu\delta\psi\,\mathrm{d}\Sigma_\mu$. For variations supported in the interior this term vanishes.

The remaining coefficients of $\delta\psi^\dagger$ and $\delta\psi$ are

$$\mathrm{i}\psi_t + \mathrm{i}\alpha\psi_x - m\beta\psi, \qquad -\mathrm{i}\psi_t^\dagger - \mathrm{i}\psi_x^\dagger\alpha - m\psi^\dagger\beta.$$

They are adjoints of each other. Stationarity gives

$$\mathrm{i}\psi_t = \mathsf{H}\psi, \qquad \mathsf{H} = -\mathrm{i}\alpha\partial_x + m\beta. \tag{14.2}$$

The operator H maps Schwartz columns to Schwartz columns. The density need not be pointwise real. The product rule gives $L - \overline{L} = \mathrm{i}\partial_t(\psi^\dagger\psi) + \mathrm{i}\partial_x(\psi^\dagger\alpha\psi)$. Its imaginary part contributes only a boundary term to the action. Thus compact variation agrees with variation of the real action in the real and imaginary components.

14.2 The Cauchy problem in L^2

For rough data, a solution means $\psi \in C(\mathbb{R}; L^2(\mathbb{R}, \mathbb{C}^2))$ satisfying (14.2) in distributions, componentwise against compact smooth spacetime tests, with the prescribed value at zero in the L^2 norm. The Fourier convention remains that of Lecture 4. The symbol and frequency are

$$H(k) = \alpha k + m\beta = \begin{pmatrix} -k & m \\ m & k \end{pmatrix}, \qquad H(k)^2 = \nu(k)^2 I, \qquad \nu(k) = \sqrt{k^2 + m^2}. \tag{14.3}$$

Theorem 14.3 (Dirac evolution). *For every $f \in L^2(\mathbb{R}, \mathbb{C}^2)$ and $m \geq 0$, there is exactly one solution in the stated class, given by*

$$\widehat{\psi}(t, k) = U(t, k)\widehat{f}(k), \qquad U(t, k) = \cos(t\nu)I - \mathrm{i}S_t(\nu)H(k). \tag{14.4}$$

Here $S_t(\nu) = \sin(t\nu)/\nu$ for $\nu > 0$ and $S_t(0) = t$. The evolution is unitary: $\|\psi(t)\|_2 = \|f\|_2$. Both components solve Klein–Gordon distributionally. If f vanishes outside a closed set K, then $\psi(t)$ vanishes almost everywhere outside $\{x : \operatorname{dist}(x, K) \leq |t|\}$.

Proof. Hermiticity of $H(k)$ and its square give $U^\dagger U = (\cos^2(t\nu) + \nu^2 S_t(\nu)^2)I = I$. Also $\mathrm{i}U_t = H(k)U$ and $U(0, k) = I$. At $m = k = 0$, the symbol vanishes and $U = I$. Pointwise continuity and the bound $\|U\widehat{f}\| \leq \|\widehat{f}\|$ permit dominated convergence in L^2. Plancherel proves time continuity and norm preservation. Applying the same argument to a difference of data proves continuous dependence, uniformly for all times.

Multiply $\widehat{f}$ by a smooth compact frequency cutoff equal to one on $[-R, R]$, as in Lecture 4. The inverse transform is a smooth solution: all differentiated inverse integrals converge absolutely on compact time intervals. Unitarity makes these solutions converge to ψ uniformly in time in L^2. Passing to the limit against compact tests proves the distributional equation.

For uniqueness, cut off the Fourier transform of a difference with zero data. The passage from compact spatial tests to Schwartz tests uses the L^2 cutoff argument of Lecture 4 and justifies this transformed equation. On its compact frequency support, $H(k)$ is a bounded Hermitian operator on L^2. Its distributional equation therefore gives a continuous L^2 derivative: integrate $-\mathrm{i}Hw$ in the norm and use the scalar zero-derivative argument of Lecture 4 to identify w with that primitive. Thus w is C^1 in L^2, and $\frac{\mathrm{d}}{\mathrm{d}t}\|w\|_2^2 = 2\operatorname{Re}(w, -\mathrm{i}Hw) = 0$. The zero datum forces $w = 0$. Removing the cutoff gives uniqueness.

Constant-coefficient distributional derivatives commute. Apply $\mathrm{i}\partial_t + \mathsf{H}$ to $(\mathrm{i}\partial_t - \mathsf{H})\psi = 0$. Since $\mathsf{H}^2 = -\partial_x^2 + m^2$, this gives $(\partial_t^2 - \partial_x^2 + m^2)\psi = 0$. For arbitrary L^2 data, the initial

velocity specified by this equation is the distribution $-\mathrm{i}\mathsf{H}f$, which need not belong to L^2. This componentwise assertion is therefore not the energy-class Klein–Gordon statement of Lecture 4.

For propagation, start with smooth compact data. Their solution is smooth by the Fourier formula. Define its charge density and current by

$$\rho_D = j^0 = \psi^\dagger\psi, \qquad j^1 = \bar\psi\gamma^1\psi = \psi^\dagger\alpha\psi, \qquad |j^1| \le \rho_D. \tag{14.5}$$

The inequality follows from $\rho_D + j^1 = 2|\psi_2|^2$ and $\rho_D - j^1 = 2|\psi_1|^2$. Using the Dirac and adjoint equations gives

$$\partial_t\rho_D = -\psi_x^\dagger\alpha\psi - \psi^\dagger\alpha\psi_x = -\partial_x j^1.$$

The mass terms cancel. Coordinate FTC on $I_t = [a - R + t, a + R - t]$, for $0 \le t < R$, yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{I_t}\rho_D\,\mathrm{d}x = -(\rho_D + j^1)(t, a + R - t) - (\rho_D - j^1)(t, a - R + t) \le 0.$$

This is the moving-interval calculation of Lecture 5, now with a density that controls the field itself. Zero initial charge on the base interval forces the solution to vanish on every smaller interval. Approximate arbitrary L^2 data by compact smooth data. Unitarity gives L^2 convergence at the initial and final times, so the local squared norms and their inequality converge. The support conclusion follows by choosing a base interval disjoint from K, as in Lecture 5. Time reversal handles $t < 0$. ∎

The conserved total charge is $Q_D = \int\rho_D\,\mathrm{d}x = \|f\|_2^2$. For rough solutions the currents are L^1 functions, and their continuity identity follows by the same L^2 approximation. The construction never requires pointwise values of an arbitrary L^2 spinor.

14.3 Modes and the sign of energy

A plane-wave mode has the form $\psi(t,x) = ue^{-\mathrm{i}\lambda t + \mathrm{i}kx}$, where $u \in \mathbb{C}^2$ and $H(k)u = \lambda u$. For $\nu > 0$, the eigenvalues are $\lambda = \nu$ and $\lambda = -\nu$, since $H^2 = \nu^2 I$ and $\operatorname{tr} H = 0$. For a normalised eigenvector, the density and current are

$$\rho_D = 1, \qquad j^1 = \frac{k}{\lambda} = \pm\frac{k}{\nu}.$$

Indeed, $H\alpha + \alpha H = 2kI$, so its expectation in u gives $2\lambda u^\dagger\alpha u = 2k$. For a concrete pair, take $k = 3$ and $m = 4$:

$$H(3) = \begin{pmatrix} -3 & 4 \\ 4 & 3 \end{pmatrix}, \qquad \nu(3) = 5, \qquad u_+ = \frac{1}{\sqrt5}\begin{pmatrix}1\\2\end{pmatrix}, \qquad u_- = \frac{1}{\sqrt5}\begin{pmatrix}-2\\1\end{pmatrix}.$$

Here $H(3)u_+ = 5u_+$ and $H(3)u_- = -5u_-$. The modes $u_+e^{-5\mathrm{i}t+3\mathrm{i}x}$ and $u_-e^{5\mathrm{i}t+3\mathrm{i}x}$ have currents $3/5$ and $-3/5$, respectively. These are densities of modes. A nonzero plane wave has infinite total charge on $\mathbb{R}$ and is not among the theorem's L^2 data.

For Schwartz data the Hamiltonian is the real functional

$$H[\psi] = \int_{\mathbb{R}}\psi^\dagger\mathsf{H}\psi\,\mathrm{d}x = \int_{\mathbb{R}}\widehat{\psi}^\dagger H(k)\widehat{\psi}\,\mathrm{d}k. \tag{14.6}$$

For Schwartz columns ψ, h, integration of $\int\psi^\dagger\mathsf{H}h\,\mathrm{d}x$ by parts has boundary term $-\mathrm{i}[\psi^\dagger\alpha h]_{-R}^{R}$ on $[-R, R]$. It vanishes for Schwartz columns, so H is symmetric on them. Its Hamiltonian

has both signs even on finite-charge data. Since $U(t,k)$ commutes with $H(k)$, its Fourier formula also proves conservation of this energy for Schwartz data. For $m > 0$, the smooth projectors $P_\pm(k) = \frac{1}{2}(I \pm H(k)/\nu)$ select the two eigenlines. Choose a nonzero smooth compact Fourier packet in either eigenline. Its inverse transform is Schwartz, and its energy is $\pm\int \nu|\widehat{\psi}|^2\,\mathrm{d}k$, while its charge is strictly positive. Positive charge therefore does not imply positive classical energy. The planned QFT course revisits this energy question together with quantisation and fermionic statistics.

14.4 The real symplectic form

The time boundary term of the action gives the momentum row $\pi_\psi = \mathrm{i}\psi^\dagger$ and the potential $\Theta = \mathrm{i}\int \psi^\dagger\delta\psi\,\mathrm{d}x$. With the sign of Lecture 8, its antisymmetric variation is

$$\Omega = -\delta\Theta = -\mathrm{i}\int \delta\psi^\dagger \wedge \delta\psi\,\mathrm{d}x = \int \delta\psi \wedge \delta\pi_\psi. \tag{14.7}$$

The components are summed. Write $\psi = a + \mathrm{i}b$, with real two-component columns a and b. Direct expansion gives

$$\Theta = \int (b\cdot\delta a - a\cdot\delta b)\,\mathrm{d}x + \frac{\mathrm{i}}{2}\delta\|\psi\|_2^2,$$

$$\Omega = 2\sum_{r=1}^{2}\int \delta a_r \wedge \delta b_r\,\mathrm{d}x.$$

Thus the imaginary part of the potential is exact and its exterior derivative vanishes. The symplectic form is real on the underlying real L^2 space. Explicitly,

$$\Omega(h_1,h_2) = 2\,\mathrm{Im}\int h_1^\dagger h_2\,\mathrm{d}x, \qquad \Omega(h,\mathrm{i}h) = 2\|h\|_2^2.$$

It is closed because it is constant, and the last identity proves nondegeneracy. Unitary evolution preserves the complex inner product and hence this form, so it is independent of the constant time slice. On Schwartz paths, $X_H = -\mathrm{i}\mathsf{H}\psi$ satisfies $\iota_{X_H}\Omega = \delta H$ and is the Dirac evolution. No anticommuting variables or additional constraint algorithm were used.

14.5 The three answers in Part I

The constructions of Part I can now be compared on their stated domains.

Field	Action	Cauchy problem	Canonical structure
Scalar	Compact variations give Klein–Gordon.	$H^1\times L^2$ data give unique causal evolution.	Slice-independent form on compact-data solutions.
Maxwell	Gauge invariance gives an identity.	Conserved compact source and compatible compact data in Lorenz gauge.	Vacuum annulus gauge quotient.
Dirac in 1 + 1	Independent adjoint variation gives a first-order equation.	L^2 spinors give unique unitary causal evolution.	Real symplectic form on the complex L^2 solution space.

The four-component Dirac equation in 3 + 1 dimensions is Lorentz covariant under transformations preserving spacetime orientation and the future direction. This statement is *cited*: see Thaller, *The Dirac Equation* (1992), §3.3.8, pp. 100–101, Chapter 3. Its spin transformations use the double cover $SL(2,\mathbb{C})$ of that Lorentz group, recalled in *Lie Groups*, Lecture 24, and *Symmetries*, Lecture 17.

Summary. In 1+1 dimensions the two-component Dirac equation has unique unitary causal evolution for every L^2 datum. Its components solve Klein–Gordon distributionally, while its positive charge and indefinite classical energy are distinct conserved quantities on their stated domains. The action induces a real, nondegenerate, slice-independent form on the underlying real solution space.

Self-check. At $m = 0$, solve the two component transport equations from their initial profiles. Which component carries current to the right?

Answer. The components are $\psi_1(t,x) = f_1(x+t)$ and $\psi_2(t,x) = f_2(x-t)$. The second component travels right with $j^1 = |\psi_2|^2$ when the first is zero.

Self-check. At $m = 0$, take $f = (\mathbf{1}_{[0,1]}, 0)^T$. Is the time derivative at zero an L^2 function? Does componentwise Klein–Gordon put this datum in L4's class?

Answer. The first time derivative is $\delta_0 - \delta_1$, a distribution rather than an L^2 function. The first component solves Klein–Gordon distributionally, but this is not an $H^1 \times L^2$ Cauchy datum.

Self-check. For the worked negative-energy mode, compare the charge density, energy density and total charge. Can finite-charge states also have negative energy?

Answer. The densities are 1 and −5. The plane wave has infinite total charge. A nonzero smooth Fourier packet in the negative eigenline has finite positive charge and strictly negative energy.

Self-check. If a real tangent vector h pairs to zero with every tangent vector under Ω, which test vector proves $h = 0$?

Answer. Choose ih, viewed as another vector in the underlying real space. The pairing is $2\|h\|_2^2$, so $h = 0$.

Part II

Nonlinear Fields, Gauge Fields, Integrable Fields, Gravity

Part II tests the action, Cauchy and canonical descriptions on nonlinear scalar fields, local gauge symmetry, integrable evolution and gravity. The scalar theory begins with local existence and energy, then relates vacua and kink fluctuations to symmetry and stability.

Lecture 15

Nonlinear local evolution

A nonlinear equation need not have an explicit solution formula. We use the linear evolution to construct a nonlinear field by successive approximation. The construction gives a time interval shared by all data in a bounded ball.

Objectives. *We solve the one-dimensional ϕ^4 equation near a vacuum or a stationary kink. We prove the embedding and forcing estimates needed for contraction, then identify the only obstruction to continuing an energy solution.*

By the end of this lecture you will be able to:

- Bound a one-dimensional H^1 function in the supremum norm.
- Construct the nonlinear energy solution by a Duhamel contraction.
- Prove continuous dependence and the finite-lifespan blow-up alternative.

15.1 A stationary background and a perturbation

The potential and field equation are

$$V(\phi) = \frac{1}{2}(1-\phi^2)^2, \qquad \phi_{tt} - \phi_{xx} + V'(\phi) = 0, \qquad V'(u) = 2u(u^2-1).$$

Definition 15.1 (Stationary background). A *stationary background* is a time-independent solution b satisfying $-b'' + V'(b) = 0$. We use

$$b(x) \in \{1, -1, \tanh x\}.$$

The constant minima 1 and -1 are the *vacua*. The profile $b(x) = \tanh x$, connecting these vacua, is the *kink*.

For a vacuum the stationary equation is immediate. For the kink, $b' = 1 - b^2$ and $b'' = -2b(1-b^2) = V'(b)$. All three backgrounds obey $|b| \le 1$. Figure 15.1 shows their profiles. None belongs to $L^2(\mathbb{R})$.

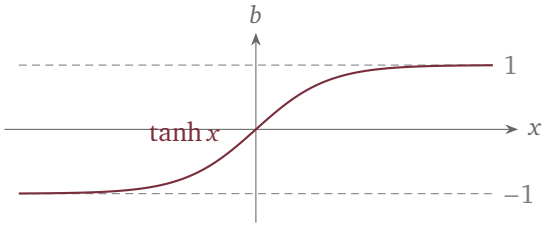


Figure 15.1: The two constant vacua and the kink $b(x) = \tanh x$. The kink approaches -1 and 1 at the two spatial ends.

Set $\phi = b + v$. Subtract the stationary equation before placing any term in L^2. Using a linear Klein–Gordon operator of mass one gives

$$\begin{aligned} v_{tt} - v_{xx} + v &= N_b(v), \\ N_b(v) &= v - [V'(b+v) - V'(b)] = (3 - 6b^2)v - 6bv^2 - 2v^3. \end{aligned} \tag{15.1}$$

There is no background forcing term. The reference mass one is a choice for the integral equation. The physical linearisation at either vacuum has $V''(\pm 1) = 4$, hence mass two.

Use the energy space $\mathcal{E} = H^1(\mathbb{R}) \times L^2(\mathbb{R})$ of Lecture 4, with the equivalent Hilbert norm

$$\|(f,g)\|_{\mathrm{lin}}^2 = \|f\|_{H^1}^2 + \|g\|_2^2.$$

An *energy solution* has $v \in C([0,T]; H^1) \cap C^1([0,T]; L^2)$, the prescribed data, and satisfies (15.1) distributionally on $(0,T) \times \mathbb{R}$.

15.2 The one-dimensional estimate

Lemma 15.2 (Embedding and nonlinear bound). *Each $v \in H^1(\mathbb{R})$ has a continuous bounded representative, with*

$$\|v\|_\infty \le \frac{1}{\sqrt{2}} \|v\|_{H^1}. \tag{15.2}$$

For $\|v\|_{H^1}, \|w\|_{H^1} \le R$, the force is in L^2 and obeys

$$\|N_b(v) - N_b(w)\|_2 \le L_R \|v - w\|_{H^1}, \qquad L_R = 3 + 6\sqrt{2}R + 3R^2. \tag{15.3}$$

In particular, $N_b : H^1 \to L^2$ is locally Lipschitz and $N_b(0) = 0$.

Proof. The unitary inverse Fourier formula and Cauchy–Schwarz give

$$|v(x)| \le (2\pi)^{-1/2} \int |\widehat{v}(k)| \, \mathrm{d}k \le (2\pi)^{-1/2} \left(\int \frac{\mathrm{d}k}{1+k^2} \right)^{1/2} \|v\|_{H^1} = \frac{1}{\sqrt{2}} \|v\|_{H^1}.$$

Thus $\widehat{v} \in L^1$. Its inverse is continuous by dominated convergence and represents the same L^2 function.

Put $M = R/\sqrt{2}$. Both $|v|$ and $|w|$ are at most M. Factor the differences of their squares and cubes in (15.1). Since $|3 - 6b^2| \le 3$, the pointwise bound is

$$|N_b(v) - N_b(w)| \le (3 + 12M + 6M^2)|v - w|.$$

Taking L^2 norms proves (15.3). Setting $w = 0$ also proves that every term in $N_b(v)$ is in L^2. ∎

15.3 The forced linear formula

Let $U(t) : \mathcal{E} \to \mathcal{E}$ be Lecture 4's mass-one evolution, including its velocity. Its Fourier frequency is $\nu(k) = \sqrt{1 + k^2}$. In Fourier variables the pair $(\nu \widehat{f}, \widehat{g})$ rotates by angle $t\nu$. Consequently

$$\|U(t)z\|_{\mathrm{lin}} = \|z\|_{\mathrm{lin}}, \qquad U(t+s) = U(t)U(s).$$

The rotation identity and the group law follow from the sine and cosine addition formulas. Strong continuity is the one proved in Lecture 4.

Definition 15.3 (Duhamel formula and mild solution)**.** For $F \in C([0,T];L^2)$, its zero-data *Duhamel pair* is

$$(\mathcal{D}F)(t) = \int_0^t U(t-s)(0,F(s))\,\mathrm{d}s. \tag{15.4}$$

A *mild solution* with data $z_0 = (v_0, g_0) \in \mathcal{E}$ is a continuous pair $z = (v,p) : [0,T] \to \mathcal{E}$ satisfying

$$z(t) = U(t)z_0 + \mathcal{D}(N_b(v))(t). \tag{15.5}$$

The integral is a norm limit of Riemann sums in the complete energy space. Continuity of its integrand makes these sums Cauchy by uniform continuity. Its first component is the Fourier sine formula

$$\widehat{(\mathcal{D}F)_1}(t,k) = \int_0^t \frac{\sin((t-s)\nu(k))}{\nu(k)}\widehat{F}(s,k)\,\mathrm{d}s,$$

understood as an L^2 identity. The sign is positive because the right-hand side of (15.1) is $+N_b(v)$.

Lemma 15.4 (Forced energy estimate)**.** *For continuous L^2 forcing, $\mathcal{D}F = (w,p)$ is continuous into $\mathcal{E}$, $w_t = p$ in L^2, and $w_{tt} - w_{xx} + w = F$ weakly, with zero initial data. Moreover,*

$$\|\mathcal{D}F(t)\|_{\mathrm{lin}} \le \int_0^t \|F(s)\|_2\,\mathrm{d}s. \tag{15.6}$$

Every energy solution of this forced equation is its homogeneous solution plus this Duhamel pair.

Proof. Isometry of U and the triangle inequality for Riemann sums prove the bound. Strong continuity of U gives continuity of the integral.

To justify the equation, use the smooth compact Fourier cutoffs P_j of Lecture 4 and set $F_j = P_jF$. These converge uniformly in $C([0,T];L^2)$: the compact set $F([0,T])$ has finite nets, and bounded operators converging strongly converge uniformly on each such compact set. For bounded frequencies, $1 - \partial_x^2$ is a bounded L^2 operator. Differentiating the sine and cosine integrals gives $(w_j)_t = p_j$ and $(p_j)_t = \partial_x^2 w_j - w_j + F_j$. Thus the cutoff pair satisfies the forced equation.

The bound applied to $F_j - F$ gives convergence of the pairs uniformly in $\mathcal{E}$. Passing to the limit in $w_j(t) = \int_0^t p_j(s)\,\mathrm{d}s$ proves $w_t = p$ in L^2. Against every compact smooth test ψ in $(0,T)\times\mathbb{R}$, the cutoff equation is

$$\int w_j(\psi_{tt} - \psi_{xx} + \psi)\,\mathrm{d}t\,\mathrm{d}x = \int F_j\psi\,\mathrm{d}t\,\mathrm{d}x.$$

Uniform L^2 convergence passes both integrals to the limit. The initial data also converge and are zero.

Finally, subtract this constructed pair and the homogeneous solution from any energy solution with the same forcing and data. The difference has zero data and solves the homogeneous equation. Lecture 4's uniqueness proves it is zero. Its frequency-cutoff proof applies on $[0,T]$ as well: it integrates the cutoff energy from the continuous initial traces at zero. ∎

Lemma 15.2 makes $N_b(v)$ continuous into L^2 for every energy solution. Lemma 15.4 therefore proves that mild and energy solutions of the nonlinear equation are the same, with $p = v_t$.

15.4 Contraction and continuation

The *lifespan* T_* is the supremum of the times to which the unique solution with the given data can be continued in the energy class.

Theorem 15.5 (Local nonlinear evolution). *For each allowed b and every $z_0 = (v_0, g_0) \in \mathcal{E}$, there is a unique energy solution on some interval $[0, T]$. A common choice for all $\|z_0\|_{\mathrm{lin}} \le B$, with $B \ge 1$, is*

$$R = 2B, \qquad T = \frac{1}{2L_R}. \tag{15.7}$$

On this interval the solution norm is at most R, and the solution map is Lipschitz in $C([0,T];\mathcal{E})$. The maximal solution satisfies

$$T_* < \infty \quad \Longrightarrow \quad \|(v(t), v_t(t))\|_{\mathrm{lin}} \longrightarrow \infty \quad \text{as } t \uparrow T_*. \tag{15.8}$$

Time reversal gives the same local and continuation statements for negative times.

Proof. The path space $X_T = C([0,T];\mathcal{E})$ is complete in the supremum of the linear norm, by completeness of $\mathcal{E}$. Its closed ball $\mathcal{B}_R$ is nonempty. Define Φ on this ball by the right-hand side of (15.5). The preceding lemmas give

$$\|\Phi z\|_{X_T} \le B + TL_R R = R, \qquad \|\Phi z - \Phi \widetilde{z}\|_{X_T} \le TL_R \|z - \widetilde{z}\|_{X_T} = \frac{1}{2}\|z - \widetilde{z}\|_{X_T}.$$

The Banach contraction principle from *Analysis* I, Lecture 1, says that a contraction of a nonempty closed subset of a Banach space has a unique fixed point. It supplies the mild solution here, and hence the energy solution.

For two data in the same ball, subtract their integral equations and take the supremum norm. This gives

$$\|z - \widetilde{z}\|_{X_T} \le \|z_0 - \widetilde{z}_0\|_{\mathrm{lin}} + \frac{1}{2}\|z - \widetilde{z}\|_{X_T},$$

so the difference is at most twice the initial difference. Every other energy solution stays inside the construction ball for a sufficiently short initial interval, by continuity. It agrees there with the fixed point. The same argument can be restarted at any common time. The set of agreement is closed by continuity and open by this local uniqueness, so the solutions agree throughout their common interval. Repeated short intervals also give continuous dependence on every compact interval on which the solutions remain in a common bounded ball.

Unique local solutions glue to a maximal one. Suppose $T_* < \infty$ and there were times $t_j \uparrow T_*$ with solution norm at most B. Replace B by $\max\{B, 1\}$. Restarting at t_j gives a solution on $[t_j, t_j + 1/(2L_{2B})]$, with the same positive duration for every j. For large j this extends beyond T_* and agrees on the overlap by uniqueness, a contradiction. Thus every bounded ball is eventually left, which proves (15.8). ∎

Example 15.6 (A common contraction time near a vacuum). Take $b = 1$ and data with $\|(v_0, g_0)\|_{\mathrm{lin}} \le 1$. The force is $N_1(v) = -3v - 6v^2 - 2v^3$. The construction uses

$$R = 2, \qquad L_2 = 15 + 12\sqrt{2}, \qquad T = \frac{1}{30 + 24\sqrt{2}}.$$

Its radius bound is $1 + TL_2 \cdot 2 = 2$, and its contraction factor is $TL_2 = 1/2$. These constants prove a common positive time for the entire data ball. They do not estimate the maximal lifespan sharply.

15.5 Why the dimension matters

The embedding used above fails in higher dimensions. In three dimensions, let $\eta(x) = e^{-|x|^2}$ and $u_\lambda(x) = \lambda^{1/2}\eta(\lambda x)$ for $\lambda \geq 1$. With $I = (\pi/2)^{3/2}$, substitution gives

$$\|u_\lambda\|_{H^1}^2 = I(\lambda^{-2} + 3) \leq 4I, \qquad \|u_\lambda\|_\infty = \lambda^{1/2} \longrightarrow \infty.$$

Thus an H^1 bound cannot replace the supremum bound in that proof. This does not disprove cubic local existence in three dimensions: $H^1(\mathbb{R}^3) \subset L^6(\mathbb{R}^3)$ controls a cubic in L^2, as developed on Sheet 15. Higher dimensions require estimates suited to the nonlinearity.

In two dimensions the failure is logarithmic. For $\Lambda \geq e$, the Fourier profile

$$\widehat{u}_\Lambda(k) = \frac{\mathbf{1}_{\{1 \leq |k| \leq \Lambda\}}}{\sqrt{\log \Lambda}\,|k|^2}$$

gives, by polar integration and the unitary inverse transform,

$$\|u_\Lambda\|_{H^1}^2 = 2\pi\left[1 + \frac{1 - \Lambda^{-2}}{2\log\Lambda}\right] \leq 3\pi, \qquad u_\Lambda(0) = \sqrt{\log \Lambda}.$$

The norm calculation integrates $(1 + r^2)/r^3$ over $1 \leq r \leq \Lambda$. The value at zero uses $\int_1^\Lambda dr/r = \log\Lambda$. The one-dimensional proof therefore cannot be applied unchanged even in two dimensions. Lecture 16 stays in one dimension and uses energy to control the continuation norm.

Summary. Subtracting a stationary vacuum or kink gives an $H^1 \times L^2$ perturbation equation without a background force. In one dimension the Fourier embedding makes its cubic force locally Lipschitz from H^1 to L^2. The forced Klein–Gordon estimate gives a contraction on energy paths, uniformly on bounded data balls. Mild and weak energy solutions coincide. Uniqueness permits continuation, and any finite maximal lifespan forces the perturbation norm to tend to infinity.

Self-check. Why is $V'(b + v)$ not used alone as the L^2 force around the kink? What happens when $v = 0$ in the correct perturbation equation?

Answer. The stationary term must cancel b''. The correct difference is $V'(b + v) - V'(b)$, so $N_b(0) = 0$ and the zero perturbation remains zero. The construction needs no assumption that $b \in L^2$.

Self-check. For the unit data ball in the worked example, what happens to the proved contraction factor if the chosen time is doubled?

Answer. It becomes $2TL_2 = 1$. The stated estimate no longer gives a strict contraction, even though a longer actual solution may exist.

Self-check. Does a finite maximal lifespan give only an unbounded sequence of energy-space norms, or does the norm tend to infinity?

Answer. The energy-space norm tends to infinity as $t \uparrow T_*$. A bounded subsequence would supply a uniform restart duration and extend the solution beyond T_*.

Self-check. Does the three-dimensional concentration example prove that the cubic wave equation has no $H^1 \times L^2$ local theory?

Answer. No. It disproves the $H^1 \to L^\infty$ estimate used here. The separate embedding $H^1 \to L^6$ can control the cubic in L^2.

Lecture 16

Energy and global existence

Lecture 15 constructs the nonlinear field for a short time. Its continuation theorem reduces global existence to a bound on the perturbation's energy-space norm. The conserved nonlinear energy supplies that bound on every finite time interval.

Objectives. *We prove energy conservation at the regularity of the constructed solution. We then exclude finite-time termination for perturbations of either vacuum and of the kink. A focusing example shows why the sign of the potential matters.*

By the end of this lecture you will be able to:

- Prove conservation by spatial smoothing of a weak energy solution.
- Bound the full perturbation norm on every bounded time interval.
- Use the continuation alternative and distinguish positivity from conservation.

16.1 Energy on an affine background

Retain $V(s) = \frac{1}{2}(1-s^2)^2$ and $b \in \{1, -1, \tanh x\}$ from Lecture 15. Write $\phi = b + v$ and $p = v_t$, where $v \in C(I; H^1(\mathbb{R})) \cap C^1(I; L^2(\mathbb{R}))$ on an interval I containing zero.

Definition 16.1 (Nonlinear energy). The *nonlinear energy* of $\phi = b + v$ with velocity p is

$$E_{\mathrm{en}}[\phi, p] = \frac{1}{2}\int_{\mathbb{R}} \left[p^2 + \phi_x^2 + (1-\phi^2)^2\right] \mathrm{d}x, \qquad \phi_x = b' + v_x. \tag{16.1}$$

Write $E_{\mathrm{en}}(t)$ for its value along the field.

This integral is finite and continuous in time. Indeed, $b' \in L^2$ and $1 - b^2 \in L^2$ for each allowed background. For the kink both are $\operatorname{sech}^2 x$, and for a vacuum both vanish. The one-dimensional embedding of Lemma 15.2 makes v bounded. The factorisation

$$1 - (b+v)^2 = (1-b^2) - 2bv - v^2$$

therefore belongs to L^2, continuously in time. No L^2 condition on b or ϕ is required. The same argument gives $V'(\phi) = 2\phi(\phi^2 - 1) \in C(I; L^2)$.

16.2 Conservation at energy regularity

Lemma 16.2 (Conservation for energy solutions). *Every energy solution of Lecture 15 satisfies $E_{\mathrm{en}}(t) = E_{\mathrm{en}}(s)$ for $s, t \in I$.*

Proof. We smooth the given solution only in space and estimate its nonlinear residual. Fix a compact time interval $K \subset I$. Choose $\vartheta \in C_c^\infty(\mathbb{R})$ nonnegative, with integral one, and set

$\vartheta_\varepsilon(x) = \varepsilon^{-1}\vartheta(x/\varepsilon)$. A subscript ε denotes convolution in x. In particular, $\phi_\varepsilon = b_\varepsilon + v_\varepsilon$ is well-defined because b is bounded.

First we justify convergence in the affine energy class. The background may fail to lie in L^2, but its translation difference obeys

$$\|b(\,\cdot - h) - b\|_2 \le |h|\|b'\|_2.$$

This follows from $b(x-h)-b(x) = -\int_0^h b'(x-s)\,\mathrm{d}s$ and the norm triangle inequality. Averaging over h gives

$$\|b_\varepsilon - b\|_2 \le \varepsilon\|b'\|_2 \int |y|\vartheta(y)\,\mathrm{d}y \longrightarrow 0, \qquad (b_\varepsilon - b)' = (b')_\varepsilon - b' \longrightarrow 0 \text{ in } L^2.$$

The last limit is the approximate-identity theorem from *Analysis* I, Lecture 5. Applying that theorem to v, v_x and p gives

$$\phi_\varepsilon - \phi \longrightarrow 0 \text{ in } C(K;H^1), \qquad p_\varepsilon \longrightarrow p \text{ in } C(K;L^2). \tag{16.2}$$

Uniformity in time follows from the compact-image, finite-net argument used in Lemma 15.4. Convolution with ϑ_ε is a contraction on L^2 and H^1, by the integral triangle inequality.

The local energy norm and the embedding bound ϕ uniformly on $K \times \mathbb{R}$. The nonnegative mollifier gives the same bound for ϕ_ε. If both absolute values are at most M, factorisation gives

$$\|(1-\phi_\varepsilon^2) - (1-\phi^2)\|_2 \le 2M\|\phi_\varepsilon - \phi\|_2.$$

Thus (16.2) proves uniform convergence of all three energy terms: $E_{\mathrm{en}}[\phi_\varepsilon, p_\varepsilon] \to E_{\mathrm{en}}$ on K.

Next, convolution of the distributional field equation gives

$$\partial_t^2\phi_\varepsilon - \partial_x^2\phi_\varepsilon + V'(\phi_\varepsilon) = R_\varepsilon, \qquad R_\varepsilon = V'(\phi_\varepsilon) - (V'(\phi))_\varepsilon. \tag{16.3}$$

This equation has enough time regularity to differentiate its energy. For fixed ε, its right side before adding $V'(\phi_\varepsilon)$ gives $\partial_t p_\varepsilon = \partial_x^2\phi_\varepsilon - (V'(\phi))_\varepsilon \in C(K;L^2)$. Here $\partial_x^2\phi_\varepsilon = \vartheta_\varepsilon' * (b' + v_x)$ is continuous in L^2. The norm-primitive argument of Lecture 4 turns this distributional equality into a norm derivative. Also $\partial_t(\partial_x\phi_\varepsilon) = \partial_x p_\varepsilon \in C(K;L^2)$. The embedding of Lecture 15 makes $\phi_\varepsilon = b_\varepsilon + v_\varepsilon$ continuous in L^∞, since b_ε is time-independent and $v_\varepsilon \in C(K;H^1)$. Its time difference quotients converge to p_ε in L^2. Factoring the difference of two squares therefore gives $\partial_t(1-\phi_\varepsilon^2) = -2\phi_\varepsilon p_\varepsilon$ in L^2. Consequently the three squared norms in the energy are differentiable.

Their spatial terms cancel by integration by parts. A cutoff equal to one on $[-R,R]$, with derivative bounded by C/R, leaves an error at most $(C/R)\|p_\varepsilon\|_2\|\partial_x\phi_\varepsilon\|_2$, which tends to zero. This uses the derivative of the field, not an L^2 norm of the background. The resulting energy identity is

$$\frac{\mathrm{d}}{\mathrm{d}t}E_{\mathrm{en}}[\phi_\varepsilon, p_\varepsilon] = \int_{\mathbb{R}} p_\varepsilon R_\varepsilon\,\mathrm{d}x. \tag{16.4}$$

Finally, V' is Lipschitz on the bounded range $[-M,M]$. The nonlinear residual satisfies

$$\|R_\varepsilon\|_2 \le C_M\|\phi_\varepsilon - \phi\|_2 + \|V'(\phi) - (V'(\phi))_\varepsilon\|_2 \longrightarrow 0 \quad \text{uniformly on } K.$$

The second term converges uniformly because $V'(\phi) \in C(K;L^2)$. The velocities p_ε are uniformly bounded in $C(K;L^2)$. Integrate (16.4) between any $s,t \in K$ and let $\varepsilon \downarrow 0$. The right side tends to zero, while the two endpoint energies converge to $E_{\mathrm{en}}(t)$ and $E_{\mathrm{en}}(s)$. This proves conservation on every such interval. ∎

16.3 The continuation norm stays finite

Theorem 16.3 (Global nonlinear evolution). *For $b \in \{1, -1, \tanh x\}$ and every $(v_0, g_0) \in H^1(\mathbb{R}) \times L^2(\mathbb{R})$, Lecture 15's solution is global in both time directions. Its nonlinear energy is the constant $E_0 = E_{\mathrm{en}}[b + v_0, g_0]$. For every $t \in \mathbb{R}$ it obeys*

$$\begin{aligned} \|v_t(t)\|_2 &\le \sqrt{2E_0}, \\ \|v(t)\|_2 &\le \|v_0\|_2 + |t|\sqrt{2E_0}, \\ \|v_x(t)\|_2 &\le \sqrt{2E_0} + \|b'\|_2. \end{aligned} \tag{16.5}$$

Proof. Prove the bounds first on the maximal local solution. Every term of (16.1) is nonnegative. Conservation therefore bounds $\|\phi_t\|_2$ and $\|\phi_x\|_2$ by $\sqrt{2E_0}$. The first equals $\|v_t\|_2$. The identity $v(t) = v_0 + \int_0^t v_t(s)\,ds$ holds in L^2, since v is C^1 there. Taking norms gives the middle bound, with reversed integration for $t < 0$. Finally, $v_x = \phi_x - b'$ gives the last bound.

These estimates bound all three terms of $\|(v, v_t)\|_{\mathrm{lin}}^2 = \|v\|_2^2 + \|v_x\|_2^2 + \|v_t\|_2^2$ on every bounded time interval. The blow-up alternative of Theorem 15.5 excludes a finite positive endpoint. Its time-reversed version excludes a finite negative endpoint. Uniqueness is the local uniqueness already proved there. ∎

Example 16.4 (A perturbation of the vacuum). Take $b = 1$, $v_0(x) = e^{-x^2}$ and $g_0 = 0$. Since $V(1 + v) = 2v^2 + 2v^3 + \frac{1}{2}v^4$, Gaussian integration gives

$$E_0 = \frac{5}{2}\sqrt{\frac{\pi}{2}} + 2\sqrt{\frac{\pi}{3}} + \frac{\sqrt{\pi}}{4}, \qquad \|v_0\|_2 = \left(\frac{\pi}{2}\right)^{1/4}.$$

The middle estimate in (16.5) becomes

$$\|v(t)\|_2 \le \left(\frac{\pi}{2}\right)^{1/4} + |t|\sqrt{2E_0}.$$

The spatial derivative and velocity are bounded by $\sqrt{2E_0}$ because $b' = 0$. Hence all parts of the existence norm stay finite on each finite time interval, even though the displayed L^2 estimate permits growth. For the kink, the same proof includes the fixed offset $\|b'\|_2$ in the spatial derivative bound.

These are continuation estimates. They allow the perturbation norm to grow with time. Lecture 18 needs further information before drawing a stability conclusion near a kink.

16.4 A focusing sign can conserve energy and still blow up

For this guard, change the spatial domain to $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ and take the focusing equation

$$u_{tt} - u_{xx} = u^3, \qquad V_{\mathrm{foc}}(u) = -\frac{1}{4}u^4.$$

Fix $T > 0$. The spatially constant periodic field

$$u(t, x) = \frac{\sqrt{2}}{T - t}, \qquad 0 \le t < T, \qquad u_{tt} = \frac{2\sqrt{2}}{(T-t)^3} = u^3$$

is a classical solution. Its initial data are smooth and have finite periodic energy-space norm. Its energy is nevertheless zero at every time:

$$E_{\mathrm{foc}}(t) = 2\pi\left(\frac{1}{2}u_t^2 - \frac{1}{4}u^4\right) = 0, \qquad \int_{\mathbb{T}} (u^2 + u_x^2 + u_t^2)\,dx = 4\pi\left[(T-t)^{-2} + (T-t)^{-4}\right] \longrightarrow \infty.$$

The negative potential cancels the kinetic energy exactly. Conservation alone supplies no bound on this solution's existence norm. The domain is essential: these constant spatial data are finite-energy periodic data, and are not claimed to belong to $H^1(\mathbb{R})$.

Summary. For the one-dimensional ϕ^4 field near a vacuum or the kink, spatial mollification proves energy conservation for the actual weak energy solution. The nonlinear commutator tends to zero on every compact time interval. Positive energy controls the field's spatial derivative and velocity. Integrating the velocity controls the perturbation's L^2 norm for finite time. These bounds exclude the finite-lifespan alternative and give global existence. A focusing potential can lose precisely the needed norm control.

Self-check. Why is the nonlinear energy finite even though the vacuum and kink backgrounds are not in $L^2(\mathbb{R})$?

Answer. The energy uses $b' + v_x$, v_t and $1 - (b+v)^2$. The last quantity is $(1 - b^2) - 2bv - v^2$, whose terms are in L^2 by boundedness of b and the one-dimensional embedding for v.

Self-check. Why does spatial smoothing not give an exact solution of the original nonlinear equation? Which error must vanish in the energy argument?

Answer. Convolution does not commute with the cubic force. The smoothed equation has residual $R_\varepsilon = V'(\phi_\varepsilon) - (V'(\phi))_\varepsilon$. Its L^2 norm tends to zero uniformly on compact time intervals, so its pairing with the uniformly bounded smoothed velocity tends to zero.

Self-check. Why does the estimate for $\|v_x\|_2$ contain $\|b'\|_2$ for a kink, while the corresponding vacuum estimate does not?

Answer. Energy bounds ϕ_x, and $v_x = \phi_x - b'$. The triangle inequality adds $\|b'\|_2$. A vacuum has $b' = 0$. The kink has a nonzero spatial derivative even when its perturbation vanishes.

Self-check. Does the zero conserved energy of the focusing periodic example force its solution to vanish? Which step of the global proof fails there?

Answer. No. The nonzero kinetic term is cancelled by the negative potential. Energy is no longer a sum of nonnegative terms, so it does not control $\|u_t\|_2$ or the existence norm. The explicit solution blows up at time T.

Lecture 17

Vacua and Goldstone modes

The curvature of a potential at equilibrium gives an oscillator's squared frequency. For a field, the same curvature gives squared masses. A continuous symmetry can force some of these curvatures to vanish. We identify the directions it forces and distinguish them from other zero curvatures.

Objectives. *We prove the orbit–Hessian Goldstone lemma for a compact group acting orthogonally on real scalar fields. We compute the radial and phase masses of the global $O(2)$ model with the course normalisation.*

By the end of this lecture you will be able to:

- Find the subgroup preserving a chosen vacuum and the orbit dimension.
- Prove that orbit tangents lie in the vacuum Hessian's kernel.
- Count symmetry-forced massless modes and identify accidental zeros.

17.1 Vacua and the linearised field

Let $V : \mathbb{R}^N \to \mathbb{R}$ be C^2 and attain its minimum value $V_{\min}$.

Definition 17.1 (Vacuum and vacuum set)**.** A *vacuum* is a constant field $\phi(x) = u_0$ with $V(u_0) = V_{\min}$. The *vacuum set* is

$$\mathcal{V} = \{u \in \mathbb{R}^N : V(u) = V_{\min}\}.$$

When this set is a smooth submanifold, it is called the *vacuum manifold*.

The set of minima need not be a manifold: $V(u_1, u_2) = u_1^2 u_2^2$ has two crossing axes as its minima. Each compact group orbit considered below is a smooth submanifold of $\mathbb{R}^N$ contained in the vacuum set. As in Lecture 15, a nonzero constant background on $\mathbb{R}^d$ need not belong to L^2. The perturbation, rather than the background, supplies the linear evolution data.

Use the canonical real-scalar density and equation of Lecture 1:

$$L = \frac{1}{2}\sum_{a=1}^{N} \partial_\mu \phi^a \partial^\mu \phi^a - V(\phi), \qquad \Box\phi + \nabla V(\phi) = 0. \tag{17.1}$$

Fix a vacuum $u_0 \in \mathcal{V}$. Its gradient vanishes. Differentiating this equation at $\phi = u_0$ in the direction h gives the *linearised equation*

$$\Box h + \mathsf{M} h = 0, \qquad \mathsf{M} = D^2 V(u_0). \tag{17.2}$$

Definition 17.2 (Mass matrix and massless mode)**.** The Hessian M in (17.2) is the *mass matrix*, whose eigenvalues are the squared masses. An eigenvector with eigenvalue zero is a *massless mode* of the linearised field. Modes are counted with their linear multiplicities.

The matrix is real symmetric. For every $w \in \mathbb{R}^N$, the second derivative of $s \mapsto V(u_0 + sw)$ at zero is $w^{\mathsf{T}}\mathsf{M}w \geq 0$. Thus its eigenvalues are $m_a^2 \geq 0$. An orthogonal change of field components diagonalises the equation into $(\Box + m_a^2)h_a = 0$. The Fourier frequencies are $\nu_a(k) = \sqrt{|k|^2 + m_a^2}$. These equations have the energy-class evolution of Lecture 4. The number of massless components is therefore $\dim \ker \mathsf{M}$.

17.2 The symmetry orbit and its Hessian

Let G be a compact finite-dimensional Lie group and let $\mathcal{R} : G \to O(N)$ be a smooth group homomorphism. It defines the linear action $u \mapsto \mathcal{R}(g)u$. Assume that

$$V(\mathcal{R}(g)u) = V(u) \qquad (g \in G,\ u \in \mathbb{R}^N). \tag{17.3}$$

Orthogonality also preserves the kinetic term of (17.1), so this is a global internal symmetry of the field theory.

Definition 17.3 (Unbroken subgroup and orbit directions)**.** The *unbroken subgroup* at u_0 is its stabiliser

$$H = \{g \in G : \mathcal{R}(g)u_0 = u_0\}.$$

The *vacuum orbit* is $\mathcal{O}_{u_0} = \{\mathcal{R}(g)u_0 : g \in G\}$. For X in the Lie algebra $\mathfrak{g} = T_eG$, let $T_X = d\mathcal{R}_e(X)$. Its orbit direction is T_Xu_0. The continuous direction is *unbroken* when $T_Xu_0 = 0$ and *broken* otherwise.

Differentiating $\mathcal{R}(g)^{\mathsf{T}}\mathcal{R}(g) = I$ at the identity gives $T_X^{\mathsf{T}} = -T_X$. The stabiliser is closed, since it is the inverse image of u_0 under the continuous orbit map. Recall the closed-subgroup quotient theorem from *Topology and Geometry*, Part I, Lecture 8: a closed subgroup is a Lie subgroup, and G/H is a smooth manifold of dimension $\dim G - \dim H$. The action need not be faithful; elements acting trivially belong to H. We also recall from Part I, Lecture 9, that $\exp(sX)$ is the one-parameter subgroup with initial tangent X. Differentiating its representation gives the matrix equation $Y' = T_XY$, hence $\mathcal{R}(\exp(sX)) = \exp(sT_X)$ by uniqueness of that linear ODE.

Lemma 17.4 (Orbit–Hessian Goldstone lemma)**.** *Under these hypotheses, the orbit is a smooth embedded submanifold $\mathcal{O}_{u_0} \cong G/H$ contained in $\mathcal{V}$, with*

$$\begin{aligned} T_{u_0}\mathcal{O}_{u_0} &= \{T_Xu_0 : X \in \mathfrak{g}\} \subset \ker D^2V(u_0), \\ \dim T_{u_0}\mathcal{O}_{u_0} &= \dim G - \dim H. \end{aligned} \tag{17.4}$$

Consequently the linearised field has at least $\dim G - \dim H$ massless modes. It has exactly that many if and only if the Hessian kernel equals the orbit tangent space.

Proof. We first identify the tangent space and its dimension. The map $G/H \to \mathbb{R}^N$ given by $gH \mapsto \mathcal{R}(g)u_0$ is well-defined and injective: two group elements have the same image exactly when they differ by an element of the stabiliser on the right. Its derivative at the identity coset sends the class of X to T_Xu_0. The kernel of $X \mapsto T_Xu_0$ is the Lie algebra $\mathfrak{h}$ of H. Indeed, a curve in H gives zero derivative of its action on u_0. Conversely, if $T_Xu_0 = 0$, then $\mathcal{R}(\exp(sX))u_0 = \exp(sT_X)u_0 = u_0$ for every s. This one-parameter subgroup lies in H, so $X \in \mathfrak{h}$.

The derivative on G/H is therefore injective. Translation by group elements gives injectivity at every coset. The inverse function theorem in local coordinates gives an immersed image with these tangent spaces. Since G/H is compact, its continuous injective map into the Hausdorff space $\mathbb{R}^N$ is a homeomorphism onto its image. Together with the injective differential, this makes the image embedded. Rank–nullity gives its dimension $\dim \mathfrak{g} - \dim \mathfrak{h}$. Invariance puts every orbit point in $\mathcal{V}$.

Next, differentiate (17.3) in a field direction w:

$$\nabla V(\mathcal{R}(g)u) \cdot \mathcal{R}(g)w = \nabla V(u) \cdot w.$$

Since this holds for every w, orthogonality gives the gradient identity

$$\nabla V(\mathcal{R}(g)u) = \mathcal{R}(g)\nabla V(u). \tag{17.5}$$

At u_0 the right side is zero. Differentiate the left side along $g(s) = \exp(sX)$ at zero. The chain rule gives $D^2V(u_0)T_X u_0 = 0$. This proves the full kernel inclusion, not only a zero quadratic form on tangent vectors.

Finally, the spectral decomposition above identifies the Hessian nullity with the number of massless linearised components. The tangent-space inclusion gives the lower bound. Equality of dimensions holds precisely when these two subspaces coincide. ∎

A *Goldstone mode* is a massless direction forced by this orbit argument. The count concerns independent continuous directions, the quotient $\mathfrak{g}/\mathfrak{h}$. Breaking a discrete symmetry does not by itself give a positive-dimensional tangent space. A Goldstone direction is tangent to a curve of vacua. The straight line with that initial tangent need not stay in the vacuum set.

17.3 The global $O(2)$ model

Take the two real components of Lecture 2, $\Phi = \phi_1 + \mathrm{i}\phi_2$, without a factor $1/\sqrt{2}$. Use the scalar part of the course's Higgs normalisation with $A_{\mathrm{em}} = 0$:

$$L = \frac{1}{2}\partial_\mu \overline{\Phi}\, \partial^\mu \Phi - \frac{\lambda}{8}(|\Phi|^2 - v^2)^2, \qquad \lambda > 0, \quad v > 0. \tag{17.6}$$

This is a global scalar model. Its symmetry is $O(2)$, acting orthogonally on (ϕ_1, ϕ_2), and its vacuum manifold is the circle $|\Phi| = v$. Choose $u_0 = (v, 0)$. The unbroken subgroup consists of I and $\mathrm{diag}(1, -1)$. It has dimension zero, whereas $O(2)$ has dimension one. The orbit lemma therefore forces one massless direction.

Figure 17.1 separates the radial direction from the tangent to the circle at the chosen vacuum. The tangent vector is an infinitesimal variation; the full orbit follows the circle.

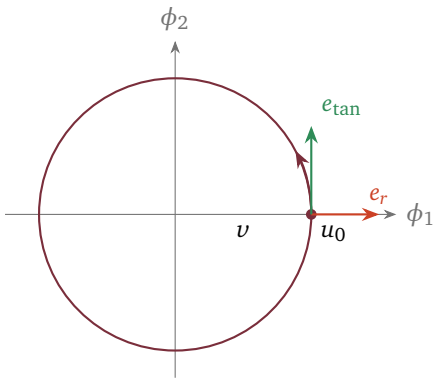


Figure 17.1: The vacuum circle and the chosen point $u_0 = (v, 0)$. The orbit tangent is vertical there; the radial direction is horizontal.

The gradient and Hessian of the potential, writing $u = (\phi_1, \phi_2)$, are

$$\nabla V(u) = \frac{\lambda}{2}(|u|^2 - v^2)u, \qquad D^2V(u) = \frac{\lambda}{2}\big[(|u|^2 - v^2)I + 2uu^{\mathsf{T}}\big].$$

At the chosen vacuum the mass matrix and tangent are

$$\mathsf{M} = \begin{pmatrix} \lambda v^2 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Tu_0 = \begin{pmatrix} 0 \\ v \end{pmatrix}, \qquad T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{17.7}$$

There is one radial mass $m_h = \sqrt{\lambda}\, v$ and one massless tangential mode. The Hessian kernel is exactly the orbit tangent.

The phase coordinate requires its own kinetic normalisation. Locally near u_0, with $|h| < v$ and phase ϑ near zero, write $\Phi = (v + h)e^{\mathrm{i}\vartheta}$. The density becomes

$$L = \frac{1}{2}\partial_\mu h \partial^\mu h + \frac{1}{2}(v + h)^2 \partial_\mu \vartheta \partial^\mu \vartheta - \frac{\lambda}{8}(2vh + h^2)^2.$$

Keeping terms of total degree two in the perturbations and their derivatives gives the quadratic density. With the canonically normalised phase field $\xi = v\vartheta$, it is

$$L^{(2)} = \frac{1}{2}\partial_\mu h \partial^\mu h + \frac{1}{2}\partial_\mu \xi \partial^\mu \xi - \frac{1}{2}\lambda v^2 h^2. \tag{17.8}$$

Its equations are $(\Box + \lambda v^2)h = 0$ and $\Box \xi = 0$. The radial potential starts with $\frac{1}{2}\lambda v^2 h^2$ because $(\lambda/8)(2vh + h^2)^2$ has that quadratic coefficient. Changing $\lambda/8$ to $\lambda/4$ would change the radial squared mass by a factor of two.

17.4 An accidental zero curvature

A zero Hessian eigenvalue need not be a Goldstone direction. Take the one-component potential $V(s) = s^4$ and the trivial compact group $G = \{e\}$. Its unique vacuum is $s = 0$ and its orbit is a point, but

$$V'(0) = 0, \qquad V''(0) = 0, \qquad \ker D^2V(0) = \mathbb{R}.$$

The linearised equation is massless even though no continuous symmetry is broken. The full restoring force is $V'(s) = 4s^3$, which is nonzero for $s \neq 0$. There is no nonconstant curve of vacua through zero. The potential is not flat in a neighbourhood of its minimum.

Thus orbit directions give a lower bound, not a complete classification of Hessian zeros without the kernel-equality hypothesis. Lecture 21 must retain this distinction when it couples the phase to a gauge field. Lecture 18 first studies perturbations about a nonconstant kink.

Summary. For a C^2 invariant potential, a compact Lie group with a smooth linear orthogonal action gives a vacuum orbit G/H. The orbit has dimension $\dim G - \dim H$, and every tangent lies in the Hessian kernel. With canonical kinetic terms this forces at least that many massless linearised modes. The count is exact when the kernel has no additional directions. The global $O(2)$ model with potential $(\lambda/8)(|\Phi|^2 - v^2)^2$ has radial mass $\sqrt{\lambda}\, v$ and one canonically normalised massless phase. A vanishing Hessian alone does not establish a broken symmetry or a continuum of vacua.

Self-check. Lecture 15 has $V(s) = \frac{1}{2}(1 - s^2)^2$ and the discrete symmetry $s \mapsto -s$. At the vacuum $s = 1$, must there be a Goldstone mode? Compute the mass.

Answer. The orbit has two points and dimension zero, so the lemma forces no massless mode. The curvature is $V''(1) = 4$, giving mass two.

Self-check. At the rotated $O(2)$ vacuum $u_\theta = v(\cos\theta, \sin\theta)$, identify the radial and tangential eigenvectors and their squared masses.

Answer. The radial unit vector is $(\cos\theta, \sin\theta)$, with squared mass λv^2. The tangent unit vector is $(-\sin\theta, \cos\theta)$, with squared mass zero. The Hessian is $\lambda u_\theta u_\theta^{\mathsf{T}}$.

Self-check. Why is $\xi = v\vartheta$, rather than ϑ, the canonically normalised phase variable? Does this rescaling give it a mass?

Answer. The phase kinetic term at the vacuum is $\frac{1}{2}v^2 \partial_\mu \vartheta \partial^\mu \vartheta$. The rescaling makes its coefficient $1/2$. There is still no quadratic potential for ξ, so it remains massless.

Self-check. Which part of the orbit–Hessian argument needs a minimum, rather than only a critical point? What does $V(s) = s^4$ show about the converse?

Answer. Gradient covariance gives the kernel inclusion at any critical point. The minimum makes the Hessian positive semidefinite (nonnegative squared masses) and puts the orbit in the vacuum set; at a mere critical point the orbit consists of critical points. The quartic example has an additional zero eigenvalue without a symmetry orbit direction, so a zero Hessian does not imply symmetry breaking.

Lecture 18

Kink fluctuations

The kink is a stationary field with finite energy. We compute its energy and the quadratic change produced by a perturbation. The resulting operator factors into an adjoint pair, which determines its sign and its zero mode. The zero mode records the freedom to translate the kink.

Objectives. *We derive the kink's charge and energy, identify the domain of its fluctuation operator, and prove nonnegativity and the exact kernel. We distinguish this linear result from nonlinear orbital stability.*

By the end of this lecture you will be able to:

- Compute the kink energy and normalise its translation mode.
- Prove the adjoint factorisation on the stated Sobolev domain.
- Explain why a nonnegative Hessian alone does not prove orbital stability.

18.1 The kink and its energy

Recall the potential and stationary kink from Lecture 15:

$$V(u) = \frac{1}{2}(1-u^2)^2, \qquad b(x) = \tanh x, \qquad -b'' + V'(b) = 0.$$

We derive the energy and charge here, without assuming the corresponding calculation from *Topology and Geometry*, Part II.

Definition 18.1 (Static energy and kink charge). For a real C^1 profile ϕ with finite displayed energy integral, its *static energy* is

$$E_{\mathrm{stat}}[\phi] = \int_{\mathbb{R}} \left(\frac{1}{2}\phi_x^2 + V(\phi) \right) \mathrm{d}x.$$

If the endpoint limits exist and are vacua, its *topological charge* is $Q = [\phi(+\infty) - \phi(-\infty)]/2$.

Since $b' = \operatorname{sech}^2 x$ and $1 - b^2 = \operatorname{sech}^2 x$, the kink's two energy terms are equal. Substitution $z = \tanh x$ gives

$$Q[b] = 1, \qquad E_{\mathrm{stat}}[b] = \int_{\mathbb{R}} \operatorname{sech}^4 x \, \mathrm{d}x = \int_{-1}^{1} (1 - z^2) \, \mathrm{d}z = \frac{4}{3}. \tag{18.1}$$

This is a minimum among finite-energy C^1 profiles with limits -1 and 1. Indeed, write $W(u) = u - u^3/3$, so $V = \frac{1}{2}(W')^2$. Completing the square on finite intervals and taking the endpoint limits yields

$$E_{\mathrm{stat}}[\phi] = \frac{1}{2} \int_{\mathbb{R}} [\phi_x - (1 - \phi^2)]^2 \, \mathrm{d}x + W(1) - W(-1) \geq \frac{4}{3}.$$

The cross term is integrable by Cauchy–Schwarz and finite energy. The kink saturates the bound because $b' = 1 - b^2$.

18.2 The fluctuation operator

For a real compact smooth perturbation h, expand the energy of $b+\varepsilon h$. The first variation has boundary term $[b'h]_{-R}^{R} = 0$ on an interval containing the compact support of h. Integration by parts and the stationary equation make its remaining integral zero. The second variation is

$$\frac{\mathrm{d}^2}{\mathrm{d}\varepsilon^2} E_{\mathrm{stat}}[b+\varepsilon h]\Big|_{\varepsilon=0} = \int_{\mathbb{R}} \left[h_x^2 + V''(b)h^2\right] \mathrm{d}x, \qquad V''(b) = 4 - 6\,\mathrm{sech}^2 x. \tag{18.2}$$

The operator suggested by integration by parts is

$$H_K = -\partial_x^2 + 4 - 6\,\mathrm{sech}^2 x. \tag{18.3}$$

The coefficient is negative near the origin, as shown in Figure 18.1. Its pointwise sign does not determine the sign of the full quadratic expression, which also contains h_x^2.

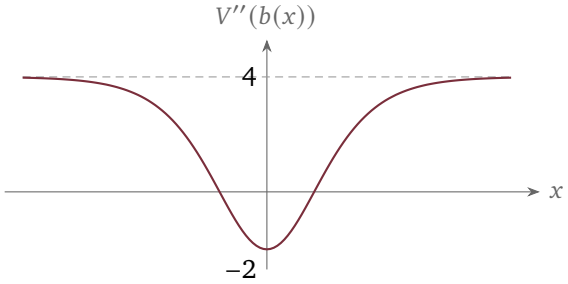


Figure 18.1: The fluctuation potential $4 - 6\,\mathrm{sech}^2 x$ has value -2 at the centre and tends to 4 at both spatial ends.

Linearising the field equation at b gives $h_{tt} + H_K h = 0$. The same operator therefore describes the energy Hessian and the linearised motion. We now assign it an operator domain.

For adjoints, use the complex Hilbert space $L^2(\mathbb{R})$ with inner product $(u, v) = \int \overline{u}v$, antilinear in the first slot. Physical perturbations are real. In addition to Lecture 4's H^1, define

$$H^2(\mathbb{R}) = \{u \in L^2 : (1+k^2)\widehat{u} \in L^2\}, \qquad \|u\|_{H^2} = \|(1+k^2)\widehat{u}\|_2.$$

By the Fourier derivative rule, this is exactly the class with first and second weak derivatives in L^2.

We need approximation and integration by parts on these spaces. For $s = 0, 1, 2$, with $H^0 = L^2$, compact smooth functions are dense in H^s. To see this, multiply u by a smooth cutoff $\chi(x/R)$. The product rule expresses the error in each derivative as a tail of an L^2 derivative of u plus terms bounded by C/R or C/R^2 times lower derivatives. Each tends to zero in L^2. Then convolve the compactly supported result with a compact smooth unit-mass mollifier. Convolution commutes with weak derivatives, and translation continuity in L^2 gives convergence of each derivative. Translation continuity follows from Plancherel and dominated convergence applied to $e^{ikh}\widehat{u}(k)$ as $h \to 0$. This proves the claimed density and, by passage from compact tests,

$$(u', v) = -(u, v') \qquad (u, v \in H^1). \tag{18.4}$$

The quadratic form from (18.2) consequently extends to H^1. It is also the energy Hessian there: the remaining terms in the exact quartic expansion are cubic and quartic in h. They are integrable and continuous in H^1 by Lecture 15's $H^1 \to L^\infty$ bound.

Definition 18.2 (First-order operator and adjoint)**.** Let $A : H^1 \subset L^2 \to L^2$ be given by

$$Au = u' + 2\tanh x\, u. \tag{18.5}$$

The domain of its Hilbert adjoint $A^\dagger$ consists of the $v \in L^2$ for which there exists $g \in L^2$ satisfying $(Au, v) = (u, g)$ for every $u \in H^1$. Its value is $A^\dagger v = g$.

Density of H^1 in L^2 makes g unique. A *formal differential identity* only compares expressions on smooth tests. An identity of unbounded operators must also identify their domains.

Proposition 18.3 (Kink factorisation and kernel)**.** *The Hilbert adjoint has domain H^1 and expression $A^\dagger = -\partial_x + 2\tanh x$. The operator product has domain H^2, and*

$$H_K = A^\dagger A : H^2(\mathbb{R}) \subset L^2(\mathbb{R}) \longrightarrow L^2(\mathbb{R}), \qquad (u, H_K u) = \|Au\|_2^2 \geq 0. \tag{18.6}$$

Its kernel is

$$\ker H_K = \operatorname{span}\{\operatorname{sech}^2 x\}. \tag{18.7}$$

The quadratic-form factorisation also holds on H^1:

$$\mathfrak{q}[u] = \int_{\mathbb{R}} \left[|u'|^2 + (4 - 6\operatorname{sech}^2 x)|u|^2\right] \mathrm{d}x = \|u' + 2\tanh x\, u\|_2^2. \tag{18.8}$$

Proof. First, integration by parts (18.4) shows that every $v \in H^1$ is in $D(A^\dagger)$, with the stated expression. Conversely, if $v \in D(A^\dagger)$ with value g, testing against $u \in C_c^\infty$ gives $-v' + 2\tanh x\, v = g$ distributionally. The bounded coefficient implies $v' \in L^2$, hence $v \in H^1$. This identifies the actual adjoint domain.

Multiplication by $w = 2\tanh x$ preserves H^1, since w and w' are bounded and $(wu)' = w'u + wu'$ by approximation. Thus $u \in H^2$ implies $Au \in H^1$. Conversely, if $u \in H^1$ and $Au \in H^1$, then $u' = Au - wu \in H^1$, so $u \in H^2$. This proves $D(A^\dagger A) = H^2$.

On a compact smooth test, the product rule gives

$$(-\partial_x + w)(\partial_x + w)u = -u'' + (w^2 - w')u = -u'' + (4 - 6\operatorname{sech}^2 x)u.$$

All terms depend continuously on u in H^2, so density proves the operator identity on that domain. The adjoint relation now gives $(u, H_K u) = (Au, Au)$. Likewise, expansion and compact integration by parts give $\|u' + wu\|_2^2 = \|u'\|_2^2 + \int (w^2 - w')|u|^2$. Density in H^1 proves (18.8).

Finally, $H_K u = 0$ forces $Au = 0$. An H^1 function has a locally absolutely continuous representative: subtracting the primitive of its weak derivative leaves a distribution with derivative zero, hence a constant on each interval. The integrating factor therefore gives

$$(\cosh^2 x\, u(x))' = \cosh^2 x\, Au(x) = 0.$$

It follows that $u = C\operatorname{sech}^2 x$. This profile and its first two derivatives decay exponentially, so it belongs to H^2 and directly satisfies $Au = H_K u = 0$. Thus it gives the whole kernel. ∎

The nonnegativity assertion means the quadratic inequality in (18.6) on the stated domain. The proof uses the adjoint and the form identity, without requiring a spectral theorem.

18.3 Normalising the translation mode

Equation (18.1) already gives the squared norm of the kernel profile. Hence its positive unit vector is

$$\psi_0(x) = \frac{\sqrt{3}}{2}\operatorname{sech}^2 x, \qquad \|\psi_0\|_2 = 1. \tag{18.9}$$

For the translated family $b_{x_0}(x) = \tanh(x - x_0)$,

$$\partial_{x_0} b_{x_0}(x)\big|_{x_0=0} = -\operatorname{sech}^2 x = -\frac{2}{\sqrt{3}}\psi_0(x).$$

Thus translation generates exactly the one-dimensional kernel. Differentiating the stationary equation along this family also gives $H_K b' = 0$, in agreement with the factorisation.

There are nonzero positive directions as well. For $u(x) = \operatorname{sech} x$, $Au = \operatorname{sech} x \tanh x$. The same substitution $z = \tanh x$ gives

$$(u, H_K u) = \|Au\|_2^2 = \int_{\mathbb{R}} \operatorname{sech}^2 x \tanh^2 x \, \mathrm{d}x = \int_{-1}^{1} z^2 \, \mathrm{d}z = \frac{2}{3}.$$

The negative coefficient near $x = 0$ has not produced a negative form value.

18.4 What the linear statement establishes

No exponentially growing normal mode of the form $h(t, x) = e^{\gamma t}u(x)$, with $\gamma > 0$ and nonzero $u \in H^2$, can solve the linearised equation. Substitution and the factorisation give

$$0 = (u, (\gamma^2 + H_K)u) = \gamma^2\|u\|_2^2 + \|Au\|_2^2,$$

which is impossible for such u. This is a statement about the linear equation. It is not a bound on every linearised displacement: the exact solution $h(t) = t\psi_0$ has growing H^1 norm, although its quadratic energy $\frac{1}{2}(\|h_t\|_2^2 + \mathfrak{q}[h]) = 1/2$ is constant. It describes drift in the translation direction to first order.

Each $b_{x_0} - b$ belongs to H^1: write the difference as an integral of translated b', and use the same argument with b'' for its derivative. On the affine class $\phi - b \in H^1$ and $\pi = \phi_t \in L^2$, define distance to the static kink family by

$$d_{\mathrm{orb}}(\phi, \pi) = \inf_{x_0 \in \mathbb{R}} \left(\|\phi - b_{x_0}\|_{H^1} + \|\pi\|_2 \right).$$

Orbital stability would require that for every $\varepsilon > 0$ there is $\delta > 0$ such that data with this distance below δ have distance below ε for all future times. The factorisation does not prove this nonlinear assertion. It does not estimate the higher-order energy terms or choose a translating centre during nonlinear evolution. Theorem 16.3 supplies global energy-class evolution, but its continuation estimates allow growth and do not give this distance bound. Lecture 27 returns to exact coherent solutions and their parameters.

Summary. The kink has charge +1 and static energy 4/3. Its energy Hessian and linearised equation use $H_K = -\partial_x^2 + 4 - 6\operatorname{sech}^2 x$. The actual Hilbert adjoint of $A = \partial_x + 2\tanh x$ has domain H^1, and $A^\dagger A = H_K$ has domain H^2. The factorisation gives a nonnegative quadratic form and the exact translation kernel, with unit mode $\sqrt{3}\operatorname{sech}^2 x/2$. These linear conclusions do not establish nonlinear orbital stability.

Self-check. Reverse the factor order. Is $AA^\dagger$ equal to H_K?

Answer. No. The derivative of $2\tanh x$ changes sign in the product, giving $AA^\dagger = -\partial_x^2 + 4 - 2\operatorname{sech}^2 x$. The order is essential.

Self-check. Why is H^1 the form domain used here, while H^2 is the operator domain?

Answer. The form needs one derivative in L^2. Applying the differential operator requires a second derivative in L^2. The product domain proof shows that $u \in H^1$ and $Au \in H^1$ are exactly $u \in H^2$.

Self-check. Normalise $b'(x)$ in L^2 and compare it with the derivative of $b_{x_0}(x)$ with respect to x_0 at zero.

Answer. Since $\|b'\|_2^2 = 4/3$, its positive unit normalisation is $\psi_0 = (\sqrt{3}/2)b'$. The translation derivative is $-b' = -(2/\sqrt{3})\psi_0$.

Self-check. Does the exact linear solution $h = t\psi_0$ contradict nonnegativity? Does it by itself disprove nonlinear orbital stability?

Answer. Neither. Its quadratic energy is the constant $1/2$, while the form gives no restoring term in the translation direction. The solution belongs to the linearised equation. Nonlinear orbital stability concerns distance to the entire translated kink family.

Lecture 19

Noether's second theorem

A symmetry depending on an arbitrary function forces relations between field equations. Maxwell's identity in Lecture 9 is one example. We prove the general relation and identify its covariant form for a noncommuting gauge field.

Objectives. *We derive the formal-adjoint Noether identity for finite-order gauge operators. We apply it to pure Yang–Mills theory and compute Maxwell charge as an outward surface flux.*

By the end of this lecture you will be able to:

- Compute a gauge operator's formal adjoint, including coefficient derivatives.
- Derive the pure Yang–Mills covariant identity off shell.
- Distinguish ordinary and covariant Gauss laws and orient the charge flux.

19.1 Gauge parameters and compact variations

Let U be the interior of a compact region $D \subset \mathbb{R}^{1+d}$. Let $\phi = (\phi^1, \dots, \phi^N)$ be smooth on a neighbourhood of D. Assume its compact first variation has smooth finite-order Euler expressions $E_a[\phi]$:

$$\delta S_D[\phi]h = \int_D E_a[\phi]h^a \,\mathrm{vol} \qquad \big(h \in C_c^\infty(U;\mathbb{R}^N)\big).$$

These expressions may depend on derivatives of any finite order. For a multi-index $I = (I_0, \dots, I_d)$, write $|I| = \sum I_\mu$ and $\partial_I = \partial_0^{I_0} \cdots \partial_d^{I_d}$.

Definition 19.1 (Gauge operator and gauge symmetry). For a fixed integer $s_R \geq 0$, a *gauge operator* at ϕ is a linear map $R_\phi : C_c^\infty(U;\mathbb{R}^r) \to C_c^\infty(U;\mathbb{R}^N)$ given by

$$R_\phi(\varepsilon)^a = \sum_{|I| \leq s_R} R_\alpha^{aI}[\phi]\partial_I \varepsilon^\alpha. \tag{19.1}$$

Its smooth coefficients may depend on the point, the field and finitely many field derivatives. The variations are a *gauge symmetry* if $\delta S_D[\phi]R_\phi(\varepsilon) = 0$ for every compactly supported parameter ε, at every smooth field.

The coefficients are evaluated at the fixed field when the operator acts on the parameter. Their dependence on the field does not remove their ordinary coordinate derivatives in integration by parts. The compact parameter and its derivatives keep the variation away from the boundary of D.

19.2 The formal adjoint identity

Definition 19.2 (Formal adjoint and Noether identity)**.** The *formal adjoint* of (19.1), at the fixed field, is

$$(R_\phi^* E)_\alpha = \sum_{|I|\le s_R} (-1)^{|I|} \partial_I \big(R_\alpha^{aI}[\phi] E_a\big). \tag{19.2}$$

A differential relation among Euler expressions holding before imposing $E = 0$ is a *Noether identity*.

For example, in one coordinate, $R(\varepsilon) = a(x)\varepsilon_x$ has $R^*E = -\partial_x(aE)$, including the term $-a_x E$. Formal adjunction holds the field fixed but differentiates the evaluated coefficient as a function of the point.

Theorem 19.3 (Noether's second theorem)**.** *Under the preceding smoothness and first-variation hypotheses, a gauge symmetry gives the off-shell identity*

$$R_\phi^*(E[\phi])(x) = 0 \qquad (x \in U) \tag{19.3}$$

for every smooth field. This identity is asserted on U. No first-order assumption on the action or its Euler expressions is required.

Proof. Fix the field and move each parameter derivative by coordinate integration by parts. For each multi-index, repeated one-dimensional FTC gives

$$\int_D E_a R_\alpha^{aI} \partial_I \varepsilon^\alpha \,\mathrm{vol} = (-1)^{|I|} \int_D \partial_I (E_a R_\alpha^{aI}) \varepsilon^\alpha \,\mathrm{vol}.$$

At every step the discarded total derivative contains a derivative of ε as a factor and has compact support in U. Extend it by zero and integrate over $\mathbb{R}^{1+d}$. Fubini and the ordinary fundamental theorem make its integral zero. No boundary regularity theorem for D is needed. Summing the finite list of terms proves the pairing identity

$$0 = \delta S_D[\phi] R_\phi(\varepsilon) = \int_D E_a R_\phi(\varepsilon)^a \,\mathrm{vol} = \int_D (R_\phi^* E)_\alpha \varepsilon^\alpha \,\mathrm{vol}.$$

The parameter components are arbitrary smooth compact tests. The smooth-test fundamental lemma of Lecture 9 therefore gives $(R_\phi^* E)_\alpha = 0$ on U for every component. The field was arbitrary, so this is an off-shell identity. ∎

The theorem concerns infinitesimal invariance under arbitrary functions. It does not require the field-dependent coefficients to be constant, nor a finite-dimensional parameter space as in Lecture 2.

19.3 The pure Yang–Mills identity

We now work on $\mathbb{R}^{1+3}$. We recall the needed matrix conventions from *Topology and Geometry*, Part II, Lectures 21 and 23, and verify the identities used here. Let $A = A_\mu \mathrm{d}x^\mu$ be a smooth one-form with values in $\mathfrak{su}(n_c)$, the anti-Hermitian traceless $n_c \times n_c$ matrices, where $n_c \ge 2$. The curvature, covariant derivative and real invariant pairing are

$$\begin{aligned} F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \\ D_\mu X &= \partial_\mu X + [A_\mu, X], \qquad \langle X, Y\rangle = -2\,\mathrm{tr}(XY). \end{aligned} \tag{19.4}$$

Here $[X,Y]=XY-YX$. In form notation $F=\mathrm{d}A+A\wedge A$; matrix multiplication accompanies the wedge product. On forms, $D\omega=\mathrm{d}x^\mu\wedge(\partial_\mu\omega+[A_\mu,\omega])$, with the commutator taken in each coefficient. The metric and orientation are the Minkowski ones of Lecture 9.

The pairing is nondegenerate because $\langle X,X\rangle=2\operatorname{tr}(X^\dagger X)>0$ for $X\neq 0$. In a real orthonormal basis for this pairing, it is the sum of products of real components. Thus the component form of Theorem 19.3 applies to these matrix-valued fields and parameters. Trace cyclicity gives $\langle[A,X],Y\rangle+\langle X,[A,Y]\rangle=0$. Thus the product rule takes the covariant form

$$\partial_\mu\langle X,Y\rangle=\langle D_\mu X,Y\rangle+\langle X,D_\mu Y\rangle.$$

For compactly supported parameters this makes the formal adjoint of $R(\varepsilon)_\mu=D_\mu\varepsilon$ equal to $R^*E=-D_\mu E^\mu$.

Fix a coupling $g>0$. The *pure Yang–Mills action* is

$$S_D[A]=-\frac{1}{4g^2}\int_D\langle F_{\mu\nu},F^{\mu\nu}\rangle\,\mathrm{vol}. \tag{19.5}$$

Spacetime indices are raised with η. Under a compact variation a_μ, direct differentiation gives $\delta F_{\mu\nu}=D_\mu a_\nu-D_\nu a_\mu$. Antisymmetry and the covariant product rule then give

$$\begin{aligned}\delta S_D[A]a&=-\frac{1}{g^2}\int_D\langle D_\mu a_\nu,F^{\mu\nu}\rangle\,\mathrm{vol}\\&=\frac{1}{g^2}\int_D\langle a_\nu,D_\mu F^{\mu\nu}\rangle\,\mathrm{vol},\qquad E^\nu=\frac{1}{g^2}D_\mu F^{\mu\nu}.\end{aligned} \tag{19.6}$$

The discarded divergence is $-g^{-2}\partial_\mu\langle a_\nu,F^{\mu\nu}\rangle$, which has compact support. The equations $D_\mu F^{\mu\nu}=0$ are written $D{\star}F=0$ in form notation.

The gauge variation $a_\mu=D_\mu\varepsilon$ gives $\delta F_{\mu\nu}=[F_{\mu\nu},\varepsilon]$. To verify this, expand two covariant derivatives and use the matrix Jacobi identity:

$$\begin{aligned}[D_\mu,D_\nu]X&=[\partial_\mu A_\nu-\partial_\nu A_\mu,X]+[A_\mu,[A_\nu,X]]-[A_\nu,[A_\mu,X]]\\&=[F_{\mu\nu},X].\end{aligned}$$

The variation of $\operatorname{tr}(F_{\mu\nu}F^{\mu\nu})$ is the trace of $[F_{\mu\nu}F^{\mu\nu},\varepsilon]$, hence zero. Thus the action has this gauge symmetry, and Theorem 19.3 gives the pure Yang–Mills identity. It can also be checked directly:

$$2D_\mu D_\nu F^{\nu\mu}=[D_\mu,D_\nu]F^{\nu\mu}=[F_{\mu\nu},F^{\nu\mu}]=0. \tag{19.7}$$

The first equality interchanges the dummy indices and uses antisymmetry. For each index pair, $F^{\nu\mu}$ is a scalar multiple of $F_{\mu\nu}$ in our diagonal metric, so the final commutator is zero. This is an identity for every smooth connection, without its equations. For coupled matter, the gauge operator also acts on the matter fields; their Euler expressions then enter the Noether identity. Sheet 19 treats that extension.

19.4 Maxwell charge on a sphere

Return to the Maxwell normalisation of Lecture 9. At fixed time, its Hodge table pulls $\star F_{\mathrm{em}}$ back to $E_x\mathrm{d}y\wedge\mathrm{d}z+E_y\mathrm{d}z\wedge\mathrm{d}x+E_z\mathrm{d}x\wedge\mathrm{d}y$. Give $S_r^2=\partial B_r(0)$ the outward boundary orientation of spatial $\mathbb{R}^3$. The ball divergence formula proved by coordinate FTC in Lecture 5 gives

$$\int_{S_r^2}\star F_{\mathrm{em}}=\int_{S_r^2}\mathbf{E}\cdot n\,\mathrm{d}S=\int_{B_r}\operatorname{div}\mathbf{E}\,\mathrm{d}^3x=\int_{B_r}\rho\,\mathrm{d}^3x. \tag{19.8}$$

The form is pulled back to the sphere in a constant-time slice. For compactly supported charge and a smooth Maxwell field, the right side is the total charge Q once the sphere contains the support. Consequently $Q = \lim_{r\to\infty} \int_{S^2_r} \star F_{\mathrm{em}}$. Figure 19.1 records the orientation.

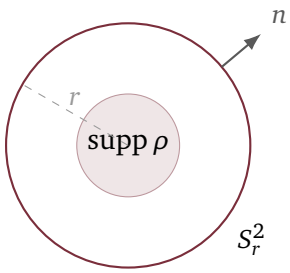


Figure 19.1: A spatial section of a sphere enclosing the charge support. The arrow gives the outward normal used for the electric flux.

Example 19.4 (A smooth static charge with a Coulomb tail)**.** Choose a smooth radial density $\rho(x) = \rho_0(|x|)$ supported in $B_L(0)$, with total charge $Q \neq 0$. Take $\mathbf{J} = 0$ and $\mathbf{B} = 0$, and define

$$M(r) = 4\pi \int_0^r s^2 \rho_0(s)\, \mathrm{d}s, \qquad \mathbf{E}(x) = \frac{M(r)}{4\pi r^3} x \quad (r = |x| > 0).$$

The identity $\mathbf{E}(x) = x \int_0^1 a^2 \rho(ax)\, \mathrm{d}a$ extends the field smoothly through $x = 0$. Differentiating that expression gives $\operatorname{div} \mathbf{E} = \int_0^1 \partial_a [a^3 \rho(ax)] \mathrm{d}a = \rho(x)$. The curl vanishes because the field is radial. These time-independent fields therefore solve all Maxwell equations. Their scalar potential can be chosen as

$$\phi_{\mathrm{em}}(r) = \int_r^\infty \frac{M(s)}{4\pi s^2}\, \mathrm{d}s, \qquad \mathbf{A} = 0.$$

It has $-\nabla \phi_{\mathrm{em}} = \mathbf{E}$ and is smooth at the origin since the electric field is smooth there. For $r > L$, it is exactly $Q/(4\pi r)$. The outward flux is $4\pi r^2 E_r = M(r)$, equal to Q outside the source. The potential has a non-L^2 Coulomb tail. This static charge persists for all times, so it is outside Lecture 10's compact-spacetime source class. It is a regular extended charge, with no point singularity at the origin.

19.5 Covariant divergence cannot be replaced by divergence

The Yang–Mills identity says $\partial_\mu E^\mu = -[A_\mu, E^\mu]$, which need not vanish. For example, in a spatial coordinate patch take $\mathfrak{su}(2)$ generators $T_a = -\mathrm{i}\sigma_a/2$, with $[T_1, T_2] = T_3$ cyclically, and static potentials $A_1 = T_1$, $A_2 = xT_2$, $A_0 = A_3 = 0$. For $g = 1$, direct differentiation gives

$$E^1 = -x^2 T_1, \qquad E^2 = -xT_2 + 2T_3, \qquad \partial_\mu E^\mu = -2xT_1.$$

The commutator $[A_2, E^2] = 2xT_1$ restores the zero covariant divergence. This is a local off-shell example, not a finite-energy assertion on all space.

The temporal Euler equation gives the non-abelian Gauss law. With $\Pi_i = g^{-2} F_{0i}$, the diagonal metric gives

$$E^0 = D_i \Pi_i = \partial_i \Pi_i + [A_i, \Pi_i] = 0. \tag{19.9}$$

The commutator must be retained. Lecture 20 studies its Hamiltonian evolution.

Summary. For a finite-order parameter-dependent symmetry, repeated coordinate integration by parts gives $R^*E = 0$ off shell. Derivatives act on the operator's evaluated coefficients as well as on E. For pure Yang–Mills theory the formal adjoint is the negative covariant divergence, giving $D_\mu D_\nu F^{\nu\mu} = 0$. Maxwell charge is the outward flux of $\star F_{\mathrm{em}}$ through a large sphere. The non-abelian Gauss equation contains a commutator, so ordinary divergence alone does not express the constraint.

Self-check. In one coordinate, let $R(\varepsilon) = a(x)\varepsilon_{xx} + b(x)\varepsilon_x$. What is R^*E for compactly supported parameters?

Answer. The result is $\partial_x^2(aE) - \partial_x(bE)$. Both derivatives act on the coefficients. Dropping them would change the integration-by-parts identity when a or b varies.

Self-check. Does $R^*E = 0$ require solving $E = 0$ first? What changes if gauge variations also act on matter fields?

Answer. It holds at every smooth field, before imposing any equation. For matter fields, the first variation and gauge operator have additional components, so the identity includes the corresponding matter Euler expressions.

Self-check. For the static radial Maxwell example, reverse the orientation of the large sphere. What flux is obtained? Why is there no factor 4π multiplying Q?

Answer. The reversed flux is $-Q$. Our field outside the source is $Qn/(4\pi r^2)$, so the area $4\pi r^2$ cancels the denominator. This uses the rationalised Maxwell normalisation.

Self-check. Choose a smooth compact cutoff $\vartheta = 1$ near the origin and put $f = y\vartheta$. Take $A_1 = \vartheta T_1$, $A_2 = A_3 = 0$ and $\Pi_1 = f_y T_2$, $\Pi_2 = -f_x T_2$, $\Pi_3 = 0$. Compare the ordinary and covariant Gauss expressions at the origin.

Answer. Mixed derivatives give $\partial_i \Pi_i = 0$ everywhere. The commutator gives $D_i \Pi_i = \vartheta f_y T_3$, equal to T_3 at the origin. All arrays are smooth and compactly supported. The ordinary equation alone would accept these data even though the non-abelian Gauss constraint fails.

Lecture 20

The non-abelian Gauss law

Maxwell evolution preserves Gauss's law because the divergence of a curl vanishes. In Yang–Mills theory, derivatives become covariant derivatives and no longer commute. We prove that their additional commutators cancel, so the non-abelian constraint also propagates.

Objectives. *We derive the temporal-gauge equations with the course normalisation, prove Gauss propagation for smooth fields and compute the constraint algebra. A noncommuting homogeneous example makes the cancellations explicit.*

By the end of this lecture you will be able to:

- Propagate the non-abelian Gauss constraint under smooth evolution.
- Derive the gauge variations and the signed constraint bracket.
- State the scope of the cited Cauchy theorem and residual gauge freedom.

20.1 Connection, curvature and pairing

Let I be an open time interval containing zero. We work on $I \times \mathbb{T}^3$, with all coefficients smooth and periodic. Fix $G = \mathrm{SU}(n_c)$, $n_c \geq 2$, and its real Lie algebra $\mathfrak{g} = \mathfrak{su}(n_c)$ of traceless anti-Hermitian matrices.

Definition 20.1 (Connection and curvature)**.** A *connection* here is the globally defined matrix-valued one-form $A = A_\mu \mathrm{d}x^\mu$, with $A_\mu : I \times \mathbb{T}^3 \to \mathfrak{g}$. Its *curvature* and adjoint covariant derivative are

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \qquad D_\mu X = \partial_\mu X + [A_\mu, X].$$

The *temporal gauge* is $A_0 = 0$.

The bracket is the matrix commutator. The positive real pairing on $\mathfrak{g}$ is $\langle X, Y\rangle = -2\,\mathrm{tr}(XY)$. Choose an orthonormal basis T_a, so $\mathrm{tr}(T_a T_b) = -\frac{1}{2}\delta_{ab}$. For SU(2), the Pauli matrices of Lecture 14 give

$$T_a = -\frac{\mathrm{i}}{2}\sigma_a, \qquad [T_a, T_b] = \epsilon_{abc} T_c, \qquad \langle T_a, T_b\rangle = \delta_{ab}.$$

Cyclic invariance of trace proves $\langle [X, Y], Z\rangle = \langle X, [Y, Z]\rangle$. Consequently

$$\partial_i \langle X, Y\rangle = \langle D_i X, Y\rangle + \langle X, D_i Y\rangle. \tag{20.1}$$

On the fundamental cube $Q_{\mathrm{per}} = [0, 2\pi]^3$, the boundary term is $\int_{\partial Q_{\mathrm{per}}} \langle X, Y\rangle n_i \,\mathrm{d}S$. Its opposite faces cancel by periodicity, giving covariant integration by parts. Expansion and the matrix Jacobi identity also give $[D_i, D_j]X = [F_{ij}, X]$.

20.2 Temporal Hamiltonian evolution

With coupling $g > 0$, the Yang–Mills action and its scalar density are

$$\begin{aligned} S &= \frac{1}{g^2}\int \operatorname{tr}(F \wedge \star F), \\ L &= \frac{1}{2g^2}\sum_i \langle F_{0i}, F_{0i}\rangle - \frac{1}{4g^2}\sum_{i,j}\langle F_{ij}, F_{ij}\rangle. \end{aligned} \tag{20.2}$$

Spatial indices run from one to three. The double sum counts each spatial curvature pair twice. The action is taken on compact time subintervals. The signature and Hodge star are those of Lecture 9.

Before setting $A_0 = 0$, note that $F_{0i} = \partial_t A_i - D_i A_0$. Thus the momentum and scalar-potential variation are

$$\begin{aligned} \Pi_i &= g^{-2} F_{0i}, \\ \delta_{A_0} S &= -\int \langle \Pi_i, D_i \delta A_0\rangle \operatorname{vol} = \int \langle D_i \Pi_i, \delta A_0\rangle \operatorname{vol}. \end{aligned} \tag{20.3}$$

Here the variation has compact time support, and the spatial face terms from (20.1) cancel. The A_0 equation is the Gauss constraint

$$\mathcal{D} := \sum_i D_i \Pi_i = 0. \tag{20.4}$$

It must be derived before removing A_0 from the density.

In temporal gauge the spatial Hamiltonian is

$$H = \int_{\mathbb{T}^3} \left[\frac{g^2}{2}\sum_i \langle \Pi_i, \Pi_i\rangle + \frac{1}{4g^2}\sum_{i,j}\langle F_{ij}, F_{ij}\rangle \right] \mathrm{d}^3x. \tag{20.5}$$

The canonical coordinates are the real components (A_i^a, Π_i^a). We use Lecture 7's functional-derivative definition, now on smooth periodic components with their C^r seminorms for every integer $r \geq 0$. Admissible gradients are smooth maps into that same space. The bracket is the sum of the canonical brackets over i and a. The constant-matrix closure and Jacobi proof applies unchanged to these finitely many components. All functionals below have such gradients.

For the magnetic part H_B of H, use $\delta F_{ij} = D_i \delta A_j - D_j \delta A_i$ and antisymmetry to obtain

$$\delta H_B = \frac{1}{g^2}\int \langle F_{ij}, D_i \delta A_j\rangle \, \mathrm{d}^3x = -\frac{1}{g^2}\int \langle D_i F_{ij}, \delta A_j\rangle \, \mathrm{d}^3x.$$

Thus $H_{A_j} = -g^{-2} D_i F_{ij}$. The periodic face terms cancel as in (20.1). Hamilton's equations are therefore

$$\partial_t A_i = g^2 \Pi_i, \qquad \partial_t \Pi_i = g^{-2}\sum_j D_j F_{ji}. \tag{20.6}$$

These equations are meaningful without imposing (20.4). The propagation theorem explains when they also preserve the missing A_0 equation.

20.3 Propagation and the constraint bracket

For a fixed smooth periodic parameter $\varepsilon : \mathbb{T}^3 \to \mathfrak{g}$, define the smeared constraint

$$C[\varepsilon] = \int \langle \Pi_i, D_i \varepsilon\rangle \, \mathrm{d}^3x = -\int \langle \mathcal{D}, \varepsilon\rangle \, \mathrm{d}^3x. \tag{20.7}$$

Its parameter is independent of time. The sign agrees with Lecture 11's abelian constraint generator.

Theorem 20.2 (Non-abelian Gauss law)**.** *Every smooth solution of* (20.6) *on* $I \times \mathbb{T}^3$ *has* $\partial_t \mathcal{D} = 0$*. In particular, zero initial Gauss constraint remains zero. The constraint generator gives*

$$\delta_\varepsilon A_i = D_i \varepsilon, \qquad \delta_\varepsilon \Pi_i = [\Pi_i, \varepsilon]. \tag{20.8}$$

For fixed smooth periodic ε *and* ζ*, the brackets satisfy*

$$\{C[\varepsilon], C[\zeta]\} = -C[[\varepsilon, \zeta]], \qquad \{C[\varepsilon], H\} = 0. \tag{20.9}$$

Thus the Gauss constraints are first class.

Proof. Differentiate the constraint, including the time dependence of D_i:

$$\partial_t \mathcal{D} = \sum_i D_i \partial_t \Pi_i + \sum_i [\partial_t A_i, \Pi_i] = g^{-2} \sum_{i,j} D_i D_j F_{ji} + g^2 \sum_i [\Pi_i, \Pi_i].$$

The last commutators vanish individually. Antisymmetry of F and interchange of the two summed indices give

$$2 \sum_{i,j} D_i D_j F_{ji} = \sum_{i,j} [D_i, D_j] F_{ji} = \sum_{i,j} [F_{ij}, F_{ji}] = 0.$$

This proves propagation without assuming the constraint during the calculation.

Next, vary $C[\varepsilon]$ at a fixed time. Invariance of the pairing gives

$$\delta C[\varepsilon] = \int \left[\langle \delta \Pi_i, D_i \varepsilon \rangle + \langle \Pi_i, [\delta A_i, \varepsilon] \rangle \right] \mathrm{d}^3 x,$$
$$C_{\Pi_i} = D_i \varepsilon, \qquad C_{A_i} = [\varepsilon, \Pi_i].$$

Hamiltonian vector fields therefore give (20.8). These are statements about the gradients and smeared observables, rather than brackets of inadmissible point-evaluation functionals.

Substitute these gradients into the canonical bracket. Pairing invariance and the derivation rule for D_i give

$$\begin{aligned} \{C[\varepsilon], C[\zeta]\} &= \int \left[\langle [\varepsilon, \Pi_i], D_i \zeta \rangle - \langle D_i \varepsilon, [\zeta, \Pi_i] \rangle \right] \mathrm{d}^3 x \\ &= - \int \langle \Pi_i, [\varepsilon, D_i \zeta] + [D_i \varepsilon, \zeta] \rangle \, \mathrm{d}^3 x = -C[[\varepsilon, \zeta]]. \end{aligned}$$

Finally, the propagation calculation is an identity after substitution of the Hamiltonian vector field at any phase point. It needs no existence claim for a path through that point. Pairing this zero directional derivative of $\mathcal{D}$ with the fixed parameter gives $\{C[\varepsilon], H\} = 0$ strongly. The first bracket generally vanishes only on the constraint surface. This is the first-class condition of Lecture 11, rather than strong commutativity of non-abelian constraints. ∎

20.4 Noncommuting homogeneous data

Take SU(2), $g = 1$ and spatially constant data on the torus:

$$A_1 = T_1, \quad A_2 = T_1 + T_2, \quad A_3 = 0, \qquad \Pi_1 = T_1 + T_2, \quad \Pi_2 = T_1, \quad \Pi_3 = 0.$$

The curvature is nonzero: $F_{12} = T_3$. Both momenta displayed are nonzero. The two nonzero Gauss terms cancel: $[A_1, \Pi_1] = T_3$ and $[A_2, \Pi_2] = -T_3$. For spatially constant fields the evolution reduces to

$$\partial_t A_i = \Pi_i, \qquad \partial_t \Pi_i = \sum_j [A_j, [A_j, A_i]].$$

At the stated instant, the two nonzero derivatives are

$$\partial_t \Pi_1 = [T_1 + T_2, -T_3] = -T_1 + T_2,$$
$$\partial_t \Pi_2 = [T_1, T_3] = -T_2, \qquad \partial_t \Pi_3 = 0.$$

The kinetic contributions $[\partial_t A_i, \Pi_i]$ vanish. The other contributions are genuinely nonzero: $[A_1, \partial_t \Pi_1] = T_3$ and $[A_2, \partial_t \Pi_2] = -T_3$. Thus the Gauss derivative cancels as claimed. The Hamiltonian density is $3/2 + 1/2 = 2$, so this is not a zero-field example.

20.5 Cited Cauchy theory

The theorem above is conditional on smooth evolution. It does not construct an energy-class Yang–Mills solution. For comparison, the following local result on $\mathbb{R}^3$, rather than the torus, is *cited*. Tao, *Local well-posedness of the Yang–Mills equation in the temporal gauge below the energy norm*, Theorem 1.1, p. 2, arXiv:math/0005064, proves: for $s > 3/4$, sufficiently small data $(A(0), \partial_t A(0)) \in H^s \times H^{s-1}$ satisfying $D_i \partial_t A_i(0) = 0$ have a solution on $[-1, 1]$. It lies in a Banach space X constructed in that paper, with $X \subset C_t H^s \cap C_t^1 H^{s-1}$, is unique in X, and depends continuously on the data into X. Here $\|u\|_{H^r} = \|\langle k \rangle^r \widehat{u}\|_2$, summed over components, with negative exponents interpreted distributionally. Tao uses the opposite metric signature; reversing the whole metric leaves $D_\mu F^{\mu\nu} = 0$ unchanged because both curvature indices are raised.

Smooth-data existence on $\mathbb{R}^{1+3}$ is treated by Eardley–Moncrief (1982), *Communications in Mathematical Physics* 83, 171–191, Part I. Finite-energy theory on $\mathbb{R}^{1+3}$ is treated by Klainerman–Machedon (1995), *Annals of Mathematics* 142, 39–119, Finite energy solutions. These analytic results are cited; none is inferred from constraint propagation.

20.6 Residual gauge and global gauge choices

The finite gauge transformation has the convention $A' = h^{-1}Ah + h^{-1}\mathrm{d}h$, with adjoint fields transforming by $\Pi_i' = h^{-1}\Pi_i h$. In temporal gauge $A_0' = h^{-1}\partial_t h$. It preserves $A_0 = 0$ exactly when $\partial_t h = 0$. The generators above therefore describe time-independent residual gauge transformations.

The Coulomb condition $\partial_i A_i = 0$ also does not select a unique representative globally. On the torus, take $h(x) = \exp(2x^1 T_3)$ in SU(2). It is periodic since $\exp(4\pi T_3) = I$. It sends $A = 0$ to $A_1' = 2T_3$, $A_2' = A_3' = 0$. Both potentials satisfy the Coulomb condition and have zero curvature. The periodic object here is group-valued h, rather than a single-valued real gauge parameter of Lecture 12. This is a *Gribov copy*: distinct gauge-equivalent connections obeying the same gauge condition.

A stronger global obstruction is *cited*: Singer, *Some remarks on the Gribov ambiguity*, *Communications in Mathematical Physics* 60 (1978), 7–12, doi:10.1007/BF01609471, treats principal bundles over the four-sphere with compact non-abelian structure group and rules out a continuous choice of one connection on each orbit in that setting. This global obstruction does not rule out local gauge slices. Lecture 21 studies the coupling to a scalar field near a vacuum.

Summary. For smooth Yang–Mills evolution on the torus in temporal gauge, $\Pi_i = g^{-2}F_{0i}$ and the constraint $D_i\Pi_i = 0$ propagates. Antisymmetry of the curvature cancels the commutator of covariant derivatives. The smeared generator gives $\delta A_i = D_i\varepsilon$ and $\delta\Pi_i = [\Pi_i, \varepsilon]$, with $\{C[\varepsilon], C[\zeta]\} = -C[[\varepsilon, \zeta]]$. Existence at low regularity requires the separately cited analytic theorems. Temporal gauge retains time-independent gauge freedom, and global Coulomb uniqueness cannot be assumed.

Self-check. Why does differentiating $D_i\Pi_i$ give a term in addition to $D_i\partial_t\Pi_i$? What happens to it in temporal Hamiltonian evolution?

Answer. The connection in D_i also varies, giving $[\partial_t A_i, \Pi_i]$. Since $\partial_t A_i = g^2\Pi_i$, each such term is $g^2[\Pi_i, \Pi_i] = 0$.

Self-check. For the homogeneous data, what remains of $\partial_t\mathcal{D}$ if the contribution from $i = 2$ is mistakenly omitted?

Answer. The uncancelled contribution is T_3. Both original Gauss terms and both magnetic derivative terms must be retained.

Self-check. Does first class mean that $\{C[\varepsilon], C[\zeta]\}$ vanishes identically? How does the abelian limit compare with Lecture 11?

Answer. The bracket is $-C[[\varepsilon, \zeta]]$, so it vanishes on the constraint surface but need not vanish off it. For an abelian algebra the parameter commutator is zero, giving the strongly commuting constraints of Lecture 11.

Self-check. Does Tao's stated theorem give uniqueness among every solution in $C_t H^s \cap C_t^1 H^{s-1}$ for arbitrary large data? Which gauges preserve $A_0 = 0$ after it has been imposed?

Answer. The stated result assumes sufficiently small compatible data and gives uniqueness in its specified space X. Residual finite transformations satisfy $\partial_t h = 0$.

Lecture 21

The abelian Higgs mechanism

A Maxwell field has two propagating polarisations. A massive vector has three. Coupling the two real components of a complex scalar to the gauge potential supplies the additional mode without changing the total count. We derive this statement near a nonzero scalar vacuum.

Objectives. *We use local amplitude and phase coordinates to split the quadratic action into a radial scalar and a massive vector. We count the vector's independent components and examine the zero-coupling limit and the obstruction at a vortex.*

By the end of this lecture you will be able to:

- Derive both masses with the course's scalar normalisation.
- Prove the Proca constraint and identify its three polarisations.
- Distinguish a local unitary gauge from a gauge smooth through a vortex zero.

21.1 The charged scalar model

Work on Minkowski spacetime $\mathbb{R}^{1+3}$, with smooth classical fields. Let $q > 0$, $v > 0$ and $\lambda > 0$ be constants. The complex field is $\Phi = \phi_1 + \mathrm{i}\phi_2$, as in Lecture 17, without a factor $1/\sqrt{2}$.

Definition 21.1 (Abelian Higgs model)**.** The *abelian Higgs model* has density

$$L = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\overline{D_\mu\Phi}D^\mu\Phi - \frac{\lambda}{8}(|\Phi|^2 - v^2)^2, \qquad D_\mu = \partial_\mu - \mathrm{i}q(A_{\mathrm{em}})_\mu, \tag{21.1}$$

where $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}$. Its gauge transformations are

$$\Phi \longmapsto e^{\mathrm{i}q\chi}\Phi, \qquad A_{\mathrm{em}} \longmapsto A_{\mathrm{em}} + \mathrm{d}\chi. \tag{21.2}$$

The product rule gives $D'_\mu(e^{\mathrm{i}q\chi}\Phi) = e^{\mathrm{i}q\chi}D_\mu\Phi$. Together with $\mathrm{d}^2\chi = 0$, this proves gauge invariance of the density. The scalar potential has the vacuum circle $|\Phi| = v$. We expand near the constant vacuum $\Phi = v$, $A_{\mathrm{em}} = 0$.

21.2 Local amplitude and phase

On a sufficiently small patch where $\Phi \neq 0$, choose a smooth real phase and write

$$\Phi = (v + h)e^{\mathrm{i}\theta}, \qquad |h| < v. \tag{21.3}$$

Here $v + h$ is positive. The derivative is

$$D_\mu\Phi = e^{\mathrm{i}\theta}\big[\partial_\mu h + \mathrm{i}(v + h)(\partial_\mu\theta - q(A_{\mathrm{em}})_\mu)\big].$$

The mixed terms cancel on multiplication by the complex conjugate. The density is therefore exactly

$$\begin{aligned} L = &-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\partial_\mu h \partial^\mu h \\ &+ \frac{1}{2}(v+h)^2(\partial_\mu\theta - q(A_{\mathrm{em}})_\mu)(\partial^\mu\theta - q(A_{\mathrm{em}})^\mu) - \frac{\lambda}{8}(2vh+h^2)^2. \end{aligned} \tag{21.4}$$

The phase transforms as $\theta \mapsto \theta + q\chi$. Thus $\mathrm{d}\theta - qA_{\mathrm{em}}$ is gauge invariant.

Definition 21.2 (Gauge-invariant vector and unitary gauge)**.** On this phase patch, the *Stueckelberg vector* is the one-form

$$B = A_{\mathrm{em}} - q^{-1}\mathrm{d}\theta. \tag{21.5}$$

The *unitary gauge* is a choice with $\theta = 0$ and Φ positive real.

The function $\chi = -\theta/q$ reaches unitary gauge on the patch. The vector B is unchanged by gauge transformations, and $\mathrm{d}B = \mathrm{d}A_{\mathrm{em}}$ there. With $F_B = \mathrm{d}B$, the exact local density becomes

$$L = -\frac{1}{4}(F_B)_{\mu\nu}(F_B)^{\mu\nu} + \frac{1}{2}\partial_\mu h\partial^\mu h + \frac{1}{2}q^2(v+h)^2 B_\mu B^\mu - \frac{\lambda}{8}(2vh+h^2)^2. \tag{21.6}$$

The phase has disappeared from this gauge-invariant expression.

Proposition 21.3 (Quadratic Higgs split)**.** *Near the chosen vacuum, the terms of degree two in h, B and their derivatives are*

$$\begin{gathered} L^{(2)} = \frac{1}{2}\partial_\mu h\partial^\mu h - \frac{1}{2}m_h^2 h^2 - \frac{1}{4}(F_B)_{\mu\nu}(F_B)^{\mu\nu} + \frac{1}{2}m_A^2 B_\mu B^\mu, \\ m_h = \sqrt{\lambda}\,v, \qquad m_A = qv. \end{gathered} \tag{21.7}$$

The radial field obeys $(\Box + m_h^2)h = 0$. The vector is a Proca field*: it obeys*

$$\partial_\mu (F_B)^{\mu\nu} + m_A^2 B^\nu = 0. \tag{21.8}$$

It satisfies $\partial_\mu B^\mu = 0$ and has three independent polarisations.

Proof. Replace h and B by εh and εB in (21.6) and retain the coefficient of ε^2. The scalar potential contributes $\frac{1}{2}\lambda v^2 h^2$. The vector mass term contributes $\frac{1}{2}q^2v^2 B_\mu B^\mu$. The terms involving hB^2 and h^3 have degree three, so they do not enter the quadratic action. This proves the split and both masses.

Varying the quadratic density with compactly supported variations gives the stated equations by coordinate integration by parts, as in Lecture 9. For the vector, the kinetic variation gives $\partial_\mu(F_B)^{\mu\nu}$ and the mass variation gives $m_A^2 B^\nu$. Taking the divergence, symmetry of mixed derivatives and antisymmetry of F_B give $m_A^2\partial_\nu B^\nu = 0$. Since $m_A > 0$, the divergence vanishes. Equation (21.8) then reduces to $(\Box + m_A^2)B^\nu = 0$.

For a plane wave $B^\mu = \epsilon^\mu e^{-\mathrm{i}\nu t + \mathrm{i}k\cdot x}$, let $k^\mu = (\nu, k)$, where $\nu = \sqrt{|k|^2 + m_A^2} > 0$. The divergence constraint is $\nu\epsilon^0 - k\cdot\epsilon = 0$. It determines ϵ^0 from three arbitrary spatial components. In the rest frame $k = 0$, $\nu = m_A$ and $\epsilon^0 = 0$. The three spatial axes give three independent polarisations. Real solutions are the real parts of these modes. ∎

For $k = (0, 0, \kappa)$, two unit polarisation vectors are the spatial x and y directions. A third, longitudinal one is

$$\epsilon_L^\mu = \frac{1}{m_A}(\kappa, 0, 0, \nu), \qquad k_\mu \epsilon_L^\mu = 0, \qquad \epsilon_{L,\mu}\epsilon_L^\mu = -1.$$

At rest it becomes the third spatial unit vector. The divergence constraint is not a freedom to remove this component by gauge: B itself is gauge invariant.

21.3 The canonical count in the quadratic vector sector

We apply the canonical bracket of Lecture 11 directly to the Proca density. Write $B = -\beta\,\mathrm{d}t + \mathbf{b}\cdot\mathrm{d}x$ and $m = m_A$. For this calculation use Schwartz data and smearings, so face terms vanish. The vector density is

$$L_B = \frac{1}{2}|\mathbf{b}_t + \nabla\beta|^2 - \frac{1}{2}|\operatorname{curl}\mathbf{b}|^2 + \frac{1}{2}m^2(\beta^2 - |\mathbf{b}|^2).$$

The momenta are $\pi_\beta = 0$ and $\mathbf{\Pi}_b = \mathbf{b}_t + \nabla\beta$. On a cube Q_R, integration of $-\mathbf{\Pi}_b\cdot\nabla\beta$ leaves $-\int_{\partial Q_R}\beta\mathbf{\Pi}_b\cdot n\,\mathrm{d}S$ as its boundary term. Schwartz decay makes this tend to zero as $R\to\infty$. The Hamiltonian is

$$H_B = \int\left[\frac{1}{2}|\mathbf{\Pi}_b|^2 + \frac{1}{2}|\operatorname{curl}\mathbf{b}|^2 + \frac{1}{2}m^2|\mathbf{b}|^2 + \beta\operatorname{div}\mathbf{\Pi}_b - \frac{1}{2}m^2\beta^2\right]\mathrm{d}^3x.$$

Preserving $\pi_\beta = 0$ gives $\Gamma_m = \operatorname{div}\mathbf{\Pi}_b - m^2\beta = 0$. For real scalar test functions f and g, set $P[f] = \int f\pi_\beta$ and $C_m[g] = \int g\Gamma_m$. The bracket is

$$\{P[f], C_m[g]\} = m^2\int fg\,\mathrm{d}^3x.$$

The primary–secondary bracket is the invertible multiplication by m^2. This pair is second class, unlike the commuting Maxwell constraints of Lecture 11. Its equations solve for $\pi_\beta = 0$ and $\beta = m^{-2}\operatorname{div}\mathbf{\Pi}_b$. As in Lecture 11, adding $\int\zeta\pi_\beta$ to the Hamiltonian initially leaves $\beta_t = \zeta$ arbitrary. The spatial momentum equation is $\partial_t\mathbf{\Pi}_b = -\operatorname{curl}\operatorname{curl}\mathbf{b} - m^2\mathbf{b}$. Preserving Γ_m therefore fixes $\beta_t = -\operatorname{div}\mathbf{b}$, rather than imposing another spatial constraint.

Pulling the canonical form to these constraint values leaves $\int\delta\mathbf{b}\wedge\delta\mathbf{\Pi}_b$: the scalar term contains $\delta\pi_\beta = 0$. Thus all three spatial canonical pairs remain. This is the constraint count of the quadratic Proca sector, computed here. Together with the radial scalar it gives $3 + 1 = 4$ linearised modes. The uncoupled description has the same count: two Maxwell modes from Lecture 12 and two real scalar modes from Lecture 17.

21.4 The zero-coupling limit in the original variables

Example 21.4 (Returning to the global scalar model)**.** Keep $v > 0$ and $\lambda > 0$ fixed and let $q \downarrow 0$. Before introducing B, the quadratic density is

$$L^{(2)} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\partial_\mu h\partial^\mu h + \frac{1}{2}v^2(\partial\theta - qA_{\mathrm{em}})^2 - \frac{1}{2}\lambda v^2h^2.$$

The squared one-form denotes contraction with $\eta^{\mu\nu}$. Setting $q = 0$ gives a free Maxwell field, the radial scalar of mass $\sqrt{\lambda}\,v$, and the massless scalar $\xi = v\theta$. This is the canonically normalised phase of Lecture 17. The count is now $2 + 1 + 1 = 4$.

The expression in $(A_{\mathrm{em}}, \theta)$ is regular at $q = 0$. The replacement $B = A_{\mathrm{em}} - q^{-1}\mathrm{d}\theta$ is singular there, so it cannot be used to take that limit with θ fixed. For $q > 0$, the phase contributes to the massive vector's longitudinal mode. At $q = 0$, it is an independent massless scalar again.

21.5 A vortex zero obstructs a smooth unitary gauge

A nonzero complex field has a real phase only locally in general. Around a zero, phase winding can prevent a global branch. Consider the local planar model $\Phi(x, y) = x + \mathrm{i}y$. On a circle about the origin its phase winds once, as shown in Figure 21.1. This is a model of the zero, not a claim of finite energy for that profile on the whole plane.

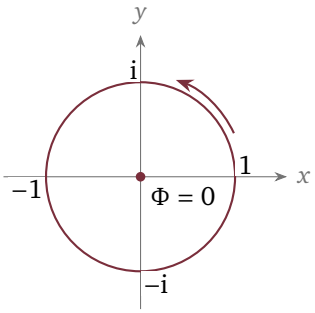


Figure 21.1: For $\Phi = x + \mathrm{i}y$, the field phase winds once around the zero. The four labels show its values on the unit circle.

On the punctured plane the circle-valued gauge function

$$g(x, y) = \frac{x - \mathrm{i}y}{\sqrt{x^2 + y^2}}$$

is smooth and makes $g\Phi = \sqrt{x^2 + y^2}$ positive. On each phase patch it is $e^{\mathrm{i}q\chi}$ with $\chi = -\arg(x + \mathrm{i}y)/q$. It has no single-valued real logarithm on the punctured plane, since its phase winds once around the origin. Any unit-modulus function u with $u\Phi \in \mathbb{R}_{>0}$ must equal $\overline{\Phi}/|\Phi| = g$, so no alternative unitary gauge avoids this obstruction. The function g cannot extend continuously through the zero: along the positive and negative x axes its limits are 1 and -1. The associated potential transformation contains $-q^{-1}\mathrm{d}\arg(x + \mathrm{i}y)$, which is singular at the origin. Consequently a circle-valued unitary gauge on the punctured plane does not supply a smooth unitary gauge through a vortex zero. More generally, a smooth circle-valued extension through a zero is contained in a small phase arc on a sufficiently small disk, by continuity. It therefore has a real phase there and zero winding on small circles. At a vortex zero with nonzero phase winding, the forced unitary gauge has the opposite nonzero winding, a contradiction. Lecture 22 keeps the phase winding and the core rather than applying (21.3) globally.

Summary. With kinetic coefficient $1/2$ and potential coefficient $\lambda/8$, the local Higgs expansion gives radial mass $\sqrt{\lambda}\,v$ and vector mass qv. The gauge-invariant vector obeys Proca's divergence constraint and has three polarisations. Its quadratic constraint pair is second class, leaving three spatial canonical pairs and one radial scalar mode. The original phase variables have a regular $q \to 0$ limit with a massless phase. A vortex zero prevents extending a local unitary gauge smoothly through its core.

Self-check. For $q = 2$, $v = 3$ and $\lambda = 9$, compute the two masses. Which coefficient in the scalar potential fixes the radial value?

Answer. The vector mass is $qv = 6$, and the radial mass is $\sqrt{\lambda}\, v = 9$. The expansion of $(\lambda/8)(2vh + h^2)^2$ has quadratic term $\frac{1}{2}\lambda v^2 h^2$.

Self-check. Under $\theta \mapsto \theta + q\chi$ and $A_{\mathrm{em}} \mapsto A_{\mathrm{em}} + \mathrm{d}\chi$, compute the changes in $\mathrm{d}\theta - qA_{\mathrm{em}}$ and in B.

Answer. Both changes vanish. The two added terms in $\mathrm{d}\theta - qA_{\mathrm{em}}$ are $q\mathrm{d}\chi$ and $-q\mathrm{d}\chi$. The cancellation in B is $\mathrm{d}\chi - q^{-1}q\mathrm{d}\chi = 0$.

Self-check. Which variables exhibit the regular $q = 0$ limit, and what replaces the third massive-vector polarisation in that description?

Answer. The variables $(A_{\mathrm{em}}, \theta)$ are regular. The massless Maxwell field has two polarisations, and $\xi = v\theta$ supplies the independent massless scalar. The radial mode remains massive.

Self-check. Does $g = (x - \mathrm{i}y)/\sqrt{x^2 + y^2}$ provide a smooth unitary gauge on the whole plane for $\Phi = x + \mathrm{i}y$?

Answer. It works only on the punctured plane. Its limits from the two x directions differ, so it has no continuous extension to the zero. The local polar decomposition cannot be used there.

Lecture 22

The critical vortex

A winding scalar phase cannot be made constant through its vortex core. At critical coupling, its static energy can be written as nonnegative squares plus a flux term. We keep the boundary contribution and specify the sector in which that flux has the required value.

Objectives. *We prove the critical energy bound with its signs and boundary assumptions. We derive the unit vortex's radial equations and regularity conditions. A smooth trial field shows why the same bound fails away from critical coupling.*

By the end of this lecture you will be able to:

- Derive the square completion for arbitrary smooth planar fields.
- Control the boundary current and identify the flux in the prescribed sector.
- Derive both radial equations with the one-form potential convention.

22.1 Planar energy and the imposed sector

Take the static model of Lecture 21, independent of the third spatial coordinate, with zero time and third potential components. Its energy is energy per unit length. On the oriented plane $\mathrm{d}x_1 \wedge \mathrm{d}x_2$, set $q = v = 1$ and write $A_{\mathrm{em}} = A_1\mathrm{d}x_1 + A_2\mathrm{d}x_2$, $D_j = \partial_j - \mathrm{i}A_j$, and $B = \partial_1 A_2 - \partial_2 A_1$.

Definition 22.1 (Planar Higgs energy)**.** For $\lambda > 0$, smooth real A_{em} and smooth complex Φ on $\mathbb{R}^2$, the energy is

$$E_\lambda = \int_{\mathbb{R}^2} e_\lambda \, \mathrm{d}^2x, \qquad e_\lambda = \frac{1}{2}\left[B^2 + |D_1\Phi|^2 + |D_2\Phi|^2\right] + \frac{\lambda}{8}(1 - |\Phi|^2)^2. \tag{22.1}$$

The *critical coupling* here is $\lambda = 1$.

Recall winding from *Topology and Geometry*, Part I, Lecture 3. A nonvanishing field on a circle has a phase lift along one traversal. Its phase increment is $2\pi n$ for an integer n, independent of the lift. The circle is traversed counterclockwise.

Definition 22.2 (Admissible winding sector)**.** For a fixed integer $n > 0$, the *admissible winding-n sector* consists of the smooth fields above with finite energy, $B \in L^1(\mathbb{R}^2)$, and $|\Phi(x)| \to 1$ uniformly as $|x| \to \infty$. On every sufficiently large circle, the phase loop has winding n.

Uniform convergence makes the field nonzero outside a disk, so these loops are defined and have the same winding by radial homotopy. The vacuum behaviour, integrable magnetic field and specified positive winding are hypotheses. Finite energy alone has not supplied them. For example, the constant vacuum has finite energy and winding zero.

22.2 The local identity and its boundary current

Define the real current one-form $K = K_j \mathrm{d}x_j$ by

$$K_j = \frac{\mathrm{i}}{4}\big(\Phi\,\overline{D_j\Phi} - \overline{\Phi} D_j\Phi\big) = \frac{1}{2}\,\mathrm{Im}(\overline{\Phi} D_j\Phi). \tag{22.2}$$

Write $\operatorname{curl} K = \partial_1 K_2 - \partial_2 K_1$. With $Z = \overline{D_1\Phi} D_2\Phi$, direct multiplication gives

$$|(D_1 + \mathrm{i}D_2)\Phi|^2 = |D_1\Phi|^2 + |D_2\Phi|^2 - 2\,\mathrm{Im}\,Z.$$

The commutator is $[D_1, D_2]\Phi = -\mathrm{i}B\Phi$. Using it when differentiating (22.2) gives

$$\begin{aligned} 2\operatorname{curl} K &= \mathrm{Im}\big[\overline{D_1\Phi} D_2\Phi - \overline{D_2\Phi} D_1\Phi + \overline{\Phi}[D_1, D_2]\Phi\big] \\ &= 2\,\mathrm{Im}\,Z - B|\Phi|^2. \end{aligned}$$

Expanding the magnetic square now proves the pointwise identity

$$e_1 = \frac{1}{2}|(D_1 + \mathrm{i}D_2)\Phi|^2 + \frac{1}{2}\left[B - \frac{1 - |\Phi|^2}{2}\right]^2 + \frac{B}{2} + \operatorname{curl} K. \tag{22.3}$$

This calculation holds for arbitrary smooth fields, before imposing any equation. It recalls the completion of *Topology and Geometry*, Lecture 27; that Part II lecture is not assumed here.

We next control the two boundary integrals. Let $a(r) = \int_{|x|=r} |D\Phi|^2\,\mathrm{d}s$, where $|D\Phi|^2 = |D_1\Phi|^2 + |D_2\Phi|^2$. Finite energy gives $\int_1^\infty a(r)\,\mathrm{d}r < \infty$. On each interval $[e^j, e^{j+1}]$, logarithmic averaging provides a radius r_j with

$$r_j a(r_j) \le \int_{e^j}^{e^{j+1}} a(r)\,\mathrm{d}r \longrightarrow 0.$$

The modulus of Φ has a global bound M and is at least $1/2$ on these sufficiently large circles. Cauchy–Schwarz therefore gives

$$\left|\int_{|x|=r_j} K\right| \le \frac{M}{2}\sqrt{2\pi r_j a(r_j)} \longrightarrow 0.$$

Along a lifted phase ϑ on such a circle, $d\vartheta - A_{\mathrm{em}} = \mathrm{Im}(\overline{\Phi} D\Phi)/|\Phi|^2$. Hence

$$\left|2\pi n - \int_{|x|=r_j} A_{\mathrm{em}}\right| \le 2\sqrt{2\pi r_j a(r_j)} \longrightarrow 0.$$

Stokes' theorem from *Topology and Geometry*, Part I, Lecture 12, applies to these smooth one-forms on each compact disk, with counterclockwise boundary orientation. Since $B \in L^1$, it follows that

$$\int_{\mathbb{R}^2} B\,\mathrm{d}^2x = \lim_{j\to\infty} \int_{|x|=r_j} A_{\mathrm{em}} = 2\pi n. \tag{22.4}$$

The same subsequence makes the boundary current in the energy identity vanish.

Theorem 22.3 (Critical vortex bound)**.** *In the admissible winding-n sector at $q = v = \lambda = 1$, $E_1 \ge \pi n$. Equality holds exactly when*

$$(D_1 + \mathrm{i}D_2)\Phi = 0, \qquad B = \frac{1}{2}(1 - |\Phi|^2). \tag{22.5}$$

These are the Bogomolny equations.

Proof. Integrate (22.3) over the disks of radii r_j. Stokes gives the boundary integral of K. The proved limits of this boundary term and the magnetic flux give

$$E_1 = \pi n + \frac{1}{2}\int_{\mathbb{R}^2} \left\{ |(D_1 + \mathrm{i}D_2)\Phi|^2 + \left[B - \frac{1 - |\Phi|^2}{2} \right]^2 \right\} \mathrm{d}^2x.$$

The square integrals increase to their full-plane values. They are nonnegative, so the bound follows. Their sum is zero exactly when both continuous squared quantities vanish everywhere, which is (22.5). ∎

22.3 The unit radial vortex

Use polar coordinates (r, θ), with θ increasing counterclockwise. For real profiles f and α, the winding-one ansatz is

$$\Phi = f(r)e^{\mathrm{i}\theta}, \qquad A_{\mathrm{em}} = \alpha(r)\,\mathrm{d}\theta.$$

This is a one-form. Its component along the unit angular vector is α/r, not α. The Cartesian components are $A_1 = -\alpha \sin\theta/r$ and $A_2 = \alpha\cos\theta/r$. Differentiation gives

$$B = \frac{\alpha'}{r}, \qquad (D_1 + \mathrm{i}D_2)\Phi = e^{2\mathrm{i}\theta}\left[f' - \frac{1-\alpha}{r} f \right].$$

Thus both first-order equations reduce to

$$f' = \frac{1-\alpha}{r} f, \qquad \frac{\alpha'}{r} = \frac{1}{2}(1 - f^2). \tag{22.6}$$

For a smooth core, use the local forms $f(r) = rF(r^2)$ and $\alpha(r) = r^2 G(r^2)$ with smooth F and G. Indeed, then $\Phi = (x_1 + \mathrm{i}x_2)F(r^2)$ and $A_{\mathrm{em}} = G(r^2)(-x_2\mathrm{d}x_1 + x_1\mathrm{d}x_2)$ are smooth Cartesian fields. Writing $s_0 = f'(0)$, the equations give the regular expansions

$$f(r) = s_0 r - \frac{s_0}{8} r^3 + O(r^5), \qquad \alpha(r) = \frac{1}{4}r^2 - \frac{s_0^2}{8}r^4 + O(r^6).$$

The smooth core forms give $f(0) = \alpha(0) = 0$. We choose the positive vacuum branch $f(r) \to 1$ at infinity; the sector alone fixes its modulus. The flux identity (22.4) with $n = 1$ and radial Stokes give $\alpha(\infty) = 1$, since $\int B\,\mathrm{d}^2x = 2\pi[\alpha(\infty) - \alpha(0)] = 2\pi$.

Existence is a separate result. Taubes's Theorem 1 gives a smooth critical vortex for any prescribed finite multiset of zeros, whose multiplicities sum to n, unique up to gauge with the vacuum asymptotics. This is *cited*, not proved here: C. H. Taubes, *Communications in Mathematical Physics* **72** (1980), 277–292, Theorem 1, pp. 280–281, doi:10.1007/BF01197552. Its equations (2.6) have exactly the normalisation in (22.5).

Figure 22.1 illustrates the radial boundary problem. The numerically fitted initial slope is $s_0 \approx 0.603288$. The calculation illustrates the profiles and does not replace the cited existence theorem.

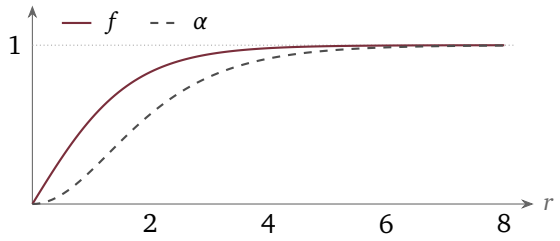


Figure 22.1: Numerical radial profiles: solid f and dashed α approach one. The calculation uses the regular origin series at $r = 10^{-4}$ and $f(18) = 1$; its normalized RMS collocation residual is below 10^{-8}.

22.4 The coupling is essential

For general positive λ, the completion has the additional term $(\lambda-1)(1-|\Phi|^2)^2/8$. A negative coefficient can destroy the critical bound. Consider the smooth winding-one trial

$$f(r) = \frac{r}{\sqrt{r^2+4}}, \qquad \alpha(r) = \frac{r^2}{r^2+4}.$$

Its Cartesian fields are $\Phi = (x_1 + \mathrm{i}x_2)/\sqrt{r^2+4}$ and $A_{\mathrm{em}} = (-x_2\mathrm{d}x_1 + x_1\mathrm{d}x_2)/(r^2+4)$, so they are smooth at the core. Its magnetic field is $8/(r^2+4)^2$ and its flux is 2π. The radial energy is

$$E_\lambda = 2\pi \int_0^\infty \left[\frac{1}{2}\left(\frac{\alpha'}{r}\right)^2 + \frac{1}{2}f'^2 + \frac{(1-\alpha)^2 f^2}{2r^2} + \frac{\lambda}{8}(1-f^2)^2\right] r\,\mathrm{d}r.$$

For this trial, $f' = (1-\alpha)f/r$. Substituting $s = r^2+4$ evaluates the integral exactly:

$$E_\lambda = \pi \int_4^\infty \left(\frac{32}{s^4} + \frac{16}{s^3} + \frac{2\lambda}{s^2}\right) \mathrm{d}s = \frac{2\pi}{3} + \frac{\pi\lambda}{2}.$$

At $\lambda = 1/4$ this is $19\pi/24 < \pi$. At the critical coupling it is $\pi + \pi/6$, since the trial does not satisfy the second Bogomolny equation away from the origin. A winding-one trial is not automatically an energy minimiser.

Summary. At $q = v = \lambda = 1$, the energy density is two nonnegative squares plus $B/2 + \mathrm{curl}\, K$. In the stated asymptotic sector, finite covariant-gradient energy supplies a sequence of large circles where the boundary current vanishes. The winding and integrable magnetic field give flux $2\pi n$. Thus $E \geq \pi n$, with equality exactly for the Bogomolny equations. The unit radial ansatz uses the one-form $\alpha\mathrm{d}\theta$ and has the stated regular core expansions. Existence is cited from Taubes. The bound need not hold away from critical coupling.

Self-check. Why is the flux contribution πn, rather than $2\pi n$, in this normalisation?

Answer. The local completion contains $B/2$. The total flux is $2\pi n$, so its energy contribution is $(1/2)(2\pi n) = \pi n$.

Self-check. For $A_{\mathrm{em}} = \alpha(r)\mathrm{d}\theta$, compute its unit angular component and the magnetic field. What core coefficient follows from the second radial equation?

Answer. They are α/r and $B = \alpha'/r$. Writing $\alpha(r) = a_2 r^2 + O(r^4)$ and $f(0) = 0$ gives $2a_2 = 1/2$, so $a_2 = 1/4$.

Self-check. Does finite energy alone select the positive winding sector and justify removing every large-circle boundary term?

Answer. No. The sector imposes the vacuum asymptotics, winding and magnetic integrability. The proof then produces a subsequence with vanishing boundary current. It does not assert vanishing along every radius from finite energy alone.

Self-check. Evaluate the smooth trial's energy at $\lambda = 1/4$. Does its unit flux force the critical lower bound at that coupling?

Answer. Its energy is $2\pi/3 + \pi/8 = 19\pi/24 < \pi$. The extra term in the off-critical completion is negative, so the critical bound does not apply even though the flux is 2π.

Lecture 23

KdV as a Hamiltonian field

The Korteweg–de Vries equation uses one real field and a third spatial derivative. We describe it by a Hamiltonian and a bracket whose fixed operator is spatial differentiation. This bracket also has conserved quantities that generate zero motion.

Objectives. *We prove the Gardner bracket's Jacobi identity on a specified class of functionals. We derive KdV and its conservation laws, construct a travelling pulse and identify the obstruction to inverting its bracket operator.*

By the end of this lecture you will be able to:

- Compute the Gardner Hamiltonian vector field with the course signs.
- Prove conservation along a Schwartz solution and identify a Casimir.
- Derive the negative travelling pulse and its speed.

23.1 Fields and functional derivatives

Write $\mathcal{S} = \mathcal{S}(\mathbb{R}, \mathbb{R})$ for the real Schwartz space of Lecture 7. Its elements are the field data $u(x)$, and its seminorms are $\sup_x (1 + |x|)^N |\partial_x^j u(x)|$ for integers $N, j \geq 0$. The integral of the field has constant functional derivative. We therefore allow constants in the gradient space for this lecture.

Definition 23.1 (Gradient space). The *gradient space* is the direct sum

$$\mathcal{B} = \mathbb{R} \oplus \mathcal{S} = \{a_0 + f : a_0 \in \mathbb{R},\ f \in \mathcal{S}\}.$$

Its topology is given by $|a_0|$ together with all Schwartz seminorms of f. For $a \in \mathcal{B}$ and $h \in \mathcal{S}$, use the pairing $\int_{\mathbb{R}} a(x)h(x)\,\mathrm{d}x$.

The decomposition is unique because a Schwartz function tends to zero at both ends. The pairing is finite and continuous: its absolute value is at most $|a_0|\|h\|_1 + \|f\|_2\|h\|_2$. The complete space $\mathcal{B}$ is a direct sum of $\mathbb{R}$ and the complete Schwartz space. The straight-segment calculus proved in Lecture 7 therefore applies to maps with this target as well.

Definition 23.2 (Functionals for the Gardner bracket). A smooth functional $F : \mathcal{S} \to \mathbb{R}$ is *admissible here* if it has a smooth gradient map $F_u : \mathcal{S} \to \mathcal{B}$ satisfying

$$DF(u)h = \int_{\mathbb{R}} F_u[u]h\,\mathrm{d}x \qquad (h \in \mathcal{S}). \tag{23.1}$$

Smoothness means jointly continuous iterated directional derivatives, as in Lecture 7. Denote this class by $\mathcal{A}_{\mathrm{KdV}}$.

The gradient is unique by testing against compactly supported smooth h and using the fundamental lemma. This argument applies even when the gradient is not in L^2. For $f \in \mathcal{S}$, the smeared field $u(f) = \int f u dx$ has gradient f. The functional $M[u] = \int u dx$ instead has gradient one in $\mathcal{B}$. It would lie outside the stricter Schwartz-gradient class of Lecture 7.

23.2 The Gardner bracket

Define the continuous linear operator $D_x : \mathcal{B} \to \mathcal{S}$ by $D_x a = \partial_x a$. If $a = a_0 + f$ and $b = b_0 + g$ lie in $\mathcal{B}$, then $ab \to a_0 b_0$ at both spatial ends. Integration by parts on $[-R, R]$ therefore gives, after taking the limit,

$$\int_{\mathbb{R}} a D_x b \, \mathrm{d}x = -\int_{\mathbb{R}} (D_x a) b \, \mathrm{d}x.$$

The two endpoint limits agree. They need not individually be zero.

Definition 23.3 (Gardner bracket and Hamiltonian vector field)**.** For $F, G \in \mathcal{A}_{\mathrm{KdV}}$, the *Gardner bracket* is

$$\{F, G\}[u] = \int_{\mathbb{R}} F_u[u] D_x G_u[u] \, \mathrm{d}x. \tag{23.2}$$

Its *Hamiltonian vector field* is $X_F(u) = D_x F_u[u] \in \mathcal{S}$.

The integral pairs an element of $\mathcal{B}$ with an element of $\mathcal{S}$. It gives $X_G F = \{F, G\}$. In particular, the smeared bracket is $\{u(f), u(g)\} = \int f g' \mathrm{d}x$ for $f, g \in \mathcal{S}$.

Lemma 23.4 (Closure and Jacobi)**.** *The class $\mathcal{A}_{\mathrm{KdV}}$ is closed under products and the Gardner bracket. The bracket is bilinear, antisymmetric, obeys the product rule and satisfies Jacobi.*

Proof. Let $H_F(u) = D(F_u)(u)$ denote the derivative of the gradient map, with $H_F(u) : \mathcal{S} \to \mathcal{B}$. Mixed scalar derivatives on $u + sh + tk$ commute, so

$$\int (H_F h) k \, \mathrm{d}x = \int h (H_F k) \, \mathrm{d}x \qquad (h, k \in \mathcal{S}).$$

Write $a = F_u$, $b = G_u$ and $a_K = K_u$, at the same field. Differentiation of the bracket, this symmetry and the skew integration by parts just proved give

$$\begin{aligned} D\{F, G\}(u)h &= \int \left[(H_F h) D_x b + a D_x (H_G h) \right] \mathrm{d}x \\ &= \int h \left[H_F D_x b - H_G D_x a \right] \mathrm{d}x. \end{aligned}$$

The new gradient is $H_F D_x b - H_G D_x a \in \mathcal{B}$. It is smooth: repeated differentiation gives finite sums of higher gradient derivatives evaluated on smooth Schwartz directions, with jointly continuous results in $\mathcal{B}$. This is the composition argument of Lecture 7, with the complete target $\mathcal{B}$. Products have gradient $F G_u + G F_u$, also smoothly in $\mathcal{B}$. These formulas prove closure and the product rule. Bilinearity and antisymmetry follow from the definition and integration by parts.

For Jacobi, substitution of this gradient into the cyclic sum gives

$$\begin{aligned} \int \big[&(H_F D_x b) D_x a_K - (H_F D_x a_K) D_x b \\ &+ (H_G D_x a_K) D_x a - (H_G D_x a) D_x a_K \\ &+ (H_K D_x a) D_x b - (H_K D_x b) D_x a \big] \, \mathrm{d}x = 0. \end{aligned}$$

Each line cancels by Hessian symmetry. All the Hessian directions $D_x a$, $D_x b$ and $D_x a_K$ belong to $\mathcal{S}$. The argument never evaluates a Hessian on a constant direction outside the field's tangent space. ∎

23.3 Hamiltonian evolution and conserved integrals

The *mass*, *momentum* and *Hamiltonian* are the functionals

$$\begin{aligned} M[u] &= \int_{\mathbb{R}} u \,\mathrm{d}x, \qquad P[u] = \frac{1}{2}\int_{\mathbb{R}} u^2 \,\mathrm{d}x, \\ H[u] &= \int_{\mathbb{R}} \left(u^3 + \frac{1}{2}u_x^2\right) \mathrm{d}x. \end{aligned} \tag{23.3}$$

They are smooth polynomial functionals, with gradients $M_u = 1$, $P_u = u$ and $H_u = 3u^2 - u_{xx}$. For the last gradient, the derivative term on $[-R, R]$ obeys

$$\int_{-R}^{R} u_x h_x \,\mathrm{d}x = [u_x h]_{-R}^{R} - \int_{-R}^{R} u_{xx} h \,\mathrm{d}x.$$

Schwartz decay removes the displayed boundary term. Their gradient maps are continuous polynomial differential maps into $\mathcal{B}$, so all three belong to the stated admissible class.

Proposition 23.5 (KdV as a Hamiltonian field)**.** *The Hamilton equation $u_t = X_H(u)$ for (23.3) is*

$$u_t = D_x \frac{\delta H}{\delta u} = 6uu_x - u_{xxx}, \qquad u_t - 6uu_x + u_{xxx} = 0. \tag{23.4}$$

For every solution $u \in C^1(I; \mathcal{S})$ on an open interval I, all three functionals M, P and H are constant in time.

Proof. Differentiating $H_u = 3u^2 - u_{xx}$ gives the stated equation. Along the assumed path, the chain rule gives $\dot F = \{F, H\}$. The mass derivative is $\int D_x H_u \mathrm{d}x = 0$ by decay. Antisymmetry gives $\dot H = \{H, H\} = 0$. For momentum, integration by parts gives

$$\{H, P\} = \int (3u^2 - u_{xx})u_x \,\mathrm{d}x = \int \partial_x \left(u^3 - \frac{1}{2}u_x^2\right) \mathrm{d}x = 0.$$

Hence $\dot P = \{P, H\} = 0$ as well. ∎

The Hamiltonian need not be positive. These are identities along the stated Schwartz solutions. Lecture 24 cites the existence theory used in the later KdV lectures. The momentum's vector field is $X_P = u_x$, which generates the spatial translation $u(x) \mapsto u(x + \varepsilon)$.

23.4 The travelling pulse

Figure 23.1 shows a translated negative pulse at two times. Its shape and speed follow from the travelling-wave equation.

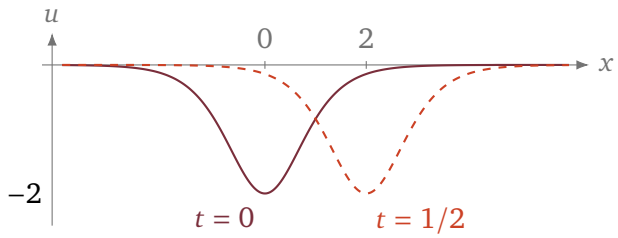


Figure 23.1: The pulse for $\kappa = 1$ and $x_0 = 0$, at $t = 0$ and $t = 1/2$. Its minimum moves from $x = 0$ to $x = 2$ while its profile stays fixed.

Example 23.6 (A decaying travelling wave)**.** Find a negative decaying profile $u(t,x) = U(z)$ with $z = x - v_{\mathrm{tr}}t - x_0$ and $v_{\mathrm{tr}} > 0$. The resulting family, parametrised by $\kappa > 0$ and $x_0 \in \mathbb{R}$, is

$$u(t,x) = -2\kappa^2 \operatorname{sech}^2\big(\kappa(x - 4\kappa^2 t - x_0)\big). \tag{23.5}$$

To derive it, substitute into KdV and integrate once, using decay of U and its derivatives:

$$U''' - v_{\mathrm{tr}}U' - 6UU' = 0, \qquad U'' - v_{\mathrm{tr}}U - 3U^2 = 0.$$

Multiplication by U' and another integration give

$$(U')^2 = v_{\mathrm{tr}}U^2 + 2U^3.$$

At the negative minimum, $U' = 0$ and $U \neq 0$, hence $U = -v_{\mathrm{tr}}/2$. Place this minimum at $z = 0$. Set $w(z) = -2U(z)/v_{\mathrm{tr}}$. On $z > 0$, take the decreasing branch from $w(0) = 1$. Separation gives

$$(w')^2 = v_{\mathrm{tr}}w^2(1 - w), \qquad \operatorname{artanh}\sqrt{1-w} = \frac{\sqrt{v_{\mathrm{tr}}}}{2}z.$$

Thus $w = \operatorname{sech}^2(\sqrt{v_{\mathrm{tr}}}z/2)$. Its even extension gives (23.5), with $v_{\mathrm{tr}} = 4\kappa^2$. Direct substitution into the second-order equation verifies the extension at the minimum, where the separated first-order equation changes branch. The profile and all its derivatives decay exponentially, so its translations form a smooth Schwartz path.

The substitution $y = \tanh(\kappa z)$ makes its invariant integrals polynomial. It gives

$$M = -4\kappa, \qquad P = \frac{8}{3}\kappa^3, \qquad H = -\frac{32}{5}\kappa^5.$$

For example, $P = 2\kappa^3 \int_{-1}^{1}(1-y^2)\mathrm{d}y = 8\kappa^3/3$. These values are independent of the translated centre.

23.5 The Casimir and the derivative's domain

A *Casimir* is an admissible functional whose bracket with every admissible functional is zero. The mass is a Casimir because

$$\{M, F\} = \int_{\mathbb{R}} D_x F_u \,\mathrm{d}x = 0, \qquad X_M = D_x 1 = 0.$$

Its nonzero gradient lies in $\mathcal{B}$ and generates zero motion.

The domain matters when discussing invertibility. On $\mathcal{S}$, the derivative has zero kernel: a constant Schwartz function must be zero. Its range is precisely

$$\mathcal{S}_0 = \{f \in \mathcal{S} : \int f \,\mathrm{d}x = 0\}.$$

Every derivative has zero integral. Conversely, for $f \in \mathcal{S}_0$, the primitive $h(x) = \int_{-\infty}^{x} f(y)\mathrm{d}y$ is Schwartz. On the positive half-line use $h(x) = -\int_{x}^{\infty} f(y)\mathrm{d}y$ to estimate its rapid decay; the negative half-line uses the original formula. Its higher derivatives are derivatives of f. Thus $D_x : \mathcal{S} \to \mathcal{S}$ is injective but not onto. On $\mathcal{B}$, its kernel is the one-dimensional space of constants, and its range remains $\mathcal{S}_0$.

The bracket therefore has a Casimir and no inverse on the full gradient space. A finite-dimensional Liouville–Arnold theorem cannot be applied to this infinite-dimensional degenerate bracket. Lecture 27 uses different structures to organise its conserved quantities.

Summary. The Gardner bracket is $\{F, G\} = \int F_u D_x G_u$. Smooth gradients in $\mathbb{R} \oplus \mathcal{S}$ include the mass derivative one. Closure and Jacobi use Hessian symmetry on the genuine Schwartz directions $D_x F_u$. The Hamiltonian $\int (u^3 + u_x^2/2)$ gives KdV. Mass, momentum and energy are constant along its C^1 Schwartz solutions. The negative travelling pulse has speed $4\kappa^2$. The mass Casimir and the derivative's range prevent a global inverse for the bracket operator.

Self-check. Why is $M[u] = \int u$ admissible here although its gradient is not Schwartz? Why does it generate zero evolution?

Answer. Its gradient is one in $\mathcal{B} = \mathbb{R} \oplus \mathcal{S}$. The pairing with every Schwartz variation is finite. The operator D_x kills this gradient, so $X_M = 0$ and M is a Casimir.

Self-check. The Hessian of an admissible functional acts only on Schwartz tangent vectors. Why are all its arguments in the Jacobi proof allowed?

Answer. The arguments are derivatives of gradients, such as $D_x G_u$. Differentiation removes the constant part and leaves a Schwartz function. No Hessian is applied to the constant function one as a tangent direction.

Self-check. For $\kappa = 2$, give the travelling pulse's minimum and speed. What happens to its centre if x_0 increases?

Answer. The minimum is -8 and the speed is 16. Its centre is $x_0 + 16t$, so increasing x_0 shifts the profile to the right.

Self-check. Can e^{-x^2} be the derivative of a Schwartz function? Compare the kernel of D_x on $\mathcal{S}$ with its kernel on $\mathcal{B}$.

Answer. No. Its integral is $\sqrt{\pi} \neq 0$, whereas a Schwartz derivative has zero integral. The kernel on $\mathcal{S}$ is zero. The kernel on $\mathcal{B}$ consists of constants. These are different domain statements.

Lecture 24

The Lax pair of KdV

Lecture 23 obtained KdV from a Hamiltonian. The same equation can be written as the evolution of a differential operator. We compute the commutator that gives this description and specify the solution class used in the following scattering lectures.

Objectives. *We prove the Lax identity by applying both compositions to a test function. We state the cited existence and decay results that supply smooth Schwartz solutions. We distinguish this differential identity from a spectral theorem.*

By the end of this lecture you will be able to:

- Expand the ordered commutator and cancel all differentiated test terms.
- Recover KdV from equality on every compactly supported smooth test.
- State the decay hypothesis and the limit of the formal operator argument.

24.1 Differential operators on test functions

Let $I \subset \mathbb{R}$ be an open interval and let $u : I \times \mathbb{R} \to \mathbb{R}$ be smooth. Write $D = \partial_x$. A test function here belongs to $C_c^\infty(\mathbb{R}, \mathbb{C})$, the smooth complex functions with compact support.

Definition 24.1 (Lax representation)**.** The *commutator* of differential operators is $[B, L] = B \circ L - L \circ B$. A *Lax representation* of an evolution equation is an identity $L_t = [B, L]$ equivalent to that equation. For KdV the proposed operators are

$$L = -D^2 + u, \qquad B = -4D^3 + 3(uD + Du). \tag{24.1}$$

Here u denotes multiplication, and Du means $D \circ (u\,\cdot)$. The time derivative L_t differentiates the coefficients with the test function held fixed.

Both operators map C_c^∞ into itself, so both compositions have a common domain. The product rule gives

$$B\psi = -4\psi''' + 6u\psi' + 3u_x\psi, \qquad L_t\psi = u_t\psi. \tag{24.2}$$

Thus the term Du in (24.1) includes both $u\psi'$ and $u_x\psi$. Omitting either contribution changes the equation. Primes on a test function denote spatial derivatives.

24.2 The commutator gives KdV

Proposition 24.2 (KdV Lax identity). *For smooth real u on $I \times \mathbb{R}$, the identity*

$$L_t\psi = [B, L]\psi \qquad \text{for every } \psi \in C_c^\infty(\mathbb{R}, \mathbb{C}),\ t \in I$$

holds if and only if

$$u_t - 6uu_x + u_{xxx} = 0 \quad \text{on } I \times \mathbb{R}. \tag{24.3}$$

More precisely, $[B, L]$ is multiplication by $6uu_x - u_{xxx}$ on these tests.

Proof. At a fixed time, distribute the commutator over $L = -D^2 + u$ and $B = -4D^3 + 6uD + 3u_x$. Constant-coefficient derivatives commute. For the other terms, repeated product differentiation gives

$$\begin{aligned} [-4D^3, u]\psi &= -4u_{xxx}\psi - 12u_{xx}\psi' - 12u_x\psi'', \\ [6uD, -D^2]\psi &= 6u_{xx}\psi' + 12u_x\psi'', \\ [3u_x, -D^2]\psi &= 3u_{xxx}\psi + 6u_{xx}\psi', \\ [6uD, u]\psi &= 6uu_x\psi, \qquad [3u_x, u]\psi = 0. \end{aligned} \tag{24.4}$$

For example, the first line uses $(u\psi)''' = u_{xxx}\psi + 3u_{xx}\psi' + 3u_x\psi'' + u\psi'''$; the last summand cancels the other ordered composition. The second line follows by expanding $(u\psi')''$. Adding the displayed terms gives

$$[B, L]\psi = (6uu_x - u_{xxx})\psi + (-12 + 6 + 6)u_{xx}\psi' + (-12 + 12)u_x\psi''.$$

Both derivative coefficients vanish. This proves the asserted multiplication formula.

If u satisfies KdV, this formula equals $u_t\psi = L_t\psi$. Conversely, the assumed operator equality gives $(u_t - 6uu_x + u_{xxx})\psi = 0$ for every test. At any chosen (t, x_*), select a test equal to one near x_*. The coefficient must vanish at that point. The point was arbitrary, so KdV holds everywhere. No spatial decay is required for this proof. ∎

The ordering is part of the statement. Replacing $[B, L]$ by $[L, B]$ negates its scalar coefficient. Dunajski, *Solitons, Instantons, and Twistors*, §2.2.4, pp. 31–32, writes $L_t = [L, A]$ with $A = -B$; that convention gives the same identity.

Example 24.3 (Determining the coefficients). Start with $B_{r_3,r_1,r_0} = r_3D^3 + r_1uD + r_0u_x$, where r_3, r_1 and r_0 are real constants. The same product rule gives

$$\begin{aligned} [B_{r_3,r_1,r_0}, L]\psi = (3r_3 + 2r_1)u_x\psi'' + (3r_3 + r_1 + 2r_0)u_{xx}\psi' \\ + \left[(r_3 + r_0)u_{xxx} + r_1uu_x\right]\psi. \end{aligned}$$

Cancellation for arbitrary smooth u forces $r_1 = -3r_3/2$ and $r_0 = -3r_3/4$. Choosing $r_3 = -4$ fixes $r_1 = 6$ and $r_0 = 3$. The remaining coefficient is then exactly $6uu_x - u_{xxx}$. This calculation determines the relative coefficients rather than assuming them.

24.3 The standing Schwartz solution class

Recall the real Schwartz space $\mathcal{S} = \mathcal{S}(\mathbb{R}, \mathbb{R})$ from Lecture 23. Its seminorms are $p_{N,j}(f) = \sup_x (1 + |x|)^N |\partial_x^j f(x)|$ for integers $N, j \geq 0$. For the time-dependent scattering statements in Lectures 24–26, use

$$u \in C^\infty(\mathbb{R}; \mathcal{S}), \qquad u_t - 6uu_x + u_{xxx} = 0. \tag{24.5}$$

Smoothness here is in the Schwartz topology. In particular, $\sup_{t\in K} p_{N,j}(u(t)) < \infty$ for every compact time interval $K \subset \mathbb{R}$ and every N, j. This is stronger than membership in $\mathcal{S}$ at each separate time. For $w = \partial_t^k u$, continuity into $\mathcal{S}$ gives

$$|\partial_x^j w(t,x) - \partial_x^j w(s,y)| \le p_{0,j}(w(t) - w(s)) + |\partial_x^j w(s,x) - \partial_x^j w(s,y)|.$$

The first term tends to zero as $t \to s$, uniformly in x; the second tends to zero as $x \to y$. Thus all mixed derivatives are jointly continuous, and the field is jointly smooth in (t,x), so Proposition 24.2 applies at every time.

Existence for arbitrary real $u_0 \in \mathcal{S}$ in this class is a *cited consequence* of Sobolev existence and weighted persistence. The relevant statements, in the source convention $v_t + vv_x + v_{xxx} = 0$, are the following.

Bona–Scott, *Solutions of the Korteweg–de Vries equation in fractional order Sobolev spaces*, Duke Math. J. 43 (1976), Theorem 3, p. 95, gives the global solution map in $H^s(\mathbb{R})$ for every $s \ge 2$. The map takes values continuously in H^s on each finite time interval. Here $\|f\|_{H^s}^2 = \int (1+k^2)^s |\widehat{f}(k)|^2 \mathrm{d}k$, with the Fourier convention of Lecture 4. Uniqueness identifies the solutions at different regularities, so Schwartz data give one solution in every such Sobolev space. The source treats positive times. To obtain negative times for datum v_0, solve forward from $v_0(-x)$ and reflect that solution by $(t,x) \mapsto (-t,-x)$. Its value at zero is then $v_0(x)$.

Nahas–Ponce, *On the persistence properties of solutions of nonlinear dispersive equations in weighted Sobolev spaces*, Theorem 2, author PDF p. 6, proves that a Sobolev solution continuous in H^m, $m \ge 1$, retains continuous membership in $L^2(|x|^m \mathrm{d}x)$ on the same time interval whenever its initial datum has that weight. This theorem includes KdV. The smooth Sobolev solutions above agree with the local solution to which the persistence theorem applies, by uniqueness. Using every large integer m therefore supplies every polynomial L^2 weight on every finite interval. Finally set $v = -6u$ to obtain exactly (24.3).

We spell out why these cited statements give the uniform Schwartz hypothesis. Put $\langle x\rangle = (1+x^2)^{1/2}$. Lemma 3 of Nahas–Ponce, author PDF p. 5, gives, for $N, n \ge 0$ integers,

$$\|\langle x\rangle^{N+1} f\|_{H^{n+1}} \le C_{N,n} \|\langle x\rangle^{2N+2} f\|_2^{1/2} \|f\|_{H^{2n+2}}^{1/2}. \tag{24.6}$$

Indeed, take its parameters $a = 2n+2$, $b = 2N+2$ and $\theta = 1/2$. Leibniz's rule and the one-dimensional embedding $H^1 \subset C_b^0$ then give

$$p_{N,n}(f) \le C'_{N,n} \|\langle x\rangle^{N+1} f\|_{H^{n+1}}.$$

To see the derivative step, write $f = \langle x\rangle^{-N-1}(\langle x\rangle^{N+1} f)$ and differentiate n times. Multiplication by $(1+|x|)^N$ leaves bounded coefficients. Every derivative of the parenthesised function through order n is bounded by its H^{n+1} norm.

Apply these estimates to $f = u(t) - u(s)$. Weighted persistence with $m = 4N+4$ makes the weighted norm on the right of (24.6) tend to zero as $t \to s$. Sobolev continuity controls the other factor. Thus u is continuous in every Schwartz seminorm, uniformly bounded on compact time intervals. The map $f \mapsto 6ff' - f'''$ is a continuous polynomial map $\mathcal{S} \to \mathcal{S}$. The PDE first gives $u \in C^1(\mathbb{R}; \mathcal{S})$: integrate this continuous right-hand side in the complete Schwartz space and use the known Sobolev equation to identify the integral with $u(t) - u(s)$. Repeated differentiation of the polynomial right-hand side then gives C^∞ time dependence. This derives (24.5); it does not reprove the cited Cauchy and persistence theorems.

Example 24.4 (The Lax derivative at a Gaussian datum)**.** Take $u_0(x) = e^{-x^2}$. The cited result supplies a solution through this datum. Its initial time derivative follows without finding that solution:

$$\begin{aligned} u_t(0,x) &= 6u_0u_0' - u_0''' \\ &= (8x^3 - 12x)e^{-x^2} - 12xe^{-2x^2}. \end{aligned}$$

At $x = 1$ the value is $-4e^{-1} - 12e^{-2} \neq 0$. For a test equal to one near this point, the value of $[B(0), L(0)]\psi$ is the same number. The reversed commutator has the opposite value. This example tests both terms in the nonlinear evolution.

24.4 The boundary of the operator statement

The test-function calculation also gives formal symmetry properties. For compactly supported smooth complex f and g, integration by parts uses the endpoint primitives

$$\begin{aligned} \overline{f}Lg - \overline{Lf}g &= D(\overline{f'}g - \overline{f}g'), \\ \overline{f}Bg + \overline{Bf}g &= D\Big[-4(\overline{f}g'' - \overline{f'}g' + \overline{f''}g) + 6u\overline{f}g\Big]. \end{aligned}$$

Both primitives vanish outside a compact set. Hence $(f, Lg) = (Lf, g)$ and $(f, Bg) = -(Bf, g)$ on this test domain, where the L^2 inner product is antilinear in its first slot. These identities mean that L is *formally symmetric* and B is *formally skew-symmetric* there.

They do not identify closed operator domains or construct an evolution operator on L^2. The operators contain derivatives and are unbounded on L^2. A finite-matrix argument that solves $U_t = BU$ and writes $L(t) = U(t)L(0)U(t)^{-1}$ needs additional analytic results in this setting. Proposition 24.2 proves equality of differential operators on tests. It makes no claim that the spectrum stays fixed. Lecture 26 treats the time evolution of bound-state data under its stated additional differentiability hypothesis. Lecture 25 first constructs the scattering solutions for each fixed real Schwartz potential.

Summary. For $L = -\partial_x^2 + u$ and $B = -4\partial_x^3 + 6u\partial_x + 3u_x$, the commutator on compactly supported smooth tests is multiplication by $6uu_x - u_{xxx}$. Thus $L_t = [B, L]$ is equivalent to KdV for smooth u. Cited Sobolev existence and weighted persistence, together with the displayed interpolation estimate, supply $C^\infty(\mathbb{R}; \mathcal{S})$ solutions for real Schwartz data. The test-function identity alone is not an isospectrality theorem.

Self-check. What does Du mean in $B = -4D^3 + 3(uD + Du)$? Compute its action on a test.

Answer. It means differentiation after multiplication: $(Du)\psi = (u\psi)'$. Thus $B\psi = -4\psi''' + 6u\psi' + 3u_x\psi$. Replacing Du by multiplication by u_x loses one $3u\psi'$ term.

Self-check. Which two coefficients in the commutator cancel the differentiated tests? What remains when the order of the commutator is reversed?

Answer. The coefficients are $(-12 + 12)u_x$ for ψ'' and $(-12 + 6 + 6)u_{xx}$ for ψ'. The reversed order leaves multiplication by $u_{xxx} - 6uu_x$. It therefore cannot replace $[B, L]$ with L_t unchanged.

Self-check. Why does Schwartz membership at every separate time not state the full standing hypothesis? What supplies the stronger conclusion here?

Answer. Separate membership does not assert continuity or uniform compact-time bounds in any Schwartz seminorm. Cited continuous Sobolev and weighted norms, applied to time differences in (24.6), give that continuity. Compactness then gives the uniform bounds.

Self-check. Does the proved Lax identity establish constancy of every eigenvalue of an unbounded operator on L^2?

Answer. No. The proof takes place on a common test-function domain. A spectral conclusion also requires control of operator domains and of the evolution or differentiability of the relevant eigenfunctions. Those analytic conclusions do not follow from the algebra alone.

Lecture 25

Scattering data

The operator of Lecture 24 replaces a field by a differential equation with a spectral parameter. At each nonzero real parameter, we construct solutions with prescribed behaviour at infinity. Their comparison gives complex scattering amplitudes. Bound states require additional data.

Objectives. *We construct the Jost solutions by a convergent Volterra series. We prove the scattering flux identity with the course's Wronskian order. Translation shows which information reflection probabilities and eigenvalues lose.*

By the end of this lecture you will be able to:

- Prove convergence and both asymptotic conditions for the Jost solutions.
- Extract the scattering coefficients and prove that their denominator is nonzero.
- State the full right scattering data and the scope of the cited inverse result.

25.1 Solutions prescribed at infinity

Fix a real potential $u \in \mathcal{S}(\mathbb{R}, \mathbb{R})$, the Schwartz class of Lecture 24. Time is fixed throughout this lecture. Write $L = -\partial_x^2 + u$ and consider

$$Lf = k^2 f, \qquad k \in \mathbb{R} \setminus \{0\}.$$

The solutions may be complex. Primes denote derivatives with respect to x.

Definition 25.1 (Jost solutions)**.** A *right Jost solution* is a C^2 solution $f_+(x,k)$ with

$$f_+(x,k) - e^{\mathrm{i}kx} \to 0, \qquad f_+'(x,k) - \mathrm{i}k e^{\mathrm{i}kx} \to 0 \quad (x \to +\infty).$$

A *left Jost solution* $f_-(x,k)$ instead satisfies

$$f_-(x,k) - e^{-\mathrm{i}kx} \to 0, \qquad f_-'(x,k) + \mathrm{i}k e^{-\mathrm{i}kx} \to 0 \quad (x \to -\infty).$$

The derivative conditions are part of the normalisation. Define the *Wronskian* by $W(f,g) = fg' - f'g$. For two solutions at the same k, differentiation gives $W' = fg'' - f''g = 0$, so their Wronskian is constant.

Theorem 25.2 (Jost construction and scattering identity). *For real Schwartz u and real $k \neq 0$, both Jost solutions exist uniquely. They satisfy $f_+(x,-k) = \overline{f_+(x,k)}$. There are unique coefficients $a(k)$ and $b(k)$ such that*

$$f_-(x,k) = a(k) f_+(x,-k) + b(k) f_+(x,k). \tag{25.1}$$

With the displayed Wronskian order,

$$a(k) = \frac{W(f_-, f_+)}{2ik}, \qquad |a(k)|^2 - |b(k)|^2 = 1. \tag{25.2}$$

In particular $a(k) \neq 0$.

Proof. Volterra construction. The right integral equation is

$$f(x) = e^{ikx} + \int_x^\infty \frac{\sin k(y-x)}{k}\, u(y) f(y)\,dy. \tag{25.3}$$

To solve it, put $m(x) = e^{-ikx} f(x)$ and define

$$(Tm)(x) = \int_x^\infty \frac{e^{2ik(y-x)} - 1}{2ik}\, u(y) m(y)\,dy, \qquad M_u(x) = \int_x^\infty |u(y)|\,dy.$$

The kernel has absolute value at most $1/|k|$. Set $m_0 = 1$ and $m_{j+1} = T m_j$. Induction gives

$$|m_j(x)| \le \frac{M_u(x)^j}{j!\,|k|^j}. \tag{25.4}$$

Indeed, the next bound follows by integrating $|u(y)| M_u(y)^j$; since $M_u' = -|u|$, its integral from x to infinity is $M_u(x)^{j+1}/(j+1)$. Because $M_u(x) \le \|u\|_1$, the series $m = \sum_{j \ge 0} m_j$ converges uniformly on the whole line. Dominated convergence gives $m = 1 + Tm$. It is continuous and bounded by $C_k = \exp(\|u\|_1/|k|)$. Thus $f = e^{ikx} m$ solves (25.3) and is bounded by C_k.

The kernel is smooth, and uf is continuous and integrable. Differentiating the integral, with its lower endpoint included, gives

$$f'(x) = ike^{ikx} - \int_x^\infty \cos k(y-x)\, u(y) f(y)\,dy.$$

A second derivative gives

$$\begin{aligned} f''(x) &= -k^2 e^{ikx} + u(x) f(x) - k \int_x^\infty \sin k(y-x) u(y) f(y)\,dy \\ &= (u(x) - k^2) f(x). \end{aligned}$$

These differentiations follow from dominated convergence and FTC, so $f \in C^2$ and $Lf = k^2 f$. The same formulas give the actual tails

$$|f(x) - e^{ikx}| \le C_k M_u(x)/|k|, \qquad |f'(x) - ike^{ikx}| \le C_k M_u(x).$$

Both vanish as $x \to +\infty$.

Uniqueness and the left solution. Any solution with the two right asymptotic conditions satisfies the same integral equation. To see this, solve backward from a finite $R > x$:

$$f(x) = f(R)\cos k(R-x) - \frac{f'(R)}{k}\sin k(R-x) + \int_x^R \frac{\sin k(y-x)}{k}\, u(y) f(y)\,dy.$$

The endpoint expression tends to e^{ikx} as $R \to \infty$. The solution is bounded on $[x, \infty)$, so the integral converges absolutely. For two solutions, their modified difference d therefore satisfies $d = Td$. On any half-line $[x_*, \infty)$ it is bounded, and iterating the same estimate gives

$$|d(x)| \le \|d\|_{L^\infty([x_*,\infty))} \frac{M_u(x)^j}{j!|k|^j} \longrightarrow 0 \quad (j \to \infty).$$

Thus $d = 0$; the choice of x_* was arbitrary. Apply the constructed right solution to the reflected potential $u(-x)$, then replace x by $-x$. This constructs the unique left solution with both required asymptotic conditions. Conjugation and uniqueness give $f_+(x,-k) = \overline{f_+(x,k)}$, and the analogous identity for f_-.

Comparison and flux. Evaluate the constant Wronskians at the appropriate ends:

$$W(\overline{f_+}, f_+) = 2ik, \qquad W(f_-, \overline{f_-}) = 2ik.$$

The derivative tails proved above justify both limits. The first Wronskian is nonzero, so $\overline{f_+}$ and f_+ form a basis of the two-dimensional solution space. Matching a value and its first derivative at any finite point determines the two coefficients in (25.1); ordinary linear ODE uniqueness makes them independent of x. Taking $W(f_-, f_+)$ gives the formula for a. Taking the Wronskian of (25.1) with its conjugate gives

$$2ik = W(f_-, \overline{f_-}) = 2ik\big(|a|^2 - |b|^2\big).$$

Division by $2ik \neq 0$ proves (25.2) and $|a| \ge 1$. ∎

25.2 Amplitudes and the free example

Definition 25.3 (Reflection and transmission amplitudes). For real $k \neq 0$, the *right reflection amplitude* and *transmission amplitude* are

$$r(k) = b(k)/a(k), \qquad \tau(k) = 1/a(k). \tag{25.5}$$

Dividing (25.1) by a gives an incoming unit-amplitude wave e^{-ikx} at $+\infty$, a reflected wave re^{ikx} there, and a transmitted wave τe^{-ikx} at $-\infty$. Here $k > 0$ fixes that description of the incoming direction. The flux identity says $|\tau|^2 + |r|^2 = 1$. These are scattering solutions of the auxiliary operator, not finite-energy field configurations. The complex amplitude r includes phase information; its squared modulus is the reflection probability.

Example 25.4 (The free potential). For $u = 0$, the Volterra operator is zero. Thus $f_+ = e^{ikx}$ and $f_- = e^{-ikx}$. Their Wronskian is $2ik$, so $a = 1$, $b = 0$, $r = 0$ and $\tau = 1$. There are no nonzero square-integrable solutions at a negative energy: the solutions are combinations of $e^{\kappa x}$ and $e^{-\kappa x}$, and decay at both ends forces both coefficients to vanish. The nonzero travelling pulse of Lecture 23 is a different potential. Its scattering and reconstruction require the discrete data used in Lecture 26.

The proof has fixed $k \neq 0$ throughout. Its majorant contains $1/|k|$, so it gives no uniform conclusion at $k = 0$. It does not assert that a Jost construction at zero is impossible.

25.3 The discrete data and the cited inverse result

A *bound state* at energy $-\kappa^2$, with $\kappa > 0$, is a nonzero $f \in H^2(\mathbb{R})$ satisfying $Lf = -\kappa^2 f$. The space H^2 uses the Fourier norm of Lecture 24. The following spectral and inverse statements

are *cited*, not consequences of the elementary real-k theorem. For real Schwartz u, there are finitely many simple negative eigenvalues $-\kappa_j^2$. The cited Jost extension gives the real right-normalised state

$$f_j(x) = f_+(x, \mathrm{i}\kappa_j), \qquad f_j(x) \sim e^{-\kappa_j x} \ (x \to +\infty), \qquad c_j = \left(\int_{\mathbb{R}} f_j(x)^2 \mathrm{d}x \right)^{-1} > 0. \tag{25.6}$$

The asymptotic coefficient is one; an arbitrary L^2 normalisation would change c_j. Our c_j is the coefficient in the Marchenko kernel of Lecture 26 (Dunajski's β_j). For a unit-L^2 state, the magnitude of its right-tail coefficient is instead $\sqrt{c_j}$, the convention used for the constants in Drazin–Johnson. Indeed, the unit state is $f_j/\|f_j\|_2 = \sqrt{c_j} f_j$. The *full right scattering data* are

$$\big(r(k) \ (k \in \mathbb{R} \setminus \{0\}); \ \kappa_j, c_j \ (1 \le j \le N) \big).$$

Within the real Schwartz class these data determine u uniquely. This is the meaning of completeness used here; it is not a claim that arbitrary functions and positive numbers are admissible scattering data. No absence of a bounded solution at zero energy is assumed.

For a precise primary formulation, see Egorova–Grunert–Teschl, *Nonlinearity* **22** (2009), 1431–1457, author preprint, Section 4, pp. 11–16, especially (4.10), (4.26)–(4.30) and Theorems 4.3–4.4. Set both background potentials to zero. Their class (1.2) includes real Schwartz potentials, and their right norming parameter obeys $(\gamma_j^+)^2 = c_j$. In this specialisation the right integral kernel depends only on the right reflection amplitude and these discrete data. Its unique solution recovers u. The foundational reference is Deift–Trubowitz, *Communications on Pure and Applied Mathematics* **32** (1979), 121–251, doi:10.1002/cpa.3160320202. The reconstruction theorem will remain a cited input in Lecture 26.

25.4 Translation loses no energy levels but changes data

For a real shift h, let $u_h(x) = u(x - h)$. Direct substitution and the endpoint normalisations give

$$f_{+,h}(x, k) = e^{\mathrm{i}kh} f_+(x - h, k), \qquad f_{-,h}(x, k) = e^{-\mathrm{i}kh} f_-(x - h, k).$$

Substitute them into (25.1) and compare coefficients:

$$a_h = a, \qquad b_h = e^{-2\mathrm{i}kh} b, \qquad r_h = e^{-2\mathrm{i}kh} r. \tag{25.7}$$

Thus $|r_h|^2 = |r|^2$. Translation of a bound state preserves its differential equation and L^2 integrability, so the negative energy levels are unchanged. Right normalisation at the shifted potential gives

$$f_{j,h}(x) = e^{-\kappa_j h} f_j(x - h), \qquad \int f_{j,h}^2 = e^{-2\kappa_j h} \int f_j^2, \qquad c_{j,h} = e^{2\kappa_j h} c_j.$$

Any nonzero Schwartz potential differs from each nonzero translate of itself: otherwise it would be periodic with nonzero period and decay at infinity, forcing it to vanish. Consequently $|r|^2$ and the eigenvalues cannot determine the potential. The full complex reflection amplitude and the norming constants retain the positional information that those reduced data discard.

Summary. For real Schwartz potentials and real $k \neq 0$, the factorial Volterra bound constructs unique Jost solutions with both function and derivative tails. The constant Wronskian gives $|a|^2 - |b|^2 = 1$, so $r = b/a$ is defined. The full right data also contain the right-normalised bound-state constants. Their inverse uniqueness is cited with its class and conventions. Translation preserves reflection probabilities and eigenvalues. Its formulas identify the phase and norming information lost by those reduced data. The elementary proof does not cover $k = 0$.

Self-check. Why does the Volterra construction converge without a smallness condition on $\|u\|_1/|k|$?

Answer. The ordered integrations give $M_u(x)^j/(j!|k|^j)$. Its sum is an exponential, which is finite for every fixed nonzero k. A geometric-series estimate would impose an unnecessary smallness condition.

Self-check. Why is it legitimate to divide by $a(k)$? What happens to the free value of a if the Wronskian order in its formula is reversed?

Answer. The flux identity gives $|a|^2 = 1 + |b|^2 \geq 1$. For the free potential the correct Wronskian is $2ik$ and $a = 1$. Reversing its order would give $a = -1$, contrary to the fixed scattering relation.

Self-check. A right-normalised bound state is multiplied by a nonzero real number d. How does its inverse squared norm change? Is it still right-normalised?

Answer. Its inverse squared norm becomes c_j/d^2. Its coefficient of $e^{-\kappa_j x}$ becomes d, so it has the required normalisation only when $d = 1$.

Self-check. Can reflection probabilities and negative energy levels distinguish a nonzero Schwartz potential from its translate by $h \neq 0$?

Answer. No. The eigenvalues and $|r|^2$ stay fixed. The complex reflection amplitude changes by e^{-2ikh} and each right norming constant changes by $e^{2\kappa_j h}$. The omitted information is needed for inverse uniqueness.

Lecture 26

Evolution of the scattering data

KdV moves the potential while its scattering data obey simple time laws. We prove the bound-state law under an isolated-branch hypothesis. The normalization at spatial infinity requires a separate estimate.

Objectives. *We prove constancy of an isolated negative eigenvalue and evolution of its right norming constant. We solve the rank-one inverse equation and recover the travelling pulse. The reflection phase is stated with a derivation sketch.*

By the end of this lecture you will be able to:

- Justify the bound-state commutator pairing on its actual domain.
- Derive the time law for the right-normalized norming constant.
- Solve the one-state Marchenko equation and locate the negative-data pole.

26.1 The isolated branch hypothesis

Use the real solution $u \in C^\infty(\mathbb{R}; \mathcal{S}(\mathbb{R}))$ of KdV from Lecture 24, equation (24.5). Recall

$$L = -\partial_x^2 + u, \qquad B = -4\partial_x^3 + 6u\partial_x + 3u_x, \qquad L_t = [B, L].$$

For each fixed time, L is self-adjoint on $H^2(\mathbb{R}, \mathbb{C})$: the free operator has this domain and real bounded multiplication is a bounded self-adjoint perturbation. Here the L^2 inner product is antilinear in its first argument. This uses Kato, V, Theorem 4.3, p. 287 (Kato–Rellich): bounded real multiplication has relative bound zero with respect to the free operator.

Definition 26.1 (Differentiable isolated bound-state branch)**.** On an open interval I, a *differentiable isolated branch* consists of $\lambda \in C^1(I; (-\infty, 0))$ and a nonzero real $g \in C^1(I; H^2(\mathbb{R}))$ such that

$$L(t)g(t) = \lambda(t)g(t).$$

At every $t \in I$, require $g(t) \neq 0$ and let $\lambda(t)$ be an isolated simple eigenvalue of $L(t)$. Simple means that its eigenspace has dimension one.

This is an additional hypothesis on the branch, rather than a consequence of the differential identity alone. The local perturbation input is Kato, *Perturbation Theory for Linear Operators*, second edition, IV, Theorem 3.16, pp. 212–213, and supplementary note II.3, pp. 568–569: an isolated finite spectral subspace persists locally, and a simple eigenpair has the parameter regularity of the reduced operator. For the present common domain, the resolvent identity differentiates because $u(t)$ is C^1 in the bounded-multiplication norm. The contour projection is therefore C^1, also into H^2 by the graph norm, and its one-dimensional range supplies the local branch. These are *cited perturbation results*; they do not assert a global evolution operator for B. An original scan is available at Kato, original text. We prove the evolution on any interval carrying the stated branch.

26.2 Constancy of the eigenvalue

The eigenfunction has all Sobolev derivatives. Indeed, $g_{xx} = (u-\lambda)g$ and bounded derivatives of u successively give $g \in H^m$ for every integer m. The same equation and its time derivative bootstrap the assumed $C^1(H^2)$ dependence to $C^1(H^m)$ on compact time intervals. In one dimension, H^1 functions have continuous representatives tending to zero at both ends. This applies to every derivative of g.

Differentiate the eigenvalue equation and pair with g. Self-adjointness cancels the g_t terms, leaving

$$\lambda' \|g\|_2^2 = (g, u_t g) = (g, [B, L]g) = \lambda(g, Bg) - (Lg, Bg) = 0.$$

The Lax identity here is the pointwise differential formula proved in Lecture 24, now applied to the smooth g. Both compositions belong to L^2: in particular $Bg \in H^2$ because $g \in H^5$ and the coefficients have bounded derivatives. Thus moving L across the inner product is legitimate. It follows that $\lambda = -\kappa^2$ for a constant $\kappa > 0$ on I.

26.3 The right normalization at negative energy

Lecture 25 constructs Jost solutions at real nonzero k. For the negative energy just obtained, we give the required extension. Write $f(x,t) = e^{-\kappa x} m(x,t)$ on a right half-line. The right normalization is $m(x,t) \to 1$ as $x \to +\infty$. The corresponding Volterra equation is

$$m(x,t) = 1 + \int_x^\infty h_\kappa(y-x) u(y,t) m(y,t)\, \mathrm{d}y, \qquad h_\kappa(s) = \frac{1 - e^{-2\kappa s}}{2\kappa}. \tag{26.1}$$

Its kernel satisfies $0 \le h_\kappa(s) \le (2\kappa)^{-1}$ for $s \ge 0$. Successive integration on ordered simplices bounds its nth iterate by $U(x)^n/n!$, where $U(x) = (2\kappa)^{-1} \int_x^\infty |u(y,t)| \mathrm{d}y$. The series converges uniformly on each right half-line. It gives a unique bounded solution there, with $|m| \le e^{U(x)}$ and $|m-1| \le e^{U(x)} - 1$. Differentiating in x proves $f_{xx} = (u + \kappa^2) f$. Ordinary differential-equation uniqueness extends f to the whole line.

The estimates can be uniform for t in any compact $J \subset I$. Set

$$U_J(x) = \frac{1}{2\kappa} \int_x^\infty \sup_{t \in J} |u(y,t)| \mathrm{d}y, \qquad V_J(x) = \frac{1}{2\kappa} \int_x^\infty \sup_{t \in J} |u_t(y,t)| \mathrm{d}y.$$

Both tend to zero by the uniform Schwartz bounds. Termwise time differentiation of the Volterra series is valid: the differentiated terms have the convergent majorant $V_J(R) e^{U_J(R)}$ on $x \ge R$. Equivalently $m_t = T m_t + T_t m$, where T is the integral operator in (26.1). Iterating this equation gives

$$\sup_{t \in J} |m_t(x,t)| \le e^{2U_J(x)} V_J(x) \longrightarrow 0.$$

The spatial derivatives needed in B also have controlled limits. Differentiation of the bounded kernel gives

$$\begin{aligned} m_x &= -\int_x^\infty e^{-2\kappa(y-x)} u(y,t) m(y,t) \mathrm{d}y, \\ m_{xx} &= um + 2\kappa m_x, \\ m_{xxx} &= u_x m + u m_x + 2\kappa m_{xx}. \end{aligned}$$

Each tends to zero uniformly on J as $x \to +\infty$. Consequently $e^{\kappa x} B f \to 4\kappa^3$ and $e^{\kappa x} f_t = m_t \to 0$. These conclusions use the Volterra estimate, not merely $C^1(L^2)$ dependence of an eigenfunction.

The constructed f is the right-normalized bound state. To see this, fix a compact time interval $J \subset I$ and choose R with $U_J(R) < \log 2$. Then $|m-1| \le e^{U_J(R)} - 1 < 1$ on $[R,\infty)\times J$, so $f(R,t) \neq 0$ for every $t \in J$. A second independent solution there is $f(x)\int_R^x f(y)^{-2}\mathrm{d}y$, which grows like $e^{\kappa x}$. Since $g \in L^2$, its growing coefficient must vanish. Thus g is a nonzero multiple of f. The ratio at the fixed point R is C^1 in time. It follows that f is a real $C^1(H^m)$ branch for every m. This also proves that the right normalization fixes its sign.

Definition 26.2 (Right norming constant)**.** As in Lecture 25, equation (25.6), the *right norming constant* is

$$c(t) = \left(\int_{\mathbb{R}} f(x,t)^2 \mathrm{d}x\right)^{-1} > 0. \tag{26.2}$$

26.4 Evolution of the bound-state data

Theorem 26.3 (Bound-state evolution)**.** *Under the standing Schwartz solution and isolated-branch hypotheses, κ is constant. The right-normalized bound state and its norming constant satisfy*

$$f_t = (B - 4\kappa^3)f, \qquad c(t) = c(t_0)e^{8\kappa^3(t-t_0)} \quad (t, t_0 \in I).$$

Proof. Constancy of κ was proved above. Differentiate $(L+\kappa^2)f = 0$ and use $L_t = [B,L]$. The result is

$$(L+\kappa^2)(f_t - Bf) = 0.$$

Both terms are in H^2, by the proved $C^1(H^m)$ regularity. Simplicity makes $f_t - Bf = \gamma(t)f$. Multiplying by $e^{\kappa x}$ and taking the proved right limits gives $\gamma = -4\kappa^3$.

Let $N = \int f^2$. All derivatives of f tend to zero at the endpoints. Integration by parts gives

$$\int_{\mathbb{R}} fBf\,\mathrm{d}x = \left[-4\left(ff'' - \frac{1}{2}(f')^2\right) + 3uf^2\right]_{-\infty}^{+\infty} = 0.$$

The $C^1(L^2)$ chain rule now yields $N' = 2\int ff_t = -8\kappa^3 N$. Since $c = N^{-1}$, we obtain $c' = 8\kappa^3 c$ and the displayed time law. ∎

The reflection coefficient has the corresponding law $r(k,t) = r(k,t_0)e^{8\mathrm{i}k^3(t-t_0)}$ for real $k \neq 0$. Its derivation is *sketched* here. With the scattering basis of (25.1), comparison at the two infinities gives

$$(f_+)_t = Bf_+ - 4\mathrm{i}k^3 f_+, \qquad (f_-)_t = Bf_- + 4\mathrm{i}k^3 f_-.$$

The signs follow from $-4\partial_x^3 e^{\pm\mathrm{i}kx} = \pm 4\mathrm{i}k^3 e^{\pm\mathrm{i}kx}$. Differentiate $f_- = af_+(\cdot,-k) + bf_+(\cdot,k)$. Independence of the right basis then gives $a_t = 0$ and $b_t = 8\mathrm{i}k^3 b$. The omitted step is justification of time differentiation and asymptotic comparison for the real-energy Jost functions; Sheet 26 supplies those estimates. It is separate from the proved negative-energy argument.

26.5 One-state reconstruction

We use the *cited right Marchenko inverse theorem* for real rapidly decaying potentials: the reflection coefficient, negative energies and positive right norming constants determine the potential through

$$K(x,y)+F(x+y)+\int_x^\infty K(x,z)F(z+y)\mathrm{d}z=0,\qquad y\geq x,$$
$$F(s)=\sum_j c_j e^{-\kappa_j s}+\frac{1}{2\pi}\int_{\mathbb{R}} r(k)e^{\mathrm{i}ks}\mathrm{d}k,\qquad u(x)=-2\frac{\mathrm{d}}{\mathrm{d}x}K(x,x).$$

The derivative in the reconstruction is the total derivative along the diagonal. The precise cited input is the zero-background specialization of Egorova–Grunert–Teschl, Theorems 4.3–4.4, author preprint, Section 4, recalled in Lecture 25. Their $(\gamma_j^+)^2$ is our c_j. See Dunajski, *Solitons, Instantons, and Twistors*, §2.2.3, equations (2.2.8)–(2.2.10), pp. 30–31; the rank-one calculation is §2.3.1, p. 34. We use the inverse theorem as an input, rather than claiming its proof from the real-energy construction.

Example 26.4 (A single positive norming constant)**.** Take reflectionless data $r=0$, one $\kappa>0$ and $c>0$. Then $F(s)=ce^{-\kappa s}$. The integral equation forces $K(x,y)$ to be proportional to $e^{-\kappa y}$. Writing $K(x,y)=\zeta_M(x)e^{-\kappa y}$ and evaluating the integral gives

$$\zeta_M+ce^{-\kappa x}+\frac{c}{2\kappa}e^{-2\kappa x}\zeta_M=0,\qquad K(x,y)=-\frac{ce^{-\kappa(x+y)}}{1+\frac{c}{2\kappa}e^{-2\kappa x}}.$$

The denominator is strictly positive. Differentiation along $y=x$ gives

$$u=-\frac{4\kappa ce^{-2\kappa x}}{\left(1+\frac{c}{2\kappa}e^{-2\kappa x}\right)^2}=-2\kappa^2\operatorname{sech}^2\bigl(\kappa(x-x_c)\bigr),\qquad x_c=\frac{1}{2\kappa}\log\frac{c}{2\kappa}.$$

Prescribe $c(t)=c(t_0)e^{8\kappa^3(t-t_0)}$ and reconstruct these one-state reflectionless data at each time. The centre then satisfies $x_c(t)=x_c(t_0)+4\kappa^2(t-t_0)$. The resulting family solves KdV by the direct substitution verified in Lecture 23. Thus the explicit solution realizes the proved norming law; this calculation needs no general theorem that the KdV flow preserves the number of bound states.

For this potential the right-normalized bound state is $f=(e^{-\kappa x_c}/2)\operatorname{sech}(\kappa(x-x_c))$. Its right tail is $e^{-\kappa x}$, and $\int f^2=e^{-2\kappa x_c}/(2\kappa)$. Thus its norming constant is indeed $2\kappa e^{2\kappa x_c}=c$. Translation changes c while leaving the eigenvalue and zero reflection unchanged. The norming constant therefore contains location information.

A negative value of c cannot represent (26.2). It also makes the algebraic reconstruction singular: the denominator vanishes at $x_*=(2\kappa)^{-1}\log(-c/(2\kappa))$. Its derivative there is $2\kappa\neq 0$, while the numerator is nonzero. The kernel has an actual pole, so the reconstructed potential is singular.

Summary. For a differentiable isolated negative branch of a Schwartz KdV solution, the eigenvalue $-\kappa^2$ is constant. A right-tail Volterra estimate fixes the temporal normalization: $f_t=(B-4\kappa^3)f$ and $c'=8\kappa^3 c$. The real-energy reflection phase was sketched with its analytic step identified. Conditional on the right inverse theorem, one positive norming constant reconstructs the negative pulse moving at speed $4\kappa^2$. A negative norming coefficient produces a pole.

Self-check. Why does the commutator pairing vanish? Which domain condition is needed?

Answer. Both terms reduce to $\lambda(g, Bg)$ by $Lg = \lambda g$ and self-adjointness. Moving L requires $Bg \in H^2$, supplied by the eigenfunction's H^5 regularity. A formal test-domain identity alone would not supply this step.

Self-check. Why does the right-normalized state satisfy $f_t = Bf - 4\kappa^3 f$ rather than $f_t = Bf$?

Answer. The leading right tail of Bf is $4\kappa^3 e^{-\kappa x}$. The Volterra estimate gives $e^{\kappa x} f_t \to 0$. The subtracting scalar term preserves the fixed unit right-tail coefficient.

Self-check. If $\kappa = 1$ and $c(t_0) = 2$, find the pulse centre and its motion.

Answer. The centre is zero at t_0. Since $c(t) = 2e^{8(t-t_0)}$, the centre is $x_c(t) = 4(t - t_0)$. The pulse minimum remains -2.

Self-check. At $\kappa = 1$ and $c = -2$, where does the Marchenko denominator vanish? Can these data describe a smooth bound-state potential in the stated class?

Answer. The denominator is $1 - e^{-2x}$ and vanishes at zero. The kernel numerator there is nonzero, so the reconstruction has a pole. The data also contradict $c = (\int f^2)^{-1} > 0$.

Lecture 27

Zero curvature and monodromy

The KdV Lax identity of Lecture 24 describes compatibility of linear operators. A related construction uses two first-order matrix equations. On a spatial circle their compatibility controls transport around one full period and gives conserved traces.

Objectives. *We prove the evolution law for periodic matrix transport with its basepoint term. We verify a matrix representation of sine-Gordon in physical coordinates and compute the transport of its vacuum.*

By the end of this lecture you will be able to:

- Derive monodromy evolution for a C^1 periodic flat matrix pair.
- Verify the converted sine-Gordon pair and the vacuum traces.
- Distinguish conserved traces from independence and Poisson commutativity.

27.1 Compatibility of the two linear equations

Fix an open interval $I \subset \mathbb{R}$, a period $L_{\mathrm{per}} > 0$ and a matrix size $n \geq 1$. For each fixed parameter λ in a set $\Lambda \subset \mathbb{C} \setminus \{0\}$, let $U(t,x;\lambda)$ and $V(t,x;\lambda)$ be complex $n \times n$ matrices, C^1 in $(t,x) \in I \times \mathbb{R}$ and periodic in x. The parameter is held constant when differentiating.

Definition 27.1 (Zero-curvature equation)**.** The pair is *flat*, or satisfies the *zero-curvature equation*, if

$$U_t - V_x + [U,V] = 0, \qquad [U,V] = UV - VU. \tag{27.1}$$

The associated linear equations for a column Ψ are $\Psi_x = U\Psi$ and $\Psi_t = V\Psi$. The variable λ is called a *spectral parameter* when such a family is used to represent the nonlinear field equation.

Indeed, formal differentiation gives $\Psi_{xt} - \Psi_{tx} = (U_t - V_x + [U,V])\Psi$. Thus (27.1) is the matrix compatibility condition, rather than an equality imposed only on a particular column. The connection convention of Lecture 19 is $A = -U\mathrm{d}x - V\mathrm{d}t$: its parallel equation $(\mathrm{d}+A)\Psi = 0$ has the stated signs. Direct expansion of $F = \mathrm{d}A + A \wedge A$ gives $F_{xt} = U_t - V_x + [U,V]$.

27.2 Transport and its time derivative

For a fixed λ, consider the initial matrix problem

$$Y_x(t,x) = U(t,x)Y(t,x), \qquad Y(t,0) = \mathbf{1}_n, \quad 0 \leq x \leq L_{\mathrm{per}}. \tag{27.2}$$

We suppress λ during the proof. Later spatial intervals multiply on the left, as Figure 27.1 shows. This order follows from $Y_x = UY$.

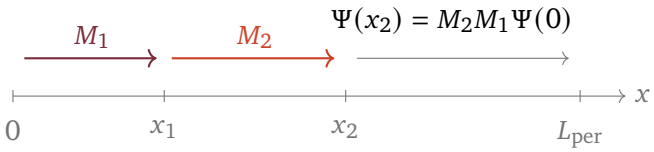


Figure 27.1: Transport first applies M_1 and then M_2 to a column. The later factor M_2 is therefore on the left.

Lemma 27.2 (Regularity of transport). *Problem* (27.2) *has a unique invertible solution. The matrices Y and Y_t are continuous, and*

$$(Y_t)_x = U_t Y + U Y_t, \qquad Y_t(t,0) = 0. \tag{27.3}$$

Proof. Iteration of $Y = \mathbf{1}_n + \int_0^x U(t,s)Y(t,s)\mathrm{d}s$ gives

$$Y = \mathbf{1}_n + \sum_{j\geq 1} \int_{0<s_j<\cdots<s_1<x} U(t,s_1)\cdots U(t,s_j)\,\mathrm{d}s_j \cdots \mathrm{d}s_1.$$

Use a submultiplicative matrix norm. On a compact time interval $K \subset I$, let $M \geq 1$ bound $\|U\|$, and let M_t bound $\|U_t\|$, for $0 \leq x \leq L_{\mathrm{per}}$. The jth term is bounded by $M^j L_{\mathrm{per}}^j / j!$. Its time derivative is bounded by $j M_t M^{j-1} L_{\mathrm{per}}^j / j!$. Both scalar series converge. Uniform convergence therefore permits termwise time differentiation. The integral equation gives the spatial derivative and, after time differentiation, (27.3).

For uniqueness, the difference of two solutions satisfies the homogeneous Volterra equation. Iterating j times bounds its supremum by its own finite supremum times $M^j L_{\mathrm{per}}^j / j!$. This factor tends to zero, so the difference vanishes. The same construction solves $Z_x = -ZU$, $Z(t,0) = \mathbf{1}_n$. Differentiating ZY gives zero, so $ZY = \mathbf{1}_n$. A square matrix with a left inverse is invertible. Hence $Z = Y^{-1}$. ∎

Definition 27.3 (Monodromy matrix). The *monodromy matrix* based at zero is

$$T(t;\lambda) = Y(t, L_{\mathrm{per}}; \lambda) = \mathcal{P} \exp \int_0^{L_{\mathrm{per}}} U(t,x;\lambda)\,\mathrm{d}x. \tag{27.4}$$

The symbol $\mathcal{P}$ denotes the ordered series just proved, with later x on the left. It does not mean the exponential of an ordinary integral when matrices at different positions do not commute.

A vector transported once around the spatial circle returns to the same basepoint with value $T\Psi(0)$. That value need not equal its initial value.

27.3 The monodromy evolves by a commutator

Theorem 27.4 (Monodromy evolution). *Under the stated C^1 and periodicity assumptions, flatness implies*

$$T_t = [V(t,0;\lambda), T]. \tag{27.5}$$

For every $\lambda \in \Lambda$ and positive integer r, the trace $\operatorname{tr}(T(t;\lambda)^r)$ is constant on I.

Proof. Define $W = Y_t - VY + YV(t,0)$. The transport lemma justifies its spatial derivative. Using flatness gives

$$\begin{aligned} W_x &= UY_t + (U_t - V_x - VU)Y + UYV(t,0) \\ &= U\big[Y_t - VY + YV(t,0)\big] = UW. \end{aligned}$$

Its value at zero is $0 - V(t,0) + V(t,0) = 0$. Uniqueness of the homogeneous matrix problem gives $W = 0$. At $x = L_{\mathrm{per}}$, this identity reads

$$T_t = V(t, L_{\mathrm{per}})T - TV(t,0). \tag{27.6}$$

Periodicity identifies the two endpoint values of V and proves (27.5). The right multiplication by $V(t,0)$ is required because the normalisation $Y(t,0) = \mathbf{1}_n$ has zero time derivative.

The product rule and cyclic invariance of finite matrix trace give

$$\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{tr}(T^r) = r\operatorname{tr}(T^{r-1}T_t) = r\operatorname{tr}\big(T^{r-1}V(t,0)T - T^rV(t,0)\big) = 0.$$

This holds for each fixed parameter and every $t \in I$. ∎

No analytic dependence on λ was needed. The result concerns finite matrix transport with C^1 coefficients, so the unbounded-operator issue in Lecture 24 does not enter this proof.

27.4 The sine-Gordon representation in physical coordinates

A *classical sine-Gordon field* is a real C^2 function satisfying

$$\phi_{tt} - \phi_{xx} + \sin\phi = 0. \tag{27.7}$$

For the circle application, assume ϕ and its derivatives are periodic with period L_{per}. The statements below are conditional on such a field; no Cauchy theorem for this nonlinear equation is asserted here.

Recall the Pauli matrices of Lecture 14:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For $\lambda \in \mathbb{C} \setminus \{0\}$, use the light-cone coordinates $\xi = (x+t)/2$ and $\eta = (x-t)/2$. Then $\partial_\xi = \partial_x + \partial_t$ and $\partial_\eta = \partial_x - \partial_t$, so $\phi_{\xi\eta} = \phi_{xx} - \phi_{tt}$. The light-cone pair is

$$U_{\mathrm{lc}} = \frac{\mathrm{i}\lambda}{2}\sigma_3 + \frac{\mathrm{i}}{2}\phi_\xi\sigma_1, \qquad V_{\mathrm{lc}} = -\frac{\mathrm{i}}{2\lambda}(\cos\phi\,\sigma_3 + \sin\phi\,\sigma_2). \tag{27.8}$$

This is Dunajski, *Solitons, Instantons, and Twistors*, §3.3, equation (3.3.11), p. 49, with $\lambda = 2\lambda_D$. We state this light-cone pair and verify its converted pair directly. Since $\partial_x = (\partial_\xi + \partial_\eta)/2$ and $\partial_t = (\partial_\xi - \partial_\eta)/2$, that pair is

$$\begin{aligned} \phi_\xi &= \phi_x + \phi_t, \qquad N = \cos\phi\,\sigma_3 + \sin\phi\,\sigma_2, \\ U &= \frac{\mathrm{i}}{4}\big(\lambda\sigma_3 + \phi_\xi\sigma_1 - \lambda^{-1}N\big), \qquad V = \frac{\mathrm{i}}{4}\big(\lambda\sigma_3 + \phi_\xi\sigma_1 + \lambda^{-1}N\big). \end{aligned} \tag{27.9}$$

In particular, $U = (U_{\mathrm{lc}} + V_{\mathrm{lc}})/2$ and $V = (U_{\mathrm{lc}} - V_{\mathrm{lc}})/2$. Using the light-cone matrices unchanged as the (x,t) pair would be incorrect.

To verify (27.9), put $N_\phi = \cos\phi\,\sigma_2 - \sin\phi\,\sigma_3$. Then $N_t + N_x = \phi_\xi N_\phi$, and the derivative terms are

$$U_t - V_x = \frac{\mathrm{i}}{4}(\phi_{tt} - \phi_{xx})\sigma_1 - \frac{\mathrm{i}\phi_\xi}{4\lambda}N_\phi.$$

The Pauli commutators $[\sigma_3, \sigma_2] = -2\mathrm{i}\sigma_1$, $[\sigma_1, \sigma_3] = -2\mathrm{i}\sigma_2$ and $[\sigma_1, \sigma_2] = 2\mathrm{i}\sigma_3$ give

$$[U, V] = \frac{\mathrm{i}}{4}\sin\phi\,\sigma_1 + \frac{\mathrm{i}\phi_\xi}{4\lambda}N_\phi.$$

The terms involving N_ϕ cancel. Thus

$$U_t - V_x + [U, V] = \frac{\mathrm{i}}{4}(\phi_{tt} - \phi_{xx} + \sin\phi)\sigma_1. \tag{27.10}$$

This proves flatness if and only if the field satisfies sine-Gordon. For a periodic classical solution, U and V are C^1 and periodic. Theorem 27.4 therefore applies at every nonzero λ.

27.5 The vacuum on a circle

Example 27.5 (Vacuum monodromy)**.** For $\phi = 0$, both matrices are constant and diagonal:

$$U_0 = \frac{\mathrm{i}}{4}(\lambda - \lambda^{-1})\sigma_3, \qquad V_0 = \frac{\mathrm{i}}{4}(\lambda + \lambda^{-1})\sigma_3.$$

They commute. Solving $Y_x = U_0 Y$, $Y(0) = \mathbf{1}_2$, gives

$$T(\lambda) = \begin{pmatrix} e^{\mathrm{i}L_{\mathrm{per}}(\lambda-\lambda^{-1})/4} & 0 \\ 0 & e^{-\mathrm{i}L_{\mathrm{per}}(\lambda-\lambda^{-1})/4} \end{pmatrix}.$$

For every positive integer r its trace is

$$\operatorname{tr} T(\lambda)^r = 2\cos\left(\frac{rL_{\mathrm{per}}}{4}(\lambda - \lambda^{-1})\right).$$

These expressions are independent of time, as the theorem predicts. The determinant is one. At $\lambda = 1$, the matrix is the identity and every trace power is two. The parameter value $\lambda = 0$ remains excluded.

27.6 Conservation does not prove complete integrability

The theorem gives conserved functions labelled by λ and r. Different labels need not give independent functions of the field. For example, a family $F_\lambda = e^\lambda H$, for real λ, contains only multiples of one Hamiltonian, regardless of how many parameters are used.

Conserved quantities also need not have zero mutual Poisson brackets. For the canonical pairs (q_1, p_1) and (q_2, p_2) with Hamiltonian $H = p_1$, both q_2 and p_2 are conserved, while $\{q_2, p_2\} = 1$. A family is *in involution* when all its mutual brackets vanish. No such bracket calculation for the monodromy traces has been proved here. Nor have we proved that these quantities supply enough independent data to determine the general field evolution. Conservation, independence, involution and completeness are separate claims. A finite-dimensional Liouville–Arnold theorem alone does not establish any corresponding assertion for this field theory.

Summary. For a C^1 periodic flat matrix pair, the normalised spatial transport has monodromy $T_t = [V(t,0), T]$. Its trace powers are constant for every fixed spectral parameter. The converted sine-Gordon pair satisfies flatness exactly when $\phi_{tt} - \phi_{xx} + \sin\phi = 0$. The vacuum gives an explicit diagonal monodromy. These conserved traces do not by themselves establish independence, involution or completeness.

Self-check. A transport first acts by M_1 and then by M_2 on a column. Which product gives its final value, and which spatial ODE fixes that order?

Answer. The final value is $M_2 M_1 \Psi(0)$. The normalised transport solves $Y_x = UY$, so later spatial factors multiply on the left in its iterated integral series.

Self-check. Before using periodicity, what is T_t? Why can the term at $x = 0$ not be omitted?

Answer. It is $V(t, L_{\mathrm{per}})T - TV(t,0)$. The basepoint normalisation has $Y_t(t,0) = 0$. The term $-YV(t,0)$ in Y_t enforces that condition.

Self-check. Convert $\phi_{\xi\eta} = \sin\phi$ to (t,x) for $\xi = (x+t)/2$ and $\eta = (x-t)/2$.

Answer. The mixed derivative is $\phi_{xx} - \phi_{tt}$. Hence the equation becomes $\phi_{tt} - \phi_{xx} + \sin\phi = 0$. The two matrices must also be converted by their half sum and half difference.

Self-check. At the sine-Gordon vacuum, what are $T(1)$ and $\operatorname{tr} T(1)^r$? Does conservation for all λ prove independence of the charges?

Answer. They are $\mathbf{1}_2$ and 2 for every positive integer r. No. Parameter labels may give dependent functions, and conservation alone also does not establish their mutual Poisson commutativity.

Lecture 28

Gravity as a field theory

The metric can itself be a dynamical field. Its action involves curvature, so it depends on second derivatives. We vary this action with the same inverse-metric convention used for matter in Lecture 3. Coordinate symmetry then gives a differential identity by Lecture 19.

Objectives. *We derive the Einstein equation using the Palatini identity, whose proof is sketched. We identify the boundary term and the covariant conservation identity. We separate the action calculation from the cited Cauchy theory and the further problem of constructing a reduced phase space.*

By the end of this lecture you will be able to:

- Derive the Einstein equation with the course's metric and action signs.
- Obtain the contracted Bianchi identity from compact coordinate variations.
- Explain why fixing the boundary metric alone leaves a boundary variation.

28.1 Metric and curvature conventions

Let $D \subset \mathbb{R}^4$ be a compact coordinate region, with interior U. All fields below are smooth on a neighbourhood of D. Indices run from zero to three, and repeated upper and lower indices are summed.

Definition 28.1 (Lorentz metric and volume). A *Lorentz metric* here is a symmetric matrix field $g_{\mu\nu}$ with one positive and three negative eigenvalues at each point. Its inverse is $g^{\mu\nu}$, defined by $g_{\mu\rho}g^{\rho\nu} = \delta^{\nu}_{\mu}$. With the coordinate orientation, its volume form is

$$\mathrm{vol}_g = \sqrt{|\det(g_{\mu\nu})|}\,\mathrm{d}^4x, \qquad \mathrm{d}^4x = \mathrm{d}t \wedge \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3.$$

We recall only the coordinate geometry used in the variation. The *Levi–Civita connection* is the torsion-free, metric-compatible connection with coefficients

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2}g^{\rho\lambda}(\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu}). \tag{28.1}$$

For example, $\nabla_\mu X^\nu = \partial_\mu X^\nu + \Gamma^{\nu}_{\mu\rho}X^\rho$ and $\nabla_\mu w_\nu = \partial_\mu w_\nu - \Gamma^{\rho}_{\mu\nu}w_\rho$. The product rule extends this to tensors, with one such term per index. Metric compatibility is $\nabla g = 0$. Our *Riemann curvature*, *Ricci tensor*, *scalar curvature* and *Einstein tensor* are

$$\begin{gathered} R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu\Gamma^{\rho}_{\nu\sigma} - \partial_\nu\Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}, \\ \mathrm{Ric}_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}, \qquad R_{\mathrm{sc}} = g^{\mu\nu}\,\mathrm{Ric}_{\mu\nu}, \qquad G_{\mu\nu} = \mathrm{Ric}_{\mu\nu} - \frac{1}{2}R_{\mathrm{sc}}g_{\mu\nu}. \end{gathered} \tag{28.2}$$

These are the recalled curvature conventions of *Topology and Geometry*, Lectures 16–17. In particular, Ricci and Einstein are symmetric.

28.2 Inverse-metric variation and the Palatini input

Vary the inverse metric by a symmetric array $k^{\mu\nu} = \delta g^{\mu\nu} \in C_c^\infty(U)$. For sufficiently small variation parameter, compactness preserves signature. As in Lecture 3, differentiating the inverse and using Jacobi's determinant formula gives

$$h_{\mu\nu} := \delta g_{\mu\nu} = -g_{\mu\alpha} g_{\nu\beta} k^{\alpha\beta}, \qquad \delta \mathrm{vol}_g = -\frac{1}{2}\tau \mathrm{vol}_g, \qquad \tau = g_{\mu\nu} k^{\mu\nu}. \tag{28.3}$$

Indeed, $\delta \det g = (\det g) g^{\mu\nu} h_{\mu\nu}$. The determinant keeps its sign for small variations, so the absolute value in the volume introduces no extra sign.

The variation of the connection is a tensor: it is the derivative of a difference of connections. Differentiating metric compatibility gives $\nabla_\mu h_{\nu\lambda} = \delta\Gamma^\rho_{\mu\nu} g_{\rho\lambda} + \delta\Gamma^\rho_{\mu\lambda} g_{\nu\rho}$. Add the two cyclic versions and subtract the third. Symmetry in the lower connection indices then gives

$$\delta\Gamma^\rho_{\mu\nu} = \frac{1}{2} g^{\rho\lambda} (\nabla_\mu h_{\lambda\nu} + \nabla_\nu h_{\lambda\mu} - \nabla_\lambda h_{\mu\nu}). \tag{28.4}$$

We use the *Palatini identity*

$$\delta \mathrm{Ric}_{\mu\nu} = \nabla_\rho \delta\Gamma^\rho_{\nu\mu} - \nabla_\nu \delta\Gamma^\rho_{\rho\mu}. \tag{28.5}$$

Its proof is *sketched* here. At a chosen point take normal coordinates, where the background connection coefficients vanish. Varying (28.2) leaves the difference of the two derivatives of $\delta\Gamma$ at that point; the varied quadratic terms vanish. The resulting expressions on both sides are tensors, so their equality holds in any coordinates. Sheet 28 supplies the complete tensor argument.

Contracting (28.5), using $\nabla g = 0$, gives a divergence:

$$\begin{aligned} g^{\mu\nu} \delta \mathrm{Ric}_{\mu\nu} &= \nabla_\mu v^\mu, \\ v^\mu &= g^{\alpha\beta} \delta\Gamma^\mu_{\alpha\beta} - g^{\mu\alpha} \delta\Gamma^\beta_{\alpha\beta} = \nabla^\mu \tau - \nabla_\nu k^{\mu\nu}. \end{aligned} \tag{28.6}$$

For the last equality, substitution of (28.3) into (28.4) gives $g^{\alpha\beta}\delta\Gamma^\mu_{\alpha\beta} = -\nabla_\nu k^{\mu\nu} + \frac{1}{2}\nabla^\mu\tau$ and $g^{\mu\alpha}\delta\Gamma^\beta_{\alpha\beta} = -\frac{1}{2}\nabla^\mu\tau$.

28.3 The Einstein–Hilbert action

Define the curvature integral by $\mathcal{I}_D[g] = \int_D R_{\mathrm{sc}} \mathrm{vol}_g$. Matter fields are held fixed in the metric variation. Assume their action has the Hilbert response of Lecture 3,

$$\delta_g S_{\mathrm{m},D} = \frac{1}{2} \int_D T_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g, \tag{28.7}$$

with a smooth symmetric tensor $T_{\mu\nu}$.

Theorem 28.2 (Einstein–Hilbert variation)**.** *Using the Palatini identity with the scope stated above, every compactly supported inverse-metric variation satisfies*

$$\delta \mathcal{I}_D[g]k = \int_D G_{\mu\nu} k^{\mu\nu} \operatorname{vol}_g. \tag{28.8}$$

For $G_{\mathrm{N}} > 0$, the Einstein–Hilbert action with matter *is*

$$S_D[g, \mathit{matter}] = -\frac{1}{16\pi G_{\mathrm{N}}} \mathcal{I}_D[g] + S_{\mathrm{m},D}. \tag{28.9}$$

Its stationarity under all these metric variations is equivalent on U to

$$G_{\mu\nu} = 8\pi G_{\mathrm{N}} T_{\mu\nu}. \tag{28.10}$$

Proof. The scalar curvature varies as $\delta R_{\mathrm{sc}} = \mathrm{Ric}_{\mu\nu}\, k^{\mu\nu} + g^{\mu\nu}\delta \mathrm{Ric}_{\mu\nu}$. Equations (28.3) and (28.6) therefore give

$$\delta(R_{\mathrm{sc}} \operatorname{vol}_g) = \big(G_{\mu\nu} k^{\mu\nu} + \nabla_\mu v^\mu\big) \operatorname{vol}_g. \tag{28.11}$$

The coordinate divergence formula is

$$\nabla_\mu v^\mu \sqrt{|\det g|} = \partial_\mu \big(\sqrt{|\det g|}\, v^\mu\big).$$

It follows by tracing (28.1) and applying Jacobi's formula: $\Gamma^\rho_{\rho\mu} = \partial_\mu \log \sqrt{|\det g|}$. The vector density on the right has compact support in U because k and all its derivatives do. Extend it by zero outside D. Coordinate FTC and Fubini make its integral zero, without a boundary regularity assumption on D. This proves (28.8).

Combining this result with (28.7) gives

$$\delta_g S_D = \int_D \left(-\frac{G_{\mu\nu}}{16\pi G_{\mathrm{N}}} + \frac{1}{2} T_{\mu\nu} \right) k^{\mu\nu} \operatorname{vol}_g.$$

The smooth-test fundamental lemma applied to each independent symmetric component gives (28.10). Conversely, this equation makes the displayed variation zero. ∎

The plus sign in (28.8) accompanies the minus sign in (28.9). Reversing only the action's gravitational prefactor would reverse the relative sign of the sourced equation.

28.4 Coordinate symmetry gives the Bianchi identity

A *diffeomorphism* is a smooth invertible map with smooth inverse. Let ξ be a smooth vector field compactly supported in U, and let F_ε be its local flow. Pulling the metric back by this flow produces its *Lie variation*. For the inverse metric it is

$$\begin{aligned} k^{\mu\nu} = \$_\xi g^{\mu\nu} &= \xi^\rho \partial_\rho g^{\mu\nu} - g^{\rho\nu}\partial_\rho \xi^\mu - g^{\mu\rho}\partial_\rho \xi^\nu \\ &= -\nabla^\mu \xi^\nu - \nabla^\nu \xi^\mu. \end{aligned} \tag{28.12}$$

Metric compatibility gives the last equality. The curvature volume form is natural under pullback. Change of variables therefore makes $\mathcal{I}_D[F_\varepsilon^* g]$ independent of ε: the flow is the identity near the boundary. This is a gauge symmetry in the sense of Lecture 19. Its theorem permits second-order metric densities such as $R_{\mathrm{sc}} \operatorname{vol}_g$.

Substitute (28.12) into (28.8). Symmetry of G and covariant integration by parts give

$$0 = -2\int_D G_{\mu\nu}\nabla^\mu\xi^\nu \,\mathrm{vol}_g = 2\int_D (\nabla^\mu G_{\mu\nu})\xi^\nu \,\mathrm{vol}_g.$$

The discarded divergence has current $-2G^\mu{}_\nu\xi^\nu$ with compact support, so the same coordinate argument removes it. Arbitrary compact ξ and the fundamental lemma imply

$$\nabla^\mu G_{\mu\nu} \equiv 0. \tag{28.13}$$

This *contracted Bianchi identity* holds for every smooth metric, without imposing the Einstein equation. It is precisely the formal-adjoint identity of Lecture 19 for the gauge operator (28.12). For a diffeomorphism-invariant matter action, the same argument including the matter variations gives $\nabla_\mu T^{\mu\nu} = 0$ on the matter equations; this matter extension is stated here.

28.5 A nonzero constant-rescaling check

Example 28.3 (Constant inverse-metric rescaling)**.** For a constant ε, take $k^{\mu\nu} = \varepsilon g^{\mu\nu}$ locally. This is a local check, not a compactly supported variation. Under $g^{\mu\nu} \mapsto (1+s\varepsilon)g^{\mu\nu}$, the covariant metric is multiplied by $(1+s\varepsilon)^{-1}$. Its constant factor cancels from (28.1), so the connection and Ricci tensor do not change. Thus

$$R_{\mathrm{sc}}(s) = (1+s\varepsilon)R_{\mathrm{sc}}(0), \qquad \mathrm{vol}_g(s) = (1+s\varepsilon)^{-2}\mathrm{vol}_g(0).$$

Differentiating their product at $s=0$ gives $-\varepsilon R_{\mathrm{sc}}\mathrm{vol}_g$. On the other side, $g^{\mu\nu}G_{\mu\nu} = -R_{\mathrm{sc}}$ in four dimensions, so $G_{\mu\nu}k^{\mu\nu}\mathrm{vol}_g$ has the same value. Here $\tau = 4\varepsilon$ is constant and $\nabla_\nu k^{\mu\nu} = 0$, so the boundary vector vanishes pointwise.

This need not be a zero-curvature test. For the explicit metric $g = \mathrm{diag}(e^{2x^1}, -1, -1, -1)$, its nonzero connection entries are $\Gamma^0_{01} = \Gamma^0_{10} = 1$ and $\Gamma^1_{00} = e^{2x^1}$. The coordinate curvature formulas give

$$\mathrm{Ric} = \mathrm{diag}(e^{2x^1}, -1, 0, 0), \qquad R_{\mathrm{sc}} = 2, \qquad \mathrm{vol}_g = e^{x^1}\mathrm{d}^4x.$$

Both variation densities are therefore $-2\varepsilon e^{x^1}\mathrm{d}^4x$. This tests a nonzero value and fixes the volume sign.

28.6 Fixing the boundary value is not enough

The compact-support hypothesis in the theorem removes derivatives of the variation near the boundary. Fixing only its boundary value is weaker. Figure 28.1 shows a variation that vanishes on a flat face but has a nonzero normal derivative there.

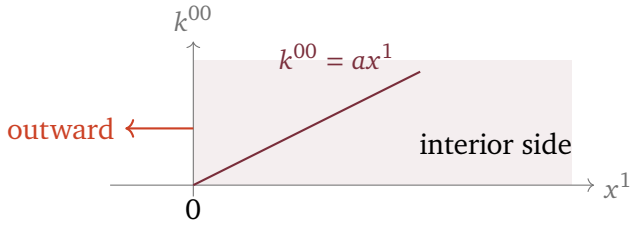


Figure 28.1: Near $x^1 = 0$, a fixed positive tangential amplitude gives $k^{00} = ax^1$. Its value on the face is zero, but its slope is $a \neq 0$.

Use the flat metric $g = \eta$ on a rectangular region with $x^1 \geq 0$. Choose a smooth nonnegative, nonzero function $a(t, x^2, x^3)$ compactly supported away from the tangential faces. Let $\chi(x^1)$ be smooth, equal to one near zero and zero before the opposite face. Set $k^{00} = x^1 a\chi$ and all other components to zero. The entire array k vanishes on the boundary, but (28.6) gives

$$\tau = x^1 a\chi, \qquad v^1 = -\partial_1(x^1 a\chi) = -a \quad \text{at } x^1 = 0.$$

The outward coordinate flux on that lower face is $-v^1 = a$. Its integral is nonzero, and all other faces contribute zero. Thus even fixing the metric on the whole boundary does not remove the Einstein–Hilbert boundary variation. The *Gibbons–Hawking–York term* is a boundary addition designed for the appropriate fixed-boundary-metric variational problem. Its formula and derivation belong to Sheet 28.

28.7 The three field-theory questions for gravity

The action question has the answer (28.9), with its variation proved using the explicitly sketched Palatini input. The Cauchy question requires constraints as well as evolution. For a precise cited case, take smooth data (Σ, γ, K) on a compact three-manifold without boundary, satisfying the vacuum constraints below. Here γ is a positive spatial metric and K is the symmetric tensor specifying its first normal change. Choose the convention $K_{ij} = -\frac{1}{2}\partial_t \gamma_{ij}$ in coordinates with unit future normal ∂_t on the initial slice. The vacuum constraints are

$$R_\gamma + (\mathrm{tr}_\gamma K)^2 - |K|^2_\gamma = 0, \qquad \nabla^j_\gamma \big(K_{ij} - (\mathrm{tr}_\gamma K)\gamma_{ij}\big) = 0.$$

The curvature and covariant derivative here are those of γ. Choquet-Bruhat, *General Relativity and the Einstein Equations*, Chapter VI, Theorem 7.4, pp. 164–165, and Theorems 8.3–8.4, p. 168, provide a local vacuum development of such data and geometric local uniqueness. A development is a spacetime solution realising γ and K on its embedded initial slice. Two developments have isometric neighbourhoods of that slice. This is a cited Cauchy result, not a consequence of the action calculation. It is a short-time statement, with no assertion of an infinite regular lifespan. The source uses the opposite spacetime signature; reversing the metric leaves the vacuum equation Ric = 0 unchanged.

The phase-space question asks for gravitational data satisfying the constraints, modulo the appropriate diffeomorphisms. This is the idea of a *reduced gravitational phase space*. Its construction and any global manifold structure are not asserted here.

Summary. With inverse-metric variation, $\delta \mathrm{vol}_g = -\frac{1}{2} g_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g$. The Palatini input gives a divergence plus $G_{\mu\nu} k^{\mu\nu}$. Compact support removes that divergence and the stated action gives $G_{\mu\nu} = 8\pi G_{\mathrm{N}} T_{\mu\nu}$. Coordinate symmetry yields $\nabla^\mu G_{\mu\nu} \equiv 0$ off shell. Fixing a boundary metric alone leaves a normal-derivative term. The Cauchy theory with constraints is cited, and the reduced phase space is named.

Self-check. For $k^{\mu\nu} = \varepsilon g^{\mu\nu}$ in four dimensions, compute $\delta \mathrm{vol}_g$ and $G_{\mu\nu} k^{\mu\nu}$.

Answer. The trace is 4ε, so $\delta \mathrm{vol}_g = -2\varepsilon \mathrm{vol}_g$. The Einstein trace is $-R_{\mathrm{sc}}$, so the contraction is $-\varepsilon R_{\mathrm{sc}}$. These are local rescaling identities, not a claim of compact support.

Self-check. Which signs combine to give a positive $8\pi G_{\mathrm{N}}$ in the Einstein equation?

Answer. The curvature-integral variation is $+\int G_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g$. Its action coefficient is $-(16\pi G_{\mathrm{N}})^{-1}$, and the Hilbert matter variation is $+\frac{1}{2}\int T_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g$. Setting their sum to zero gives the stated equation.

Self-check. Does the Bianchi identity require the Einstein equation? Which compact variation produces it?

Answer. No. It follows from invariance at every smooth metric. The inverse-metric Lie variation is $k^{\mu\nu} = -\nabla^\mu \xi^\nu - \nabla^\nu \xi^\mu$ for arbitrary compact ξ. Its formal adjoint gives $\nabla^\mu G_{\mu\nu} \equiv 0$.

Self-check. For the flat-face variation $k^{00} = x^1 a\chi$, why does $k = 0$ on the boundary not make its boundary contribution vanish?

Answer. Its normal derivative remains nonzero: $v^1 = -a$ at the face $x^1 = 0$. The outward flux there is a. Compact support in the interior would remove both the variation and its derivatives near the boundary.